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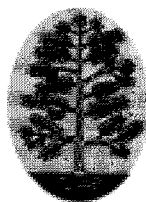
J.-P. Demailly T. Peternell
G. Tian A. N. Tyurin

Transcendental Methods in Algebraic Geometry

Lectures given at the 3rd Session of the
Centro Internazionale Matematico Estivo
(C.I.M.E.)

held in Cetraro, Italy, July 4–12, 1994

Editors: F. Catanese, C. Ciliberto



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Authors

Jean-Pierre Demailly
Institut Fourier
Université de Grenoble I
F-38402 Saint-Martin-d'Hères, France

Thomas Peternell
Mathematisches Institut
Universität Bayreuth
Universitätsstrasse 30
D-95447 Bayreuth, Germany

Gang Tian
Massachusetts Institute of Technology
1105 Massachusetts Avenue
Cambridge, MA 02139, USA

Andrej N. Tyurin
Steklov Institute of Mathematics
ul. Vavilov 42
117996 Moscow, Russia

Editors

Fabrizio Catanese
Dipartimento di Matematica
Università di Pisa
Via Buonarroti, 2
I-56127 Pisa, Italy

Ciro Ciliberto
Dipartimento di Matematico
Università di Tor Vergata
V. della Ricerca Scientifica
I-00133 Rome, Italy

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Preface

The Third 1994 C.I.M.E. Session “Transcendental Methods in Algebraic Geometry” took place from July 4 to July 12 in the beautiful location of the Grand Hotel San Michele, Cetraro (Cosenza).

Already in the prehistory of algebraic geometry we find the theory of elliptic and Abelian integrals, which is directly linked with Riemann’s topological approach to creating the concept of a manifold. Later on, from Poincaré’s use of potential theory for the study of function theory on complex tori to Hodge’s theory of harmonic integrals and the vanishing theorems of Kodaira and others, we see that the transcendental approach puts many algebraic geometric questions on a firm basis.

In doing so, it establishes deep and surprising results, which can often be stated simply, invigorating a century-long tradition of manifold and fruitful relations with other disciplines. In a surprising way we see close analogies displayed between apparently distant methodologies, thus concretely augmenting the unified edifice of mathematics. It was one of the purposes of the 1994 course to look at the recent developments relating algebraic geometry to complex analysis, complex differential geometry, and differential topology as further manifestations of the core of algebraic geometry: a core which, although nourished by a myriad of subtle and intricate problems, has as its lifeblood the crucial interplay with a host of other subjects, be they physics, topology, algebra, analysis, differential geometry, or arithmetic.

From this point of view, the courses given by Demailly, Peternell, Tian, and Tyurin covered a very wide spectrum, each offering not only a broad view of recent developments and new results published here for the first time, but also opening wide perspectives still in the earliest stages of exploration. The beautiful texts of the four courses reproduced here give us ample justification for dispensing with further historical and mathematical description.

We would just like to recall that, as in the ancient Greek dramas, unity of place (lecturers and participants brought close together in the “golden cage” of San Michele), unity of action (there were only courses and problem sessions), and unity of time (one of the features of C.I.M.E.) contributed to the success of the course. This success was in large part due not only to the excellent lecturers but also to the brightness and knowledge of the participants: the variety of their cultural interests was for us very impressive, as well as their devotion to science amidst such tempting scenery.

The organizers: Fabrizio Catanese and Ciro Ciliberto

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L^2 Vanishing Theorems for Positive Line Bundles and Adjunction Theory

Jean-Pierre Demailly

Université de Grenoble I, Institut Fourier

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0. Introduction

Transcendental methods of algebraic geometry have been extensively studied since a very long time, starting with the work of Abel, Jacobi and Riemann in the nineteenth century. More recently, in the period 1940-1970, the work of Hodge, Hirzebruch, Kodaira, Atiyah revealed still deeper relations between complex analysis, topology, PDE theory and algebraic geometry. In the last ten years, gauge theory has proved to be a very efficient tool for the study of many important questions: moduli spaces, stable sheaves, non abelian Hodge theory, low dimensional topology ...

Our main purpose here is to describe a few analytic tools which are useful to study questions such as linear series and vanishing theorems for algebraic vector bundles. One of the early success of analytic methods in this context is Kodaira's use of the Bochner technique in relation with the theory of harmonic forms, during the decade 1950-60. The idea is to represent cohomology classes by harmonic forms and to prove vanishing theorems by means of suitable a priori curvature estimates. The prototype of such results is the Akizuki-Kodaira-Nakano theorem (1954): if X is a nonsingular projective algebraic variety and L is a holomorphic line bundle on X with positive curvature, then $H^q(X, \Omega_X^p \otimes L) = 0$ for $p+q > \dim X$ (throughout the

paper we set $\Omega_X^p = \Lambda^p T_X^*$ and $K_X = \Lambda^n T_X^*$, $n = \dim X$, viewing these objects either as holomorphic bundles or as locally free $\mathcal{O}_X$ -modules). It is only much later that an algebraic proof of this result has been proposed by Deligne-Illusie, via characteristic p methods, in 1986.

A refinement of the Bochner technique used by Kodaira led about ten years later to fundamental L^2 estimates due to Hörmander [Hör65], concerning solutions of the Cauchy-Riemann operator. Not only vanishing theorems are proved, but more precise information of a quantitative nature is obtained about solutions of $\bar{\partial}$ -equations. The best way of expressing these L^2 estimates is to use a geometric setting first considered by Andreotti-Vesentini [AV65]. More explicitly, suppose that we have a holomorphic line bundle L is equipped with a hermitian metric of weight $e^{-2\varphi}$, where φ is a (locally defined) plurisubharmonic function; then explicit bounds on the L^2 norm $\int_X |f|^2 e^{-2\varphi}$ of solutions is obtained. The result is still more useful if the plurisubharmonic weight φ is allowed to have singularities. Following Nadel [Nad89], one defines the *multiplier ideal sheaf* $\mathcal{I}(\varphi)$ to be the sheaf of germs of holomorphic functions f such that $|f|^2 e^{-2\varphi}$ is locally summable. Then $\mathcal{I}(\varphi)$ is a coherent algebraic sheaf over X and $H^q(X, K_X \otimes L \otimes \mathcal{I}(\varphi)) = 0$ for all $q \geq 1$ if the curvature of L is positive (as a current). This important result can be seen as a generalization of the Kawamata-Viehweg vanishing theorem ([Kaw82], [Vie82]), which is one of the cornerstones of higher dimensional algebraic geometry (especially of Mori's minimal model program).

In the dictionary between analytic geometry and algebraic geometry, the ideal $\mathcal{I}(\varphi)$ plays a very important role, since it directly converts an analytic object into an algebraic one, and, simultaneously, takes care of the singularities in a very efficient way. Another analytic tool used to deal with singularities is the theory of positive currents introduced by Lelong [Lel57]. Currents can be seen as generalizations of algebraic cycles, and many classical results of intersection theory still apply to currents. The concept of Lelong number of a current is the analytic analogue of the concept of multiplicity of a germ of algebraic variety. Intersections of cycles correspond to wedge products of currents (whenever these products are defined). A convenient measure of local positivity of a holomorphic line can be defined in this context: the *Seshadri constant* of a line bundle at a point is the largest possible Lelong number for a singular metric of positive curvature assuming an isolated singularity at the given point (see [Dem90]). Seshadri constants can also be given equivalent purely algebraic definitions. We refer to Ein-Lazarsfeld [EL92] and Ein-Küchle-Lazarsfeld [EKL94] for very interesting new results concerning Seshadri constants.

One of our main motivations has been the study of the following conjecture of Fujita: if L is an ample (i.e. positive) line bundle on a projective n -dimensional algebraic variety X , then $K_X + (n+2)L$ is very ample. A major result obtained by Reider [Rei88] is a proof of the Fujita conjecture in the case of surfaces (the case of curves is easy). Reider's approach is based on Bogomolov's inequality for stable vector bundles and the results obtained are almost optimal. Unfortunately, it seems difficult to extend Reider's original method to higher dimensions. In the analytic approach, which works for arbitrary dimensions, one tries to construct a suitable (singular) hermitian metric on L such that the ideal $\mathcal{I}(\varphi)$ has a given 0-dimensional subscheme of X as its zero variety. As we showed in [Dem93b], this can be done essentially by solving a complex Monge-Ampère equation

$(id' d'' \varphi)^n = \text{linear combination of Dirac measures,}$

via the Aubin-Calabi-Yau theorem ([Aub78], [Yau78]). The solution φ then assumes logarithmic poles and the difficulty is to force the singularity to be an isolated pole; this is the point where intersection theory of currents is useful. In this way, we can prove e.g. that $2K_X + L$ is very ample under suitable numerical conditions for L . Alternative algebraic techniques have been developed recently by Kollár [Kol92], Ein-Lazarsfeld [EL93], Fujita [Fuj93] and [Siu94a, b]. The basic idea is to apply the Kawamata-Viehweg vanishing theorem, and to use the Riemann-Roch formula instead of the Monge-Ampère equation. The proofs proceed with careful inductions on dimension, together with an analysis of the base locus of the linear systems involved. Although the results obtained in low dimensions are slightly more precise than with the analytic method, it is still not clear whether the range of applicability of the methods are exactly the same. Because it fits well with our approach, we have included here a simple algebraic method due to Y.T. Siu [Siu94a], showing that $2K_X + mL$ is very ample for $m \geq 2 + \binom{3n+1}{n}$.

Our final concern in these notes is a proof of the effective Matsusaka big theorem obtained by [Siu93]. Siu's result is the existence of an effective value m_0 depending only on the intersection numbers L^n and $L^{n-1} \cdot K_X$, such that mL is very ample for $m \geq m_0$. The basic idea is to combine results on the very ampleness of $2K_X + mL$ together with the theory of holomorphic Morse inequalities ([Dem85b]). The Morse inequalities are used to construct sections of $m'L - K_X$ for m' large. Again this step can be made algebraic (following suggestions by F. Catanese and R. Lazarsfeld), but the analytic formulation apparently has a wider range of applicability.

These notes are essentially written with the idea of serving as an analytic toolbox for algebraic geometry. Although efficient algebraic techniques exist, our feeling is that the analytic techniques are very flexible and offer a large variety of guidelines for more algebraic questions (including applications to number theory which are not discussed here). We made a special effort to use as little prerequisites and to be as self-contained as possible; hence the rather long preliminary sections dealing with basic facts of complex differential geometry. The reader wishing to have a presentation of the algebraic approach to vanishing theorems and linear series is referred to the excellent notes written by R. Lazarsfeld [Laz93]. In the last years, there has been a continuous and fruitful interplay between the algebraic and analytic viewpoints on these questions, and I have greatly benefitted from observations and ideas contained in the works of J. Kollár, L. Ein, R. Lazarsfeld and Y.T. Siu. I would like to thank them for their interest in my work and for their encouragements.

1. Preliminary Material

1.A. Dolbeault Cohomology and Sheaf Cohomology

Let X be a $\mathbb{C}$ -analytic manifold of dimension n . We denote by $A^{p,q}T_X^*$ the bundle of differential forms of bidegree (p, q) on X , i.e., differential forms which can be written as

$$u = \sum_{|I|=p, |J|=q} u_{I,J} dz_I \wedge d\bar{z}_J.$$

Here $(z_1, \dots, z_n)$ denote arbitrary local holomorphic coordinates, $I = (i_1, \dots, i_p)$, $J = (j_1, \dots, j_q)$ are multiindices (increasing sequences of integers in the range $[1, \dots, n]$, of lengths $|I| = p$, $|J| = q$), and

$$dz_I := dz_{i_1} \wedge \dots \wedge dz_{i_p}, \quad d\bar{z}_J := d\bar{z}_{j_1} \wedge \dots \wedge d\bar{z}_{j_q}.$$

Let $\mathcal{E}^{p,q}$ be the sheaf of germs of complex valued differential (p, q) -forms with C^∞ coefficients. Recall that the exterior derivative d splits as $d = d' + d''$ where

$$\begin{aligned} d'u &= \sum_{|I|=p, |J|=q, 1 \leq k \leq n} \frac{\partial u_{I,J}}{\partial z_k} dz_k \wedge dz_I \wedge d\bar{z}_J, \\ d''u &= \sum_{|I|=p, |J|=q, 1 \leq k \leq n} \frac{\partial u_{I,J}}{\partial \bar{z}_k} d\bar{z}_k \wedge dz_I \wedge d\bar{z}_J \end{aligned}$$

are of type $(p+1, q)$, $(p, q+1)$ respectively. The well-known Dolbeault-Grothendieck lemma asserts that any d'' -closed form of type (p, q) with $q > 0$ is locally d'' -exact (this is the analogue for d'' of the usual Poincaré lemma for d , see e.g. Hörmander 1966). In other words, the complex of sheaves $(\mathcal{E}^{p,\bullet}, d'')$ is exact in degree $q > 0$; in degree $q = 0$, $\text{Ker } d''$ is the sheaf $\mathcal{O}_X^p$ of germs of holomorphic forms of degree p on X .

More generally, if F is a holomorphic vector bundle of rank r over X , there is a natural d'' operator acting on the space $C^\infty(X, A^{p,q}T_X^* \otimes F)$ of smooth (p, q) -forms with values in F ; if $s = \sum_{1 \leq \lambda \leq r} s_\lambda e_\lambda$ is a (p, q) -form expressed in terms of a local holomorphic frame of F , we simply define $d''s := \sum d''s_\lambda \otimes e_\lambda$, observing that the holomorphic transition matrices involved in changes of holomorphic frames do not affect the computation of d'' . It is then clear that the Dolbeault-Grothendieck lemma still holds for F -valued forms. For every integer $p = 0, 1, \dots, n$, the *Dolbeault Cohomology* groups $H^{p,q}(X, F)$ are defined to be the cohomology groups of the complex of global (p, q) forms (graded by q):

$$(1.1) \quad H^{p,q}(X, F) = H^q(C^\infty(X, A^{p,\bullet}T_X^* \otimes F)).$$

Now, let us recall the following fundamental result from sheaf theory (De Rham-Weil isomorphism theorem): let $(\mathcal{L}^\bullet, d)$ be a resolution of a sheaf $\mathcal{A}$ by acyclic sheaves, i.e. a complex of sheaves $(\mathcal{L}^\bullet, d)$ such that there is an exact sequence of sheaves

$$0 \longrightarrow \mathcal{A} \xrightarrow{j} \mathcal{L}^0 \xrightarrow{\delta^0} \mathcal{L}^1 \longrightarrow \dots \longrightarrow \mathcal{L}^q \xrightarrow{\delta^q} \mathcal{L}^{q+1} \longrightarrow \dots,$$

and $H^s(X, \mathcal{L}^q) = 0$ for all $q \geq 0$ and $s \geq 1$. Then there is a functorial isomorphism

$$(1.2) \quad H^q(\Gamma(X, \mathcal{L}^\bullet)) \longrightarrow H^q(X, \mathcal{A}).$$

We apply this to the following situation: let $\mathcal{E}(F)^{p,q}$ be the sheaf of germs of C^∞ sections of $\Lambda^{p,q} T_X^* \otimes F$. Then $(\mathcal{E}(F)^{p,\bullet}, d'')$ is a resolution of the locally free $\mathcal{O}_X$ -module $\Omega_X^p \otimes \mathcal{O}(F)$ (Dolbeault-Grothendieck lemma), and the sheaves $\mathcal{E}(F)^{p,q}$ are acyclic as modules over the soft sheaf of rings C^∞ . Hence by (1.2) we get

(1.3) Dolbeault Isomorphism Theorem (1953). *For every holomorphic vector bundle F on X , there is a canonical isomorphism*

$$H^{p,q}(X, F) \simeq H^q(X, \Omega_X^p \otimes \mathcal{O}(F)).$$

□

If X is projective algebraic and F is an algebraic vector bundle, Serre's GAGA theorem [Ser56] shows that the algebraic sheaf cohomology group $H^q(X, \Omega_X^p \otimes \mathcal{O}(F))$ computed with algebraic sections over Zariski open sets is actually isomorphic to the analytic cohomology group. These results are the most basic tools to attack algebraic problems via analytic methods. Another important tool is the theory of plurisubharmonic functions and positive currents originated by K. Oka and P. Lelong in the decades 1940-1960.

1.B. Plurisubharmonic Functions

Plurisubharmonic functions have been introduced independently by Lelong and Oka in the study of holomorphic convexity. We refer to [Lel67, 69] for more details.

(1.4) Definition. *A function $u : \Omega \longrightarrow [-\infty, +\infty[$ defined on an open subset $\Omega \subset \mathbb{C}^n$ is said to be plurisubharmonic (psh for short) if*

- a) *u is upper semicontinuous ;*
- b) *for every complex line $L \subset \mathbb{C}^n$, $u|_{\Omega \cap L}$ is subharmonic on $\Omega \cap L$, that is, for all $a \in \Omega$ and $\xi \in \mathbb{C}^n$ with $|\xi| < d(a, \mathbb{C} \setminus \Omega)$, the function u satisfies the mean value inequality*

$$u(a) \leq \frac{1}{2\pi} \int_0^{2\pi} u(a + e^{i\theta} \xi) d\theta.$$

The set of psh functions on Ω is denoted by $\text{Psh}(\Omega)$.

We list below the most basic properties of psh functions. They all follow easily from the definition.

(1.5) Basic properties.

- a) Every function $u \in \text{Psh}(\Omega)$ is subharmonic, namely it satisfies the mean value inequality on euclidean balls or spheres:

$$u(a) \leq \frac{1}{\pi^n r^{2n}/n!} \int_{B(a,r)} u(z) d\lambda(z)$$

for every $a \in \Omega$ and $r < d(a, \mathbb{C}\Omega)$. Either $u \equiv -\infty$ or $u \in L^1_{\text{loc}}$ on every connected component of Ω .

- b) For any decreasing sequence of psh functions $u_k \in \text{Psh}(\Omega)$, the limit $u = \lim u_k$ is psh on Ω .
- c) Let $u \in \text{Psh}(\Omega)$ be such that $u \not\equiv -\infty$ on every connected component of Ω . If (ρ_ε) is a family of smoothing kernels, then $u \star \rho_\varepsilon$ is C^∞ and psh on

$$\Omega_\varepsilon = \{x \in \Omega; d(x, \mathbb{C}\Omega) > \varepsilon\},$$

the family $(u \star \rho_\varepsilon)$ is increasing in ε and $\lim_{\varepsilon \rightarrow 0} u \star \rho_\varepsilon = u$.

- d) Let $u_1, \dots, u_p \in \text{Psh}(\Omega)$ and $\chi : \mathbb{R}^p \rightarrow \mathbb{R}$ be a convex function such that $\chi(t_1, \dots, t_p)$ is increasing in each t_j . Then $\chi(u_1, \dots, u_p)$ is psh on Ω . In particular $u_1 + \dots + u_p$, $\max\{u_1, \dots, u_p\}$, $\log(e^{u_1} + \dots + e^{u_p})$ are psh on Ω . $\square$

(1.6) Lemma. *A function $u \in C^2(\Omega, \mathbb{R})$ is psh on Ω if and only if the hermitian form $Hu(a)(\xi) = \sum_{1 \leq j, k \leq n} \partial^2 u / \partial z_j \partial \bar{z}_k(a) \xi_j \bar{\xi}_k$ is semipositive at every point $a \in \Omega$.*

Proof. This is an easy consequence of the following standard formula

$$\frac{1}{2\pi} \int_0^{2\pi} u(a + e^{i\theta} \xi) d\theta - u(a) = \frac{2}{\pi} \int_0^1 \frac{dt}{t} \int_{|\zeta| < t} Hu(a + \zeta \xi)(\xi) d\lambda(\zeta),$$

where $d\lambda$ is the Lebesgue measure on $\mathbb{C}$. Lemma 1.6 is a strong evidence that plurisubharmonicity is the natural complex analogue of linear convexity. $\square$

For non smooth functions, a similar characterization of plurisubharmonicity can be obtained by means of a regularization process.

(1.7) Theorem. *If $u \in \text{Psh}(\Omega)$, $u \not\equiv -\infty$ on every connected component of Ω , then for all $\xi \in \mathbb{C}^n$*

$$Hu(\xi) = \sum_{1 \leq j, k \leq n} \frac{\partial^2 u}{\partial z_j \partial \bar{z}_k} \xi_j \bar{\xi}_k \in \mathcal{D}'(\Omega)$$

is a positive measure. Conversely, if $v \in \mathcal{D}'(\Omega)$ is such that $Hv(\xi)$ is a positive measure for every $\xi \in \mathbb{C}^n$, there exists a unique function $u \in \text{Psh}(\Omega)$ which is locally integrable on Ω and such that v is the distribution associated to u . $\square$

In order to get a better geometric insight of this notion, we assume more generally that u is a function on a complex n -dimensional manifold X . If $\Phi : X \rightarrow Y$ is a holomorphic mapping and if $v \in C^2(Y, \mathbb{R})$, we have $d'd''(v \circ \Phi) = \Phi^* d'd''v$, hence

$$H(v \circ \Phi)(a, \xi) = Hv(\Phi(a), \Phi'(a) \cdot \xi).$$

In particular Hu , viewed as a hermitian form on T_X , does not depend on the choice of coordinates $(z_1, \dots, z_n)$. Therefore, the notion of psh function makes sense on any complex manifold. More generally, we have

(1.8) Proposition. *If $\Phi : X \rightarrow Y$ is a holomorphic map and $v \in \text{Psh}(Y)$, then $v \circ \Phi \in \text{Psh}(X)$.* $\square$

(1.9) Example. It is a standard fact that $\log |z|$ is psh (i.e. subharmonic) on $\mathbb{C}$. Thus $\log |f| \in \text{Psh}(X)$ for every holomorphic function $f \in H^0(X, \mathcal{O}_X)$. More generally

$$\log (|f_1|^{\alpha_1} + \cdots + |f_q|^{\alpha_q}) \in \text{Psh}(X)$$

for every $f_j \in H^0(X, \mathcal{O}_X)$ and $\alpha_j \geq 0$ (apply Property 1.5 d with $u_j = \alpha_j \log |f_j|$). We will be especially interested in the singularities obtained at points of the zero variety $f_1 = \cdots = f_q = 0$, when the α_j are rational numbers. $\square$

(1.10) Definition. *A psh function $u \in \text{Psh}(X)$ will be said to have analytic singularities if u can be written locally as*

$$u = \frac{\alpha}{2} \log (|f_1|^2 + \cdots + |f_N|^2) + v,$$

where $\alpha \in \mathbb{R}_+$, v is a locally bounded function and the f_j are holomorphic functions. If X is algebraic, we say that u has algebraic singularities if u can be written as above on sufficiently small Zariski open sets, with $\alpha \in \mathbb{Q}_+$ and f_j algebraic.

We then introduce the ideal $\mathcal{J} = \mathcal{J}(u/\alpha)$ of germs of holomorphic functions h such that $|h| \leq C e^{u/\alpha}$ for some constant C , i.e.

$$|h| \leq C (|f_1| + \cdots + |f_N|).$$

This is a globally defined ideal sheaf on X , locally equal to the integral closure $\overline{\mathcal{I}}$ of the ideal sheaf $\mathcal{I} = (f_1, \dots, f_N)$, thus $\mathcal{J}$ is coherent on X . If $(g_1, \dots, g_{N'})$ are local generators of $\mathcal{J}$, we still have

$$u = \frac{\alpha}{2} \log (|g_1|^2 + \cdots + |g_{N'}|^2) + O(1).$$

If X is projective algebraic and u has analytic singularities with $\alpha \in \mathbb{Q}_+$, then u automatically has algebraic singularities. From an algebraic point of view, the singularities of u are in 1:1 correspondence with the “algebraic data” $(\mathcal{J}, \alpha)$. Later on, we will see another important method for associating an ideal sheaf to a psh function.

(1.11) Exercise. Show that the above definition of the integral closure of an ideal $\mathcal{I}$ is equivalent to the following more algebraic definition: $\overline{\mathcal{I}}$ consists of all germs h satisfying an integral equation

$$h^d + a_1 h^{d-1} + \cdots + a_{d-1} h + a_d = 0, \quad a_k \in \mathcal{I}^k.$$

Hint. One inclusion is clear. To prove the other inclusion, consider the normalization of the blow-up of X along the (non necessarily reduced) zero variety $V(\mathcal{I})$. $\square$

1.C. Positive Currents

The reader can consult [Fed69] for a more thorough treatment of current theory. Let us first recall a few basic definitions. A *current* of degree q on an oriented differentiable manifold M is simply a differential q -form Θ with distribution coefficients. The space of currents of degree q over M will be denoted by $\mathcal{D}'^q(M)$. Alternatively, a current of degree q can be seen as an element Θ in the dual space $\mathcal{D}'_p(M) := (\mathcal{D}^p(M))'$ of the space $\mathcal{D}^p(M)$ of smooth differential forms of degree $p = \dim M - q$ with compact support; the duality pairing is given by

$$(1.12) \quad \langle \Theta, \alpha \rangle = \int_M \Theta \wedge \alpha, \quad \alpha \in \mathcal{D}^p(M).$$

A basic example is the *current of integration* $[S]$ over a compact oriented submanifold S of M :

$$(1.13) \quad \langle [S], \alpha \rangle = \int_S \alpha, \quad \deg \alpha = p = \dim_{\mathbb{R}} S.$$

Then $[S]$ is a current with measure coefficients, and Stokes' formula shows that $d[S] = (-1)^{q-1}[\partial S]$, in particular $d[S] = 0$ if S has no boundary. Because of this example, the integer p is said to be the dimension of Θ when $\Theta \in \mathcal{D}'_p(M)$. The current Θ is said to be *closed* if $d\Theta = 0$.

On a complex manifold X , we have similar notions of bidegree and bidimension; as in the real case, we denote by

$$\mathcal{D}'^{p,q}(X) = \mathcal{D}'_{n-p,n-q}(X), \quad n = \dim X,$$

the space of currents of bidegree (p, q) and bidimension $(n-p, n-q)$ on X . According to [Lel57], a current Θ of bidimension (p, p) is said to be (*weakly*) *positive* if for every choice of smooth $(1, 0)$ -forms $\alpha_1, \dots, \alpha_p$ on X the distribution

$$(1.14) \quad \Theta \wedge i\alpha_1 \wedge \bar{\alpha}_1 \wedge \dots \wedge i\alpha_p \wedge \bar{\alpha}_p \quad \text{is a positive measure.}$$

(1.15) Exercise. If Θ is positive, show that the coefficients $\Theta_{I,J}$ of Θ are complex measures, and that, up to constants, they are dominated by the trace measure

$$\sigma_{\Theta} = \Theta \wedge \frac{1}{p!} \beta^p = 2^{-p} \sum \Theta_{I,I}, \quad \beta = \frac{i}{2} d' d'' |z|^2 = \frac{i}{2} \sum_{1 \leq j \leq n} dz_j \wedge d\bar{z}_j,$$

which is a positive measure.

Hint. Observe that $\sum \Theta_{I,I}$ is invariant by unitary changes of coordinates and that the (p, p) -forms $i\alpha_1 \wedge \bar{\alpha}_1 \wedge \dots \wedge i\alpha_p \wedge \bar{\alpha}_p$ generate $\Lambda^{p,p} T_{\mathbb{C}^n}^*$ as a $\mathbb{C}$ -vector space. $\square$

A current $\Theta = i \sum_{1 \leq j, k \leq n} \Theta_{jk} dz_j \wedge d\bar{z}_k$ of bidegree $(1, 1)$ is easily seen to be positive if and only if the complex measure $\sum \lambda_j \bar{\lambda}_k \Theta_{jk}$ is a positive measure for every n -tuple $(\lambda_1, \dots, \lambda_n) \in \mathbb{C}^n$.

(1.16) Example. If u is a (not identically $-\infty$) psh function on X , we can associate with u a (closed) positive current $\Theta = i\partial\bar{\partial}u$ of bidegree $(1, 1)$. Conversely, every

closed positive current of bidegree $(1, 1)$ can be written under this form on any open subset $\Omega \subset X$ such that $H_{DR}^2(\Omega, \mathbb{R}) = H^1(\Omega, \mathcal{O}) = 0$, e.g. on small coordinate balls (exercise to the reader). $\square$

It is not difficult to show that a product $\Theta_1 \wedge \dots \wedge \Theta_q$ of positive currents of bidegree $(1, 1)$ is positive whenever the product is well defined (this is certainly the case if all Θ_j but one at most are smooth; much finer conditions will be discussed in Section 2).

We now discuss another very important example of closed positive current. In fact, with every closed analytic set $A \subset X$ of pure dimension p is associated a current of integration

$$(1.17) \quad \langle [A], \alpha \rangle = \int_{A_{\text{reg}}} \alpha, \quad \alpha \in \mathcal{D}^{p,p}(X),$$

obtained by integrating over the regular points of A . In order to show that (1.17) is a correct definition of a current on X , one must show that A_{reg} has locally finite area in a neighborhood of A_{sing} . This result, due to [Lel57] is shown as follows. Suppose that 0 is a singular point of A . By the local parametrization theorem for analytic sets, there is a linear change of coordinates on $\mathbb{C}^n$ such that all projections

$$\pi_I : (z_1, \dots, z_n) \mapsto (z_{i_1}, \dots, z_{i_p})$$

define a finite ramified covering of the intersection $A \cap \Delta$ with a small polydisk Δ in $\mathbb{C}^n$ onto a small polydisk Δ_I in $\mathbb{C}^p$. Let n_I be the sheet number. Then the p -dimensional area of $A \cap \Delta$ is bounded above by the sum of the areas of its projections counted with multiplicities, i.e.

$$\text{Area}(A \cap \Delta) \leq \sum n_I \text{Vol}(\Delta_I).$$

The fact that $[A]$ is positive is also easy. In fact

$$i\alpha_1 \wedge \bar{\alpha}_1 \wedge \dots \wedge i\alpha_p \wedge \bar{\alpha}_p = |\det(\alpha_{jk})|^2 iw_1 \wedge \bar{w}_1 \wedge \dots \wedge iw_p \wedge \bar{w}_p$$

if $\alpha_j = \sum \alpha_{jk} dw_k$ in terms of local coordinates $(w_1, \dots, w_p)$ on A_{reg} . This shows that all such forms are ≥ 0 in the canonical orientation defined by $iw_1 \wedge \bar{w}_1 \wedge \dots \wedge iw_p \wedge \bar{w}_p$. More importantly, Lelong [Lel57] has shown that $[A]$ is d -closed in X , even at points of A_{sing} . This last result can be seen today as a consequence of the Skoda-El Mir extension theorem. For this we need the following definition: a *complete pluripolar* set is a set E such that there is an open covering (Ω_j) of X and psh functions u_j on Ω_j with $E \cap \Omega_j = u_j^{-1}(-\infty)$. Any (closed) analytic set is of course complete pluripolar (take u_j as in Example 1.9).

(1.18) Theorem (Skoda [Sko81], El Mir [EM84], Sibony [Sib85]). *Let E be a closed complete pluripolar set in X , and let Θ be a closed positive current on $X \setminus E$ such that the coefficients $\Theta_{I,J}$ of Θ are measures with locally finite mass near E . Then the trivial extension $\tilde{\Theta}$ obtained by extending the measures $\Theta_{I,J}$ by 0 on E is still closed on X .*

Lelong's result $d[A] = 0$ is obtained by applying the Skoda-El Mir theorem to $\Theta = [A_{\text{reg}}]$ on $X \setminus A_{\text{sing}}$.

Proof of Theorem 1.18. The statement is local on X , so we may work on a small open set Ω such that $E \cap \Omega = v^{-1}(-\infty)$, $v \in \text{Psh}(\Omega)$. Let $\chi : \mathbb{R} \rightarrow \mathbb{R}$ be a convex increasing function such that $\chi(t) = 0$ for $t \leq -1$ and $\chi(0) = 1$. By shrinking Ω and putting $v_k = \chi(k^{-1}v \star \rho_{\varepsilon_k})$ with $\varepsilon_k \rightarrow 0$ fast, we get a sequence of functions $v_k \in \text{Psh}(\Omega) \cap C^\infty(\Omega)$ such that $0 \leq v_k \leq 1$, $v_k = 0$ in a neighborhood of $E \cap \Omega$ and $\lim v_k(x) = 1$ at every point of $\Omega \setminus E$. Let $\theta \in C^\infty([0, 1])$ be a function such that $\theta = 0$ on $[0, 1/3]$, $\theta = 1$ on $[2/3, 1]$ and $0 \leq \theta \leq 1$. Then $\theta \circ v_k = 0$ near $E \cap \Omega$ and $\theta \circ v_k \rightarrow 1$ on $\Omega \setminus E$. Therefore $\tilde{\Theta} = \lim_{k \rightarrow +\infty} (\theta \circ v_k) \Theta$ and

$$d' \tilde{\Theta} = \lim_{k \rightarrow +\infty} \Theta \wedge d'(\theta \circ v_k)$$

in the weak topology of currents. It is therefore sufficient to verify that $\Theta \wedge d'(\theta \circ v_k)$ converges weakly to 0 (note that $d'' \tilde{\Theta}$ is conjugate to $d' \tilde{\Theta}$, thus $d'' \tilde{\Theta}$ will also vanish).

Assume first that $\Theta \in \mathcal{D}'^{n-1, n-1}(X)$. Then $\Theta \wedge d'(\theta \circ v_k) \in \mathcal{D}'^{n, n-1}(\Omega)$, and we have to show that

$$\langle \Theta \wedge d'(\theta \circ v_k), \bar{\alpha} \rangle = \langle \Theta, \theta'(v_k) d' v_k \wedge \bar{\alpha} \rangle \xrightarrow{k \rightarrow +\infty} 0, \quad \forall \alpha \in \mathcal{D}^{1,0}(\Omega).$$

As $\gamma \mapsto \langle \Theta, i\gamma \wedge \bar{\gamma} \rangle$ is a non-negative hermitian form on $\mathcal{D}^{1,0}(\Omega)$, the Cauchy-Schwarz inequality yields

$$|\langle \Theta, i\beta \wedge \bar{\gamma} \rangle|^2 \leq \langle \Theta, i\beta \wedge \bar{\beta} \rangle \langle \Theta, i\gamma \wedge \bar{\gamma} \rangle, \quad \forall \beta, \gamma \in \mathcal{D}^{1,0}(\Omega).$$

Let $\psi \in \mathcal{D}(\Omega)$, $0 \leq \psi \leq 1$, be equal to 1 in a neighborhood of $\text{Supp } \alpha$. We find

$$|\langle \Theta, \theta'(v_k) d' v_k \wedge \bar{\alpha} \rangle|^2 \leq \langle \Theta, \psi i d' v_k \wedge d'' v_k \rangle \langle \Theta, \theta'(v_k)^2 i \alpha \wedge \bar{\alpha} \rangle.$$

By hypothesis $\int_{\Omega \setminus E} \Theta \wedge i \alpha \wedge \bar{\alpha} < +\infty$ and $\theta'(v_k)$ converges everywhere to 0 on Ω , thus $\langle \Theta, \theta'(v_k)^2 i \alpha \wedge \bar{\alpha} \rangle$ converges to 0 by Lebesgue's dominated convergence theorem. On the other hand

$$\begin{aligned} i d' d'' v_k^2 &= 2 v_k i d' d'' v_k + 2 i d' v_k \wedge d'' v_k \geq 2 i d' v_k \wedge d'' v_k, \\ 2 \langle \Theta, \psi i d' v_k \wedge d'' v_k \rangle &\leq \langle \Theta, \psi i d' d'' v_k^2 \rangle. \end{aligned}$$

As $\psi \in \mathcal{D}(\Omega)$, $v_k = 0$ near E and $d\Theta = 0$ on $\Omega \setminus E$, an integration by parts yields

$$\langle \Theta, \psi i d' d'' v_k^2 \rangle = \langle \Theta, v_k^2 i d' d'' \psi \rangle \leq C \int_{\Omega \setminus E} \|\Theta\| < +\infty$$

where C is a bound for the coefficients of $i d' d'' \psi$. Thus $\langle \Theta, \psi i d' v_k \wedge d'' v_k \rangle$ is bounded, and the proof is complete when $\Theta \in \mathcal{D}'^{n-1, n-1}$.

In the general case $\Theta \in \mathcal{D}'^{p,p}$, $p < n$, we simply apply the result already proved to all positive currents $\Theta \wedge \gamma \in \mathcal{D}'^{n-1, n-1}$ where $\gamma = i\gamma_1 \wedge \bar{\gamma}_1 \wedge \dots \wedge i\gamma_{n-p-1} \wedge \bar{\gamma}_{n-p-1}$ runs over a basis of forms of $\Lambda^{n-p-1, n-p-1} T_\Omega^*$ with constant coefficients (Lemma 1.4). Then we get $d(\tilde{\Theta} \wedge \gamma) = d\tilde{\Theta} \wedge \gamma = 0$ for all such γ , hence $d\tilde{\Theta} = 0$. $\square$

(1.19) Corollary. *Let Θ be a closed positive current on X and let E be a complete pluripolar set. Then $\mathbf{1}_E\Theta$ and $\mathbf{1}_{X \setminus E}\Theta$ are closed positive currents. In fact, $\tilde{\Theta} = \mathbf{1}_{X \setminus E}\Theta$ is the trivial extension of $\Theta|_{X \setminus E}$ to X , and $\mathbf{1}_E\Theta = \Theta - \tilde{\Theta}$. $\square$*

As mentioned above, any current $\Theta = \text{id}' d'' u$ associated with a psh function u is a closed positive $(1, 1)$ -current. In the special case $u = \log |f|$ where $f \in H^0(X, \mathcal{O}_X)$ is a non zero holomorphic function, we have the important

(1.20) Lelong-Poincaré equation. *Let $f \in H^0(X, \mathcal{O}_X)$ be a non zero holomorphic function, $Z_f = \sum m_j Z_j$, $m_j \in \mathbb{N}$, the zero divisor of f and $[Z_f] = \sum m_j [Z_j]$ the associated current of integration. Then*

$$\frac{i}{\pi} \partial \bar{\partial} \log |f| = [Z_f].$$

Proof (sketch). It is clear that $\text{id}' d'' \log |f| = 0$ in a neighborhood of every point $x \notin \text{Supp}(Z_f) = \bigcup Z_j$, so it is enough to check the equation in a neighborhood of every point of $\text{Supp}(Z_f)$. Let A be the set of singular points of $\text{Supp}(Z_f)$, i.e. the union of the pairwise intersections $Z_j \cap Z_k$ and of the singular loci $Z_{j, \text{sing}}$; we thus have $\dim A \leq n - 2$. In a neighborhood of any point $x \in \text{Supp}(Z_f) \setminus A$ there are local coordinates $(z_1, \dots, z_n)$ such that $f(z) = z_1^{m_j}$ where m_j is the multiplicity of f along the component Z_j which contains x and $z_1 = 0$ is an equation for Z_j near x . Hence

$$\frac{i}{\pi} d' d'' \log |f| = m_j \frac{i}{\pi} d' d'' \log |z_1| = m_j [Z_j]$$

in a neighborhood of x , as desired (the identity comes from the standard formula $\frac{i}{\pi} d' d'' \log |z| = \text{Dirac measure } \delta_0 \text{ in } \mathbb{C}$). This shows that the equation holds on $X \setminus A$. Hence the difference $\frac{i}{\pi} d' d'' \log |f| - [Z_f]$ is a closed current of degree 2 with measure coefficients, whose support is contained in A . By Exercise 1.21, this current must be 0, for A has too small dimension to carry its support (A is stratified by submanifolds of real codimension ≥ 4). $\square$

(1.21) Exercise. Let Θ be a current of degree q on a real manifold M , such that both Θ and $d\Theta$ have measure coefficients ("normal current"). Suppose that $\text{Supp } \Theta$ is contained in a real submanifold A with $\text{codim}_{\mathbb{R}} A > q$. Show that $\Theta = 0$.

Hint: Let $m = \dim_{\mathbb{R}} M$ and let $(x_1, \dots, x_m)$ be a coordinate system in a neighborhood Ω of a point $a \in A$ such that $A \cap \Omega = \{x_1 = \dots = x_k = 0\}$, $k > q$. Observe that $x_j \Theta = x_j d\Theta = 0$ for $1 \leq j \leq k$, thanks to the hypothesis on supports and on the normality of Θ , hence $dx_j \wedge \Theta = d(x_j \Theta) - x_j d\Theta = 0$, $1 \leq j \leq k$. Infer from this that all coefficients in $\Theta = \sum_{|I|=q} \Theta_I dx_I$ vanish. $\square$

We now recall a few basic facts of slicing theory (the reader will profitably consult [Fed69] and [Siu74] for further developments). Let $\sigma : M \rightarrow M'$ be a submersion of smooth differentiable manifolds and let Θ be a *locally flat* current on M , that is, a current which can be written locally as $\Theta = U + dV$ where U, V have L^1_{loc} coefficients. It is a standard fact (see Federer) that every current Θ such that

both Θ and $d\Theta$ have measure coefficients is locally flat; in particular, closed positive currents are locally flat. Then, for almost every $x' \in M'$, there is a well defined *slice* $\Theta_{x'}$, which is the current on the fiber $\sigma^{-1}(x')$ defined by

$$\Theta_{x'} = U|_{\sigma^{-1}(x')} + dV|_{\sigma^{-1}(x')}.$$

The restrictions of U, V to the fibers exist for almost all x' by the Fubini theorem. The slices $\Theta_{x'}$ are currents on the fibers with the same degree as Θ (thus of dimension $\dim \Theta - \dim(\text{fibers})$). Of course, every slice $\Theta_{x'}$ coincides with the usual restriction of Θ to the fiber if Θ has smooth coefficients. By using a regularization $\Theta_\varepsilon = \Theta \star \rho_\varepsilon$, it is easy to show that the slices of a closed positive current are again closed and positive: in fact $U_{\varepsilon, x'}$ and $V_{\varepsilon, x'}$ converge to $U_{x'}$ and $V_{x'}$ in $L^1_{\text{loc}}(\sigma^{-1}(x'))$, thus $\Theta_{\varepsilon, x'}$ converges weakly to $\Theta_{x'}$ for almost every x' . Now, the basic slicing formula is

$$(1.22) \quad \int_M \Theta \wedge \alpha \wedge \sigma^* \beta = \int_{x' \in M'} \left(\int_{x'' \in \sigma^{-1}(x')} \Theta_{x'}(x'') \wedge \alpha|_{\sigma^{-1}(x')}(x'') \right) \beta(x')$$

for every smooth form α on M and β on M' , such that α has compact support and $\deg \alpha = \dim M - \dim M' - \deg \Theta$, $\deg \beta = \dim M'$. This is an easy consequence of the usual Fubini theorem applied to U and V in the decomposition $\Theta = U + dV$, if we identify locally σ with a projection map $M = M' \times M'' \rightarrow M'$, $x = (x', x'') \mapsto x'$, and use a partition of unity on the support of α .

To conclude this section, we discuss De Rham and Dolbeault cohomology theory in the context of currents. A basic observation is that the Poincaré and Dolbeault-Grothendieck lemma still hold for currents. Namely, if $(\mathcal{D}^q, d)$ and $(\mathcal{D}'(F)^{p,q}, d'')$ denote the complex of sheaves of degree q currents (resp. of (p, q) -currents with values in a holomorphic vector bundle F), we still have De Rham and Dolbeault sheaf resolutions

$$0 \rightarrow \mathbb{R} \rightarrow \mathcal{D}'^\bullet, \quad 0 \rightarrow \Omega_X^p \otimes \mathcal{O}(F) \rightarrow \mathcal{D}'(F)^{p,\bullet}.$$

Hence we get canonical isomorphisms

$$(1.23) \quad \begin{aligned} H_{\text{DR}}^q(M, \mathbb{R}) &= H^q((\Gamma(M, \mathcal{D}'^\bullet), d)), \\ H^{p,q}(X, F) &= H^q((\Gamma(X, \mathcal{D}'(F)^{p,\bullet}), d'')). \end{aligned}$$

In other words, we can attach a cohomology class $\{\Theta\} \in H_{\text{DR}}^q(M, \mathbb{R})$ to any closed current Θ of degree q , resp. a cohomology class $\{\Theta\} \in H^{p,q}(X, F)$ to any d'' -closed current of bidegree (p, q) . Replacing if necessary every current by a smooth representative in the same cohomology class, we see that there is a well defined cup product given by the wedge product of differential forms

$$\begin{aligned} H^{q_1}(M, \mathbb{R}) \times \dots \times H^{q_m}(M, \mathbb{R}) &\longrightarrow H^{q_1 + \dots + q_m}(M, \mathbb{R}), \\ (\{\Theta_1\}, \dots, \{\Theta_m\}) &\longmapsto \{\Theta_1\} \wedge \dots \wedge \{\Theta_m\}. \end{aligned}$$

In particular, if M is a compact oriented variety and $q_1 + \dots + q_m = \dim M$, there is a well defined intersection number

$$\{\Theta_1\} \cdot \{\Theta_2\} \cdot \dots \cdot \{\Theta_m\} = \int_M \{\Theta_1\} \wedge \dots \wedge \{\Theta_m\}.$$

However, as we will see in the next section, the pointwise product $\Theta_1 \wedge \dots \wedge \Theta_m$ need not exist in general.

2. Lelong Numbers and Intersection Theory

2.A. Multiplication of Currents and Monge-Ampère Operators

Let X be a n -dimensional complex manifold. We set

$$d^c = \frac{1}{2i\pi}(d' - d'').$$

It follows in particular that d^c is a real operator, i.e. $\overline{d^c u} = d^c \overline{u}$, and that $dd^c = \frac{1}{\pi}d'd''$. Although not quite standard, the $1/2i\pi$ normalization is very convenient for many purposes, since we may then forget the factor π or 2π almost everywhere (e.g. in the Lelong-Poincaré equation (1.20)).

Let u be a psh function and let Θ be a closed positive current on X . Our desire is to define the wedge product $dd^c u \wedge \Theta$ even when neither u nor Θ are smooth. In general, this product does not make sense because $dd^c u$ and Θ have measure coefficients and measures cannot be multiplied; see Kiselman [Kis84] for interesting counterexamples. Even in the algebraic setting considered here, multiplication of currents is not always possible: suppose e.g. that $\Theta = [D]$ is the exceptional divisor of a blow-up in a surface; then $D \cdot D = -1$ cannot be the cohomology class of a closed positive current $[D]^2$. Assume however that u is a *locally bounded* psh function. Then the current $u\Theta$ is well defined since u is a locally bounded Borel function and Θ has measure coefficients. According to Bedford-Taylor [BT82] we define

$$dd^c u \wedge \Theta = dd^c(u\Theta)$$

where $dd^c(\quad)$ is taken in the sense of distribution theory.

(2.1) Proposition. *If u is a locally bounded psh function, the wedge product $dd^c u \wedge \Theta$ is again a closed positive current.*

Proof. The result is local. Use a convolution $u_\nu = u \star \rho_{1/\nu}$ to get a decreasing sequence of smooth psh functions converging to u . Then write

$$dd^c(u\Theta) = \lim_{\nu \rightarrow +\infty} dd^c(u_\nu\Theta) = dd^c u_\nu \wedge \Theta$$

as a weak limit of closed positive currents. Observe that $u_\nu\Theta$ converges weakly to $u\Theta$ by Lebesgue's monotone convergence theorem. $\square$

More generally, if $u_1, \dots, u_m$ are locally bounded psh functions, we can define

$$dd^c u_1 \wedge \dots \wedge dd^c u_m \wedge \Theta = dd^c(u_1 dd^c u_2 \wedge \dots \wedge dd^c u_m \wedge \Theta)$$

by induction on m . (Chern, Levine and Nirenberg, 1969) noticed the following useful inequality. Define the *mass* of a current Θ on a compact set K to be

$$||\Theta||_K = \int_K \sum_{I,J} |\Theta_{I,J}|$$

whenever K is contained in a coordinate patch and $\Theta = \sum \Theta_{I,J} dz_I \wedge d\bar{z}_J$. Up to seminorm equivalence, this does not depend on the choice of coordinates. If K is not contained in a coordinate patch, we use a partition of unity to define a suitable seminorm $\|\Theta\|_K$. If $\Theta \geq 0$, Exercise 1.15 shows that the mass is controlled by the trace measure, i.e. $\|\Theta\|_K \leq C \int_K \Theta \wedge \beta^p$.

(2.2) Chern-Levine-Nirenberg inequality. *For all compact subsets K, L of X with $L \subset K^\circ$, there exists a constant $C_{K,L} \geq 0$ such that*

$$\|dd^c u_1 \wedge \dots \wedge dd^c u_m \wedge \Theta\|_L \leq C_{K,L} \|u_1\|_{L^\infty(K)} \dots \|u_m\|_{L^\infty(K)} \|\Theta\|_K$$

Proof. By induction, it is sufficient to prove the result for $m = 1$ and $u_1 = u$. There is a covering of L by a family of open balls $B'_j \subset B_j \subset K$ contained in coordinate patches of X . Let (p, p) be the bidimension of Θ , let $\beta = \frac{i}{2} d' d'' |z|^2$, and let $\chi \in \mathcal{D}(B_j)$ be equal to 1 on $\bar{B}'_j$. Then

$$\|dd^c u \wedge \Theta\|_{L \cap \bar{B}'_j} \leq C \int_{\bar{B}'_j} dd^c u \wedge \Theta \wedge \beta^{p-1} \leq C \int_{B_j} \chi dd^c u \wedge \Theta \wedge \beta^{p-1}.$$

As Θ and β are closed, an integration by parts yields

$$\|dd^c u \wedge \Theta\|_{L \cap \bar{B}'_j} \leq C \int_{B_j} u \Theta \wedge dd^c \chi \wedge \beta^{p-1} \leq C' \|u\|_{L^\infty(K)} \|\Theta\|_K$$

where C' is equal to C multiplied by a bound for the coefficients of the smooth form $dd^c \chi \wedge \beta^{p-1}$. $\square$

Various examples (cf. [Kis84]) show however that products of $(1,1)$ -currents $dd^c u_j$ cannot be defined in a reasonable way for arbitrary psh functions u_j . However, functions u_j with $-\infty$ poles can be admitted if the polar sets are sufficiently small.

(2.3) Proposition. *Let u be a psh function on X , and let Θ be a closed positive current of bidimension (p, p) . Suppose that u is locally bounded on $X \setminus A$, where A is an analytic subset of X of dimension $< p$ at each point. Then $dd^c u \wedge \Theta$ can be defined in such a way that $dd^c u \wedge \Theta = \lim_{\nu \rightarrow +\infty} dd^c u_\nu \wedge \Theta$ in the weak topology of currents, for any decreasing sequence $(u_\nu)_{\nu \geq 0}$ of psh functions converging to u .*

Proof. When u is locally bounded everywhere, we have $\lim u_\nu \Theta = u \Theta$ by the monotone convergence theorem and the result follows from the continuity of dd^c with respect to the weak topology.

First assume that A is discrete. Since our results are local, we may suppose that X is a ball $B(0, R) \subset \mathbb{C}^n$ and that $A = \{0\}$. For every $s \leq 0$, the function $u^{\geq s} = \max(u, s)$ is locally bounded on X , so the product $\Theta \wedge dd^c u^{\geq s}$ is well defined. For $|s|$ large, the function $u^{\geq s}$ differs from u only in a small neighborhood of the origin, at which u may have a $-\infty$ pole. Let γ be a $(p-1, p-1)$ -form with constant coefficients and set $s(r) = \liminf_{|z| \rightarrow r-0} u(z)$. By Stokes' formula, we see that the integral

$$(2.4) \quad I(s) := \int_{B(0,r)} dd^c u^{\geq s} \wedge \Theta \wedge \gamma$$

does not depend on s when $s < s(r)$, for the difference $I(s) - I(s')$ of two such integrals involves the dd^c of a current $(u^{\geq s} - u^{\geq s'}) \wedge \Theta \wedge \gamma$ with compact support in $B(0, r)$. Taking $\gamma = (dd^c |z|^2)^{p-1}$, we see that the current $dd^c u \wedge \Theta$ has finite mass on $B(0, r) \setminus \{0\}$ and we can define $\langle \mathbf{1}_{\{0\}}(dd^c u \wedge \Theta), \gamma \rangle$ to be the limit of the integrals (2.4) as r tends to zero and $s < s(r)$. In this case, the weak convergence statement is easily deduced from the locally bounded case discussed above.

In the case where $0 < \dim A < p$, we use a slicing technique to reduce the situation to the discrete case. Set $q = p - 1$. There are linear coordinates $(z_1, \dots, z_n)$ centered at any point of A , such that 0 is an isolated point of $A \cap (\{0\} \times \mathbb{C}^{n-q})$. Then there are small balls $B' = B(0, r')$ in $\mathbb{C}^q$, $B'' = B(0, r'')$ in $\mathbb{C}^{n-q}$ such that $A \cap (\overline{B'} \times \partial B'') = \emptyset$, and the projection map

$$\pi : \mathbb{C}^n \rightarrow \mathbb{C}^q, \quad z = (z_1, \dots, z_n) \mapsto z' = (z_1, \dots, z_q)$$

defines a finite proper mapping $A \cap (B' \times B'') \rightarrow B'$. These properties are preserved if we slightly change the direction of projection. Take sufficiently many projections π_m associated to coordinate systems $(z_1^m, \dots, z_n^m)$, $1 \leq m \leq N$, in such a way that the family of (q, q) -forms

$$i dz_1^m \wedge d\bar{z}_1^m \wedge \dots \wedge i dz_q^m \wedge d\bar{z}_q^m$$

defines a basis of the space of (q, q) -forms. Expressing any compactly supported smooth (q, q) -form in such a basis, we see that we need only define

$$(2.5) \quad \int_{B' \times B''} dd^c u \wedge \Theta \wedge f(z', z'') i dz_1 \wedge d\bar{z}_1 \wedge \dots \wedge i dz_q \wedge d\bar{z}_q = \int_{B'} \left\{ \int_{B''} f(z', \bullet) dd^c u(z', \bullet) \wedge \Theta(z', \bullet) \right\} i dz_1 \wedge d\bar{z}_1 \wedge \dots \wedge i dz_q \wedge d\bar{z}_q$$

where f is a test function with compact support in $B' \times B''$, and $\Theta(z', \bullet)$ denotes the slice of Θ on the fiber $\{z'\} \times B''$ of the projection $\pi : \mathbb{C}^n \rightarrow \mathbb{C}^q$. Each integral $\int_{B''}$ in the right hand side of (2.5) makes sense since the slices $(\{z'\} \times B'') \cap A$ are discrete. Moreover, the double integral $\int_{B'} \int_{B''}$ is convergent. Indeed, observe that u is bounded on any compact cylinder

$$K_{\delta, \varepsilon} = \overline{B}((1 - \delta)r') \times \left(\overline{B}(r'') \setminus \overline{B}((1 - \varepsilon)r'') \right)$$

disjoint from A . Take $\varepsilon \ll \delta \ll 1$ so small that

$$\text{Supp } f \subset \overline{B}((1 - \delta)r') \times \overline{B}((1 - \varepsilon)r'').$$

For all $z' \in \overline{B}((1 - \delta)r')$, the proof of the Chern-Levine-Nirenberg inequality 2.2 with a cut-off function $\chi(z'')$ equal to 1 on $B((1 - \varepsilon)r'')$ and with support in $B((1 - \varepsilon/2)r'')$ shows that

$$\begin{aligned} & \int_{B((1 - \varepsilon)r'')} dd^c u(z', \bullet) \wedge \Theta(z', \bullet) \\ & \leq C_\varepsilon \|u\|_{L^\infty(K_{\delta, \varepsilon})} \int_{z'' \in B((1 - \varepsilon/2)r'')} \Theta(z', z'') \wedge dd^c |z''|^2. \end{aligned}$$

This implies that the double integral is convergent. Now replace u everywhere by u_ν and observe that $\lim_{\nu \rightarrow +\infty} \int_{B''}$ is the expected integral for every z' such that $\Theta(z', \bullet)$ exists (apply the discrete case already proven). Moreover, the Chern-Levine-Nirenberg inequality yields uniform bounds for all functions u_ν , hence Lebesgue's dominated convergence theorem can be applied to $\int_{B'}$. We conclude from this that the sequence of integrals (2.5) converges when $u_\nu \downarrow u$, as expected. $\square$

(2.6) Remark. In the above proof, the fact that A is an analytic set does not play an essential role. The main point is just that the slices $(\{z'\} \times B'') \cap A$ consist of isolated points for generic choices of coordinates (z', z'') . In fact, the proof even works if the slices are totally discontinuous, in particular if they are of zero Hausdorff measure $\mathcal{H}_1$. It follows that Proposition 2.3 still holds whenever A is a closed set such that $\mathcal{H}_{2p-1}(A) = 0$. $\square$

2.B. Lelong Numbers

The concept of Lelong number is an analytic analogue of the algebraic notion of multiplicity. It is a very useful technique to extend results in the intersection theory of algebraic cycles to currents. Lelong numbers have been introduced for the first time by Lelong in [Lel57]. See also [Lel69], [Siu74], [Dem82a, 85a, 87] for further developments.

Let us first recall a few definitions. Let Θ be a closed positive current of bidegree (p, p) on a coordinate open set $\Omega \subset \mathbb{C}^n$ of a complex manifold X . The Lelong number of Θ at a point $x \in \Omega$ is defined to be the limit

$$\nu(\Theta, x) = \lim_{r \rightarrow 0+} \nu(\Theta, x, r), \quad \text{where } \nu(\Theta, x, r) = \frac{\sigma_\Theta(B(x, r))}{\pi^p r^{2p} / p!}$$

measures the ratio of the area of Θ in the ball $B(x, r)$ to the area of the ball of radius r in $\mathbb{C}^p$. As $\sigma_\Theta = \Theta \wedge \frac{1}{p!}(\pi dd^c |z|^2)^p$ by 1.15, we also get

$$(2.7) \quad \nu(\Theta, x, r) = \frac{1}{r^{2p}} \int_{B(x, r)} \Theta(z) \wedge (dd^c |z|^2)^p.$$

The main results concerning Lelong numbers are summarized in the following theorems, due respectively to Lelong, Thie and Siu.

(2.8) Theorem ([Lel57]).

- a) *For every positive current Θ , the ratio $\nu(\Theta, x, r)$ is a nonnegative increasing function of r , in particular the limit $\nu(\Theta, x)$ as $r \rightarrow 0+$ always exists.*
- b) *If $\Theta = dd^c u$ is the bidegree $(1, 1)$ -current associated with a psh function u , then*

$$\nu(\Theta, x) = \sup \{ \gamma \geq 0; u(z) \leq \gamma \log |z - x| + O(1) \text{ at } x \}.$$

In particular, if $u = \log |f|$ with $f \in H^0(X, \mathcal{O}_X)$ and $\Theta = dd^c u = [Z_f]$, we have

$$\nu([Z_f], x) = \text{ord}_x(f) = \max \{ m \in \mathbb{N}; D^\alpha f(x) = 0, |\alpha| < m \}.$$

(2.9) Theorem ([Thi67]). *In the case where Θ is a current of integration $[A]$ over an analytic subvariety A , the Lelong number $\nu([A], x)$ coincides with the multiplicity of A at x (defined e.g. as the sheet number in the ramified covering obtained by taking a generic linear projection of the germ (A, x) onto a p -dimensional linear subspace through x in any coordinate patch Ω).*

(2.10) Theorem ([Siu74]). *Let Θ be a closed positive current of bidimension (p, p) on the complex manifold X .*

- a) *The Lelong number $\nu(\Theta, x)$ is invariant by holomorphic changes of local coordinates.*
- b) *For every $c > 0$, the set $E_c(\Theta) = \{x \in X; \nu(\Theta, x) \geq c\}$ is a closed analytic subset of X of dimension $\leq p$.*

The most important result is 2.10 b), which is a deep application of Hörmander L^2 estimates (see Section 5). The earlier proofs of all other results were rather intricate in spite of their rather simple nature. We reproduce below a sketch of elementary arguments based on the use of a more general and more flexible notion of Lelong number introduced in [Dem87]. Let φ be a continuous psh function with an isolated $-\infty$ pole at x , e.g. a function of the form $\varphi(z) = \log \sum_{1 \leq j \leq N} |g_j(z)|^{\gamma_j}$, $\gamma_j > 0$, where $(g_1, \dots, g_N)$ is an ideal of germs of holomorphic functions in $\mathcal{O}_x$ with $g^{-1}(0) = \{x\}$. The *generalized Lelong number* $\nu(\Theta, \varphi)$ of Θ with respect to the weight φ is simply defined to be the mass of the measure $\Theta \wedge (dd^c \varphi)^p$ carried by the point x (the measure $\Theta \wedge (dd^c \varphi)^p$ is always well defined thanks to Prop. 2.3). This number can also be seen as the limit $\nu(\Theta, \varphi) = \lim_{t \rightarrow -\infty} \nu(\Theta, \varphi, t)$, where

$$(2.11) \quad \nu(\Theta, \varphi, t) = \int_{\varphi(z) < t} \Theta \wedge (dd^c \varphi)^p.$$

The relation with our earlier definition of Lelong numbers (as well as part a) of Theorem 2.8) comes from the identity

$$(2.12) \quad \nu(\Theta, x, r) = \nu(\Theta, \varphi, \log r), \quad \varphi(z) = \log |z - x|,$$

in particular $\nu(\Theta, x) = \nu(\Theta, \log |\bullet - x|)$. This equality is in turn a consequence of the following general formula, applied to $\chi(t) = e^{2t}$ and $t = \log r$:

$$(2.13) \quad \int_{\varphi(z) < t} \Theta \wedge (dd^c \chi \circ \varphi)^p = \chi'(t - 0)^p \int_{\varphi(z) < t} \Theta \wedge (dd^c \varphi)^p,$$

where χ is an arbitrary convex increasing function. To prove the formula, we use a regularization and thus suppose that Θ , φ and χ are smooth, and that t is a non critical value of φ . Then Stokes' formula shows that the integrals on the left and right hand side of (2.13) are equal respectively to

$$\int_{\varphi(z)=t} \Theta \wedge (dd^c \chi \circ \varphi)^{p-1} \wedge d^c(\chi \circ \varphi), \quad \int_{\varphi(z)=t} \Theta \wedge (dd^c \varphi)^{p-1} \wedge d^c \varphi,$$

and the differential form of bidegree $(p-1, p)$ appearing in the integrand of the first integral is equal to $(\chi' \circ \varphi)^p (dd^c \varphi)^{p-1} \wedge d^c \varphi$. The expected formula follows. Part b)

of Theorem 2.8 is a consequence of the Jensen-Lelong formula, whose proof is left as an exercise to the reader.

(2.14) Jensen-Lelong formula. *Let u be any psh function on X . Then u is integrable with respect to the measure $\mu_r = (dd^c\varphi)^{n-1} \wedge d^c\varphi$ supported by the pseudo-sphere $\{\varphi(z) = r\}$ and*

$$\mu_r(u) = \int_{\{\varphi < r\}} u(dd^c\varphi)^n + \int_{-\infty}^r \nu(dd^c u, \varphi, t) dt. \quad \square$$

In our case, we set $\varphi(z) = \log|z - x|$. Then $(dd^c\varphi)^n = \delta_x$ and μ_r is just the unitary invariant mean value measure on the sphere $S(x, e^r)$. For $r < r_0$, Formula 2.14 implies

$$\mu_r(u) - \mu_{r_0}(u) = \int_{r_0}^r \nu(dd^c u, x, t) \sim (r - r_0)\nu(dd^c u, x) \quad \text{as } r \rightarrow -\infty.$$

From this, using the Harnack inequality for subharmonic functions, we get

$$\liminf_{z \rightarrow x} \frac{u(z)}{\log|z - x|} = \lim_{r \rightarrow -\infty} \frac{\mu_r(u)}{r} = \nu(dd^c u, x).$$

These equalities imply statement 2.8 b).

Next, we show that the Lelong numbers $\nu(T, \varphi)$ only depend on the asymptotic behaviour of φ near the polar set $\varphi^{-1}(-\infty)$. In a precise way:

(2.15) Comparison theorem. *Let Θ be a closed positive current on X , and let $\varphi, \psi : X \rightarrow [-\infty, +\infty[$ be continuous psh functions with isolated poles at some point $x \in X$. Assume that*

$$\ell := \limsup_{z \rightarrow x} \frac{\psi(z)}{\varphi(z)} < +\infty.$$

Then $\nu(\Theta, \psi) \leq \ell^p \nu(\Theta, \varphi)$, and the equality holds if $\ell = \lim \psi/\varphi$.

Proof. (2.12) shows that $\nu(\Theta, \lambda\varphi) = \lambda^p \nu(\Theta, \varphi)$ for every positive constant λ . It is thus sufficient to verify the inequality $\nu(\Theta, \psi) \leq \nu(\Theta, \varphi)$ under the hypothesis $\limsup \psi/\varphi < 1$. For any $c > 0$, consider the psh function

$$u_c = \max(\psi - c, \varphi).$$

Fix $r \ll 0$. For $c > 0$ large enough, we have $u_c = \varphi$ on a neighborhood of $\varphi^{-1}(r)$ and Stokes' formula gives

$$\nu(\Theta, \varphi, r) = \nu(\Theta, u_c, r) \geq \nu(\Theta, u_c).$$

The hypothesis $\limsup \psi/\varphi < 1$ implies on the other hand that there exists $t_0 < 0$ such that $u_c = \psi - c$ on $\{u_c < t_0\}$. We thus get

$$\nu(\Theta, u_c) = \nu(\Theta, \psi - c) = \nu(\Theta, \psi),$$

hence $\nu(\Theta, \psi) \leq \nu(\Theta, \varphi)$. The equality case is obtained by reversing the roles of φ and ψ and observing that $\lim \varphi/\psi = 1/l$. $\square$

Part a) of Theorem 2.10 follows immediately from 2.15 by considering the weights $\varphi(z) = \log |\tau(z) - \tau(x)|$, $\psi(z) = \log |\tau'(z) - \tau'(x)|$ associated to coordinates systems $\tau(z) = (z_1, \dots, z_n)$, $\tau'(z) = (z'_1, \dots, z'_n)$ in a neighborhood of x . Another application is a direct simple proof of Thie's Theorem 2.9 when $\Theta = [A]$ is the current of integration over an analytic set $A \subset X$ of pure dimension p . For this, we have to observe that Theorem 2.15 still holds provided that x is an isolated point in $\text{Supp}(\Theta) \cap \varphi^{-1}(-\infty)$ and $\text{Supp}(\Theta) \cap \psi^{-1}(-\infty)$ (even though x is not isolated in $\varphi^{-1}(-\infty)$ or $\psi^{-1}(-\infty)$), under the weaker assumption that $\limsup_{\text{Supp}(\Theta) \ni z \rightarrow x} \psi(z)/\varphi(z) = \ell$. The reason for this is that all integrals involve currents supported on $\text{Supp}(\Theta)$. Now, by a generic choice of local coordinates $z' = (z_1, \dots, z_p)$ and $z'' = (z_{p+1}, \dots, z_n)$ on (X, x) , the germ (A, x) is contained in a cone $|z''| \leq C|z'|$. If $B' \subset \mathbb{C}^p$ is a ball of center 0 and radius r' small, and $B'' \subset \mathbb{C}^{n-p}$ is the ball of center 0 and radius $r'' = Cr'$, the projection

$$\text{pr} : A \cap (B' \times B'') \longrightarrow B'$$

is a ramified covering with finite sheet number m . When $z \in A$ tends to $x = 0$, the functions

$$\varphi(z) = \log |z| = \log(|z'|^2 + |z''|^2)^{1/2}, \quad \psi(z) = \log |z'|.$$

satisfy $\lim_{z \rightarrow x} \psi(z)/\varphi(z) = 1$. Hence Theorem 2.15 implies

$$\nu([A], x) = \nu([A], \varphi) = \nu([A], \psi).$$

Now, Formula 2.13 with $\chi(t) = e^{2t}$ yields

$$\begin{aligned} \nu([A], \psi, \log t) &= t^{-2p} \int_{\{\psi < \log t\}} [A] \wedge \left(\frac{1}{2} dd^c e^{2\psi} \right)^p \\ &= t^{-2p} \int_{A \cap \{|z'| < t\}} \left(\frac{1}{2} \text{pr}^* dd^c |z'|^2 \right)^p \\ &= m t^{-2p} \int_{\mathbb{C}^p \cap \{|z'| < t\}} \left(\frac{1}{2} dd^c |z'|^2 \right)^p = m, \end{aligned}$$

hence $\nu([A], \psi) = m$. Here, we have used the fact that pr is an étale covering with m sheets over the complement of the ramification locus $S \subset B'$, and the fact that S is of zero Lebesgue measure in B' .

(2.16) Proposition. *Under the assumptions of Proposition 2.3, we have*

$$\nu(dd^c u \wedge \Theta, x) \geq \nu(u, x) \nu(\Theta, x)$$

at every point $x \in X$.

Proof. Assume that $X = B(0, r)$ and $x = 0$. By definition

$$\nu(dd^c u \wedge \Theta, x) = \lim_{r \rightarrow 0} \int_{|z| \leq r} dd^c u \wedge \Theta \wedge (dd^c \log |z|)^{p-1}.$$

Set $\gamma = \nu(u, x)$ and

$$u_\nu(z) = \max(u(z), (\gamma - \varepsilon) \log |z| - \nu)$$

with $0 < \varepsilon < \gamma$ (if $\gamma = 0$, there is nothing to prove). Then u_ν decreases to u and

$$\int_{|z| \leq r} dd^c u \wedge \Theta \wedge (dd^c \log |z|)^{p-1} \geq \limsup_{\nu \rightarrow +\infty} \int_{|z| \leq r} dd^c u_\nu \wedge \Theta \wedge (dd^c \log |z|)^{p-1}$$

by the weak convergence of $dd^c u_\nu \wedge \Theta$; here $(dd^c \log |z|)^{p-1}$ is not smooth on $\overline{B}(0, r)$, but the integrals remain unchanged if we replace $\log |z|$ by $\chi(\log |z|/r)$ with a smooth convex function χ such that $\chi(t) = t$ for $t \geq -1$ and $\chi(t) = 0$ for $t \leq -2$. Now, we have $u(z) \leq \gamma \log |z| + C$ near 0, so $u_\nu(z)$ coincides with $(\gamma - \varepsilon) \log |z| - \nu$ on a small ball $B(0, r_\nu) \subset B(0, r)$ and we infer

$$\begin{aligned} \int_{|z| \leq r} dd^c u_\nu \wedge \Theta \wedge (dd^c \log |z|)^{p-1} &\geq (\gamma - \varepsilon) \int_{|z| \leq r_\nu} \Theta \wedge (dd^c \log |z|)^p \\ &\geq (\gamma - \varepsilon) \nu(\Theta, x). \end{aligned}$$

As $r \in]0, R[$ and $\varepsilon \in]0, \gamma[$ were arbitrary, the desired inequality follows. $\square$

We will later need an important decomposition formula of [Siu74]. We start with the following lemma.

(2.17) Lemma. *If Θ is a closed positive current of bidimension (p, p) and Z is an irreducible analytic set in X , we set*

$$m_Z = \inf\{x \in Z; \nu(\Theta, x)\}.$$

- a) *There is a countable family of proper analytic subsets (Z'_j) of Z such that $\nu(\Theta, x) = m_Z$ for all $x \in Z \setminus \bigcup Z'_j$. We say that m_Z is the generic Lelong number of Θ along Z .*
- b) *If $\dim Z = p$, then $\Theta \geq m_Z[Z]$ and $\mathbf{1}_Z \Theta = m_Z[Z]$.*

Proof. a) By definition of m_Z and $E_c(\Theta)$, we have $\nu(\Theta, x) \geq m_Z$ for every $x \in Z$ and

$$\nu(\Theta, x) = m_Z \quad \text{on } Z \setminus \bigcup_{c \in \mathbb{Q}, c > m_Z} Z \cap E_c(\Theta).$$

However, for $c > m_Z$, the intersection $Z \cap E_c(\Theta)$ is a proper analytic subset of A .

b) Left as an exercise to the reader. It is enough to prove that $\Theta \geq m_Z[Z_{\text{reg}}]$ at regular points of Z , so one may assume that Z is a p -dimensional linear subspace in $\mathbb{C}^n$. Show that the measure $(\Theta - m_Z[Z]) \wedge (dd^c |z|^2)^p$ has nonnegative mass on every ball $|z - a| < r$ with center $a \in Z$. Conclude by using arbitrary affine changes of coordinates that $\Theta - m_Z[Z] \geq 0$. $\square$

(2.18) Decomposition formula ([Siu74]). *Let Θ be a closed positive current of bidimension (p, p) . Then Θ can be written as a convergent series of closed positive currents*

$$\Theta = \sum_{k=1}^{+\infty} \lambda_k [Z_k] + R,$$

where $[Z_k]$ is a current of integration over an irreducible analytic set of dimension p , and R is a residual current with the property that $\dim E_c(R) < p$ for every $c > 0$. This decomposition is locally and globally unique: the sets Z_k are precisely the p -dimensional components occurring in the sublevel sets $E_c(\Theta)$, and $\lambda_k = \min_{x \in Z_k} \nu(\Theta, x)$ is the generic Lelong number of Θ along Z_k .

Proof of uniqueness. If Θ has such a decomposition, the p -dimensional components of $E_c(\Theta)$ are $(Z_j)_{\lambda_j \geq c}$, for $\nu(\Theta, x) = \sum \lambda_j \nu([Z_j], x) + \nu(R, x)$ is non zero only on $\bigcup Z_j \cup \bigcup E_c(R)$, and is equal to λ_j generically on Z_j (more precisely, $\nu(\Theta, x) = \lambda_j$ at every regular point of Z_j which does not belong to any intersection $Z_j \cup Z_k$, $k \neq j$ or to $\bigcup E_c(R)$). In particular Z_j and λ_j are unique.

Proof of existence. Let $(Z_j)_{j \geq 1}$ be the countable collection of p -dimensional components occurring in one of the sets $E_c(\Theta)$, $c \in \mathbb{Q}_+^*$, and let $\lambda_j > 0$ be the generic Lelong number of Θ along Z_j . Then Lemma 2.17 shows by induction on N that $R_N = \Theta - \sum_{1 \leq j \leq N} \lambda_j [Z_j]$ is positive. As R_N is a decreasing sequence, there must be a limit $R = \lim_{N \rightarrow +\infty} R_N$ in the weak topology. Thus we have the asserted decomposition. By construction, R has zero generic Lelong number along Z_j , so $\dim E_c(R) < p$ for every $c > 0$. $\square$

It is very important to note that some components of lower dimension can actually occur in $E_c(R)$, but they cannot be subtracted because R has bidimension (p, p) . A typical case is the case of a bidimension $(n-1, n-1)$ current $\Theta = dd^c u$ with $u = \log(|f_j|^{\gamma_1} + \dots + |f_N|^{\gamma_N})$ and $f_j \in H^0(X, \mathcal{O}_X)$. In general $\bigcup E_c(\Theta) = \bigcap f_j^{-1}(0)$ has dimension $< n-1$.

Corollary 2.19. *Let $\Theta_j = dd^c u_j$, $1 \leq j \leq p$, be closed positive $(1, 1)$ -currents on a complex manifold X . Suppose that there are analytic sets $A_2 \supset \dots \supset A_p$ in X with $\text{codim } A_j \geq j$ at every point such that each u_j , $j \geq 2$, is locally bounded on $X \setminus A_j$. Let $\{A_{p,k}\}_{k \geq 1}$ be the irreducible components of A_p of codimension p exactly and let $\nu_{j,k} = \min_{x \in A_{p,k}} \nu(\Theta_j, x)$ be the generic Lelong number of Θ_j along $A_{p,k}$. Then $\Theta_1 \wedge \dots \wedge \Theta_p$ is well-defined and*

$$\Theta_1 \wedge \dots \wedge \Theta_p \geq \sum_{k=1}^{+\infty} \nu_{1,k} \dots \nu_{p,k} [A_{p,k}].$$

Proof. By induction on p , Prop. 2.3 shows that $\Theta_1 \wedge \dots \wedge \Theta_p$ is well defined. Moreover, Prop. 2.16 implies

$$\nu(\Theta_1 \wedge \dots \wedge \Theta_p, x) \geq \nu(\Theta_1, x) \dots \nu(\Theta_p, x) \geq \nu_{1,k} \dots \nu_{p,k}$$

at every point $x \in A_{p,k}$. The desired inequality is then a consequence of Siu's decomposition theorem. $\square$

3. Hermitian Vector Bundles, Connections and Curvature

The goal of this section is to recall the most basic definitions of hermitian differential geometry related to the concepts of connection, curvature and first Chern class of a line bundle.

Let F be a complex vector bundle of rank r over a smooth differentiable manifold M . A *connection* D on F is a linear differential operator of order 1

$$D : C^\infty(M, \Lambda^q T_M^* \otimes F) \rightarrow C^\infty(M, \Lambda^{q+1} T_M^* \otimes F)$$

such that

$$(3.1) \quad D(f \wedge u) = df \wedge u + (-1)^{\deg f} f \wedge Du$$

for all forms $f \in C^\infty(M, \Lambda^p T_M^*)$, $u \in C^\infty(M, \Lambda^q T_M^* \otimes F)$. On an open set $\Omega \subset M$ where F admits a trivialization $\theta : F|_\Omega \xrightarrow{\sim} \Omega \times \mathbb{C}^r$, a connection D can be written

$$Du \simeq_\theta du + \Gamma \wedge u$$

where $\Gamma \in C^\infty(\Omega, \Lambda^1 T_M^* \otimes \text{Hom}(\mathbb{C}^r, \mathbb{C}^r))$ is an arbitrary matrix of 1-forms and d acts componentwise. It is then easy to check that

$$D^2 u \simeq_\theta (d\Gamma + \Gamma \wedge \Gamma) \wedge u \quad \text{on } \Omega.$$

Since D^2 is a globally defined operator, there is a global 2-form

$$(3.2) \quad \Theta(D) \in C^\infty(M, \Lambda^2 T_M^* \otimes \text{Hom}(F, F))$$

such that $D^2 u = \Theta(D) \wedge u$ for every form u with values in F .

Assume now that F is endowed with a C^∞ hermitian metric along the fibers and that the isomorphism $F|_\Omega \simeq \Omega \times \mathbb{C}^r$ is given by a C^∞ frame (e_λ) . We then have a canonical sesquilinear pairing

$$(3.3) \quad C^\infty(M, \Lambda^p T_M^* \otimes F) \times C^\infty(M, \Lambda^q T_M^* \otimes F) \longrightarrow C^\infty(M, \Lambda^{p+q} T_M^* \otimes \mathbb{C})$$

$$(u, v) \longmapsto \{u, v\}$$

given by

$$\{u, v\} = \sum_{\lambda, \mu} u_\lambda \wedge \bar{v}_\mu \langle e_\lambda, e_\mu \rangle, \quad u = \sum_\lambda u_\lambda \otimes e_\lambda, \quad v = \sum_\mu v_\mu \otimes e_\mu.$$

The connection D is said to be *hermitian* if it satisfies the additional property

$$d\{u, v\} = \{Du, v\} + (-1)^{\deg u} \{u, Dv\}.$$

Assuming that (e_λ) is orthonormal, one easily checks that D is hermitian if and only if $\Gamma^* = -\Gamma$. In this case $\Theta(D)^* = -\Theta(D)$, thus

$$\text{i}\Theta(D) \in C^\infty(M, \Lambda^2 T_M^* \otimes \text{Herm}(F, F)).$$

(3.4) Special case. For a bundle F of rank 1, the connection form Γ of a hermitian connection D can be seen as a 1-form with purely imaginary coefficients $\Gamma = \text{i}A$ (A

real). Then we have $\Theta(D) = d\Gamma = idA$. In particular $i\Theta(F)$ is a closed 2-form. The *First Chern class* of F is defined to be the cohomology class

$$c_1(F)_{\mathbb{R}} = \left\{ \frac{i}{2\pi} \Theta(D) \right\} \in H_{\text{DR}}^2(M, \mathbb{R}).$$

The cohomology class is actually independent of the connection, since any other connection D_1 differs by a global 1-form, $D_1 u = Du + B \wedge u$, so that $\Theta(D_1) = \Theta(D) + dB$. It is well-known that $c_1(F)_{\mathbb{R}}$ is the image in $H^2(M, \mathbb{R})$ of an integral class $c_1(F) \in H^2(M, \mathbb{Z})$; by using the exponential exact sequence

$$0 \rightarrow \mathbb{Z} \rightarrow \mathcal{E} \rightarrow \mathcal{E}^* \rightarrow 0,$$

$c_1(F)$ can be defined in Čech cohomology theory as the image by the coboundary map $H^1(M, \mathcal{E}^*) \rightarrow H^2(M, \mathbb{Z})$ of the cocycle $\{g_{jk}\} \in H^1(M, \mathcal{E}^*)$ defining F ; see e.g. [GH78] for details. $\square$

We now concentrate ourselves on the complex analytic case. If $M = X$ is a complex manifold X , every connection D on a complex C^∞ vector bundle F can be splitted in a unique way as a sum of a $(1, 0)$ and of a $(0, 1)$ -connection, $D = D' + D''$. In a local trivialization θ given by a C^∞ frame, one can write

$$(3.5') \quad D'u \simeq_\theta d'u + \Gamma' \wedge u,$$

$$(3.5'') \quad D''u \simeq_\theta d''u + \Gamma'' \wedge u,$$

with $\Gamma = \Gamma' + \Gamma''$. The connection is hermitian if and only if $\Gamma' = -(\Gamma'')^*$ in any orthonormal frame. Thus there exists a unique hermitian connection D corresponding to a prescribed $(0, 1)$ part D'' .

Assume now that the bundle F itself has a *holomorphic* structure. The unique hermitian connection for which D'' is the d'' operator defined in § 1 is called the *Chern connection* of F . In a local holomorphic frame (e_λ) of $E|_\Omega$, the metric is given by the hermitian matrix $H = (h_{\lambda\mu})$, $h_{\lambda\mu} = \langle e_\lambda, e_\mu \rangle$. We have

$$\{u, v\} = \sum_{\lambda, \mu} h_{\lambda\mu} u_\lambda \wedge \bar{v}_\mu = u^\dagger \wedge H \bar{v},$$

where $u^\dagger$ is the transposed matrix of u , and easy computations yield

$$\begin{aligned} d\{u, v\} &= (du)^\dagger \wedge H \bar{v} + (-1)^{\deg u} u^\dagger \wedge (dH \wedge \bar{v} + H d\bar{v}) \\ &= (du + \bar{H}^{-1} d' \bar{H} \wedge u)^\dagger \wedge H \bar{v} + (-1)^{\deg u} u^\dagger \wedge \overline{(dv + \bar{H}^{-1} d' \bar{H} \wedge v)} \end{aligned}$$

using the fact that $dH = d'H + \overline{d'\bar{H}}$ and $\bar{H}^\dagger = H$. Therefore the Chern connection D coincides with the hermitian connection defined by

$$(3.6) \quad \begin{cases} Du \simeq_\theta du + \bar{H}^{-1} d' \bar{H} \wedge u, \\ D' \simeq_\theta d' + \bar{H}^{-1} d' \bar{H} \wedge \bullet = \bar{H}^{-1} d'(\bar{H} \bullet), \quad D'' = d''. \end{cases}$$

It is clear from this relations that $D'^2 = D''^2 = 0$. Consequently D^2 is given by to $D^2 = D'D'' + D''D'$, and the curvature tensor $\Theta(D)$ is of type $(1, 1)$. Since $d'd'' + d''d' = 0$, we get

$$\begin{aligned} (D' D'' + D'' D') u &\simeq_{\theta} \overline{H}^{-1} d' \overline{H} \wedge d'' u + d'' (\overline{H}^{-1} d' \overline{H} \wedge u) \\ &= d'' (\overline{H}^{-1} d' \overline{H}) \wedge u. \end{aligned}$$

(3.7) Proposition. *The Chern curvature tensor $\Theta(F) := \Theta(D)$ is such that*

$$i\Theta(F) \in C^{\infty}(X, \Lambda^{1,1} T_X^* \otimes \text{Herm}(F, F)).$$

If $\theta : E|_{\Omega} \rightarrow \Omega \times \mathbb{C}^r$ is a holomorphic trivialization and if H is the hermitian matrix representing the metric along the fibers of $F|_{\Omega}$, then

$$i\Theta(F) \simeq_{\theta} i d'' (\overline{H}^{-1} d' \overline{H}) \quad \text{on } \Omega. \quad \square$$

Let $(z_1, \dots, z_n)$ be holomorphic coordinates on X and let $(e_{\lambda})_{1 \leq \lambda \leq r}$ be an orthonormal frame of F . Writing

$$i\Theta(F) = \sum_{1 \leq j, k \leq n, 1 \leq \lambda, \mu \leq r} c_{jk\lambda\mu} dz_j \wedge dz_k \otimes e_{\lambda}^* \otimes e_{\mu},$$

we can identify the curvature tensor to a hermitian form

$$(3.8) \quad \tilde{\Theta}(F)(\xi \otimes v) = \sum_{1 \leq j, k \leq n, 1 \leq \lambda, \mu \leq r} c_{jk\lambda\mu} \xi_j \bar{\xi}_k v_{\lambda} \bar{v}_{\mu}$$

on $T_X \otimes F$. This leads in a natural way to positivity concepts, following definitions introduced by Kodaira [Kod53], Nakano [Nak55] and Griffiths [Gri66].

(3.9) Definition. *The hermitian vector bundle F is said to be*

- a) *positive in the sense of Nakano if $\tilde{\Theta}(F)(\tau) > 0$ for all non zero tensors $\tau = \sum \tau_{j\lambda} \partial/\partial z_j \otimes e_{\lambda} \in T_X \otimes F$.*
- b) *positive in the sense of Griffiths if $\tilde{\Theta}(F)(\xi \otimes v) > 0$ for all non zero decomposable tensors $\xi \otimes v \in T_X \otimes F$;*

Corresponding semipositivity concepts are defined by relaxing the strict inequalities.

(3.10) Special case of rank 1 bundles. Assume that F is a line bundle. The hermitian matrix $H = (h_{11})$ associated to a trivialization $\theta : F|_{\Omega} \simeq \Omega \times \mathbb{C}$ is simply a positive function which we find convenient to denote by $e^{-2\varphi}$, $\varphi \in C^{\infty}(\Omega, \mathbb{R})$. In this case the curvature form $\Theta(F)$ can be identified to the $(1, 1)$ -form $2d' d'' \varphi$, and

$$\frac{i}{2\pi} \Theta(F) = \frac{i}{\pi} d' d'' \varphi = dd^c \varphi$$

is a real $(1, 1)$ -form. Hence F is semipositive (in either Nakano or Griffiths sense) if and only if φ is psh, resp. positive if and only if φ is *strictly psh*. In this setting, the Lelong-Poincaré equation can be generalized as follows: let $\sigma \in H^0(X, F)$ be a non zero holomorphic section. Then

$$(3.11) \quad dd^c \log \|\sigma\| = [Z_{\sigma}] - \frac{i}{2\pi} \Theta(F).$$

Formula (3.11) is immediate if we write $\|\sigma\| = |\theta(\sigma)|e^{-\varphi}$ and if we apply (1.20) to the holomorphic function $f = \theta(\sigma)$. As we shall see later, it is very important for applications to consider also singular hermitian metrics.

(3.12) Definition. A singular (hermitian) metric on a line bundle F is a metric which is given in any trivialization $\theta : F|_{\Omega} \xrightarrow{\simeq} \Omega \times \mathbb{C}$ by

$$\|\xi\| = |\theta(\xi)|e^{-\varphi(x)}, \quad x \in \Omega, \quad \xi \in F_x$$

where $\varphi \in L^1_{\text{loc}}(\Omega)$ is an arbitrary function, called the weight of the metric with respect to the trivialization θ .

If $\theta' : F|_{\Omega'} \rightarrow \Omega' \times \mathbb{C}$ is another trivialization, φ' the associated weight and $g \in \mathcal{O}^*(\Omega \cap \Omega')$ the transition function, then $\theta'(\xi) = g(x)\theta(\xi)$ for $\xi \in F_x$, and so $\varphi' = \varphi + \log |g|$ on $\Omega \cap \Omega'$. The curvature form of F is then given formally by the closed $(1, 1)$ -current $\frac{i}{2\pi}\Theta(F) = dd^c\varphi$ on Ω ; our assumption $\varphi \in L^1_{\text{loc}}(\Omega)$ guarantees that $\Theta(F)$ exists in the sense of distribution theory. As in the smooth case, $\frac{i}{2\pi}\Theta(F)$ is globally defined on X and independent of the choice of trivializations, and its De Rham cohomology class is the image of the first Chern class $c_1(F) \in H^2(X, \mathbb{Z})$ in $H^2_{DR}(X, \mathbb{R})$. Before going further, we discuss two basic examples.

(3.13) Example. Let $D = \sum \alpha_j D_j$ be a divisor with coefficients $\alpha_j \in \mathbb{Z}$ and let $F = \mathcal{O}(D)$ be the associated invertible sheaf of meromorphic functions u such that $\text{div}(u) + D \geq 0$; the corresponding line bundle can be equipped with the singular metric defined by $\|u\| = |u|$. If g_j is a generator of the ideal of D_j on an open set $\Omega \subset X$ then $\theta(u) = u \prod g_j^{\alpha_j}$ defines a trivialization of $\mathcal{O}(D)$ over Ω , thus our singular metric is associated to the weight $\varphi = \sum \alpha_j \log |g_j|$. By the Lelong-Poincaré equation, we find

$$\frac{i}{2\pi}\Theta(\mathcal{O}(D)) = dd^c\varphi = [D],$$

where $[D] = \sum \alpha_j [D_j]$ denotes the current of integration over D . $\square$

(3.14) Example. Assume that $\sigma_1, \dots, \sigma_N$ are non zero holomorphic sections of F . Then we can define a natural (possibly singular) hermitian metric on F^* by

$$\|\xi^*\|^2 = \sum_{1 \leq j \leq N} |\xi^* \cdot \sigma_j(x)|^2 \quad \text{for } \xi^* \in F_x^*.$$

The dual metric on F is given by

$$\|\xi\|^2 = \frac{|\theta(\xi)|^2}{|\theta(\sigma_1(x))|^2 + \dots + |\theta(\sigma_N(x))|^2}$$

with respect to any trivialization θ . The associated weight function is thus given by $\varphi(x) = \log \left(\sum_{1 \leq j \leq N} |\theta(\sigma_j(x))|^2 \right)^{1/2}$. In this case φ is a psh function, thus $i\Theta(F)$ is a closed positive current. Let us denote by Σ the linear system defined by $\sigma_1, \dots, \sigma_N$ and by $B_{\Sigma} = \bigcap \sigma_j^{-1}(0)$ its base locus. We have a meromorphic map

$$\Phi_{\Sigma} : X \setminus B_{\Sigma} \rightarrow \mathbb{P}^{N-1}, \quad x \mapsto (\sigma_1(x) : \sigma_2(x) : \dots : \sigma_N(x)).$$

Then $\frac{i}{2\pi}\Theta(F)$ is equal to the pull-back over $X \setminus B_\Sigma$ of the Fubini-Study metric $\omega_{\text{FS}} = \frac{1}{2\pi} \log(|z_1|^2 + \dots + |z_N|^2)$ of $\mathbb{P}^{N-1}$ by Φ_Σ . $\square$

(3.15) Ample and very ample line bundles. A holomorphic line bundle F over a compact complex manifold X is said to be

- a) *very ample* if the map $\Phi_{|F|} : X \rightarrow \mathbb{P}^{N-1}$ associated to the complete linear system $|F| = P(H^0(X, F))$ is a regular embedding (by this we mean in particular that the base locus is empty, i.e. $B_{|F|} = \emptyset$).
- b) *ample* if some multiple mF , $m > 0$, is very ample.

Here we use an additive notation for $\text{Pic}(X) = H^1(X, \mathcal{O}^*)$, hence the symbol mF denotes the line bundle $F^{\otimes m}$. By Example 3.15, every ample line bundle F has a smooth hermitian metric with positive definite curvature form; indeed, if the linear system $|mF|$ gives an embedding in projective space, then we get a smooth hermitian metric on $F^{\otimes m}$, and the m -th root yields a metric on F such that $\frac{i}{2\pi}\Theta(F) = \frac{1}{m}\Phi_{|mF|}^* \omega_{\text{FS}}$. Conversely, the Kodaira embedding theorem [Kod54] tells us that every positive line bundle F is ample (see Exercise 5.14 for a straightforward analytic proof of the Kodaira embedding theorem).

4. Bochner Technique and Vanishing Theorems

We first recall briefly a few basic facts of Hodge theory. Assume for the moment that M is a differentiable manifold equipped with a riemannian metric $g = \sum g_{ij} dx_i \otimes dx_j$. Given a q -form u on M with values in F , we consider the global L^2 norm

$$\|u\|^2 = \int_M |u(x)|^2 dV_g(x)$$

where $|u|$ is the pointwise hermitian norm and dV_g is the riemannian volume form. The Laplace-Beltrami operator associated to the connection D is

$$\Delta = DD^* + D^*D$$

where

$$D^* : C^\infty(M, \Lambda^q T_M^* \otimes F) \rightarrow C^\infty(M, \Lambda^{q-1} T_M^* \otimes F)$$

is the (formal) adjoint of D with respect to the L^2 inner product. Assume that M is compact. Since

$$\Delta : C^\infty(M, \Lambda^q T_M^* \otimes F) \rightarrow C^\infty(M, \Lambda^q T_M^* \otimes F)$$

is a self-adjoint elliptic operator in each degree, standard results of PDE theory show that there is an orthogonal decomposition

$$C^\infty(M, \Lambda^q T_M^* \otimes F) = \mathcal{H}^q(M, F) \oplus \text{Im } \Delta$$

where $\mathcal{H}^q(M, F) = \text{Ker } \Delta$ is the space of harmonic forms of degree q ; $\mathcal{H}^q(M, F)$ is a finite dimensional space. Assume moreover that the connection D is *integrable*, i.e. that $D^2 = 0$. It is then easy to check that there is an orthogonal direct sum

$$\operatorname{Im} \Delta = \operatorname{Im} D \oplus \operatorname{Im} D^*,$$

indeed $\langle Du, D^*v \rangle = \langle D^2u, v \rangle = 0$ for all u, v . Hence we get an orthogonal decomposition

$$C^\infty(M, \Lambda^q T_M^* \otimes F) = \mathcal{H}^q(M, F) \oplus \operatorname{Im} D \oplus \operatorname{Im} D^*,$$

and $\operatorname{Ker} \Delta$ is precisely equal to $\mathcal{H}^q(M, F) \oplus \operatorname{Im} D$. Especially, the q -th cohomology group $\operatorname{Ker} \Delta / \operatorname{Im} \Delta$ is isomorphic to $\mathcal{H}^q(M, F)$. All this can be applied for example in the case of the De Rham groups $H_{\text{DR}}^q(M, \mathbb{C})$, taking F to be the trivial bundle $F = M \times \mathbb{C}$ (notice, however, that a nontrivial bundle F usually does not admit any integrable connection):

(4.1) Hodge Fundamental Theorem. *If M is a compact riemannian manifold, there is an isomorphism*

$$H_{\text{DR}}^q(M, \mathbb{C}) \simeq \mathcal{H}^q(M, \mathbb{C})$$

from De Rham cohomology groups onto spaces of harmonic forms. $\square$

A rather important consequence of the Hodge fundamental theorem is a proof of the *Poincaré duality theorem*. Assume that the Riemannian manifold (M, g) is oriented. Then there is a (conjugate linear) Hodge star operator

$$\star : \Lambda^q T_M^* \otimes \mathbb{C} \rightarrow \Lambda^{m-q} T_M^* \otimes \mathbb{C}, \quad m = \dim_{\mathbb{R}} M$$

defined by $u \wedge \star v = \langle u, v \rangle dV_g$ for any two complex valued q -forms u, v . A standard computation shows that $\star$ commutes with Δ , hence $\star u$ is harmonic if and only if u is. This implies that the natural pairing

$$(4.2) \quad H_{\text{DR}}^q(M, \mathbb{C}) \times H_{\text{DR}}^{m-q}(M, \mathbb{C}), \quad (\{u\}, \{v\}) \mapsto \int_M u \wedge v$$

is a nondegenerate duality, the dual of a class $\{u\}$ represented by a harmonic form being $\{\star u\}$.

Let us now suppose that X is a compact complex manifold equipped with a hermitian metric $\omega = \sum \omega_{j\bar{k}} dz_j \wedge d\bar{z}_k$. Let F be a holomorphic vector bundle on X equipped with a hermitian metric, and let $D = D' + D''$ be its Chern curvature form. All that we said above for the Laplace-Beltrami operator Δ still applies to the complex Laplace operators

$$\Delta' = D' D'^* + D'^* D', \quad \Delta'' = D'' D''^* + D''^* D'',$$

with the great advantage that we always have $D'^2 = D''^2 = 0$. Especially, if X is a compact complex manifold, there are isomorphisms

$$(4.3) \quad H^{p,q}(X, F) \simeq \mathcal{H}^{p,q}(X, F)$$

between Dolbeault cohomology groups $H^{p,q}(X, F)$ and spaces $\mathcal{H}^{p,q}(X, F)$ of Δ'' -harmonic forms of bidegree (p, q) with values in F . Now, there is a generalized Hodge star operator

$$\star : \Lambda^{p,q} T_X^* \otimes F \rightarrow \Lambda^{n-p, n-q} T_X^* \otimes F^*, \quad n = \dim_{\mathbb{C}} X,$$

such that $u \wedge \star v = \langle u, v \rangle dV_g$, when the for any two F -valued (p, q) -forms, when the wedge product $u \wedge \star v$ is combined with the pairing $F \times F^* \rightarrow \mathbb{C}$. This leads to the *Serre duality theorem* [Ser55]: the bilinear pairing

$$(4.4) \quad H^{p,q}(X, F) \times H^{n-p, n-q}(X, F^*), \quad (\{u\}, \{v\}) \mapsto \int_X u \wedge v$$

is a nondegenerate duality. Combining this with the Dolbeault isomorphism, we may restate the result in the form of the duality formula

$$(4.4') \quad H^q(X, \Omega_X^p \otimes \mathcal{O}(F))^* \simeq H^{n-q}(X, \Omega_X^{n-p} \otimes \mathcal{O}(F^*)).$$

We now proceed to explain the basic ideas of the Bochner technique used to prove vanishing theorems. Great simplifications occur in the computations if the hermitian metric on X is supposed to be *Kähler*, i.e. if the associated *fundamental* $(1, 1)$ -form

$$\omega = i \sum \omega_{j\bar{k}} dz_j \wedge d\bar{z}_k$$

satisfies $d\omega = 0$. It can be easily shown that ω is Kähler if and only if there are holomorphic coordinates $(z_1, \dots, z_n)$ centered at any point $x_0 \in X$ such that the matrix of coefficients $(\omega_{j\bar{k}})$ is tangent to identity at order 2, i.e.

$$\omega_{j\bar{k}}(z) = \delta_{j\bar{k}} + O(|z|^2) \quad \text{at } x_0.$$

It follows that all order 1 operators D, D', D'' and their adjoints D^*, D'^*, D''^* admit at x_0 the same expansion as the analogous operators obtained when all hermitian metrics on X or F are constant. From this, the basic commutation relations of Kähler geometry can be checked. If A, B are differential operators acting on the algebra $C^\infty(X, \Lambda^{\bullet, \bullet} T_X^* \otimes F)$, their graded commutator (or graded Lie bracket) is defined by

$$[A, B] = AB - (-1)^{ab} BA$$

where a, b are the degrees of A and B respectively. If C is another endomorphism of degree c , the following purely formal *Jacobi identity* holds:

$$(-1)^{ca} [A, [B, C]] + (-1)^{ab} [B, [C, A]] + (-1)^{bc} [C, [A, B]] = 0.$$

(4.5) Basic commutation relations. *Let (X, ω) be a Kähler manifold and let L be the operators defined by $Lu = \omega \wedge u$ and $\Lambda = L^*$. Then*

$$\begin{aligned} [D''^*, L] &= iD', & [D'^*, L] &= -iD'', \\ [\Lambda, D''] &= -iD'^*, & [\Lambda, D'] &= iD''^*. \end{aligned}$$

Proof (sketch). The first step is to check the identity $[d''^*, L] = id'$ for constant metrics on $X = \mathbb{C}^n$ and $F = X \times \mathbb{C}$, by a brute force calculation. All three other identities follow by taking conjugates or adjoints. The case of variable metrics follows by looking at Taylor expansions up to order 1. $\square$

(4.6) Bochner-Kodaira-Nakano identity. *If (X, ω) is Kähler, the complex Laplace operators Δ' and Δ'' acting on F -valued forms satisfy the identity*

$$\Delta'' = \Delta' + [i\Theta(F), \Lambda].$$

Proof. The last equality in (4.5) yields $D''^* = -i[\Lambda, D']$, hence

$$\Delta'' = [D'', \delta''] = -i[D'', [\Lambda, D']].$$

By the Jacobi identity we get

$$[D'', [\Lambda, D']] = [\Lambda, [D', D'']] + [D', [D'', \Lambda]] = [\Lambda, \Theta(F)] + i[D', D'^*],$$

taking into account that $[D', D''] = D^2 = \Theta(F)$. The formula follows. $\square$

Assume that X is compact and that $u \in C^\infty(X, \Lambda^{p,q} T^* X \otimes F)$ is an arbitrary (p, q) -form. An integration by parts yields

$$\langle \Delta' u, u \rangle = \|D' u\|^2 + \|D'^* u\|^2 \geq 0$$

and similarly for Δ'' , hence we get the basic a priori inequality

$$(4.7) \quad \|D'' u\|^2 + \|D''^* u\|^2 \geq \int_X \langle [i\Theta(F), \Lambda] u, u \rangle dV_\omega.$$

This inequality is known as the *Bochner-Kodaira-Nakano* inequality (see [Boc48], [Kod53], [Nak55]). When u is Δ'' -harmonic, we get

$$\int_X (\langle [i\Theta(F), \Lambda] u, u \rangle + \langle T_\omega u, u \rangle) dV \leq 0.$$

If the hermitian operator $[i\Theta(F), \Lambda]$ acting on $\Lambda^{p,q} T_X^* \otimes F$ is positive on each fiber, we infer that u must be zero, hence

$$H^{p,q}(X, F) = \mathcal{H}^{p,q}(X, F) = 0$$

by Hodge theory. The main point is thus to compute the curvature form $\Theta(F)$ and find sufficient conditions under which the operator $[i\Theta(F), \Lambda]$ is positive definite. Elementary (but somewhat tedious) calculations yield the following formulae: if the curvature of F is written as in (3.8) and $u = \sum u_{J,K,\lambda} dz_I \wedge d\bar{z}_J \otimes e_\lambda$, $|J| = p$, $|K| = q$, $1 \leq \lambda \leq r$ is a (p, q) -form with values in F , then

$$(4.8) \quad \begin{aligned} \langle [i\Theta(F), \Lambda] u, u \rangle = & \sum_{j,k,\lambda,\mu,J,S} c_{jk\lambda\mu} u_{J,jS,\lambda} \overline{u_{j,kS,\mu}} \\ & + \sum_{j,k,\lambda,\mu,R,K} c_{jk\lambda\mu} u_{kR,K,\lambda} \overline{u_{jR,K,\mu}} \\ & - \sum_{j,\lambda,\mu,J,K} c_{jj\lambda\mu} u_{J,K,\lambda} \overline{u_{j,K,\mu}}, \end{aligned}$$

where the sum is extended to all indices $1 \leq j, k \leq n$, $1 \leq \lambda, \mu \leq r$ and multiindices $|R| = p - 1$, $|S| = q - 1$ (here the notation $u_{JK\lambda}$ is extended to non necessarily increasing multiindices by making it alternate with respect to permutations). It is usually hard to decide the sign of the curvature term (4.8), except in some special cases.

The easiest case is when $p = n$. Then all terms in the second summation of (4.8) must have $j = k$ and $R = \{1, \dots, n\} \setminus \{j\}$, therefore the second and third summations are equal. It follows that $[i\Theta(F), A]$ is positive on (n, q) -forms under the assumption that F is positive in the sense of Nakano. In this case X is automatically Kähler since

$$\omega = \text{Tr}_F(i\Theta(F)) = i \sum_{j,k,\lambda} c_{jk\lambda\lambda} dz_j \wedge d\bar{z}_k = i\Theta(\det F)$$

is a Kähler metric.

(4.9) Nakano vanishing theorem (1955). *Let X be a compact complex manifold and let F be a Nakano positive vector bundle on X . Then*

$$H^{n,q}(X, F) = H^q(X, K_X \otimes F) = 0 \quad \text{for every } q \geq 1. \quad \square$$

Another tractable case is the case where F is a line bundle ($r = 1$). Indeed, at each point $x \in X$, we may then choose a coordinate system which diagonalizes simultaneously the hermitians forms $\omega(x)$ and $i\Theta(F)(x)$, in such a way that

$$\omega(x) = i \sum_{1 \leq j \leq n} dz_j \wedge d\bar{z}_j, \quad i\Theta(F)(x) = i \sum_{1 \leq j \leq n} \gamma_j dz_j \wedge d\bar{z}_j$$

with $\gamma_1 \leq \dots \leq \gamma_n$. The curvature eigenvalues $\gamma_j = \gamma_j(x)$ are then uniquely defined and depend continuously on x . With our previous notation, we have $\gamma_j = c_{jj11}$ and all other coefficients $c_{jk\lambda\mu}$ are zero. For any (p, q) -form $u = \sum u_{JK} dz_J \wedge d\bar{z}_K \otimes e_1$, this gives

$$\begin{aligned} \langle [i\Theta(F), A]u, u \rangle &= \sum_{|J|=p, |K|=q} \left(\sum_{j \in J} \gamma_j + \sum_{j \in K} \gamma_j - \sum_{1 \leq j \leq n} \gamma_j \right) |u_{JK}|^2 \\ (4.10) \quad &\geq (\gamma_1 + \dots + \gamma_q - \gamma_{n-p+1} - \dots - \gamma_n) |u|^2. \end{aligned}$$

Assume that $i\Theta(F)$ is positive. It is then natural to make the special choice $\omega = i\Theta(F)$ for the Kähler metric. Then $\gamma_j = 1$ for $j = 1, 2, \dots, n$ and we obtain $\langle [i\Theta(F), A]u, u \rangle = (p + q - n) |u|^2$. As a consequence:

(4.11) Akizuki-Kodaira-Nakano vanishing theorem ([AN54]). *If F is a positive line bundle on a compact complex manifold X , then*

$$H^{p,q}(X, F) = H^q(X, \Omega_X^p \otimes F) = 0 \quad \text{for } p + q \geq n + 1. \quad \square$$

More generally, if F is a Griffiths positive (or ample) vector bundle of rank $r \geq 1$, Le Potier [LP75] proved that $H^{p,q}(X, F) = 0$ for $p + q \geq n + r$. The proof is not a direct consequence of the Bochner technique. A rather easy proof has been found by M. Schneider [Sch74], using the Leray spectral sequence associated to the projectivized bundle projection $\mathbb{P}(F) \rightarrow X$.

(4.12) Exercise. It is important for various applications to obtain vanishing theorems which are also valid in the case of semipositive line bundles. The easiest case is the following result of Girbau [Gir76]: let (X, ω) be compact Kähler; assume that F is a line bundle and that $i\Theta(F) \geq 0$ has at least $n - k$ positive eigenvalues at each point, for some integer $k \geq 0$; show that $H^{p,q}(X, F) = 0$ for $p + q \geq n + k + 1$. *Hint:* use the Kähler metric $\omega_\varepsilon = i\Theta(F) + \varepsilon\omega$ with $\varepsilon > 0$ small.

A stronger and more natural “algebraic version” of this result has been obtained by Sommese [Som78]: define F to be k -ample if some multiple mF is such that the canonical map

$$\Phi_{|mF|} : X \setminus B_{|mF|} \rightarrow \mathbb{P}^{N-1}$$

has at most k -dimensional fibers and $\dim B_{|mF|} \leq k$. If X is projective and F is k -ample, show that $H^{p,q}(X, F) = 0$ for $p + q \geq n + k + 1$.

Hint: prove the dual result $H^{p,q}(X, F^{-1}) = 0$ for $p + q \leq n - k - 1$ by induction on k . First show that F 0-ample $\Rightarrow F$ positive; then use hyperplane sections $Y \subset X$ to prove the induction step, thanks to the exact sequences

$$\begin{aligned} 0 \longrightarrow \Omega_X^p \otimes F^{-1} \otimes \mathcal{O}(-Y) &\longrightarrow \Omega_X^p \otimes F^{-1} \longrightarrow (\Omega_X^p \otimes F^{-1})|_Y \longrightarrow 0, \\ 0 \longrightarrow \Omega_Y^{p-1} \otimes F|_Y^{-1} &\longrightarrow (\Omega_X^p \otimes F^{-1})|_Y \longrightarrow \Omega_Y^p \otimes F|_Y^{-1} \longrightarrow 0. \end{aligned} \quad \square$$

5. L^2 Estimates and Existence Theorems

The starting point is the following L^2 existence theorem, which is essentially due to Hörmander [Hör65, 66], and Andreotti-Vesentini [AV65]. We will only outline the main ideas, referring e.g. to [Dem82b] for a detailed exposition of the technical situation considered here.

(5.1) Theorem. *Let (X, ω) be a Kähler manifold. Here X is not necessarily compact, but we assume that the geodesic distance δ_ω is complete on X . Let F be a hermitian vector bundle of rank r over X , and assume that the curvature operator $A = A_{F, \omega}^{p,q} = [i\Theta(F), \Lambda_\omega]$ is positive definite everywhere on $\Lambda^{p,q}T_X^* \otimes F$, $q \geq 1$. Then for any form $g \in L^2(X, \Lambda^{p,q}T_X^* \otimes F)$ satisfying $D''g = 0$ and $\int_X \langle A^{-1}g, g \rangle dV_\omega < +\infty$, there exists $f \in L^2(X, \Lambda^{p,q-1}T_X^* \otimes F)$ such that $D''f = g$ and*

$$\int_X |f|^2 dV_\omega \leq \int_X \langle A^{-1}g, g \rangle dV_\omega.$$

Proof. The assumption that δ_ω is complete implies the existence of cut-off functions ψ_ν with arbitrarily large compact support such that $|d\psi_\nu| \leq 1$ (take ψ_ν to be a function of the distance $x \mapsto \delta_\omega(x_0, x)$, which is an almost everywhere differentiable 1-Lipschitz function, and regularize if necessary). From this, it follows that very form $u \in L^2(X, \Lambda^{p,q}T_X^* \otimes F)$ such that $D''u \in L^2$ and $D''^*u \in L^2$ in the sense of distribution theory is a limit of a sequence of smooth forms u_ν with compact support, in such a way that $u_\nu \rightarrow u$, $D''u_\nu \rightarrow D''u$ and $D''^*u_\nu \rightarrow D''^*u$ in L^2 (just take u_ν to be a regularization of $\psi_\nu u$). As a consequence, the basic a priori

inequality (4.7) extends to arbitrary forms u such that $u, D''u, D''^*u \in L^2$. Now, consider the Hilbert space orthogonal decomposition

$$L^2(X, A^{p,q}T_X^* \otimes F) = \text{Ker } D'' \oplus (\text{Ker } D'')^\perp,$$

observing that $\text{Ker } D''$ is weakly (hence strongly) closed. Let $v = v_1 + v_2$ be the decomposition of a smooth form $v \in \mathcal{D}^{p,q}(X, F)$ with compact support according to this decomposition (v_1, v_2 do not have compact support in general!). Since $(\text{Ker } D'')^\perp \subset \text{Ker } D''^*$ by duality and $g, v_1 \in \text{Ker } D''$ by hypothesis, we get $D''^*v_2 = 0$ and

$$|\langle g, v \rangle|^2 = |\langle g, v_1 \rangle|^2 \leq \int_X \langle A^{-1}g, g \rangle dV_\omega \int_X \langle Av_1, v_1 \rangle dV_\omega$$

thanks to the Cauchy-Schwarz inequality. The a priori inequality (4.7) applied to $u = v_1$ yields

$$\int_X \langle Av_1, v_1 \rangle dV_\omega \leq \|D''v_1\|^2 + \|D''^*v_1\|^2 = \|D''^*v_1\|^2 = \|D''^*v\|^2.$$

Combining both inequalities, we find

$$|\langle g, v \rangle|^2 \leq \left(\int_X \langle A^{-1}g, g \rangle dV_\omega \right) \|D''^*v\|^2$$

for every smooth (p, q) -form v with compact support. This shows that we have a well defined linear form

$$w = D''^*v \longmapsto \langle v, g \rangle, \quad L^2(X, A^{p,q-1}T_X^* \otimes F) \supset D''^*(\mathcal{D}^{p,q}(F)) \longrightarrow \mathbb{C}$$

on the range of D''^* . This linear form is continuous in L^2 norm and has norm $\leq C$ with

$$C = \left(\int_X \langle A^{-1}g, g \rangle dV_\omega \right)^{1/2}.$$

By the Hahn-Banach theorem, there is an element $f \in L^2(X, A^{p,q-1}T_X^* \otimes F)$ with $\|f\| \leq C$, such that $\langle v, g \rangle = \langle D''^*v, f \rangle$ for every v , hence $D''f = g$ in the sense of distributions. The inequality $\|f\| \leq C$ is equivalent to the last estimate in the theorem. $\square$

The above L^2 existence theorem can be applied in the fairly general context of *weakly pseudoconvex* manifolds. By this, we mean a complex manifold X such that there exists a smooth psh exhaustion function ψ on X (ψ is said to be an exhaustion if for every $c > 0$ the sublevel set $X_c = \psi^{-1}(c)$ is relatively compact, i.e. $\psi(z)$ tends to $+\infty$ when z is taken outside larger and larger compact subsets of X). In particular, every compact complex manifold X is weakly pseudoconvex (take $\psi = 0$), as well as every Stein manifold, e.g. affine algebraic submanifolds of $\mathbb{C}^N$ (take $\psi(z) = |z|^2$), open balls $X = B(z_0, r)$ (take $\psi(z) = 1/(r - |z - z_0|^2)$), convex open subsets, etc. Now, a basic observation is that every weakly pseudoconvex Kähler manifold (X, ω) carries a *complete* Kähler metric: let $\psi \geq 0$ be a psh exhaustion function and set

$$\omega_\varepsilon = \omega + \varepsilon \text{id}' d'' \psi^2 = \omega + 2\varepsilon(2i\psi d' d'' \psi + \text{id}' \psi \wedge d'' \psi).$$

Then $|d\psi|_{\omega_\varepsilon} \leq 1/\varepsilon$ and $|\psi(x) - \psi(y)| \leq \varepsilon^{-1} \delta_{\omega_\varepsilon}(x, y)$. It follows easily from this estimate that the geodesic balls are relatively compact, hence $\delta_{\omega_\varepsilon}$ is complete for every $\varepsilon > 0$. Therefore, the L^2 existence theorem can be applied to each Kähler metric ω_ε , and by passing to the limit it can even be applied to the non necessarily complete metric ω . An important special case is the following

(5.2) Theorem. *Let (X, ω) be a Kähler manifold, $\dim X = n$. Assume that X is weakly pseudoconvex. Let F be a hermitian line bundle and let*

$$\gamma_1(x) \leq \dots \leq \gamma_n(x)$$

be the curvature eigenvalues (i.e. the eigenvalues of $i\Theta(F)$ with respect to the metric ω) at every point. Assume that the curvature is positive, i.e. $\gamma_1 > 0$ everywhere. Then for any form $g \in L^2(X, \Lambda^{n,q} T_X^ \otimes F)$ satisfying $D''g = 0$ and $\int_X (\gamma_1 + \dots + \gamma_q)^{-1} |g|^2 dV_\omega < +\infty$, there exists $f \in L^2(X, \Lambda^{p,q-1} T_X^* \otimes F)$ such that $D''f = g$ and*

$$\int_X |f|^2 dV_\omega \leq \int_X (\gamma_1 + \dots + \gamma_q)^{-1} |g|^2 dV_\omega.$$

Proof. Indeed, for $p = n$, Formula 4.10 shows that

$$\langle Au, u \rangle \geq (\gamma_1 + \dots + \gamma_q) |u|^2,$$

hence $\langle A^{-1}u, u \rangle \geq (\gamma_1 + \dots + \gamma_q)^{-1} |u|^2$. □

An important observation is that the above theorem still applies when the hermitian metric on F is a singular metric with positive curvature in the sense of currents. In fact, by standard regularization techniques (convolution of psh functions by smoothing kernels), the metric can be made smooth and the solutions obtained by (5.1) or (5.2) for the smooth metrics have limits satisfying the desired estimates. Especially, we get the following

(5.3) Corollary. *Let (X, ω) be a Kähler manifold, $\dim X = n$. Assume that X is weakly pseudoconvex. Let F be a holomorphic line bundle equipped with a singular metric whose local weights are denoted $\varphi \in L^1_{\text{loc}}$. Suppose that*

$$i\Theta(F) = 2\text{id}' d''\varphi \geq \varepsilon \omega$$

for some $\varepsilon > 0$. Then for any form $g \in L^2(X, \Lambda^{n,q} T_X^ \otimes F)$ satisfying $D''g = 0$, there exists $f \in L^2(X, \Lambda^{p,q-1} T_X^* \otimes F)$ such that $D''f = g$ and*

$$\int_X |f|^2 e^{-2\varphi} dV_\omega \leq \frac{1}{q\varepsilon} \int_X |g|^2 e^{-2\varphi} dV_\omega. \quad \square$$

Here we denoted somewhat incorrectly the metric by $|f|^2 e^{-2\varphi}$, as if the weight φ was globally defined on X (of course, this is so only if F is globally trivial). We will use this notation anyway, because it clearly describes the dependence of the L^2 norm on the psh weights.

We now introduce the concept of *multiplier ideal sheaf*, following A. Nadel [Nad89]. The main idea actually goes back to the fundamental works of Bombieri [Bom70] and H. Skoda [Sko72a].

(5.4) Definition. Let φ be a psh function on an open subset $\Omega \subset X$; to φ is associated the ideal subsheaf $\mathcal{I}(\varphi) \subset \mathcal{O}_\Omega$ of germs of holomorphic functions $f \in \mathcal{O}_{\Omega,x}$ such that $|f|^2 e^{-2\varphi}$ is integrable with respect to the Lebesgue measure in some local coordinates near x .

The zero variety $V(\mathcal{I}(\varphi))$ is thus the set of points in a neighborhood of which $e^{-2\varphi}$ is non integrable. Of course, such points occur only if φ has logarithmic poles. This is made precise as follows.

(5.5) Definition. A psh function φ is said to have a logarithmic pole of coefficient γ at a point $x \in X$ if the Lelong number

$$\nu(\varphi, x) := \liminf_{z \rightarrow x} \frac{\varphi(z)}{\log |z - x|}$$

is non zero and if $\nu(\varphi, x) = \gamma$.

(5.6) Lemma (Skoda [Sko72a]). Let φ be a psh function on an open set Ω and let $x \in \Omega$.

- a) If $\nu(\varphi, x) < 1$, then $e^{-2\varphi}$ is integrable in a neighborhood of x , in particular $\mathcal{I}(\varphi)_x = \mathcal{O}_{\Omega,x}$.
- b) If $\nu(\varphi, x) \geq n + s$ for some integer $s \geq 0$, then $e^{-2\varphi} \geq C|z - x|^{-2n-2s}$ in a neighborhood of x and $\mathcal{I}(\varphi)_x \subset \mathfrak{m}_{\Omega,x}^{s+1}$, where $\mathfrak{m}_{\Omega,x}$ is the maximal ideal of $\mathcal{O}_{\Omega,x}$.
- c) The zero variety $V(\mathcal{I}(\varphi))$ of $\mathcal{I}(\varphi)$ satisfies

$$E_n(\varphi) \subset V(\mathcal{I}(\varphi)) \subset E_1(\varphi)$$

where $E_c(\varphi) = \{x \in X; \nu(\varphi, x) \geq c\}$ is the c -sublevel set of Lelong numbers of φ .

Proof. a) Set $\Theta = dd^c \varphi$ and $\gamma = \nu(\Theta, x) = \nu(\varphi, x)$. Let χ be a cut-off function will support in a small ball $B(x, r)$, equal to 1 in $B(x, r/2)$. As $(dd^c \log |z|)^n = \delta_0$, we get

$$\begin{aligned} \varphi(z) &= \int_{B(x,r)} \chi(\zeta) \varphi(\zeta) (dd^c \log |\zeta - z|)^n \\ &= \int_{B(x,r)} dd^c(\chi(\zeta) \varphi(\zeta)) \wedge \log |\zeta - z| (dd^c \log |\zeta - z|)^{n-1} \end{aligned}$$

for $z \in B(x, r/2)$. Expanding $dd^c(\chi \varphi)$ and observing that $d\chi = dd^c \chi = 0$ on $B(x, r/2)$, we find

$$\varphi(z) = \int_{B(x,r)} \chi(\zeta) \Theta(\zeta) \wedge \log |\zeta - z| (dd^c \log |\zeta - z|)^{n-1} + \text{smooth terms}$$

on $B(x, r/2)$. Fix r so small that

$$\int_{B(x, r)} \chi(\zeta) \Theta(\zeta) \wedge (dd^c \log |\zeta - x|)^{n-1} \leq \nu(\Theta, x, r) < 1.$$

By continuity, there exists $\delta, \varepsilon > 0$ such that

$$I(z) := \int_{B(x, r)} \chi(\zeta) \Theta(\zeta) \wedge (dd^c \log |\zeta - z|)^{n-1} \leq 1 - \delta$$

for all $z \in B(x, \varepsilon)$. Applying Jensen's convexity inequality to the probability measure

$$d\mu_z(\zeta) = I(z)^{-1} \chi(\zeta) \Theta(\zeta) \wedge (dd^c \log |\zeta - z|)^{n-1},$$

we find

$$\begin{aligned} -\varphi(z) &= \int_{B(x, r)} I(z) \log |\zeta - z|^{-1} d\mu_z(\zeta) + O(1) \implies \\ e^{-2\varphi(z)} &\leq C \int_{B(x, r)} |\zeta - z|^{-2I(z)} d\mu_z(\zeta). \end{aligned}$$

As

$$d\mu_z(\zeta) \leq C_1 |\zeta - z|^{-(2n-2)} \Theta(\zeta) \wedge (dd^c |\zeta|^2)^{n-1} = C_2 |\zeta - z|^{-(2n-2)} d\sigma_\Theta(\zeta),$$

we get

$$e^{-2\varphi(z)} \leq C_3 \int_{B(x, r)} |\zeta - z|^{-2(1-\delta)-(2n-2)} d\sigma_\Theta(\zeta),$$

and the Fubini theorem implies that $e^{-2\varphi(z)}$ is integrable on a neighborhood of x .

b) If $\nu(\varphi, x) = \gamma$, the convexity properties of psh functions, namely, the convexity of $\log r \mapsto \sup_{|z-x|=r} \varphi(z)$ implies that

$$\varphi(z) \leq \gamma \log |z - x|/r_0 + M,$$

where M is the supremum on $B(x, r_0)$. Hence there exists a constant $C > 0$ such that $e^{-2\varphi(z)} \geq C |z - x|^{-2\gamma}$ in a neighborhood of x . The desired result follows from the identity

$$\int_{B(0, r_0)} \frac{|\sum a_\alpha z^\alpha|^2}{|z|^{2\gamma}} dV(z) = \text{Const} \int_0^{r_0} \left(\sum |a_\alpha|^2 r^{2|\alpha|} \right) r^{2n-1-2\gamma} dr,$$

which is an easy consequence of Parseval's formula. In fact, if γ has integral part $[\gamma] = n + s$, the integral converges if and only if $a_\alpha = 0$ for $|\alpha| \leq s$.

c) is just a simple formal consequence of a) and b). $\square$

(5.7) Proposition ([Nad89]). *For any psh function φ on $\Omega \subset X$, the sheaf $\mathcal{I}(\varphi)$ is a coherent sheaf of ideals over Ω .*

Proof. Since the result is local, we may assume that Ω is the unit ball in $\mathbb{C}^n$. Let E be the set of all holomorphic functions f on Ω such that $\int_\Omega |f|^2 e^{-2\varphi} d\lambda < +\infty$. By the strong noetherian property of coherent sheaves, the set E generates a coherent

ideal sheaf $\mathcal{I} \subset \mathcal{O}_\Omega$. It is clear that $\mathcal{I} \subset \mathcal{I}(\varphi)$; in order to prove the equality, we need only check that $\mathcal{I}_x + \mathcal{I}(\varphi)_x \cap \mathfrak{m}_{\Omega,x}^{s+1} = \mathcal{I}(\varphi)_x$ for every integer s , in view of the Krull lemma. Let $f \in \mathcal{I}(\varphi)_x$ be defined in a neighborhood V of x and let θ be a cut-off function with support in V such that $\theta = 1$ in a neighborhood of x . We solve the equation $d''u = g := d''(\theta f)$ by means of Hörmander's L^2 estimates 5.3, where F is the trivial line bundle $\Omega \times \mathbb{C}$ equipped with the strictly psh weight

$$\tilde{\varphi}(z) = \varphi(z) + (n+s)\log|z-x| + |z|^2.$$

We get a solution u such that $\int_\Omega |u|^2 e^{-2\varphi} |z-x|^{-2(n+s)} d\lambda < \infty$, thus $F = \theta f - u$ is holomorphic, $F \in E$ and $f_x - F_x = u_x \in \mathcal{I}(\varphi)_x \cap \mathfrak{m}_{\Omega,x}^{s+1}$. This proves our contention. $\square$

The multiplier ideal sheaves satisfy the following basic functoriality property with respect to direct images of sheaves by modifications.

(5.8) Proposition. *Let $\mu : X' \rightarrow X$ be a modification of non singular complex manifolds (i.e. a proper generically 1:1 holomorphic map), and let φ be a psh function on X . Then*

$$\mu_* (\mathcal{O}(K_{X'}) \otimes \mathcal{I}(\varphi \circ \mu)) = \mathcal{O}(K_X) \otimes \mathcal{I}(\varphi).$$

Proof. Let $n = \dim X = \dim X'$ and let $S \subset X$ be an analytic set such that $\mu : X' \setminus S' \rightarrow X \setminus S$ is a biholomorphism. By definition of multiplier ideal sheaves, $\mathcal{O}(K_X) \otimes \mathcal{I}(\varphi)$ is just the sheaf of holomorphic n -forms f on open sets $U \subset X$ such that $i^{n^2} f \wedge \bar{f} e^{-2\varphi} \in L_{\text{loc}}^1(U)$. Since φ is locally bounded from above, we may even consider forms f which are a priori defined only on $U \setminus S$, because f will be in $L_{\text{loc}}^2(U)$ and therefore will automatically extend through S . The change of variable formula yields

$$\int_U i^{n^2} f \wedge \bar{f} e^{-2\varphi} = \int_{\mu^{-1}(U)} i^{n^2} \mu^* f \wedge \overline{\mu^* f} e^{-2\varphi \circ \mu},$$

hence $f \in \Gamma(U, \mathcal{O}(K_X) \otimes \mathcal{I}(\varphi))$ iff $\mu^* f \in \Gamma(\mu^{-1}(U), \mathcal{O}(K_{X'}) \otimes \mathcal{I}(\varphi \circ \mu))$. Proposition 5.8 is proved. $\square$

(5.9) Remark. If φ has analytic singularities (according to Definition 1.10), the computation of $\mathcal{I}(\varphi)$ can be reduced to a purely algebraic problem.

The first observation is that $\mathcal{I}(\varphi)$ can be computed easily if φ has the form $\varphi = \sum \alpha_j \log |g_j|$ where $D_j = g_j^{-1}(0)$ are nonsingular irreducible divisors with normal crossings. Then $\mathcal{I}(\varphi)$ is the sheaf of functions h on open sets $U \subset X$ such that

$$\int_U |h|^2 \prod |g_j|^{-2\alpha_j} dV < +\infty.$$

Since locally the g_j can be taken to be coordinate functions from a local coordinate system $(z_1, \dots, z_n)$, the condition is that h is divisible by $\prod g_j^{m_j}$ where $m_j - \alpha_j > -1$ for each j , i.e. $m_j \geq \lfloor \alpha_j \rfloor$ (integer part). Hence

$$\mathcal{I}(\varphi) = \mathcal{O}(-\lfloor D \rfloor) = \mathcal{O}(-\sum \lfloor \alpha_j \rfloor D_j)$$

where $\lfloor D \rfloor$ denotes the integral part of the $\mathbb{Q}$ -divisor $D = \sum \alpha_j D_j$.

Now, consider the general case of analytic singularities and suppose that $\varphi \sim \frac{\alpha}{2} \log(|f_1|^2 + \dots + |f_N|^2)$ near the poles. By the remarks after Definition 1.10, we may assume that the (f_j) are generators of the integrally closed ideal sheaf $\mathcal{J} = \mathcal{J}(\varphi/\alpha)$, defined as the sheaf of holomorphic functions h such that $|h| \leq C \exp(\varphi/\alpha)$. In this case, the computation is made as follows (see also L. Bonavero's work [Bon93], where similar ideas are used in connection with "singular" holomorphic Morse inequalities).

First, one computes a smooth modification $\mu: \tilde{X} \rightarrow X$ of X such that $\mu^*\mathcal{J}$ is an invertible sheaf $\mathcal{O}(-D)$ associated with a normal crossing divisor $D = \sum \lambda_j D_j$, where (D_j) are the components of the exceptional divisor of $\tilde{X}$ (take the blow-up X' of X with respect to the ideal $\mathcal{J}$ so that the pull-back of $\mathcal{J}$ to X' becomes an invertible sheaf $\mathcal{O}(-D')$, then blow up again by Hironaka [Hir64] to make X' smooth and D' have normal crossings). Now, we have $K_{\tilde{X}} = \mu^*K_X + R$ where $R = \sum \rho_j D_j$ is the zero divisor of the Jacobian function J_μ of the blow-up map. By the direct image formula 5.8, we get

$$\mathcal{I}(\varphi) = \mu_*(\mathcal{O}(K_{\tilde{X}} - \mu^*K_X) \otimes \mathcal{I}(\varphi \circ \mu)) = \mu_*(\mathcal{O}(R) \otimes \mathcal{I}(\varphi \circ \mu)).$$

Now, $(f_j \circ \mu)$ are generators of the ideal $\mathcal{O}(-D)$, hence

$$\varphi \circ \mu \sim \alpha \sum \lambda_j \log |g_j|$$

where g_j are local generators of $\mathcal{O}(-D_j)$. We are thus reduced to computing multiplier ideal sheaves in the case where the poles are given by a $\mathbb{Q}$ -divisor with normal crossings $\sum \alpha \lambda_j D_j$. We obtain $\mathcal{I}(\varphi \circ \mu) = \mathcal{O}(-\sum \lfloor \alpha \lambda_j \rfloor D_j)$, hence

$$\mathcal{I}(\varphi) = \mu_* \mathcal{O}_{\tilde{X}}(-\sum (\rho_j - \lfloor \alpha \lambda_j \rfloor) D_j). \quad \square$$

(5.10) Exercise. Compute the multiplier ideal sheaf $\mathcal{I}(\varphi)$ associated with $\varphi = \log(|z_1|^{\alpha_1} + \dots + |z_p|^{\alpha_p})$ for arbitrary real numbers $\alpha_j > 0$.

Hint: using Parseval's formula and polar coordinates $z_j = r_j e^{i\theta_j}$, show that the problem is equivalent to determining for which p -tuples $(\beta_1, \dots, \beta_p) \in \mathbb{N}^p$ the integral

$$\int_{[0,1]^p} \frac{r_1^{2\beta_1} \dots r_p^{2\beta_p} r_1 dr_1 \dots r_p dr_p}{r_1^{2\alpha_1} + \dots + r_p^{2\alpha_p}} = \int_{[0,1]^p} \frac{t_1^{(\beta_1+1)/\alpha_1} \dots t_p^{(\beta_p+1)/\alpha_p}}{t_1 + \dots + t_p} \frac{dt_1}{t_1} \dots \frac{dt_p}{t_p}$$

is convergent. Conclude from this that $\mathcal{I}(\varphi)$ is generated by the monomials $z_1^{\beta_1} \dots z_p^{\beta_p}$ such that $\sum (\beta_p + 1)/\alpha_p > 1$. (This exercise shows that the analytic definition of $\mathcal{I}(\varphi)$ is sometimes also quite convenient for computations). $\square$

Let F be a line bundle over X with a singular metric h of curvature current $\Theta_h(F)$. If φ is the weight representing the metric in an open set $\Omega \subset X$, the ideal sheaf $\mathcal{I}(\varphi)$ is independent of the choice of the trivialization and so it is the restriction to Ω of a global coherent sheaf $\mathcal{I}(h)$ on X . We will sometimes still write $\mathcal{I}(h) = \mathcal{I}(\varphi)$ by abuse of notation. In this context, we have the following fundamental vanishing theorem, which is probably one of the most central results of analytic and algebraic

geometry (as we will see later, it contains the Kawamata-Viehweg vanishing theorem as a special case).

(5.11) Nadel vanishing theorem ([Nad89], [Dem93b]). *Let (X, ω) be a Kähler weakly pseudoconvex manifold, and let F be a holomorphic line bundle over X equipped with a singular hermitian metric h of weight φ . Assume that $i\Theta_h(F) \geq \varepsilon\omega$ for some continuous positive function ε on X . Then*

$$H^q(X, \mathcal{O}(K_X + F) \otimes \mathcal{I}(h)) = 0 \quad \text{for all } q \geq 1.$$

Proof. Let $\mathcal{L}^q$ be the sheaf of germs of (n, q) -forms u with values in F and with measurable coefficients, such that both $|u|^2 e^{-2\varphi}$ and $|d''u|^2 e^{-2\varphi}$ are locally integrable. The d'' operator defines a complex of sheaves $(\mathcal{L}^\bullet, d'')$ which is a resolution of the sheaf $\mathcal{O}(K_X + F) \otimes \mathcal{I}(\varphi)$: indeed, the kernel of d'' in degree 0 consists of all germs of holomorphic n -forms with values in F which satisfy the integrability condition; hence the coefficient function lies in $\mathcal{I}(\varphi)$; the exactness in degree $q \geq 1$ follows from Corollary 5.3 applied on arbitrary small balls. Each sheaf $\mathcal{L}^q$ is a $\mathcal{C}^\infty$ -module, so $\mathcal{L}^\bullet$ is a resolution by acyclic sheaves. Let ψ be a smooth psh exhaustion function on X . Let us apply Corollary 5.3 globally on X , with the original metric of F multiplied by the factor $e^{-\chi \circ \psi}$, where χ is a convex increasing function of arbitrary fast growth at infinity. This factor can be used to ensure the convergence of integrals at infinity. By Corollary 5.3, we conclude that $H^q(\Gamma(X, \mathcal{L}^\bullet)) = 0$ for $q \geq 1$. The theorem follows. $\square$

(5.12) Corollary. *Let (X, ω) , F and φ be as in Theorem 5.11 and let $x_1, \dots, x_N$ be isolated points in the zero variety $V(\mathcal{I}(\varphi))$. Then there is a surjective map*

$$H^0(X, K_X + F) \twoheadrightarrow \bigoplus_{1 \leq j \leq N} \mathcal{O}(K_X + L)_{x_j} \otimes (\mathcal{O}_X / \mathcal{I}(\varphi))_{x_j}.$$

Proof. Consider the long exact sequence of cohomology associated to the short exact sequence $0 \rightarrow \mathcal{I}(\varphi) \rightarrow \mathcal{O}_X \rightarrow \mathcal{O}_X / \mathcal{I}(\varphi) \rightarrow 0$ twisted by $\mathcal{O}(K_X + F)$, and apply Theorem 5.11 to obtain the vanishing of the first H^1 group. The asserted surjectivity property follows. $\square$

(5.13) Corollary. *Let (X, ω) , F and φ be as in Theorem 5.11 and suppose that the weight function φ is such that $\nu(\varphi, x) \geq n + s$ at some point $x \in X$ which is an isolated point of $E_1(\varphi)$. Then $H^0(X, K_X + F)$ generates all s -jets at x .*

Proof. The assumption is that $\nu(\varphi, y) < 1$ for y near x , $y \neq x$. By Skoda's lemma 5.6 b), we conclude that $e^{-2\varphi}$ is integrable at all such points y , hence $\mathcal{I}(\varphi)_y = \mathcal{O}_{X,y}$, whilst $\mathcal{I}(\varphi)_x \subset \mathfrak{m}_{X,x}^{s+1}$ by 5.6 a). Corollary 5.13 is thus a special case of 5.12. $\square$

The philosophy of these results (which can be seen as generalizations of the Hörmander-Bombieri-Skoda theorem [Bom70], [Sko72a, 75]) is that the problem of

constructing holomorphic sections of $K_X + F$ can be solved by constructing suitable hermitian metrics on F such that the weight φ has isolated poles at given points x_j .

(5.14) Exercise. Assume that X is compact and that L is a positive line bundle on X . Let $\{x_1, \dots, x_N\}$ be a finite set. Show that there are constants $a, b \geq 0$ depending only on L and N such that $H^0(X, mL)$ generates jets of any order s at all points x_j for $m \geq as + b$.

Hint: Apply Corollary 5.12 to $F = -K_X + mL$, with a singular metric on L of the form $h = h_0 e^{-\varepsilon \psi}$, where h_0 is smooth of positive curvature, $\varepsilon > 0$ small and $\psi(z) \sim \log |z - x_j|$ in a neighborhood of x_j .

Derive the Kodaira embedding theorem from the above result:

(5.15) Theorem (Kodaira). *If L is a line bundle on a compact complex manifold, then L is ample if and only if L is positive.* $\square$

(5.16) Exercise (solution of the Levi problem). Show that the following two properties are equivalent.

- a) X is strongly pseudoconvex, i.e. X admits a strongly psh exhaustion function.
- b) X is Stein, i.e. the global holomorphic functions $H^0(X, \mathcal{O}_X)$ separate points and yield local coordinates at any point, and X is holomorphically convex (this means that for any discrete sequence z_ν there is a function $f \in H^0(X, \mathcal{O}_X)$ such that $|f(z_\nu)| \rightarrow \infty$). $\square$

(5.17) Remark. As long as forms of bidegree (n, q) are considered, the L^2 estimates can be extended to complex spaces with arbitrary singularities. In fact, if X is a complex space and φ is a psh weight function on X , we may still define a sheaf $K_X(\varphi)$ on X , such that the sections on an open set U are the holomorphic n -forms f on the regular part $U \cap X_{\text{reg}}$, satisfying the integrability condition $i^{n^2} f \wedge \bar{f} e^{-2\varphi} \in L^1_{\text{loc}}(U)$. In this setting, the functoriality property 5.8 becomes

$$\mu_* (K_{X'}(\varphi \circ \mu)) = K_X(\varphi)$$

for arbitrary complex spaces X, X' such that $\mu : X' \rightarrow X$ is a modification. If X is nonsingular we have $K_X(\varphi) = \mathcal{O}(K_X) \otimes \mathcal{I}(\varphi)$, however, if X is singular, the symbols K_X and $\mathcal{I}(\varphi)$ must not be dissociated. The statement of the Nadel vanishing theorem becomes $H^q(X, \mathcal{O}(F) \otimes K_X(\varphi)) = 0$ for $q \geq 1$, under the same assumptions (X Kähler and weakly pseudoconvex, curvature $\geq \varepsilon \omega$). The proof can be obtained by restricting everything to X_{reg} . Although in general X_{reg} is not weakly pseudoconvex (e.g. in case $\text{codim } X_{\text{sing}} \geq 2$), X_{reg} is always Kähler complete (the complement of a proper analytic subset in a Kähler weakly pseudoconvex space is complete Kähler, see e.g. [Dem82a]). As a consequence, Nadel's vanishing theorem is essentially insensitive to the presence of singularities. $\square$

6. Numerically Effective Line Bundles

Many problems of algebraic geometry (e.g. problems of classification of algebraic surfaces or higher dimensional varieties) lead in a natural way to the study of line bundles satisfying semipositivity conditions. It turns out that semipositivity in the sense of curvature (at least, as far as smooth metrics are considered) is not a very satisfactory notion. A more flexible notion perfectly suitable for algebraic purposes is the notion of *numerical effectivity*. The goal of this section is to give a few fundamental algebraic definitions and to discuss their differential geometric counterparts. We first suppose that X is a projective algebraic manifold, $\dim X = n$.

(6.1) Definition. A holomorphic line bundle L over a projective manifold X is said to be *numerically effective*, *nef* for short, if $L \cdot C = \int_C c_1(L) \geq 0$ for every curve $C \subset X$.

If L is nef, it can be shown that $L^p \cdot Y = \int_Y c_1(L)^p \geq 0$ for any p -dimensional subvariety $Y \subset X$ (see e.g. [Har70]). In relation with this, let us recall the Nakai-Moishezon ampleness criterion: a line bundle L is ample if and only if $L^p \cdot Y > 0$ for every p -dimensional subvariety Y . From this, we easily infer

(6.2) Proposition. Let L be a line bundle on a projective algebraic manifold X , on which an ample line bundle A and a hermitian metric ω are given. The following properties are equivalent:

- a) L is nef;
- b) for any integer $k \geq 1$, the line bundle $kL + A$ is ample;
- c) for every $\varepsilon > 0$, there is a smooth metric h_ε on L such that $i\Theta_{h_\varepsilon}(L) \geq -\varepsilon\omega$.

Proof. a) $\Rightarrow$ b). If L is nef and A is ample then clearly $kL + A$ satisfies the Nakai-Moishezon criterion, hence $kL + A$ is ample.

b) $\Rightarrow$ c). Condition c) is independent of the choice of the hermitian metric, so we may select a metric h_A on A with positive curvature and set $\omega = i\Theta(A)$. If $kL + A$ is ample, this bundle has a metric h_{kL+A} of positive curvature. Then the metric $h_L = (h_{kL+A} \otimes h_A^{-1})^{1/k}$ has curvature

$$i\Theta(L) = \frac{1}{k}(i\Theta(kL + A) - i\Theta(A)) \geq -\frac{1}{k}i\Theta(A);$$

in this way the negative part can be made smaller than $\varepsilon\omega$ by taking k large enough.

c) $\Rightarrow$ a). Under hypothesis c), we get $L \cdot C = \int_C \frac{i}{2\pi} \Theta_{h_\varepsilon}(L) \geq -\frac{\varepsilon}{2\pi} \int_C \omega$ for every curve C and every $\varepsilon > 0$, hence $L \cdot C \geq 0$ and L is nef. $\square$

Let now X be an arbitrary compact complex manifold. Since there need not exist any curve in X , Property 6.2 c) is simply taken as a definition of nefness ([DPS94]):

(6.3) Definition. A line bundle L on a compact complex manifold X is said to be nef if for every $\varepsilon > 0$, there is a smooth hermitian metric h_ε on L such that $i\Theta_{h_\varepsilon}(L) \geq -\varepsilon\omega$.

In general, it is not possible to extract a smooth limit h_0 such that $i\Theta_{h_0}(L) \geq 0$. The following simple example is given in [DPS94] (Example 1.7). Let E be a non trivial extension $0 \rightarrow \mathcal{O} \rightarrow E \rightarrow \mathcal{O} \rightarrow 0$ over an elliptic curve C and let $X = P(E)$ be the corresponding ruled surface over C . Then $L = \mathcal{O}_{P(E)}(1)$ is nef but does not admit any smooth metric of nonnegative curvature. This example answers negatively a question raised by [Fuj83].

Let us now introduce the important concept of *Kodaira-Iitaka dimension* of a line bundle.

(6.4) Definition. If L is a line bundle, the *Kodaira-Iitaka dimension* $\kappa(L)$ is the supremum of the rank of the canonical maps

$$\Phi_m : X \setminus B_m \rightarrow P(V_m^*), \quad x \mapsto H_x = \{\sigma \in V_m; \sigma(x) = 0\}, \quad m \geq 1$$

with $V_m = H^0(X, mL)$ and $B_m = \bigcap_{\sigma \in V_m} \sigma^{-1}(0) = \text{base locus of } V_m$. In case $V_m = \{0\}$ for all $m \geq 1$, we set $\kappa(L) = -\infty$.

A line bundle is said to be big if $\kappa(L) = \dim X$.

The following lemma is well-known (the proof is a rather elementary consequence of the Schwarz lemma).

(6.5) Serre-Siegel lemma ([Ser54], [Sie55]). Let L be any line bundle on a compact complex manifold. Then we have

$$h^0(X, mL) \leq O(m^{\kappa(L)}) \quad \text{for } m \geq 1,$$

and $\kappa(L)$ is the smallest constant for which this estimate holds. $\square$

We now discuss the various concepts of positive cones in the space of numerical classes of line bundles, and establish a simple dictionary relating these concepts to corresponding concepts in the context of differential geometry.

Let us recall that an integral cohomology class in $H^2(X, \mathbb{Z})$ is the first Chern class of a holomorphic (or algebraic) line bundle if and only if it lies in the *Neron-Severi* group

$$\text{NS}(X) = \text{Ker} (H^2(X, \mathbb{Z}) \rightarrow H^2(X, \mathcal{O}_X))$$

(this fact is just an elementary consequence of the exponential exact sequence $0 \rightarrow \mathbb{Z} \rightarrow \mathcal{O} \rightarrow \mathcal{O}^* \rightarrow 0$). If X is compact Kähler, as we will suppose from now on in this section, this is the same as saying that the class is of type $(1, 1)$ with respect to Hodge decomposition.

Let $\text{NS}_{\mathbb{R}}(X)$ be the real vector space $\text{NS}(X) \otimes \mathbb{R} \subset H^2(X, \mathbb{R})$. We define four convex cones

$$\begin{aligned} N_{\text{amp}}(X) &\subset N_{\text{eff}}(X) \subset \text{NS}_{\mathbb{R}}(X), \\ N_{\text{nef}}(X) &\subset N_{\text{psnf}}(X) \subset \text{NS}_{\mathbb{R}}(X) \end{aligned}$$

which are, respectively, the *convex cones* generated by Chern classes $c_1(L)$ of ample and effective line bundles, resp. the *closure of the convex cones* generated by numerically effective and pseudo-effective line bundles; we say that L is effective if mL has a section for some $m > 0$, i.e. if $\mathcal{O}(mL) \simeq \mathcal{O}(D)$ for some effective divisor D ; and we say that L pseudo-effective if $c_1(L)$ is the cohomology class of some closed positive current T , i.e. if L can be equipped with a singular hermitian metric h with $T = \frac{i}{2\pi} \Theta_h(L) \geq 0$ as a current. For each of the ample, effective, nef and pseudo-effective cones, the first Chern class $c_1(L)$ of a line bundle L lies in the cone if and only if L has the corresponding property (for N_{psef} use the fact that the space of positive currents of mass 1 is weakly compact; the case of all other cones is obvious).

(6.6) Proposition. *Let (X, ω) be a compact Kähler manifold. The numerical cones satisfy the following properties.*

- a) $N_{\text{amp}} = N_{\text{amp}}^\circ \subset N_{\text{nef}}^\circ$, $N_{\text{nef}} \subset N_{\text{psef}}$.
- b) *If moreover X is projective algebraic, we have $N_{\text{amp}} = N_{\text{nef}}^\circ$ (therefore $\overline{N}_{\text{amp}} = N_{\text{nef}}$), and $\overline{N}_{\text{eff}} = N_{\text{psef}}$.*

If L is a line bundle on X and h denotes a hermitian metric on L , the following properties are equivalent:

- c) $c_1(L) \in N_{\text{amp}} \Leftrightarrow \exists \varepsilon > 0, \exists h \text{ smooth such that } i\Theta_h(L) \geq \varepsilon\omega$.
- d) $c_1(L) \in N_{\text{nef}} \Leftrightarrow \forall \varepsilon > 0, \exists h \text{ smooth such that } i\Theta_h(L) \geq -\varepsilon\omega$.
- e) $c_1(L) \in N_{\text{psef}} \Leftrightarrow \exists h \text{ possibly singular such that } i\Theta_h(L) \geq 0$.
- f) *If moreover X is projective algebraic, then*
 $c_1(L) \in N_{\text{eff}}^\circ \Leftrightarrow \kappa(L) = \dim X$
 $\Leftrightarrow \exists \varepsilon > 0, \exists h \text{ possibly singular such that } i\Theta_h(L) \geq \varepsilon\omega$.

Proof. c) and d) are already known and e) is a definition.

a) The ample cone N_{amp} is always open by definition and contained in N_{nef} , so the first inclusion is obvious (N_{amp} is of course empty if X is not projective algebraic). Let us now prove that $N_{\text{nef}} \subset N_{\text{psef}}$. Let L be a line bundle with $c_1(L) \in N_{\text{nef}}$. Then for every $\varepsilon > 0$, there is a current $T_\varepsilon = \frac{i}{2\pi} \Theta_{h_\varepsilon}(L) \geq -\varepsilon\omega$. Then $T_\varepsilon + \varepsilon\omega$ is a closed positive current and the family is uniformly bounded in mass for $\varepsilon \in]0, 1]$, since

$$\int_X (T_\varepsilon + \varepsilon\omega) \wedge \omega^{n-1} = \int_X c_1(L) \wedge \omega^{n-1} + \varepsilon \int_X \omega^n.$$

By weak compactness, some subsequence converges to a weak limit $T \geq 0$ and $T \in c_1(L)$ (the cohomology class $\{T\}$ of a current is easily shown to depend continuously on T with respect to the weak topology; use e.g. Poincaré duality to check this).

b) If X is projective, the equality $N_{\text{amp}} = N_{\text{nef}}^\circ$ is a simple consequence of 6.2 b) and of the fact that ampleness (or positivity) is an open property. It remains to show that $N_{\text{psef}} \subset \overline{N}_{\text{eff}}$. Let L be a line bundle with $c_1(L) \in N_{\text{psef}}$ and let h_L be a singular hermitian on L such that $T = \frac{i}{2\pi} \Theta(L) \geq 0$. Fix a point $x_0 \in X$ such that the Lelong number of T at x_0 is zero, and take a sufficiently positive line bundle A (replacing A by a multiple if necessary), such that $A - K_X$ has a singular metric

h_{A-K_X} of curvature $\geq \varepsilon\omega$ and such that h_{A-K_X} is smooth on $X \setminus \{x_0\}$ and has an isolated logarithmic pole of Lelong number $\geq n$ at x_0 . Then apply Corollary 5.13 to $F = mL + A - K_X$ equipped with the metric $h_L^{\otimes m} \otimes h_{A-K_X}$. Since the weight φ of this metric has a Lelong number $\geq n$ at x_0 and a Lelong number equal to the Lelong number of $T = \frac{i}{2\pi}\Theta(L)$ at nearby points, $\limsup_{x \rightarrow x_0} \nu(T, x) = \nu(T, x_0) = 0$, Corollary 5.13 implies that $H^0(X, K_X + F) = H^0(X, mL + A)$ has a section which does not vanish at x_0 . Hence there is an effective divisor D_m such that $\mathcal{O}(mL + A) = \mathcal{O}(D_m)$ and $c_1(L) = \frac{1}{m}\{D_m\} - \frac{1}{m}c_1(A) = \lim \frac{1}{m}\{D_m\}$ is in $\overline{N}_{\text{eff}}$. $\square$

f) Fix a nonsingular ample divisor A . If $c_1(L) \in N_{\text{eff}}^\circ$, there is an integer $m > 0$ such that $c_1(L) - \frac{1}{m}c_1(A)$ is still effective, hence for m, p large we have $mpL - pA = D + F$ with an effective divisor D and a numerically trivial line bundle F . This implies $\mathcal{O}(kmpL) = \mathcal{O}(kpA + kD + kF) \supset \mathcal{O}(kpA + kF)$, hence $h^0(X, kmpL) \geq h^0(X, kpA + kF) \sim (kp)^n A^n / n!$ by the Riemann-Roch formula. Therefore $\kappa(L) = n$.

If $\kappa(L) = n$, then $h^0(X, kL) \geq ck^n$ for $k \geq k_0$ and $c > 0$. The exact cohomology sequence

$$0 \longrightarrow H^0(X, kL - A) \longrightarrow H^0(X, kL) \longrightarrow H^0(A, kL|_A)$$

where $h^0(A, kL|_A) = O(k^{n-1})$ shows that $kL - A$ has non zero sections for k large. If D is the divisor of such a section, then $kL \simeq \mathcal{O}(A + D)$. Select a smooth metric on A such that $\frac{i}{2\pi}\Theta(A) \geq \varepsilon_0\omega$ for some $\varepsilon_0 > 0$, and take the singular metric on $\mathcal{O}(D)$ with weight function $\varphi_D = \sum \alpha_j \log |g_j|$ described in Example 3.13. Then the metric with weight $\varphi_L = \frac{1}{k}(\varphi_A + \varphi_D)$ on L yields

$$\frac{i}{2\pi}\Theta(L) = \frac{1}{k} \left(\frac{i}{2\pi}\Theta(A) + [D] \right) \geq (\varepsilon_0/k)\omega,$$

as desired.

Finally, the curvature condition $i\Theta_h(L) \geq \varepsilon\omega$ in the sense of currents yields by definition $c_1(L) \in N_{\text{psf}}^\circ$. Moreover, b) implies $N_{\text{psf}}^\circ = N_{\text{eff}}^\circ$. $\square$

Before going further, we need a lemma.

(6.7) Lemma. *Let X be a compact Kähler n -dimensional manifold, let L be a nef line bundle on X , and let E be an arbitrary holomorphic vector bundle. Then $h^q(X, c\mathcal{O}(E) \otimes \mathcal{O}(kL)) = o(k^n)$ as $k \rightarrow +\infty$, for every $q \geq 1$. If X is projective algebraic, the following more precise bound holds:*

$$h^q(X, \mathcal{O}(E) \otimes \mathcal{O}(kL)) = O(k^{n-q}), \quad \forall q \geq 0.$$

Proof. The Kähler case will be proved in Section 12, as a consequence of the holomorphic Morse inequalities. In the projective algebraic case, we proceed by induction on $n = \dim X$. If $n = 1$ the result is clear, as well as if $q = 0$. Now let A be a nonsingular ample divisor such that $E \otimes \mathcal{O}(A - K_X)$ is Nakano positive. Then the Nakano vanishing theorem applied to the vector bundle $F = E \otimes \mathcal{O}(kL + A - K_X)$ shows that $H^q(X, \mathcal{O}(E) \otimes \mathcal{O}(kL + A)) = 0$ for all $q \geq 1$. The exact sequence

$$0 \rightarrow \mathcal{O}(kL) \rightarrow \mathcal{O}(kL + A) \rightarrow \mathcal{O}(kL + A)|_A \rightarrow 0$$

twisted by E implies

$$H^q(X, \mathcal{O}(E) \otimes \mathcal{O}(kL)) \simeq H^{q-1}(A, \mathcal{O}(E|_A \otimes \mathcal{O}(kL + A)|_A)),$$

and we easily conclude by induction since $\dim A = n - 1$. Observe that the argument does not work any more if X is not algebraic. It seems to be unknown whether the $O(k^{n-q})$ bound still holds in that case. $\square$

(6.8) Corollary. *If L is nef, then L is big (i.e. $\kappa(L) = n$) if and only if $L^n > 0$. Moreover, if L is nef and big, then for every $\delta > 0$, L has a singular metric $h = e^{-2\varphi}$ such that $\max_{x \in X} \nu(\varphi, x) \leq \delta$ and $i\Theta_h(L) \geq \varepsilon\omega$ for some $\varepsilon > 0$. The metric h can be chosen to be smooth on the complement of a fixed divisor D , with logarithmic poles along D .*

Proof. By Lemma 6.7 and the Riemann-Roch formula, we have $h^0(X, kL) = \chi(X, kL) + o(k^n) = k^n L^n / n! + o(k^n)$, whence the first statement. If L is big, the proof made in 6.5 f) shows that there is a singular metric h_1 on L such that

$$\frac{i}{2\pi} \Theta_{h_1}(L) = \frac{1}{k} \left(\frac{i}{2\pi} \Theta(A) + [D] \right)$$

with a positive line bundle A and an effective divisor D . Now, for every $\varepsilon > 0$, there is a smooth metric h_ε on L such that $\frac{i}{2\pi} \Theta_{h_\varepsilon}(L) \geq -\varepsilon\omega$, where $\omega = \frac{i}{2\pi} \Theta(A)$. The convex combination of metrics $h'_\varepsilon = h_1^{k\varepsilon} h_\varepsilon^{1-k\varepsilon}$ is a singular metric with poles along D which satisfies

$$\frac{i}{2\pi} \Theta_{h'_\varepsilon}(L) \geq \varepsilon(\omega + [D]) - (1 - k\varepsilon)\varepsilon\omega \geq k\varepsilon^2\omega.$$

Its Lelong numbers are $\varepsilon\nu(D, x)$ and they can be made smaller than δ by choosing $\varepsilon > 0$ small. $\square$

We still need a few elementary facts about the numerical dimension of nef line bundles.

(6.9) Definition. *Let L be a nef line bundle on a compact Kähler manifold X . One defines the numerical dimension of L to be*

$$\nu(L) = \max \{k = 0, \dots, n; c_1(L)^k \neq 0 \text{ in } H^{2k}(X, \mathbb{R})\}.$$

By Corollary 6.8, we have $\kappa(L) = n$ if and only if $\nu(L) = n$. In general, we merely have an inequality.

(6.10) Proposition. *If L is a nef line bundle on a compact Kähler manifold, then $\kappa(L) \leq \nu(L)$.*

Proof. By induction on $n = \dim X$. If $\nu(L) = n$ or $\kappa(L) = n$ the result is true, so we may assume $r := \kappa(L) \leq n - 1$ and $k := \nu(L) \leq n - 1$. Fix $m > 0$ so that $\Phi = \Phi|_{mL}$ has generic rank r . Select a nonsingular ample divisor A in X such that

the restriction of $\Phi|_{mL}$ to A still has rank r (for this, just take A passing through a point $x \notin B|_{mL}$ at which $\text{rank}(d\Phi_x) = r < n$, in such a way that the tangent linear map $d\Phi_{x|T_{A,x}}$ still has rank r). Then $\kappa(L|_A) \geq r = \kappa(L)$ (we just have an equality because there might exist sections in $H^0(A, mL|_A)$ which do not extend to X). On the other hand, we claim that $\nu(L|_A) = k = \nu(L)$. The inequality $\nu(L|_A) \geq \nu(L)$ is clear. Conversely, if we set $\omega = \frac{i}{2\pi}\Theta(A) > 0$, the cohomology class $c_1(L)^k$ can be represented by a closed positive current of bidegree (k, k)

$$T = \lim_{\varepsilon \rightarrow 0} \left(\frac{i}{2\pi} \Theta_{h_\varepsilon}(L) + \varepsilon \omega \right)^k$$

after passing to some subsequence (there is a uniform bound for the mass thanks to the Kähler assumption, taking wedge products with ω^{n-k}). The current T must be non zero since $c_1(L)^k \neq 0$ by definition of $k = \nu(L)$. Then $\{[A]\} = \{\omega\}$ as cohomology classes, and

$$\int_A c_1(L|_A)^k \wedge \omega^{n-1-k} = \int_X c_1(L)^k \wedge [A] \wedge \omega^{n-1-k} = \int_X T \wedge \omega^{n-k} > 0.$$

This implies $\nu(L|_A) \geq k$, as desired. The induction hypothesis with X replaced by A yields

$$\kappa(L) \leq \kappa(L|_A) \leq \nu(L|_A) \leq \nu(L). \quad \square$$

(6.11) Remark. It may happen that $\kappa(L) < \nu(L)$: take e.g.

$$L \rightarrow X = X_1 \times X_2$$

equal to the total tensor product of an ample line bundle L_1 on a projective manifold X_1 and of a unitary flat line bundle L_2 on an elliptic curve X_2 given by a representation $\pi_1(X_2) \rightarrow U(1)$ such that no multiple kL_2 with $k \neq 0$ is trivial. Then $H^0(X, kL) = H^0(X_1, kL_1) \otimes H^0(X_2, kL_2) = 0$ for $k > 0$, and thus $\kappa(L) = -\infty$. However $c_1(L) = \text{pr}_1^* c_1(L_1)$ has numerical dimension equal to $\dim X_1$. The same example shows that the Kodaira dimension may increase by restriction to a subvariety (if $Y = X_1 \times \{\text{point}\}$, then $\kappa(L|_Y) = \dim Y$). $\square$

We now derive an algebraic version of the Nadel vanishing theorem in the context of nef line bundles. This algebraic vanishing theorem has been obtained independently by Kawamata [Kaw82] and Viehweg [Vie82], who both reduced it to the Kodaira-Nakano vanishing theorem by cyclic covering constructions. Since then, a number of other proofs have been given, one based on connections with logarithmic singularities [EV86], another on Hodge theory for twisted coefficient systems [Kol85], a third one on the Bochner technique [Dem89] (see also [EV92] for a general survey, and [Eno93] for an extension to the compact Kähler case). Since the result is best expressed in terms of multiplier ideal sheaves (avoiding then any unnecessary desingularization in the statement), we feel that the direct approach via Nadel's vanishing theorem is probably the most natural one.

If $D = \sum \alpha_j D_j \geq 0$ is an effective $\mathbb{Q}$ -divisor, we define the *multiplier ideal sheaf* $\mathcal{I}(D)$ to be equal to $\mathcal{I}(\varphi)$ where $\varphi = \sum \alpha_j |g_j|$ is the corresponding psh function defined by generators g_j of $\mathcal{O}(-D_j)$; as we saw in Remark 5.9, the computation of

$\mathcal{I}(D)$ can be made algebraically by using desingularizations $\mu : \tilde{X} \rightarrow X$ such that μ^*D becomes a divisor with normal crossings on $\tilde{X}$.

(6.12) Kawamata-Viehweg vanishing theorem. *Let X be a projective algebraic manifold and let F be a line bundle over X such that some positive multiple mF can be written $mF = L + D$ where L is a nef line bundle and D an effective divisor. Then*

$$H^q(X, \mathcal{O}(K_X + F) \otimes \mathcal{I}(m^{-1}D)) = 0 \quad \text{for } q > n - \nu(L).$$

(6.13) Special case. *If F is a nef line bundle, then*

$$H^q(X, \mathcal{O}(K_X + F)) = 0 \quad \text{for } q > n - \nu(F).$$

Proof of Theorem 6.12. First suppose that $\nu(L) = n$, i.e. that L is big. By the proof of 6.5 f), there is a singular hermitian metric on L such that the corresponding weight $\varphi_{L,0}$ has algebraic singularities and

$$i\Theta_0(L) = 2id'd''\varphi_L \geq \varepsilon_0\omega$$

for some $\varepsilon_0 > 0$. On the other hand, since L is nef, there are metrics given by weights $\varphi_{L,\varepsilon}$ such that $\frac{1}{2\pi}\Theta_\varepsilon(L) \geq \varepsilon\omega$ for every $\varepsilon > 0$, ω being a Kähler metric. Let $\varphi_D = \sum \alpha_j \log |g_j|$ be the weight of the singular metric on $\mathcal{O}(D)$ described in Example 3.13. We define a singular metric on F by

$$\varphi_F = \frac{1}{m}((1 - \delta)\varphi_{L,\varepsilon} + \delta\varphi_{L,0} + \varphi_D)$$

with $\varepsilon \ll \delta \ll 1$, δ rational. Then φ_F has algebraic singularities, and by taking δ small enough we find $\mathcal{I}(\varphi_F) = \mathcal{I}(\frac{1}{m}\varphi_D) = \mathcal{I}(\frac{1}{m}D)$. In fact, $\mathcal{I}(\varphi_F)$ can be computed by taking integer parts of $\mathbb{Q}$ -divisors (as explained in Remark 5.9), and adding $\delta\varphi_{L,0}$ does not change the integer part of the rational numbers involved when δ is small. Now

$$\begin{aligned} dd^c\varphi_F &= \frac{1}{m}((1 - \delta)dd^c\varphi_{L,\varepsilon} + \delta dd^c\varphi_{L,0} + dd^c\varphi_D) \\ &\geq \frac{1}{m}(-(1 - \delta)\varepsilon\omega + \delta\varepsilon_0\omega + [D]) \geq \frac{\delta\varepsilon}{m}\omega, \end{aligned}$$

if we choose $\varepsilon \leq \delta\varepsilon_0$. Nadel's theorem thus implies the desired vanishing result for all $q \geq 1$.

Now, if $\nu(L) < n$, we use hyperplane sections and argue by induction on $n = \dim X$. Since the sheaf $\mathcal{O}(K_X) \otimes \mathcal{I}(m^{-1}D)$ behaves functorially with respect to modifications (and since the L^2 cohomology complex is “the same” upstairs and downstairs), we may assume after blowing-up that D is a divisor with normal crossings. By Remark 5.9, the multiplier ideal sheaf $\mathcal{I}(m^{-1}D) = \mathcal{O}(-\lfloor m^{-1}D \rfloor)$ is locally free. By Serre duality, the expected vanishing is equivalent to

$$H^q(X, \mathcal{O}(-F) \otimes \mathcal{O}(\lfloor m^{-1}D \rfloor)) = 0 \quad \text{for } q < \nu(L).$$

Then select a nonsingular ample divisor A such that A meets all components D_j transversally. Select A positive enough so that $\mathcal{O}(A + F - \lfloor m^{-1}D \rfloor)$ is ample. Then

$H^q(X, \mathcal{O}(-A-F) \otimes \mathcal{O}(\lfloor m^{-1}D \rfloor)) = 0$ for $q < n$ by Kodaira vanishing, and the exact sequence $0 \rightarrow \mathcal{O}_X(-A) \rightarrow \mathcal{O}_X \rightarrow (i_A)_* \mathcal{O}_A \rightarrow 0$ twisted by $\mathcal{O}(-F) \otimes \mathcal{O}(\lfloor m^{-1}D \rfloor)$ yields an isomorphism

$$H^q(X, \mathcal{O}(-F) \otimes \mathcal{O}(\lfloor m^{-1}D \rfloor)) \simeq H^q(A, \mathcal{O}(-F|_A) \otimes \mathcal{O}(\lfloor m^{-1}D|_A \rfloor)).$$

The proof of 5.8 showed that $\nu(L|_A) = \nu(L)$, hence the induction hypothesis implies that the cohomology group on A in the right hand side is zero for $q < \nu(L)$. $\square$

(6.14) Remark. Enoki [Eno92] proved that the special case 6.13 of the Kawamata-Viehweg vanishing theorem still holds when X is a compact Kähler manifold. The idea is to replace the induction on dimension (hyperplane section argument) by a clever use of the Aubin-Calabi-Yau theorem. We conjecture that the general case 6.12 is also valid for compact Kähler manifolds. $\square$

7. Seshadri Constants and the Fujita Conjecture

In questions related to the search of effective bounds for the existence of sections in multiples of ample line bundles, one is lead to study the “local positivity” of such bundles. In fact, there is a simple way of giving a precise measurement of positivity. We more or less follow the original exposition given in [Dem90], and discuss some interesting results obtained in [EL92] and [EKL94].

(7.1) Definition. Let L be a nef line bundle over a projective algebraic manifold X . The Seshadri constant of L at a point $x \in X$ is the nonnegative real number

$$(7.1') \quad \varepsilon(L, x) = \sup \{ \varepsilon \geq 0; \mu^* L - \varepsilon E \text{ is nef} \},$$

where $\pi: \tilde{X} \rightarrow X$ is the blow-up of X at x and E is the exceptional divisor. An equivalent definition is

$$(7.1'') \quad \varepsilon(L, x) = \inf_{C \ni x} \frac{L \cdot C}{\nu(C, x)},$$

where the infimum is taken over all irreducible curves C passing through x and $\nu(C, x)$ is the multiplicity of C at x .

To get the equivalence between the two definitions, we just observe that for any irreducible curve $\tilde{C} \subset \tilde{X}$ not contained in the exceptional divisor and $C = \pi(\tilde{C})$, then (exercise to the reader!)

$$(\pi^* L - \varepsilon E) \cdot \tilde{C} = L \cdot C - \varepsilon \nu(C, x).$$

The infimum

$$(7.2) \quad \varepsilon(L) = \inf_{x \in X} \varepsilon(L, x) = \inf_C \frac{L \cdot C}{\nu(C)} \quad \text{where} \quad \nu(C) = \max_{x \in C} \nu(C, x)$$

will be called the *global Seshadri constant* of L . It is well known that L is ample if and only if $\varepsilon(L) > 0$ (Seshadri's criterion for ampleness, see [Har70] Chapter 1). It is useful to think of the Seshadri constant $\varepsilon(L, x)$ as measuring how positive L is along curves passing through x . The following exercise presents some illustrations of this intuitive idea.

(7.3) Exercise.

- a) If L is an ample line bundle such that mL is *very ample*, then $\varepsilon(L) \geq \frac{1}{m}$. (This is the elementary half of Seshadri's criterion for ampleness).
- b) For any two nef line bundles L_1, L_2 , show that

$$\varepsilon(L_1 + L_2, x) \geq \varepsilon(L_1, x) + \varepsilon(L_2, x) \quad \text{for all } x \in X. \quad \square$$

If L is a nef line bundle, we are also interested in singular metrics with isolated logarithmic poles (e.g. in view of applying Corollary 5.13): we say that a logarithmic pole x of the weight φ is *isolated* if φ is finite and continuous on $V \setminus \{x\}$ for some neighborhood V of x and we define

$$(7.4) \quad \gamma(L, x) = \sup \left\{ \begin{array}{l} \gamma \in \mathbb{R}_+ \text{ such that } L \text{ has a singular metric} \\ \text{with } i\Theta(L) \geq 0 \text{ and with an isolated} \\ \text{logarithmic pole of coefficient } \gamma \text{ at } x \end{array} \right\}.$$

If there are no such metrics, we set $\gamma(L, x) = 0$.

The numbers $\varepsilon(L, x)$ and $\gamma(L, x)$ will be seen to carry a lot of useful information about the global sections of L and its multiples kL . To make this precise, we first introduce some further definitions. Let $s(L, x)$ be the largest integer $s \in \mathbb{N}$ such that the global sections in $H^0(X, L)$ generate all s -jets $J_x^s L = \mathcal{O}_x(L)/\mathfrak{m}_x^{s+1}\mathcal{O}_x(L)$. If L_x is not generated, i.e. if all sections of L vanish at x , we set $s(L, x) = -\infty$. We also introduce the limit value

$$(7.5) \quad \sigma(L, x) = \limsup_{k \rightarrow +\infty} \frac{1}{k} s(kL, x) = \sup_{k \in \mathbb{N}^*} \frac{1}{k} s(kL, x)$$

if $s(kL, x) \neq -\infty$ for some k , and $\sigma(L, x) = 0$ otherwise. The limsup is actually equal to the sup thanks to the superadditivity property

$$s(L_1 + L_2, x) \geq s(L_1, x) + s(L_2, x).$$

The limsup is in fact a limit as soon as kL spans at x for $k \geq k_0$, e.g. when L is ample.

(7.6) Theorem. *Let L be a line bundle over X .*

- a) *If L is nef then $\varepsilon(L, x) \geq \gamma(L, x) \geq \sigma(L, x)$ for every $x \in X$.*
- b) *If L is ample then $\varepsilon(L, x) = \gamma(L, x) = \sigma(L, x)$ for every $x \in X$.*
- c) *If L is big and nef, the equality holds for all x outside some divisor D in X .*

Proof. Fix a point $x \in X$ and a coordinate system $(z_1, \dots, z_n)$ centered at x . If $s = s(kL, x)$, then $H^0(X, kL)$ generates all s -jets at x and we can find holomorphic

sections $f_1, \dots, f_N$ whose s -jets are all monomials z^α , $|\alpha| = s$. We define a global singular metric on L by

$$(7.7) \quad |\xi| = \left(\sum_{1 \leq j \leq N} |f_j(z) \cdot \xi^{-k}|^2 \right)^{-1/2k}, \quad \xi \in L_z$$

associated to the weight function $\varphi(z) = \frac{1}{2k} \log \sum |\theta(f_j(z))|^2$ in any trivialization $L|_\Omega \simeq \Omega \times \mathbb{C}$. Then φ has an isolated logarithmic pole of coefficient s/k at x , thus

$$\gamma(L, x) \geq \frac{1}{k} s(kL, x)$$

and in the limit we get $\gamma(L, x) \geq \sigma(L, x)$.

Now, suppose that L has a singular metric with an isolated log pole of coefficient $\geq \gamma$ at x . Set $\frac{i}{2\pi} \Theta(L) = dd^c \varphi$ on a neighborhood Ω of x and let C be an irreducible curve passing through x . Then all weight functions associated to the metric of L must be locally integrable along C (since φ has an isolated pole at x). We infer

$$L \cdot C = \int_C \frac{i}{2\pi} \Theta(L) \geq \int_{C \cap \Omega} dd^c \varphi \geq \gamma \nu(C, x)$$

because the last integral is larger than the Lelong number of the current $[C]$ with respect to the weight φ and we may apply our comparison theorem 2.15 with the ordinary Lelong number associated to the weight $\log |z - x|$. Therefore

$$\varepsilon(L, x) = \inf \frac{L \cdot C}{\nu(C, x)} \geq \sup \gamma = \gamma(L, x).$$

Finally, we show that $\sigma(L, x) \geq \varepsilon(L, x)$ when L is ample. This is done essentially by same arguments as in the proof of Seshadri's criterion, as explained in [Har70]. Consider the blow-up $\pi : \tilde{X} \rightarrow X$ at point x , the exceptional divisor $E = \pi^{-1}(x)$ and the line bundles $F_{p,q} = \mathcal{O}(p\pi^*L - qE)$ over $\tilde{X}$, where $p, q > 0$. Recall that $\mathcal{O}(-E)|_E$ is the canonical line bundle $\mathcal{O}_E(1)$ over $E \simeq \mathbb{P}^{n-1}$, in particular we have $E^n = \mathcal{O}_E(-1)^{n-1} = (-1)^{n-1}$. For any irreducible curve $\tilde{C} \subset \tilde{X}$, either $\tilde{C} \subset E$ and

$$F_{p,q} \cdot \tilde{C} = \mathcal{O}(-qE) \cdot \tilde{C} = q \mathcal{O}_E(1) \cdot \tilde{C} = q \deg \tilde{C}$$

or $\pi(\tilde{C}) = C$ is a curve and

$$F_{p,q} \cdot C = pL \cdot C - q\nu(C, x) \geq (p - q/\varepsilon(L, x))L \cdot C.$$

Thus $F_{p,q}$ is nef provided that $p \geq q/\varepsilon(L, x)$. Since $F_{p,q}$ is ample when p/q is large, a simple interpolation argument shows that $F_{p,q}$ is ample for $p > q/\varepsilon(L, x)$. In that case, the Kodaira-Serre vanishing theorem gives

$$H^1(\tilde{X}, kF_{p,q}) = H^1(\tilde{X}, \mathcal{O}(kp\pi^*L - kqE)) = 0$$

for k large. Hence we get a surjective map

$$H^0(\tilde{X}, kp\pi^*L) \longrightarrow H^0\left(\tilde{X}, \mathcal{O}(kp\pi^*L) \otimes (\mathcal{O}/\mathcal{O}(-kqE))\right) \simeq J_x^{kq-1}(kpL),$$

that is, $H^0(X, kpL)$ generates all $(kq-1)$ jets at x . Therefore $p > q/\varepsilon(L, x)$ implies $s(kpL, x) \geq kq-1$ for k large, so $\sigma(L, x) \geq q/p$. At the limit we get $\sigma(L, x) \geq \varepsilon(L, x)$.

Assume now that L is nef and big and that $\varepsilon(L, x) > 0$. By the proof of 6.6 f), there exist an integer $k_0 \geq 1$ and effective divisors A, D such that $k_0L \simeq A + D$ where A is ample. Then $a\pi^*A - E$ is ample for a large. Hence there are integers $a, b > 0$ such that $a\pi^*A - bE - K_{\tilde{X}}$ is ample. When $F_{p,q}$ is nef, the sum with any positive multiple $kF_{p,q}$ is still ample and the Akizuki-Kodaira-Nakano vanishing theorem gives

$$H^1(\tilde{X}, kF_{p,q} + a\pi^*A - bE) = H^1(\tilde{X}, (kp + k_0a)\pi^*L - a\pi^*D - (kq + b)E) = 0$$

when we substitute $A = k_0L - D$. As above, this implies that we have a surjective map

$$H^0(X, (kp + k_0a)L - aD) \twoheadrightarrow J_x^{kq+b-1}((kp + k_0a)L - aD)$$

when $p \geq q/\varepsilon(L, x)$. Since $\mathcal{O}(-aD) \subset \mathcal{O}$, we infer $s((kp + k_0a)L, x) \geq kq + b - 1$ at every point $x \in X \setminus D$ and at the limit $\sigma(L, x) \geq \varepsilon(L, x)$. $\square$

(7.8) Remark. Suppose that the line bundle L is ample. The same arguments show that if $\pi : \tilde{X} \rightarrow X$ is the blow-up at two points x, y and if $E_x + E_y$ is the exceptional divisor, then $F_{p,q} = p\pi^*L - qE_x - E_y$ is ample for $p > q/\varepsilon(L, x) + 1/\varepsilon(L, y)$. In that case, $H^0(X, kpL)$ generates $J_x^{kq-1}(kpL) \oplus J_y^{k-1}(kpL)$ for k large. Take $p > q/\varepsilon(L, x) + 1/\varepsilon(L)$ and let y run over $X \setminus \{x\}$. For k large, we obtain sections $f_j \in H^0(X, kpL)$ whose jets at x are all monomials z^α , $|\alpha| = kq - 1$, and with no other common zeros. Moreover, Formula (7.7) produces a metric on L which is smooth and has positive definite curvature on $X \setminus \{x\}$, and which has a log pole of coefficient $(kq - 1)/kp$ at x . Therefore the supremum $\gamma(L, x) = \sup\{\gamma\}$ is always achieved by metrics which are smooth and have positive definite curvature on $X \setminus \{x\}$. $\square$

(7.9) Remark. If Y is a p -dimensional algebraic subset of X passing through x , then

$$L^p \cdot Y \geq \varepsilon(L, x)^p \nu(Y, x)$$

where $L^p \cdot Y = \int_Y c_1(L)^p$ and $\nu(Y, x)$ is the multiplicity of Y at x . In fact, if L is ample, we can take a metric on L which is smooth on $X \setminus \{x\}$ and defined on a neighborhood Ω of x by a weight function φ with a log pole of coefficient γ at x . By the comparison theorem for Lelong numbers, we get

$$L^p \cdot Y \geq \int_{Y \cap \Omega} (dd^c \varphi)^p \geq \gamma^p \nu(Y, x)$$

and γ can be chosen arbitrarily close to $\varepsilon(L, x)$. If L is nef, we apply the inequality to $kL + M$ with M ample and take the limit as $k \rightarrow +\infty$. $\square$

The Seshadri constants $\varepsilon(L, x)$ and $\varepsilon(L) = \inf \varepsilon(L, x)$ are especially interesting because they provide effective results concerning the existence of sections of the so called *adjoint line bundle* $K_X + L$. The following proposition illustrates this observation.

(7.10) Proposition. *Let L be a big nef line bundle over X .*

- a) *If $\varepsilon(L, x) > n + s$, then $H^0(X, K_X + L)$ generates all s -jets at x .*
b) *If $\varepsilon(L) > 2n$, then $K_X + L$ is very ample.*

Proof. a) By the proof of Theorem 7.6, the line bundle $\pi^*L - qE$ is nef for $q \leq \varepsilon(L, x)$. Moreover, its n -th self intersection is equal to $L^n + (-q)^n E^n = L^n - q^n$ and as $L^n \geq \varepsilon(L, x)^n$ by remark 3.5, we see that $\pi^*L - qE$ is big for $q < \varepsilon(L, x)$. The Kawamata-Viehweg vanishing theorem 5.2 then gives

$$H^1(\tilde{X}, K_{\tilde{X}} + \pi^*L - qE) = H^1(\tilde{X}, \pi^*K_X + \pi^*L - (q - n + 1)E) = 0,$$

since $K_{\tilde{X}} = \pi^*K_X + (n - 1)E$. Thus we get a surjective map

$$\begin{array}{ccc} H^0(\tilde{X}, \pi^*K_X + \pi^*L) & \twoheadrightarrow & H^0(\tilde{X}, \pi^*\mathcal{O}(K_X + L) \otimes \mathcal{O}/\mathcal{O}(-(q - n + 1)E)) \\ \parallel & & \parallel \\ H^0(X, K_X + L) & \twoheadrightarrow & J_x^{q-n}(K_X + L) \end{array}$$

provided that $\varepsilon(L, x) > q$. The first statement is proved. To show that $K_X + L$ is very ample, we blow up at two points x, y . The line bundle $\pi^*L - nE_x - nE_y$ is ample for $1/\varepsilon(L, x) + 1/\varepsilon(L, y) < 1/n$, a sufficient condition for this is $\varepsilon(L) > 2n$. Then we see that

$$H^0(X, K_X + L) \longrightarrow (K_X + L)_x \oplus (K_X + L)_y$$

is also surjective. □

(7.11) Exercise. Derive Proposition 7.10 directly from Corollary 5.13, assuming that L is ample and using the equality $\varepsilon(L, x) = \gamma(L, x)$. □

In relation with these questions, Fujita [Fuj87, 88] made the following interesting conjecture about adjoint line bundles.

(7.12) Conjecture (Fujita). *If L is an ample line bundle, then $K_X + mL$ is generated by global sections for $m \geq n + 1$ and very ample for $m \geq n + 2$.*

Using Mori theory, Fujita proved that $K_X + mL$ is nef for $m \geq n + 1$ and ample for $m \geq n + 2$, but these results are of course much weaker than the conjecture; they can be derived rather easily by a direct application of the Kawamata-Viehweg vanishing theorem (see Section 8). Observe that if the conjecture holds, it would be actually optimal, as shown by the case of projective space: if $X = \mathbb{P}^n$, then $K_X = \mathcal{O}(-n - 1)$, hence the bounds $m = n + 1$ for global generation and $m = n + 2$ for very ampleness are sharp when $L = \mathcal{O}(1)$. The case of curves ($n = 1$) is easily settled in the affirmative:

(7.13) Exercise. If X is a curve and L is a line bundle of positive degree on X , show by Riemann-Roch that $\varepsilon(L, x) = \sigma(L, x) = \deg L$ at every point. Show in this case that Fujita's conjecture follows from Proposition 7.10. □

In the case of surfaces ($n = 2$), Fujita's conjecture has been proved by I. Reider in [Rei88], as a special case of a stronger numerical criterion for very ampleness (a deep generalization of Bombieri's work [Bom73] on pluricanonical embeddings of surfaces of general type). Reider's method is based on the Serre construction for rank 2 bundles and on the Bogomolov instability criterion. Since then, various other proofs have been obtained (Sakai [Sak88], Ein-Lazarsfeld [EL93]). We will give in the next section an algebraic proof of Reider's result based on vanishing theorems, following closely [EL93]. In higher dimensions, the global generation part of the statement has been proved by Ein-Lazarsfeld [EL93] for $n = 3$. However, up to now, there is no strong indication that the conjecture should be true in higher dimensions.

In a naive attempt to prove Fujita's conjecture using part a) of Proposition 7.10, it is natural to ask whether one has $\varepsilon(L, x) \geq 1$ when L is an ample line bundle. Unfortunately, simple examples due to R. Miranda show that $\varepsilon(L, x)$ may be arbitrarily small as soon as $\dim X \geq 2$.

(7.14) Proposition (R. Miranda). *Given $\varepsilon > 0$, there exists a rational surface X , a point $x \in X$ and an ample line bundle L on X such that $\varepsilon(L, x) \leq \varepsilon$.*

Proof. Let $C \subset \mathbb{P}^2$ be an irreducible curve of large degree d with a point x of multiplicity m . Let C' be another irreducible curve of the same degree meeting C transversally. Blow-up the points of $C \cap C'$ to obtain a rational surface X , admitting a map $\varphi : X \rightarrow \mathbb{P}^1$ (the map φ is simply given by $\varphi(z) = P'(z)/P(z)$ where $P = 0$, $P' = 0$ are equations of C and C'). The fibers of φ are the curves in the pencil spanned by C and C' . If C' is chosen general enough, then all these fibers are irreducible. As the fibers are disjoint, we have $C^2 = C \cdot C' = 0$ in X . Now, let E be one of the components of the exceptional divisor of X . Fix an integer $a \geq 2$. It follows from the Nakai-Moishezon criterion that the divisor $L = aC + E$ is ample: in fact $L^2 = 2aC \cdot E + E^2 = 2a - 1$, $L \cdot E = a - 1$, $L \cdot \Gamma = L \cdot C = 1$ if Γ is a fiber of φ ; all other irreducible curves $\Gamma \subset X$ must satisfy $\varphi(\Gamma) = \mathbb{P}^1$, hence they have non empty intersection with C and $L \cdot \Gamma \geq a$. However $\nu(C, x) = m$ by construction, hence $\varepsilon(L, x) \leq 1/m$. $\square$

Although Seshadri constants fail to be bounded below in a uniform way, the following statement (which would imply the Fujita conjecture at generic points) is expected to be true.

(7.15) Conjecture. *If L is a big nef line bundle, then $\varepsilon(L, x) \geq 1$ for x generic.*

By a generic point, we mean a point in the complement of a proper algebraic subvariety, or possibly, in a countable union of proper algebraic subvarieties. Quite recently, major progress has been made on this question by Ein-Lazarsfeld [EL92] and Ein-Küchle-Lazarsfeld [EKL94].

(7.16) Theorem ([EL92]). *Let L be an ample line bundle on a smooth projective surface X . Then $\varepsilon(L, x) \geq 1$ for all except perhaps countably many points $x \in X$, and moreover if $L^2 > 1$ then the set of exceptional points is finite. If $L^2 \geq 5$ and $L \cdot C \geq 2$ for all curves $C \subset X$, then $\varepsilon(L, x) \geq 2$ for all but finitely many $x \in X$. $\square$*

(7.17) Theorem ([EKL94]). *Let L be a big nef line bundle on a projective n -dimensional manifold X . Then $\varepsilon(L, x) \geq 1/n$ for all x outside a countable union of proper algebraic subsets.* $\square$

8. Algebraic Approach to the Fujita Conjecture

This section is devoted to a proof of various results related to the Fujita conjecture. The main ideas occurring here are inspired by a recent work of Y.T. Siu [Siu94a]. His method, which is algebraic in nature and quite elementary, consists in a combination of the Riemann-Roch formula together with Nadel's vanishing theorem (in fact, only the algebraic case is needed, thus the original Kawamata-Viehweg vanishing theorem would be sufficient). Slightly later, Angehrn and Siu [AS94], [Siu94b] introduced other closely related methods, producing better bounds for the global generation question; since their method is rather delicate, we can only refer the reader to the above references. In the sequel, X denotes a projective algebraic n -dimensional manifold. The first observation is the following well-known consequence of the Riemann-Roch formula.

(8.1) Special case of Riemann-Roch. *Let $\mathcal{J} \subset \mathcal{O}_X$ be a coherent ideal sheaf on X such that the subscheme $Y = V(\mathcal{J})$ has dimension d (with possibly some lower dimensional components). Let $[Y] = \sum \lambda_j [Y_j]$ be the effective algebraic cycle of dimension d associated to the d dimensional components of Y (taking into account multiplicities λ_j given by the ideal $\mathcal{J}$). Then for any line bundle F , the Euler characteristic*

$$\chi(Y, \mathcal{O}(F + mL)|_Y) = \chi(X, \mathcal{O}(F + mL) \otimes \mathcal{O}_X/\mathcal{J})$$

is a polynomial $P(m)$ of degree d and leading coefficient $L^d \cdot [Y]/d!$

The second fact is an elementary lemma about numerical polynomials (polynomials with rational coefficients, mapping $\mathbb{Z}$ into $\mathbb{Z}$).

(8.2) Lemma. *Let $P(m)$ be a numerical polynomial of degree $d > 0$ and leading coefficient $a_d/d!$, $a_d \in \mathbb{Z}$, $a_d > 0$. Suppose that $P(m) \geq 0$ for $m \geq m_0$. Then*

- a) *For every integer $N \geq 0$, there exists $m \in [m_0, m_0 + Nd]$ such that $P(m) \geq N$.*
- b) *For every $k \in \mathbb{N}$, there exists $m \in [m_0, m_0 + kd]$ such that $P(m) \geq a_d k^d / 2^{d-1}$.*
- c) *For every integer $N \geq 2d^2$, there exists $m \in [m_0, m_0 + N]$ such that $P(m) \geq N$.*

Proof. a) Each of the N equations $P(m) = 0, P(m) = 1, \dots, P(m) = N - 1$ has at most d roots, so there must be an integer $m \in [m_0, m_0 + dN]$ which is not a root of these.

b) By Newton's formula for iterated differences $\Delta P(m) = P(m+1) - P(m)$, we get

$$\Delta^d P(m) = \sum_{1 \leq j \leq d} (-1)^j \binom{d}{j} P(m + d - j) = a_d, \quad \forall m \in \mathbb{Z}.$$

Hence if $j \in \{0, 2, 4, \dots, 2[d/2]\} \subset [0, d]$ is the even integer achieving the maximum of $P(m_0 + d - j)$ over this finite set, we find

$$2^{d-1}P(m_0 + d - j) = \left(\binom{d}{0} + \binom{d}{2} + \dots \right) P(m_0 + d - j) \geq a_d,$$

whence the existence of an integer $m \in [m_0, m_0 + d]$ with $P(m) \geq a_d/2^{d-1}$. The case $k = 1$ is thus proved. In general, we apply the above case to the polynomial $Q(m) = P(km - (k-1)m_0)$, which has leading coefficient $a_d k^d/d!$

c) If $d = 1$, part a) already yields the result. If $d = 2$, a look at the parabola shows that

$$\max_{m \in [m_0, m_0 + N]} P(m) \geq \begin{cases} a_2 N^2/8 & \text{if } N \text{ is even,} \\ a_2(N^2 - 1)/8 & \text{if } N \text{ is odd;} \end{cases}$$

thus $\max_{m \in [m_0, m_0 + N]} P(m) \geq N$ whenever $N \geq 8$. If $d \geq 3$, we apply b) with k equal to the smallest integer such that $k^d/2^{d-1} \geq N$, i.e. $k = \lceil 2(N/2)^{1/d} \rceil$, where $\lceil x \rceil \in \mathbb{Z}$ denotes the round-up of $x \in \mathbb{R}$. Then $kd \leq (2(N/2)^{1/d} + 1)d \leq N$ whenever $N \geq 2d^2$, as a short computation shows. $\square$

We now apply Nadel's vanishing theorem pretty much in the same way as Siu [Siu94a], but with substantial simplifications in the technique and improvements in the bounds. Our method yields simultaneously a simple proof of the following basic result.

(8.3) Theorem. *If L is an ample line bundle over a projective n -fold X , then the adjoint line bundle $K_X + (n+1)L$ is nef.*

By using Mori theory and the base point free theorem ([Mor82], [Kaw84]), one can even show that $K_X + (n+1)L$ is semiample, i.e., there exists a positive integer m such that $m(K_X + (n+1)L)$ is generated by sections (see [Kaw85] and [Fuj87]). The proof rests on the observation that $n+1$ is the maximal length of extremal rays of smooth projective n -folds. Our proof of (8.3) is different and will be given simultaneously with the proof of Th. (8.4) below.

(8.4) Theorem. *Let L be an ample line bundle and let G be a nef line bundle on a projective n -fold X . Then the following properties hold.*

a) $2K_X + mL + G$ generates simultaneous jets of order $s_1, \dots, s_p \in \mathbb{N}$ at arbitrary points $x_1, \dots, x_p \in X$, i.e., there is a surjective map

$$H^0(X, 2K_X + mL + G) \twoheadrightarrow \bigoplus_{1 \leq j \leq p} \mathcal{O}(2K_X + mL + G) \otimes \mathcal{O}_{X, x_j} / \mathfrak{m}_{X, x_j}^{s_j+1},$$

$$\text{provided that } m \geq 2 + \sum_{1 \leq j \leq p} \binom{3n + 2s_j - 1}{n}.$$

$$\text{In particular } 2K_X + mL + G \text{ is very ample for } m \geq 2 + \binom{3n+1}{n}.$$

- b) $2K_X + (n+1)L + G$ generates simultaneous jets of order $s_1, \dots, s_p$ at arbitrary points $x_1, \dots, x_p \in X$ provided that the intersection numbers $L^d \cdot Y$ of L over all d -dimensional algebraic subsets Y of X satisfy

$$L^d \cdot Y > \frac{2^{d-1}}{[n/d]^d} \sum_{1 \leq j \leq p} \binom{3n + 2s_j - 1}{n}.$$

Proof. The proofs of (8.3) and (8.4 a, b) go along the same lines, so we deal with them simultaneously (in the case of (8.3), we simply agree that $\{x_1, \dots, x_p\} = \emptyset$). The idea is to find an integer (or rational number) m_0 and a singular hermitian metric h_0 on $K_X + m_0L$ with strictly positive curvature current $i\Theta_{h_0} \geq \varepsilon\omega$, such that $V(\mathcal{I}(h_0))$ is 0-dimensional and the weight φ_0 of h_0 satisfies $\nu(\varphi_0, x_j) \geq n + s_j$ for all j . As L and G are nef, $(m - m_0)L + G$ has for all $m \geq m_0$ a metric h' whose curvature $i\Theta_{h'}$ has arbitrary small negative part (see [Dem90]), e.g., $i\Theta_{h'} \geq -\frac{\varepsilon}{2}\omega$. Then $i\Theta_{h_0} + i\Theta_{h'} \geq \frac{\varepsilon}{2}\omega$ is again positive definite. An application of Cor (1.5) to $F = K_X + mL + G = (K_X + m_0L) + ((m - m_0)L + G)$ equipped with the metric $h_0 \otimes h'$ implies the existence of the desired sections in $K_X + F = 2K_X + mL + G$ for $m \geq m_0$.

Let us fix an embedding $\Phi_{|\mu L|} : X \rightarrow \mathbb{P}^N$, $\mu \gg 0$, given by sections $\lambda_0, \dots, \lambda_N \in H^0(X, \mu L)$, and let h_L be the associated metric on L of positive definite curvature form $\omega = \frac{i}{2\pi}\Theta(L)$. In order to obtain the desired metric h_0 on $K_X + m_0L$, we fix $a \in \mathbb{N}^*$ and use a double induction process to construct singular metrics $(h_{k,\nu})_{\nu \geq 1}$ on $aK_X + b_kL$ for a non increasing sequence of positive integers $b_1 \geq b_2 \geq \dots \geq b_k \geq \dots$. Such a sequence must be stationary and m_0 will just be the stationary limit $m_0 = \lim b_k/a$. The metrics $h_{k,\nu}$ are taken to satisfy the following properties:

- $\alpha)$ $h_{k,\nu}$ is an algebraic metric of the form

$$\|\xi\|_{h_{k,\nu}}^2 = \frac{|\tau_k(\xi)|^2}{\left(\sum_{1 \leq i \leq \nu, 0 \leq j \leq N} |\tau_k^{(a+1)\mu}(\sigma_i^{a\mu} \cdot \lambda_j^{(a+1)b_k - am_i})|^2\right)^{1/(a+1)\mu}},$$

defined by sections $\sigma_i \in H^0(X, (a+1)K_X + m_iL)$, $m_i < \frac{a+1}{a}b_k$, $1 \leq i \leq \nu$, where $\xi \mapsto \tau_k(\xi)$ is an arbitrary local trivialization of $aK_X + b_kL$; note that $\sigma_i^{a\mu} \cdot \lambda_j^{(a+1)b_k - am_i}$ is a section of

$$a\mu((a+1)K_X + m_iL) + ((a+1)b_k - am_i)\mu L = (a+1)\mu(aK_X + b_kL).$$

- $\beta)$ $\text{ord}_{x_j}(\sigma_i) \geq (a+1)(n + s_j)$ for all i, j ;

- $\gamma)$ $\mathcal{I}(h_{k,\nu+1}) \supset \mathcal{I}(h_{k,\nu})$ and $\mathcal{I}(h_{k,\nu+1}) \neq \mathcal{I}(h_{k,\nu})$ whenever the zero variety $V(\mathcal{I}(h_{k,\nu}))$ has positive dimension.

The weight $\varphi_{k,\nu} = \frac{1}{2(a+1)\mu} \log \sum |\tau_k^{(a+1)\mu}(\sigma_i^{a\mu} \cdot \lambda_j^{(a+1)b_k - am_i})|^2$ of $h_{k,\nu}$ is plurisubharmonic and the condition $m_i < \frac{a+1}{a}b_k$ implies $(a+1)b_k - am_i \geq 1$, thus the difference $\varphi_{k,\nu} - \frac{1}{2(a+1)\mu} \log \sum |\tau(\lambda_j)|^2$ is also plurisubharmonic. Hence $\frac{i}{2\pi}\Theta_{h_{k,\nu}}(aK_X + b_kL) = \frac{i}{\pi}d'd''\varphi_{k,\nu} \geq \frac{1}{(a+1)}\omega$. Moreover, condition $\beta)$ clearly implies $\nu(\varphi_{k,\nu}, x_j) \geq a(n + s_j)$. Finally, condition $\gamma)$ combined with the strong Noetherian property of coherent sheaves ensures that the sequence $(h_{k,\nu})_{\nu \geq 1}$ will finally produce a zero dimensional

subscheme $V(\mathcal{I}(h_{k,\nu}))$. We agree that the sequence $(h_{k,\nu})_{\nu \geq 1}$ stops at this point, and we denote by $h_k = h_{k,\nu}$ the final metric, such that $\dim V(\mathcal{I}(h_k)) = 0$.

For $k = 1$, it is clear that the desired metrics $(h_{1,\nu})_{\nu \geq 1}$ exist if b_1 is taken large enough (so large, say, that $(a+1)K_X + (b_1-1)L$ generates jets of order $(a+1)(n + \max s_j)$ at every point; then the sections $\sigma_1, \dots, \sigma_\nu$ can be chosen with $m_1 = \dots = m_\nu = b_1 - 1$). Suppose that the metrics $(h_{k,\nu})_{\nu \geq 1}$ and h_k have been constructed and let us proceed with the construction of $(h_{k+1,\nu})_{\nu \geq 1}$. We do this again by induction on ν , assuming that $h_{k+1,\nu}$ is already constructed and that $\dim V(\mathcal{I}(h_{k+1,\nu})) > 0$. We start in fact the induction with $\nu = 0$, and agree in this case that $\mathcal{I}(h_{k+1,0}) = 0$ (this would correspond to an infinite metric of weight identically equal to $-\infty$). By Nadel's vanishing theorem applied to

$$F_m = aK_X + mL = (aK_X + b_k L) + (m - b_k)L$$

with the metric $h_k \otimes (h_L)^{\otimes m - b_k}$, we get

$$H^q(X, \mathcal{O}((a+1)K_X + mL) \otimes \mathcal{I}(h_k)) = 0 \quad \text{for } q \geq 1, m \geq b_k.$$

As $V(\mathcal{I}(h_k))$ is 0-dimensional, the sheaf $\mathcal{O}_X/\mathcal{I}(h_k)$ is a skyscraper sheaf, and the exact sequence $0 \rightarrow \mathcal{I}(h_k) \rightarrow \mathcal{O}_X \rightarrow \mathcal{O}_X/\mathcal{I}(h_k) \rightarrow 0$ twisted with the invertible sheaf $\mathcal{O}((a+1)K_X + mL)$ shows that

$$H^q(X, \mathcal{O}((a+1)K_X + mL)) = 0 \quad \text{for } q \geq 1, m \geq b_k.$$

Similarly, we find

$$H^q(X, \mathcal{O}((a+1)K_X + mL) \otimes \mathcal{I}(h_{k+1,\nu})) = 0 \quad \text{for } q \geq 1, m \geq b_{k+1}$$

(also true for $\nu = 0$, since $\mathcal{I}(h_{k+1,0}) = 0$), and when $m \geq \max(b_k, b_{k+1}) = b_k$, the exact sequence $0 \rightarrow \mathcal{I}(h_{k+1,\nu}) \rightarrow \mathcal{O}_X \rightarrow \mathcal{O}_X/\mathcal{I}(h_{k+1,\nu}) \rightarrow 0$ implies

$$H^q(X, \mathcal{O}((a+1)K_X + mL) \otimes \mathcal{O}_X/\mathcal{I}(h_{k+1,\nu})) = 0 \quad \text{for } q \geq 1, m \geq b_k.$$

In particular, since the H^1 group vanishes, every section u' of $(a+1)K_X + mL$ on the subscheme $V(\mathcal{I}(h_{k+1,\nu}))$ has an extension u to X . Fix a basis $u'_1, \dots, u'_N$ of the sections on $V(\mathcal{I}(h_{k+1,\nu}))$ and take arbitrary extensions $u_1, \dots, u_N$ to X . Look at the linear map assigning the collection of jets of order $(a+1)(n + s_j) - 1$ at all points x_j

$$u = \sum_{1 \leq j \leq N} a_j u_j \mapsto \bigoplus_{x_j} J_{x_j}^{(a+1)(n+s_j)-1}(u).$$

Since the rank of the bundle of s -jets is $\binom{n+s}{n}$, the target space has dimension

$$\delta = \sum_{1 \leq j \leq p} \binom{n + (a+1)(n + s_j) - 1}{n}.$$

In order to get a section $\sigma_{\nu+1} = u$ satisfying condition β) with non trivial restriction $\sigma'_{\nu+1}$ to $V(\mathcal{I}(h_{k+1,\nu}))$, we need at least $N = \delta + 1$ independent sections $u'_1, \dots, u'_N$. This condition is achieved by applying Lemma (8.2) to the numerical polynomial

$$\begin{aligned} P(m) &= \chi(X, \mathcal{O}((a+1)K_X + mL) \otimes \mathcal{O}_X/\mathcal{I}(h_{k+1,\nu})) \\ &= h^0(X, \mathcal{O}((a+1)K_X + mL) \otimes \mathcal{O}_X/\mathcal{I}(h_{k+1,\nu})) \geq 0, \quad m \geq b_k. \end{aligned}$$

The polynomial P has degree $d = \dim V(\mathcal{I}(h_{k+1,\nu})) > 0$. We get the existence of an integer $m \in [b_k, b_k + \eta]$ such that $N = P(m) \geq \delta + 1$ with some explicit integer $\eta \in \mathbb{N}$ (for instance $\eta = n(\delta + 1)$ always works by (8.2a), but we will also use the other possibilities to find an optimal choice in each case). Then we find a section $\sigma_{\nu+1} \in H^0(X, (a+1)K_X + mL)$ with non trivial restriction $\sigma'_{\nu+1}$ to $V(\mathcal{I}(h_{k+1,\nu}))$, vanishing at order $\geq (a+1)(n + s_j)$ at each point x_j . We just set $m_{\nu+1} = m$, and the condition $m_{\nu+1} < \frac{a+1}{a}b_{k+1}$ is satisfied if $b_k + \eta < \frac{a+1}{a}b_{k+1}$. This shows that we can take inductively

$$b_{k+1} = \left\lfloor \frac{a}{a+1}(b_k + \eta) \right\rfloor + 1.$$

By definition, $h_{k+1,\nu+1} \leq h_{k+1,\nu}$, hence $\mathcal{I}(h_{k+1,\nu+1}) \supset \mathcal{I}(h_{k+1,\nu})$. We necessarily have $\mathcal{I}(h_{k+1,\nu+1}) \neq \mathcal{I}(h_{k+1,\nu})$, for $\mathcal{I}(h_{k+1,\nu+1})$ contains the ideal sheaf associated with the zero divisor of $\sigma_{\nu+1}$, whilst $\sigma_{\nu+1}$ does not vanish identically on $V(\mathcal{I}(h_{k+1,\nu}))$. Now, an easy computation shows that the iteration of $b_{k+1} = \left\lfloor \frac{a}{a+1}(b_k + \eta) \right\rfloor + 1$ stops at $b_k = a(\eta + 1) + 1$ for any large initial value b_1 . In this way, we obtain a metric h_∞ of positive definite curvature on $aK_X + (a(\eta + 1) + 1)L$, with $\dim V(\mathcal{I}(h_\infty)) = 0$ and $\nu(\varphi_\infty, x_j) \geq a(n + s_j)$ at each point x_j .

Proof of (8.3). In this case, the set $\{x_j\}$ is taken to be empty, thus $\delta = 0$. By (8.2a), the condition $P(m) \geq 1$ is achieved for some $m \in [b_k, b_k + n]$ and we can take $\eta = n$. As μL is very ample, there exists on μL a metric with an isolated logarithmic pole of Lelong number 1 at any given point x_0 (e.g., the algebraic metric defined with all sections of μL vanishing at x_0). Hence

$$F'_a = aK_X + (a(n+1) + 1)L + n\mu L$$

has a metric h'_a such that $V(\mathcal{I}(h'_a))$ is zero dimensional and contains $\{x_0\}$. By Cor (1.5), we conclude that

$$K_X + F'_a = (a+1)K_X + (a(n+1) + 1 + n\mu)L$$

is generated by sections, in particular $K_X + \frac{a(n+1)+1+n\mu}{a+1}L$ is nef. As a tends to $+\infty$, we infer that $K_X + (n+1)L$ is nef. $\square$

Proof of (8.4a). Here, the choice $a = 1$ is sufficient for our purposes. Then

$$\delta = \sum_{1 \leq j \leq p} \binom{3n + 2s_j - 1}{n}.$$

If $\{x_j\} \neq \emptyset$, we have $\delta + 1 \geq \binom{3n-1}{n} + 1 \geq 2n^2$ for $n \geq 2$. Lemma (8.2c) shows that $P(m) \geq \delta + 1$ for some $m \in [b_k, b_k + \eta]$ with $\eta = \delta + 1$. We can start in fact the induction procedure $k \mapsto k+1$ with $b_1 = \eta + 1 = \delta + 2$, because the only property needed for the induction step is the vanishing property

$$H^0(X, 2K_X + mL) = 0 \quad \text{for } q \geq 1, m \geq b_1,$$

which is true by the Kodaira vanishing theorem and the ampleness of $K_X + b_1L$ (here we use Fujita's result (8.3), observing that $b_1 > n + 1$). Then the recursion

formula $b_{k+1} = \lfloor \frac{1}{2}(b_k + \eta) \rfloor + 1$ yields $b_k = \eta + 1 = \delta + 2$ for all k , and (8.4a) follows. $\square$

Proof of (8.4b). Quite similar to (8.4a), except that we take $\eta = n$, $a = 1$ and $b_k = n + 1$ for all k . By Lemma (8.2b), we have $P(m) \geq a_d k^d / 2^{d-1}$ for some integer $m \in [m_0, m_0 + kd]$, where $a_d > 0$ is the coefficient of highest degree in P . By Lemma (8.1) we have $a_d \geq \inf_{\dim Y=d} L^d \cdot Y$. We take $k = \lfloor n/d \rfloor$. The condition $P(m) \geq \delta + 1$ can thus be realized for some $m \in [m_0, m_0 + kd] \subset [m_0, m_0 + n]$ as soon as

$$\inf_{\dim Y=d} L^d \cdot Y \lfloor n/d \rfloor^d / 2^{d-1} > \delta,$$

which is equivalent to the condition given in (8.4b). $\square$

(8.5) Corollary. *Let X be a smooth projective n -fold, let L be an ample line bundle and G a nef line bundle over X . Then $m(K_X + (n+2)L) + G$ is very ample for $m \geq \binom{3n+1}{n} - 2n$.*

Proof. Apply Th. (8.4a) with $G' = a(K_X + (n+1)L) + G$, so that

$$2K_X + mL + G' = (a+2)(K_X + (n+2)L) + (m-2n-4-a)L + G,$$

and take $m = a + 2n + 4 \geq 2 + \binom{3n+1}{n}$. $\square$

The main drawback of the above technique is that multiples of L at least equal to $(n+1)L$ are required to avoid zeroes of the Hilbert polynomial. In particular, it is not possible to obtain directly a very ampleness criterion for $2K_X + L$ in the statement of (8.4b). Nevertheless, using different ideas from Angehrn-Siu [AS94], [Siu94b] has obtained such a criterion. We derive here a slightly weaker version, thanks to the following elementary Lemma.

(8.6) Lemma. *Assume that for some integer $\mu \in \mathbb{N}^*$ the line bundle μF generates simultaneously all jets of order $\mu(n+s_j)+1$ at any point x_j in a subset $\{x_1, \dots, x_p\}$ of X . Then $K_X + F$ generates simultaneously all jets of order s_j at x_j .*

Proof. Take the algebraic metric on F defined by a basis of sections $\sigma_1, \dots, \sigma_N$ of μF which vanish at order $\mu(n+s_j)+1$ at all points x_j . Since we are still free to choose the homogeneous term of degree $\mu(n+s_j)+1$ in the Taylor expansion at x_j , we find that $x_1, \dots, x_p$ are isolated zeroes of $\bigcap \sigma_j^{-1}(0)$. If φ is the weight of the metric of F near x_j , we thus have $\varphi(z) \sim (n+s_j + \frac{1}{\mu}) \log |z - x_j|$ in suitable coordinates. We replace φ in a neighborhood of x_j by

$$\varphi'(z) = \max(\varphi(z), |z|^2 - C + (n+s_j) \log |z - x_j|)$$

and leave φ elsewhere unchanged (this is possible by taking $C > 0$ very large). Then $\varphi'(z) = |z|^2 - C + (n+s_j) \log |z - x_j|$ near x_j , in particular φ' is strictly plurisubharmonic near x_j . In this way, we get a metric h' on F with semipositive curvature everywhere on X , and with positive definite curvature on a neighborhood of $\{x_1, \dots, x_p\}$. The conclusion then follows directly from Hörmander's L^2 estimates (5.1) and (5.2). $\square$

(8.7) Theorem. *Let X be a smooth projective n -fold, and let L be an ample line bundle over X . Then $2K_X + L$ generates simultaneous jets of order $s_1, \dots, s_p$ at arbitrary points $x_1, \dots, x_p \in X$ provided that the intersection numbers $L^d \cdot Y$ of L over all d -dimensional algebraic subsets Y of X satisfy*

$$L^d \cdot Y > \frac{2^{d-1}}{[n/d]^d} \sum_{1 \leq j \leq p} \binom{(n+1)(4n+2s_j+1)-2}{n}, \quad 1 \leq d \leq n.$$

Proof. By Lemma (8.6) applied with $F = K_X + L$ and $\mu = n+1$, the desired jet generation of $2K_X + L$ occurs if $(n+1)(K_X + L)$ generates jets of order $(n+1)(n+s_j)+1$ at x_j . By Lemma (8.5) again with $F = aK_X + (n+1)L$ and $\mu = 1$, we see by backward induction on a that we need the simultaneous generation of jets of order $(n+1)(n+s_j)+1 + (n+1-a)(n+1)$ at x_j . In particular, for $2K_X + (n+1)L$ we need the generation of jets of order $(n+1)(2n+s_j-1)+1$. Theorem (8.4 b) yields the desired condition. $\square$

We now list a few immediate consequences of Theorem 8.4, in connection with some classical questions of algebraic geometry.

(8.8) Corollary. *Let X be a projective n -fold of general type with K_X ample. Then mK_X is very ample for $m \geq m_0 = \binom{3n+1}{n} + 4$.*

(8.9) Corollary. *Let X be a Fano n -fold, that is, a n -fold such that $-K_X$ is ample. Then $-mK_X$ is very ample for $m \geq m_0 = \binom{3n+1}{n}$.*

Proof. Corollaries 8.8, 8.9 follow easily from Theorem 8.4 a) applied to $L = \pm K_X$. Hence we get pluricanonical embeddings $\Phi : X \rightarrow \mathbb{P}^N$ such that $\Phi^* \mathcal{O}(1) = \pm m_0 K_X$. The image $Y = \Phi(X)$ has degree

$$\deg(Y) = \int_Y c_1(\mathcal{O}(1))^n = \int_X c_1(\pm m_0 K_X)^n = m_0^n |K_X^n|.$$

It can be easily reproved from this that there are only finitely many deformation types of Fano n -folds, as well as of n -folds of general type with K_X ample, corresponding to a given discriminant $|K_X^n|$ (from a theoretical viewpoint, this result is a consequence of Matsusaka's big theorem [Mat72] and [KoM72], but the bounds which can be obtained from it are probably extremely huge). In the Fano case, a fundamental result obtained indepently by Kollár-Miyaoka-Mori [KoMM92] and Campana [Cam92] shows that the discriminant K_X^n is in fact bounded by a constant C_n depending only on n . Therefore, one can find an explicit bound C'_n for the degree of the embedding Φ , and it follows that there are only finitely many families of Fano manifolds in each dimension. $\square$

In the case of surfaces, much more is known. We will content ourselves with a brief account of recent results. If X is a surface, the failure of an adjoint bundle $K_X + L$ to be globally generated or very ample is described in a very precise way by the following result of I. Reider [Rei88].

(8.10) Reider's Theorem. *Let X be a smooth projective surface and let L be a nef line bundle on X .*

- a) *Assume that $L^2 \geq 5$ and let $x \in X$ be a given point. Then $K_X + L$ has a section which does not vanish at x , unless there is an effective divisor $D \subset X$ passing through x such that either*

$$\begin{aligned} L \cdot D = 0 \quad \text{and} \quad D^2 = -1; \quad \text{or} \\ L \cdot D = 1 \quad \text{and} \quad D^2 = 0. \end{aligned}$$

- b) *Assume that $L^2 \geq 10$. Then any two points $x, y \in X$ (possibly infinitely near) are separated by sections of $K_X + L$, unless there is an effective divisor $D \subset X$ passing through x and y such that either*

$$\begin{aligned} L \cdot D = 0 \quad \text{and} \quad D^2 = -1 \text{ or } -2; \quad \text{or} \\ L \cdot D = 1 \quad \text{and} \quad D^2 = 0 \text{ or } -1; \quad \text{or} \\ L \cdot D = 2 \quad \text{and} \quad D^2 = 0. \end{aligned} \quad \square$$

(8.11) Corollary. *Let L be an ample line bundle on a smooth projective surface X . Then $K_X + 3L$ is globally generated and $K_X + 4L$ is very ample. If $L^2 \geq 2$ then $K_X + 2L$ is globally generated and $K_X + 3L$ is very ample.* $\square$

The case of higher order jets can be treated similarly. The most general result in this direction has been obtained by Beltrametti and Sommese [BeS93].

(8.12) Theorem ([BeS93]). *Let X be a smooth projective surface and let L be a nef line bundle on X . Let p be a positive integer such that $L^2 > 4p$. Then for every 0-dimensional subscheme $Z \subset X$ of length $h^0(Z, \mathcal{O}_Z) \leq p$ the restriction*

$$\rho_Z : H^0(X, \mathcal{O}_X(K_X + L)) \longrightarrow H^0(Z, \mathcal{O}_Z(K_X + L))$$

is surjective, unless there is an effective divisor $D \subset X$ intersecting the support $|Z|$ such that

$$L \cdot D - p \leq D^2 < \frac{1}{2} L \cdot D. \quad \square$$

Proof (Sketch). The proof the above theorems rests in an essential way on the construction of rank 2 vector bundles sitting in an exact sequence

$$0 \rightarrow \mathcal{O}_X \rightarrow E \rightarrow L \otimes \mathcal{I}_Z \rightarrow 0.$$

Arguing by induction on the length of Z , we may assume that Z is a 0-dimensional subscheme such that ρ_Z is not surjective, but such that $\rho_{Z'}$ is surjective for every proper subscheme $Z' \subset Z$. The existence of E is obtained by a classical construction of Serre (unfortunately, this construction only works in dimension 2). The numerical condition on L^2 in the hypotheses ensures that $c_1(E)^2 - 4c_2(E) > 0$, hence E is unstable in the sense of Bogomolov. The existence of the effective divisor D asserted in 8.10 or 8.12 follows. We refer to [Rei88], [BeS93] and [Laz93] for details. The

reader will find in [FdB93] a proof of the Bogomolov inequality depending only on the Kawamata-Viehweg vanishing theorem. $\square$

(8.13) Exercise. The goal of the exercise is to prove the following weaker form of Theorems 8.10 and 8.12, by a simple direct method based on Nadel's vanishing theorem:

Let L be a nef line bundle on a smooth projective surface X . Fix points $x_1, \dots, x_N$ and corresponding multiplicities $s_1, \dots, s_N$, and set $p = \sum (2 + s_j)^2$. Then $H^0(X, K_X + L)$ generates simultaneously jets of order s_j at all points x_j provided that $L^2 > p$ and $L \cdot C > p$ for all curves C passing through one of the points x_j .

- Using the Riemann-Roch formula, show that the condition $L^2 > p$ implies the existence of a section of a large multiple mL vanishing at order $> m(2 + s_j)$ at each of the points.
- Construct a sequence of singular hermitian metrics on L with positive definite curvature, such that the weights φ_ν have algebraic singularities, $\nu(\varphi_\nu, x_j) \geq 2 + s_j$ at each point, and such that for some integer $m_1 > 0$ the multiplier ideal sheaves satisfy $\mathcal{I}(m_1\varphi_{\nu+1}) \supsetneq \mathcal{I}(m_1\varphi_\nu)$ if $V(\mathcal{I}(\varphi_\nu))$ is not 0-dimensional near some x_j .

Hint: a) starts the procedure. Fix $m_0 > 0$ such that $m_0L - K_X$ is ample. Use Nadel's vanishing theorem to show that

$$H^q(X, \mathcal{O}((m + m_0)L) \otimes \mathcal{I}(\lambda m \varphi_\nu)) = 0 \quad \text{for all } q \geq 1, m \geq 0, \lambda \in [0, 1].$$

Let D_ν be the effective $\mathbb{Q}$ -divisor describing the 1-dimensional singularities of φ_ν . Then $\mathcal{I}(\lambda m \varphi_\nu) \subset \mathcal{O}(-\lfloor \lambda m D_\nu \rfloor)$ and the quotient has 0-dimensional support, hence

$$H^q(X, \mathcal{O}((m + m_0)L) \otimes \mathcal{O}(-\lfloor \lambda m D_\nu \rfloor)) = 0 \quad \text{for all } q \geq 1, m \geq 0, \lambda \in [0, 1].$$

By Riemann-Roch again prove that

$$(\star) \quad h^0(X, \mathcal{O}((m + m_0)L) \otimes \mathcal{O}/\mathcal{O}(-\lfloor \lambda m D_\nu \rfloor)) = \frac{m^2}{2}(2\lambda L \cdot D_\nu - \lambda^2 D_\nu^2) + O(m).$$

As the left hand side of $(\star)$ is increasing with λ , one must have $D_\nu^2 \leq L \cdot D_\nu$. If $V(\mathcal{I}(\varphi_\nu))$ is not 0-dimensional at x_j , then the coefficient of some component of D_ν passing through x_j is at least 1, hence

$$2L \cdot D_\nu - D_\nu^2 \geq L \cdot D_\nu \geq p + 1.$$

Show the existence of an integer $m_1 > 0$ independent of ν such that

$$h^0(X, \mathcal{O}((m + m_0)L) \otimes \mathcal{O}/\mathcal{O}(-\lfloor m D_\nu \rfloor)) > \sum_{1 \leq j \leq N} \binom{(m + m_0)(2 + s_j) + 2}{2}$$

for $m \geq m_1$, and infer the existence of a suitable section of $(m_1 + m_0)L$ which is not in $H^0(X, \mathcal{O}((m_1 + m_0)L - \lfloor m_1 D_\nu \rfloor))$. Use this section to construct $\varphi_{\nu+1}$ such that $\mathcal{I}(m_1\varphi_{\nu+1}) \supsetneq \mathcal{I}(m_1\varphi_\nu)$.

9. Regularization of Currents and Self-intersection Inequalities

Let X be a compact complex n -dimensional manifold. It will be convenient to work with currents that are not necessarily positive, but such that their negative part is locally bounded. We say that a bidimension (p, p) current T is *almost positive* if there exists a smooth form v of bidegree $(n-p, n-p)$ such that $T+v \geq 0$. Similarly, a function φ on X is said to be *almost psh* if φ is locally equal to the sum of a psh function and of a smooth function; then the $(1, 1)$ -current $dd^c\varphi$ is almost positive; conversely, if a locally integrable function φ is such that $dd^c\varphi$ is almost positive, then φ is equal a.e. to an almost psh function. If T is closed and almost positive, the Lelong numbers $\nu(T, x)$ are well defined, since the negative part always contributes to zero.

(9.1) Theorem. *Let T be a closed almost positive $(1, 1)$ -current and let α be a smooth real $(1, 1)$ -form in the same dd^c -cohomology class as T , i.e. $T = \alpha + dd^c\psi$ where ψ is an almost psh function. Let γ be a continuous real $(1, 1)$ -form such that $T \geq \gamma$. Suppose that $\mathcal{O}_{T_X}(1)$ is equipped with a smooth hermitian metric such that the curvature form satisfies*

$$\frac{i}{2\pi}\Theta(\mathcal{O}_{T_X}(1)) + \pi^*u \geq 0$$

*with $\pi : P(T^*X) \rightarrow X$ and with some nonnegative smooth $(1, 1)$ -form u on X . Fix a hermitian metric ω on X . Then for every $c > 0$, there is a sequence of closed almost positive $(1, 1)$ -currents $T_{c,k} = \alpha + dd^c\psi_{c,k}$ such that $\psi_{c,k}$ is smooth on $X \setminus E_c(T)$ and decreases to ψ as k tends to $+\infty$ (in particular, $T_{c,k}$ is smooth on $X \setminus E_c(T)$ and converges weakly to T on X), and*

$$T_{c,k} \geq \gamma - \lambda_{c,k}u - \varepsilon_k\omega$$

where

- a) $\lambda_{c,k}(x)$ is a decreasing sequence of continuous functions on X such that $\lim_{k \rightarrow +\infty} \lambda_{c,k}(x) = \min(\nu(T, x), c)$ at every point,
- b) $\lim_{k \rightarrow +\infty} \varepsilon_k = 0$,
- c) $\nu(T_{c,k}, x) = (\nu(T, x) - c)_+$ at every point $x \in X$.

Here $\mathcal{O}_{T_X}(1)$ is the canonical line bundle associated with T_X over the hyperplane bundle $P(T^*X)$. Observe that the theorem gives in particular approximants $T_{c,k}$ which are smooth everywhere on X if c is taken such that $c > \max_{x \in X} \nu(T, x)$. The equality in c) means that the procedure kills all Lelong numbers that are $\leq c$ and shifts all others downwards by c . Hence Theorem 9.1 is an analogue over manifolds of Kiselman's procedure [Kis78, 79] for killing Lelong numbers of a psh function on an open subset of $\mathbb{C}^n$.

Proof (Sketch). We refer to [Dem92] for a detailed proof. We only sketch a special case for which the main idea is simple to explain. The special case we wish to

consider is the following: X is projective algebraic, $u = \frac{i}{2\pi}\Theta(G)$ is the curvature form of a nef $\mathbb{Q}$ -divisor G , and T has the form

$$T = \alpha + dd^c\psi, \quad \psi = \log(|f_1|^2 + \dots + |f_N|^2)^{1/2},$$

where α is smooth and the f_j 's are sections of some C^∞ hermitian line bundle L on X . The lower bound for T is

$$T \geq \gamma := \alpha - \frac{i}{2\pi}\Theta(L),$$

hence T is almost positive. Somehow, the general situation can be reduced locally to this one by an approximation theorem for currents based on Hörmander's L^2 estimates, in the form given by Ohsawa-Takegoshi ([OT87], [Ohs88]); the main point is to show that any closed positive $(1,1)$ -current is locally a weak limit of effective $\mathbb{Q}$ -divisors which have roughly the same Lelong numbers as the given current, up to small errors converging to 0; the proof is then completed by means of rather tricky gluing techniques for psh functions (see [Dem92]). Now, let A be an ample divisor and let $\omega = \frac{i}{2\pi}\Theta(A)$ be a positive curvature form for A . After adding εA to G ($\varepsilon \in \mathbb{Q}^+$), which is the same as adding $\varepsilon\omega$ to u , we may assume that u is positive definite and that $\mathcal{O}_{T_X}(1) + \pi^*G$ is ample ($\mathcal{O}_{T_X}(1)$ is relatively ample, so adding something ample from X is enough to make it ample on $P(T_X^*)$). The Lelong numbers of T are given by the simple formula

$$\nu(T, x) = \min_{1 \leq i \leq N} \text{ord}_x(f_i).$$

The basic idea is to decrease the Lelong numbers of T by adding some “derivatives” of the f_i 's in the sum of squares. The derivatives have of course to be computed as global sections on X . For this, we introduce two positive integers m, p which will be selected later more carefully, and we consider the m -jet sections $J^m f_i^{\otimes p}$, viewed as sections of the m -jet vector bundle $J^m L^{\otimes p}$. The bundle $J^m L^{\otimes p}$ has a filtration whose graded terms are $S^\nu T_X^* \otimes L^{\otimes p}$, $0 \leq \nu \leq m$, hence $(J^m L^{\otimes p})^* \otimes L^{\otimes p}$ has a filtration with graded terms $S^\nu T_X$, $0 \leq \nu \leq m$. Our hypothesis $T_X \otimes \mathcal{O}(G)$ ample implies that $(J^m L^{\otimes p})^* \otimes \mathcal{O}(pL + mG)$ is ample. Hence, raising to some symmetric power S^ν where ν is a multiple of a denominator for the $\mathbb{Q}$ -divisor G , we obtain that $S^\nu(J^m L^{\otimes p})^* \otimes \mathcal{O}(\nu pL + \nu mG)$ is generated by global sections $P_1, \dots, P_{N'}$ for ν large (we view the P_j 's as some kind of “differential operators” acting on sections of L). Now, the ν -th symmetric product $S^\nu(J^m f_i^{\otimes p})$ takes values in $S^\nu(J^m L^{\otimes p})$, hence the pairing with the dual bundle yields a section

$$F_{m,i,j}^{p,\nu} := P_j \cdot S^\nu(J^m f_i^{\otimes p}) \in H^0(X, \mathcal{O}(\nu pL + \nu mG)).$$

Since the P_j generate sections and $J^m f_i^{\otimes p}$ vanishes at order $(p \text{ord}_x(f_i) - m)_+$ at any point x , we get

$$\min_j \text{ord}_x(F_{m,i,j}^{p,\nu}) = \nu(p \text{ord}_x(f_i) - m)_+.$$

We set

$$\begin{aligned} T_{c,k} &= \alpha + dd^c\psi_{c,k}, \\ \psi_{c,k} &= \frac{1}{2\nu p} \log \left(\left(\sum_i |f_i|^2 \right)^{\nu p} + \sum_{1 \leq \mu \leq m} k^{-\mu} \sum_{i,j} |F_{\mu,i,j}^{p,\nu}|^2 \right) \end{aligned}$$

Clearly, $\psi_{c,k}$ is decreasing and converges to ψ , and its Lelong numbers are

$$\nu(\psi_{c,k}, x) = \min_i \left(\text{ord}_x(f_i) - \frac{m}{p} \right)_+ = \left(\nu(\psi, x) - \frac{m}{p} \right)_+.$$

Select $c = \frac{m}{p}$ (or take a very close rational approximation if an arbitrary real number $c \in \mathbb{R}^+$ is to be considered). As $F_{m,i,j}^{p,\nu}$ takes values in $\mathcal{O}(\nu p L + \nu m G)$, an easy computation yields

$$T_{c,k} \geq \alpha - \frac{i}{2\pi} \Theta \left(L + \frac{m}{p} G \right) = \gamma - \frac{m}{p} u = \gamma - cu.$$

However, at points where $\nu(T, x) < c$, there is already a term with $\frac{\mu}{p} \simeq \nu(T, x)$ such that the contribution of the terms indexed by μ give a non zero contribution. The terms corresponding to higher indices will then not contribute much to the lower bound of $dd^c \psi_{c,k}$ since in the summation $k^{-\mu+1}$ becomes negligible against $k^{-\mu}$ as $k \rightarrow +\infty$. This gives a lower bound $T_{c,k} \geq \gamma - \lambda_{c,k} - \varepsilon_k \omega$ of the expected form. $\square$

(9.2) Corollary. *Let Θ be a closed almost positive current of bidimension (p, p) and let $\alpha_1, \dots, \alpha_q$ be closed almost positive $(1, 1)$ -currents such that $\alpha_1 \wedge \dots \wedge \alpha_q \wedge \Theta$ is well defined by application of criteria 2.3 or 2.5, when α_j is written locally as $\alpha_j = dd^c u_j$. Then*

$$\{\alpha_1 \wedge \dots \wedge \alpha_q \wedge \Theta\} = \{\alpha_1\} \cdots \{\alpha_q\} \cdot \{\Theta\}.$$

Proof. Theorem 9.1 and the monotone continuity theorem for Monge-Ampère operators show that

$$\alpha_1 \wedge \dots \wedge \alpha_q \wedge \Theta = \lim_{k \rightarrow +\infty} \alpha_1^k \wedge \dots \wedge \alpha_q^k \wedge \Theta$$

where $\alpha_j^k \in \{\alpha_j\}$ is smooth. Since the result is by definition true for smooth forms, we conclude by the weak continuity of cohomology class assignment. $\square$

Now, let X be a compact Kähler manifold equipped with a Kähler metric ω . The *degree* of a closed positive current Θ with respect to ω is by definition

$$(9.3) \quad \deg_\omega \Theta = \int_X \Theta \wedge \omega^p, \quad \text{bidim } \Theta = (p, p).$$

In particular, the degree of a p -dimensional analytic set $A \subset X$ is its volume $\int_A \omega^p$ with respect to ω . We are interested in the following problem.

(9.4) Problem. *Let T be a closed positive $(1, 1)$ -current on X . Is it possible to derive a bound for the codimension p components in the Lelong upperlevel sets $E_c(T)$ in terms of the cohomology class $\{T\} \in H_{DR}^2(X, \mathbb{R})$?*

Let $\Xi \subset X$ be an arbitrary subset. We introduce the sequence

$$0 = b_1 \leq \dots \leq b_n \leq b_{n+1}$$

of “jumping values” of $E_c(T)$ over Ξ , defined by the property that the dimension of $E_c(T)$ in a neighborhood of Ξ drops by one unit when c gets larger than b_p , namely

$$b_p = \inf \{c > 0; \text{codim}(E_c(T), x) \geq p, \forall x \in \Xi\}.$$

Then, when $c \in]b_p, b_{p+1}]$, we have $\text{codim } E_c(T) = p$ in a neighborhood of Ξ , with at least one component of codimension p meeting Ξ . Let $(Z_{p,k})_{k \geq 1}$ be the family of all these p -codimensional components (occurring in any of the sets $E_c(T)$ for $c \in]b_p, b_{p+1}]$), and let

$$\nu_{p,k} = \min_{x \in Z_{p,k}} \nu(T, x) \in]b_p, b_{p+1}]$$

be the generic Lelong number of T along $Z_{p,k}$. Then we have the following self-intersection inequality.

(9.5) Theorem. *Suppose that X is Kähler and that $\mathcal{O}_{T_X}(1)$ has a hermitian metric such that $\frac{i}{2\pi} \Theta(\mathcal{O}_{T_X}(1)) + \pi^* u \geq 0$, where u is a smooth closed semipositive $(1,1)$ -form. Let $\Xi \subset X$ be an arbitrary subset, let T be a closed positive current of bidegree $(1,1)$, and let (b_p) , $(Z_{p,k})$ be the corresponding jumping values and p -codimensional components of $E_c(T)$ meeting Ξ . Assume either that $\Xi = X$ or that the cohomology class $\{T\} \in H^{1,1}(X)$ is nef (i.e. in the closure of the Kähler cone). Then, for each $p = 1, \dots, n$, the De Rham cohomology class $(\{T\} + b_1\{u\}) \cdots (\{T\} + b_p\{u\})$ can be represented by a closed positive current Θ_p of bidegree (p,p) such that*

$$\Theta_p \geq \sum_{k \geq 1} (\nu_{p,k} - b_1) \cdots (\nu_{p,k} - b_p) [Z_{p,k}] + (T_{\text{ac}} + b_1 u) \wedge \cdots \wedge (T_{\text{ac}} + b_p u)$$

where $T_{\text{ac}} \geq 0$ is the absolutely continuous part in the Lebesgue decomposition of T (decomposition of the coefficients of T into absolutely continuous and singular measures), $T = T_{\text{ac}} + T_{\text{sing}}$.

By neglecting the second term in the right hand side and taking the wedge product with ω^{n-p} , we get the following interesting consequence:

(9.6) Corollary. *If ω is a Kähler metric on X and if $\{u\}$ is a nef cohomology class such that $c_1(\mathcal{O}_{T_X}(1)) + \pi^* \{u\}$ is nef, the degrees of the components $Z_{p,k}$ with respect to ω satisfy the estimate*

$$\begin{aligned} \sum_{k=1}^{+\infty} (\nu_{p,k} - b_1) \cdots (\nu_{p,k} - b_p) \int_X [Z_{p,k}] \wedge \omega^{n-p} \\ \leq (\{T\} + b_1\{u\}) \cdots (\{T\} + b_p\{u\}) \cdot \{\omega\}^{n-p}. \end{aligned}$$

As a special case, if D is an effective divisor and $T = [D]$, we get a bound for the degrees of the p -codimensional singular strata of D in terms of a polynomial of degree p in the cohomology class $\{D\}$; the multiplicities $(\nu_{p,k} - b_1) \cdots (\nu_{p,k} - b_p)$ are then positive integers. The case when X is $\mathbb{P}^n$ or a homogeneous manifold is especially simple: then T_X is generated by sections and we can take $u = 0$; the bound

is thus simply $\{D\}^p \cdot \{\omega\}^{n-p}$; the same is true more generally as soon as $\mathcal{O}_{T_X}(1)$ is nef. The main idea of the proof is to kill the Lelong numbers of T up to the level b_j ; then the singularities of the resulting current T_j occur only in codimension j and it becomes possible to define the wedge product $T_1 \wedge \dots \wedge T_p$ by means of Proposition 2.3. Here are the details:

Proof of Theorem 9.5. First suppose that $\Xi = X$. We argue by induction on p . For $p = 1$, Siu's decomposition formula shows that

$$T = \sum \nu_{1,k} [Z_{1,k}] + R,$$

and we have $R \geq T_{ac}$ since the other part has singular measures as coefficients. The result is thus true with $\Theta_1 = T$. Now, suppose that Θ_{p-1} has been constructed. For $c > b_p$, the current $T_{c,k} = \alpha + dd^c \psi_{c,k}$ produced by Theorem 9.1 is such that the codimension of the set of poles $\psi_{c,k}^{-1}(-\infty) = E_c(T)$ is at least p at every point $x \in X$ (recall that $\Xi = X$). Then Proposition 2.5 shows that

$$\Theta_{p,c,k} = \Theta_{p-1} \wedge (T_{c,k} + c u + \varepsilon_k \omega)$$

is well defined. If ε_k tends to zero slowly enough, $T_{c,k} + c u + \varepsilon_k \omega$ is positive by (9.1 a), so $\Theta_{p,c,k} \geq 0$. Moreover, by Corollary 9.2, the cohomology class of $\Theta_{p,c,k}$ is $\{\Theta_{p-1}\} \cdot (\{T\} + c\{u\} + \varepsilon_k\{\omega\})$, converging to $\{\Theta_{p-1}\} \cdot (\{T\} + c\{u\})$. Since the mass $\int_X \Theta_{p,c,k} \wedge \omega^{n-p}$ remains uniformly bounded, the family $(\Theta_{p,c,k})_{c \in]b_p, b_p+1], k \geq 1}$ is relatively compact in the weak topology. We define

$$\Theta_p = \lim_{c \rightarrow b_p+0} \lim_{k \rightarrow +\infty} \Theta_{p,c,k},$$

possibly after extracting some weakly convergent subsequence. Then $\{\Theta_p\} = \{\Theta_{p-1}\} \cdot (\{T\} + b_p\{u\})$, and so $\{\Theta_p\} = (\{T\} + b_1\{u\}) \cdots (\{T\} + b_p\{u\})$. Moreover, it is well-known (and easy to check) that Lelong numbers are upper semi-continuous with respect to weak limits of currents. Therefore

$$\begin{aligned} \nu(\Theta_p, x) &\geq \limsup_{c \rightarrow b_p+0} \limsup_{k \rightarrow +\infty} \nu(\Theta_{p-1} \wedge (T_{c,k} + c u + \varepsilon_k \omega), x) \\ &\geq \nu(\Theta_{p-1}, x) \times \limsup_{c \rightarrow b_p+0} \limsup_{k \rightarrow +\infty} \nu(T_{c,k}, x) \\ &\geq \nu(\Theta_{p-1}, x) (\nu(T, x) - b_p)_+ \end{aligned}$$

by application of Proposition 2.16 and (9.1 a). Hence by induction we get

$$\nu(\Theta_p, x) \geq (\nu(T, x) - b_1)_+ \dots (\nu(T, x) - b_p)_+,$$

in particular, the generic Lelong number of Θ_p along $Z_{p,k}$ is at least equal to the product $(\nu_{p,k} - b_1) \dots (\nu_{p,k} - b_p)$. This already implies

$$\Theta_p \geq \sum_{k \geq 1} (\nu_{p,k} - b_1) \dots (\nu_{p,k} - b_p) [Z_{p,k}].$$

Since the right hand side is Lebesgue singular, the desired inequality will be proved if we show in addition that

$$\Theta_{p,ac} \geq (T_{ac} + b_1 u) \wedge \dots \wedge (T_{ac} + b_p u),$$

or inductively, that $\Theta_{p,ac} \geq \Theta_{p-1,ac} \wedge (T_{ac} + b_p u)$. In order to do this, we simply have to make sure that $\lim_{k \rightarrow +\infty} T_{c,k,ac} = T_{ac}$ almost everywhere and use induction again. But our arguments are not affected if we replace $T_{c,k}$ by $T'_{c,k} = \alpha + dd^c \psi'_{c,k}$ where $\psi'_{c,k} = \max\{\psi, \psi_{c,k} - A_k\}$ and (A_k) is a sequence converging quickly to $+\infty$. Lemma 9.7 below shows that a suitable choice of A_k gives $\lim(dd^c \psi'_{c,k})_{ac} = (dd^c \psi)_{ac}$ almost everywhere. This concludes the proof in the case $\Xi = X$.

When $\Xi \neq X$, a slight difficulty appears: in fact, there may remain in $T_{c,k}$ some poles of codimension $\leq p-1$, corresponding to components of $E_c(T)$ which do not meet Ξ (since we completely forgot these components in the definition of the jumping values). It follows that the wedge product $\Theta_{p-1} \wedge (T'_{c,k} + cu + \varepsilon_k \omega)$ is no longer well defined. In this case, we proceed as follows. The assumption that T is nef implies that are smooth functions $\tilde{\psi}_k$ such that the cohomology class $\{T\}$ has a representative $\alpha + dd^c \tilde{\psi}_k \geq -\varepsilon_k \omega$. We replace $T'_{c,k}$ in the above arguments by $T'_{c,k,\nu} = \alpha + dd^c \psi'_{c,k,\nu}$ where

$$\psi'_{c,k,\nu} = \max\{\psi, \psi_{c,k} - A_k, \tilde{\psi}_k - \nu\}.$$

Then certainly $T'_{c,k,\nu} + cu + \varepsilon_k \omega \geq 0$ and we can define a closed positive current $\Theta_{p-1} \wedge (T'_{c,k,\nu} + cu + \varepsilon_k \omega)$ without any difficulty since $\psi'_{c,k,\nu}$ is locally bounded. We first extract a weak limit $\Theta_{p,c,k}$ as $\nu \rightarrow +\infty$. By monotone continuity of Monge-Ampère operators, we find

$$\Theta_{p,c,k} = \Theta_{p-1} \wedge (T'_{c,k} + cu + \varepsilon_k \omega)$$

in the neighborhood of Ξ where this product is well defined. All other arguments are the same as before.

(9.7) Lemma. *Let $\Omega \subset \mathbb{C}^n$ be an open subset and let φ be an arbitrary psh function on Ω . Set $\varphi_\nu = \max(\varphi, \psi_\nu)$ where ψ_ν is a decreasing sequence of psh functions converging to $-\infty$, each ψ_ν being locally bounded in Ω (or perhaps only in the complement of an analytic subset of codimension $\geq p$). Let Θ be a closed positive current of bidegree $(p-1, p-1)$. If $\Theta \wedge dd^c \varphi_\nu$ converges to a weak limit Θ' , then*

$$\Theta'_{ac} \geq \Theta_{ac} \wedge (dd^c \varphi)_{ac}.$$

Proof. Let (ρ_ε) (resp. $(\tilde{\rho}_\varepsilon)$) be a family of regularizing kernels on $\mathbb{C}^n$ (resp. on $\mathbb{R}^2$), and let $\max_\varepsilon(x, y) = (\max \star \tilde{\rho}_\varepsilon)(x, y)$ be a regularized max function. For $\varepsilon > 0$ small enough, the function

$$\varphi_{\nu,\varepsilon} = \max_\varepsilon(\varphi \star \rho_\varepsilon, \psi_\nu \star \rho_\varepsilon)$$

is psh and well defined on any preassigned open set $\Omega' \subset \subset \Omega$. As $\varphi_{\nu,\varepsilon}$ decreases to φ_ν when ε decreases to 0, proposition 10.2 shows that

$$\lim_{\varepsilon \rightarrow 0} \Theta \wedge dd^c \varphi_{\nu,\varepsilon} = \Theta \wedge dd^c \varphi_\nu$$

in the weak topology. Let (β_j) be a sequence of test forms which is dense in the space of test forms of bidegree $(n-p, n-p)$ and contains strongly positive forms with arbitrary large compact support in Ω . Select $\varepsilon_\nu > 0$ so small that

$$\langle \Theta \wedge dd^c \varphi_{\nu, \varepsilon_\nu} - \Theta \wedge dd^c \varphi_\nu, \beta_j \rangle \leq \frac{1}{\nu} \quad \text{for } j \leq \nu.$$

Then the sequence $\Theta \wedge dd^c \varphi_{\nu, \varepsilon_\nu}$ is locally uniformly bounded in mass and converges weakly to the same limit Θ' as $\Theta \wedge dd^c \varphi_\nu$. Moreover, at every point $x \in \Omega$ such that $\varphi(x) > -\infty$, we have $\varphi_{\nu, \varepsilon_\nu}(x) \geq \varphi(x) > \psi_\nu \star \rho_{\varepsilon_\nu}(x) + 1$ for ν large, because $\lim_{\nu \rightarrow -\infty} \psi_\nu = -\infty$ locally uniformly. Hence $\varphi_{\nu, \varepsilon_\nu} = \varphi \star \rho_{\varepsilon_\nu}$ on a neighborhood of x (which may depend on ν) and $dd^c \varphi_{\nu, \varepsilon_\nu}(x) = (dd^c \varphi) \star \rho_{\varepsilon_\nu}(x)$ for $\nu \geq \nu(x)$. By the Lebesgue density theorem, if μ is a measure of absolutely continuous part μ_{ac} , the sequence $\mu \star \rho_{\varepsilon_\nu}(x)$ converges to $\mu_{ac}(x)$ at almost every point. Therefore $\lim dd^c \varphi_{\nu, \varepsilon_\nu}(x) = (dd^c \varphi)_{ac}(x)$ almost everywhere. For any strongly positive test form $\alpha = i\alpha_1 \wedge \bar{\alpha}_1 \wedge \dots \wedge i\alpha_{n-p} \wedge \bar{\alpha}_{n-p}$ of bidegree $(n-p, n-p)$ on Ω , we get

$$\begin{aligned} \int_\Omega \Theta' \wedge \alpha &= \lim_{\nu \rightarrow +\infty} \int_\Omega \Theta \wedge dd^c \varphi_{\nu, \varepsilon_\nu} \wedge \alpha \\ &\geq \liminf_{\nu \rightarrow +\infty} \int_\Omega \Theta_{ac} \wedge dd^c \varphi_{\nu, \varepsilon_\nu} \wedge \alpha \geq \int_\Omega \Theta_{ac} \wedge dd^c \varphi_{ac} \wedge \alpha. \end{aligned}$$

Indeed, the first inequality holds because $dd^c \varphi_{\nu, \varepsilon_\nu}$ is smooth, and the last one results from Fatou's lemma. This implies $\Theta'_{ac} \geq \Theta_{ac} \wedge (dd^c \varphi)_{ac}$ and Lemma 9.7 follows. $\square$

10. Use of Monge-Ampère Equations

The goal of the next two sections is to find numerical criteria for an adjoint line bundle $K_X + L$ to be generated by sections (resp. very ample, s -jet ample). The conditions ensuring these conclusions should be ideally expressed in terms of explicit lower bounds for the intersection numbers $L^p \cdot Y$, where Y runs over p -dimensional subvarieties of X (as the form of Reider's theorem suggests in the case of surfaces). Unfortunately, the simple algebraic approach described in the proof of Theorems 8.4 and 8.5 does not seem to be applicable to get criteria for the very ampleness of $K_X + mL$ in the range $m \leq n+1$. We will now explain how this can be achieved by an alternative analytic method. The essential idea is to construct directly the psh weight function φ needed in Nadel's vanishing theorem by solving a Monge-Ampère equation (Aubin-Calabi-Yau theorem). Quite recently, Ein and Lazarsfeld [EL94] have developed purely algebraic methods which yield similar results; however, up to now, the bounds obtained with the algebraic method are not as good as with the analytic approach.

Let us first recall a special case of the well-known theorem of Aubin-Yau related to the Calabi conjecture. The special case we need is the following fundamental existence result about solutions of Monge-Ampère equations.

(10.1) Theorem ([Yau78], see also [Aub78]). *Let X be a compact complex n -dimensional manifold with a smooth Kähler metric ω . Then for any smooth volume form $f > 0$ with $\int_X f = \int_X \omega^n$, there exists a unique Kähler metric $\tilde{\omega} = \omega + dd^c \psi$ in the Kähler class $\{\omega\}$ such that $\tilde{\omega}^n = (\omega + dd^c \psi)^n = f$.*

There are several equivalent ways of formulating this result. Usually, one starts with a $(1,1)$ -form γ representing the first Chern class $c_1(X) = c_1(\Lambda^n T_X^*)$. Then,

under the normalization $\int_X f = \int_X \omega^n$, there is a unique volume form f on X which, viewed as a hermitian form on $\Lambda^n T_X$, yields $\frac{i}{2\pi} \Theta_f(\Lambda^n T_X) = \gamma$. Then Theorem 10.1 is actually equivalent to finding a Kähler metric $\tilde{\omega}$ in the Kähler class $\{\omega\}$, such that $\text{Ricci}(\tilde{\omega}) = \gamma$. We will not use this viewpoint here, and will be essentially concerned instead with the Monge-Ampère equation $(\omega + dd^c \psi)^n = f$.

There are two different ways in which the Monge-Ampère equation will be used. The most essential idea is that the Monge-Ampère equation can be used to produce weights with logarithmic singularities (as needed for the application of Corollary 5.13), when the right hand side f is taken to be a linear combination of Dirac measures (in fact, f has to be smooth so we rather find solutions ψ_ε corresponding to smooth approximations f_ε of the Dirac measures). This will be explained later.

Another useful consequence of the Monge-Ampère equation is a general version of convexity inequality due to Hovanski [Hov79] and Teissier [Tei79, 82], which is a natural generalization of the usual Hodge index theorem for surfaces. This inequality is reproved along similar lines in [BBS89], where it is applied to the study of projective n -folds of log-general type. For the sake of completeness, we include here a different and slightly simpler proof, based on the Aubin-Yau theorem 10.1 instead of the Hodge index theorem. Our proof also has the (relatively minor) advantage of working over arbitrary Kähler manifolds.

(10.2) Proposition. *The following inequalities hold in any dimension n .*

a) *If $\alpha_1, \dots, \alpha_n$ are semipositive $(1,1)$ -forms on $\mathbb{C}^n$, then*

$$\alpha_1 \wedge \alpha_2 \wedge \dots \wedge \alpha_n \geq (\alpha_1^n)^{1/n} (\alpha_2^n)^{1/n} \dots (\alpha_n^n)^{1/n}.$$

b) *If $u_1, \dots, u_n$ are nef cohomology classes of type $(1,1)$ on a Kähler manifold X of dimension n , then*

$$u_1 \cdot u_2 \cdots u_n \geq (u_1^n)^{1/n} (u_2^n)^{1/n} \dots (u_n^n)^{1/n}.$$

By a nef cohomology class of type $(1,1)$, we mean a class in the closed convex cone of $H^{1,1}(X, \mathbb{R})$ generated by Kähler classes, that is, a class $\{u\}$ admitting representatives u_ε with $u_\varepsilon \geq -\varepsilon \omega$ for every $\varepsilon > 0$. For instance, inequality b) can be applied to $u_j = c_1(L_j)$ when $L_1, \dots, L_n$ are nef line bundles over a projective manifold.

Proof. Observe that a) is a pointwise inequality between (n,n) -forms whereas b) is an inequality of a global nature for the cup product intersection form. We first show that a) holds when only two of the forms α_j are distinct, namely that

$$\alpha^p \wedge \beta^{n-p} \geq (\alpha^n)^{p/n} (\beta^n)^{(n-p)/n}$$

for all $\alpha, \beta \geq 0$. By a density argument, we may suppose $\alpha, \beta > 0$. Then there is a simultaneous orthogonal basis in which

$$\alpha = i \sum_{1 \leq j \leq n} \lambda_j dz_j \wedge d\bar{z}_j, \quad \beta = i \sum_{1 \leq j \leq n} dz_j \wedge d\bar{z}_j$$

with $\lambda_j > 0$, and a) is equivalent to

$$p!(n-p)! \sum_{j_1 < \dots < j_p} \lambda_{j_1} \dots \lambda_{j_p} \geq n! (\lambda_1 \dots \lambda_n)^{p/n}.$$

As both sides are homogeneous of degree p in (λ_j) , we may assume $\lambda_1 \dots \lambda_n = 1$. Then our inequality follows from the inequality between the arithmetic and geometric means of the numbers $\lambda_{j_1} \dots \lambda_{j_p}$. Next, we show that statements a) and b) are equivalent in any dimension n .

a) $\implies$ b). By density, we may suppose that $u_1, \dots, u_n$ are Kähler classes. Fix a positive (n, n) form f such that $\int_X f = 1$. Then Theorem 10.1 implies that there is a Kähler metric α_j representing u_j such that $\alpha_j^n = u_j^n f$. Inequality a) combined with an integration over X yields

$$u_1 \dots u_n = \int_X \alpha_1 \wedge \dots \wedge \alpha_n \geq (u_1^n)^{1/n} \dots (u_n^n)^{1/n} \int_X f.$$

b) $\implies$ a). The forms $\alpha_1, \dots, \alpha_n$ can be considered as constant $(1, 1)$ -forms on any complex torus $X = \mathbb{C}^n / \Gamma$. Inequality b) applied to the associated cohomology classes $u_j \in H^{1,1}(X, \mathbb{R})$ is then equivalent to a).

Finally we prove a) by induction on n , assuming the result already proved in dimension $n-1$. We may suppose that α_n is positive definite, say $\alpha_n = i \sum dz_j \wedge d\bar{z}_j$ in a suitable basis. Denote by $u_1, \dots, u_n$ the associated cohomology classes on the abelian variety $X = \mathbb{C}^n / \mathbb{Z}[i]^n$. Then u_n has integral periods, so some multiple of u_n is the first Chern class of a very ample line bundle $\mathcal{O}(D)$ where D is a smooth irreducible divisor in X . Without loss of generality, we may suppose $u_n = c_1(\mathcal{O}(D))$. Thus

$$u_1 \dots u_{n-1} \cdot u_n = u_1|_D \dots u_{n-1}|_D$$

and by the induction hypothesis we get

$$u_1 \dots u_n \geq (u_{1|D}^{n-1})^{1/(n-1)} \dots (u_{n-1|D}^{n-1})^{1/(n-1)}.$$

However $u_{j|D}^{n-1} = u_j^{n-1} \cdot u_n \geq (u_j^n)^{(n-1)/n} (u_n^n)^{1/n}$, since a) and b) are equivalent and a) is already proved in the case of two forms. b) follows in dimension n , and therefore a) holds in $\mathbb{C}^n$. $\square$

(10.3) Remark. In case α_j (resp. u_j) are positive definite, the equality holds in 10.2 (a,b) if and only if $\alpha_1, \dots, \alpha_n$ (resp. $u_1, \dots, u_n$) are proportional. In our inductive proof, the restriction morphism $H^{1,1}(X, \mathbb{R}) \longrightarrow H^{1,1}(D, \mathbb{R})$ is injective for $n \geq 3$ by the hard Lefschetz theorem, hence it is enough to consider the case of $\alpha^p \wedge \beta^{n-p}$. The equality between arithmetic and geometric means occurs only when all numbers $\lambda_{j_1} \dots \lambda_{j_p}$ are equal, so all λ_j must be equal and $\alpha = \lambda_1 \beta$, as desired. More generally, one can show (exercise to the reader!) that there is an inequality

$$\begin{aligned} \alpha_1 \wedge \dots \wedge \alpha_p \wedge \beta_1 \wedge \dots \wedge \beta_{n-p} &\geq \\ &\geq (\alpha_1^p \wedge \beta_1 \wedge \dots \wedge \beta_{n-p})^{1/p} \dots (\alpha_p^p \wedge \beta_1 \wedge \dots \wedge \beta_{n-p})^{1/p} \end{aligned}$$

for all $(1, 1)$ -forms $\alpha_j, \beta_k \geq 0$, and a similar inequality for products of nef cohomology classes u_j, v_k . $\square$

We now show how the Aubin-Calabi-Yau theorem can be applied to construct singular metrics on ample (or more generally big and nef) line bundles. We first suppose that L is an ample line bundle over a projective n -fold X and that L is equipped with a smooth metric of positive curvature. We consider the Kähler metric $\omega = \frac{i}{2\pi} \Theta(L)$. Any form $\tilde{\omega}$ in the Kähler class of ω can be written as $\tilde{\omega} = \omega + dd^c \psi$, i.e. is the curvature form of L after multiplication of the original metric by a smooth weight function $e^{-\psi}$. By lemma 5.1, the Monge-Ampère equation

$$(10.4) \quad (\omega + dd^c \psi)^n = f$$

can be solved for ψ , whenever f is a smooth (n, n) -form with $f > 0$ and $\int_X f = L^n$. In order to produce logarithmic poles at given points $x_1, \dots, x_N \in X$, the main idea is to let f converge to a Dirac measure at x_j ; then $\tilde{\omega}$ will be shown to converge to a closed positive $(1, 1)$ -current with non zero Lelong number at x_j .

Let $(z_1, \dots, z_n)$ be local coordinates defined on some neighborhood V_j of x_j , and let

$$(10.5) \quad \alpha_{j,\varepsilon} = dd^c(\chi(\log|z_j - x_j|/\varepsilon))$$

where $\chi : \mathbb{R} \rightarrow \mathbb{R}$ is a smooth convex increasing function such that $\chi(t) = t$ for $t \geq 0$ and $\chi(t) = -1/2$ for $t \leq -1$. Then $\alpha_{j,\varepsilon}$ is a smooth positive $(1, 1)$ -form, and $\alpha_{j,\varepsilon} = dd^c \log|z_j - x_j|$ in the complement of the ball $|z_j - x_j| \leq \varepsilon$. It follows that $\alpha_{j,\varepsilon}^n$ has support in the ball $|z_j - x_j| \leq \varepsilon$, and Stokes' formula gives

$$(10.6) \quad \int_{B(x_j, \varepsilon)} \alpha_{j,\varepsilon}^n = \int_{B(x_j, \varepsilon)} (dd^c \log|z_j - x_j|)^n = 1.$$

Hence $\alpha_{j,\varepsilon}^n$ converges weakly to the Dirac measure δ_{x_j} as ε tends to 0. For all positive numbers $\tau_j > 0$ such that $\sigma := \sum \tau_j^n < L^n = \int_X \omega^n$, Theorem 10.1 gives a solution of the Monge-Ampère equation

$$(10.7) \quad \omega_\varepsilon^n = \sum_{1 \leq j \leq N} \tau_j^n \alpha_{j,\varepsilon}^n + \left(1 - \frac{\sigma}{L^n}\right) \omega^n \quad \text{with } \omega_\varepsilon = \omega + dd^c \psi_\varepsilon,$$

since the right-hand side of the first equation is > 0 and has the correct integral value L^n over X . The solution ψ_ε is merely determined up to a constant. If γ is an arbitrary Kähler metric on X , we can normalize ψ_ε in such a way that $\int_X \psi_\varepsilon \gamma^n = 0$.

(10.8) Lemma. *There is a sequence ε_ν converging to zero such that ψ_{ε_ν} has a limit ψ in $L^1(X)$ and such that the sequence of $(1, 1)$ -forms ω_{ε_ν} converges weakly towards a closed positive current T of type $(1, 1)$. Moreover, the cohomology class of T is equal to $c_1(L)$ and $T = \omega + dd^c \psi$.*

Proof. The integral $\int_X \omega_\varepsilon \wedge \gamma^{n-1} = L \cdot \{\gamma\}^{n-1}$ remains bounded, so we can find a sequence ε_ν converging to zero such that the subsequence ω_{ε_ν} converges weakly towards a closed positive current T of bidegree $(1, 1)$. The cohomology class of a current is continuous with respect to the weak topology (this can be seen by Poincaré duality). The cohomology class of T is thus equal to $c_1(L)$. The function ψ_ε satisfies the equation $\frac{1}{\pi} \Delta \psi_\varepsilon = \text{tr}_\gamma(\omega_\varepsilon - \omega)$ where Δ is the Laplace operator associated to γ . Our normalization of ψ_ε implies

$$\psi_\varepsilon = \pi G \operatorname{tr}_\gamma(\omega_\varepsilon - \omega),$$

where G is the Green operator of Δ . As G is a compact operator from the Banach space of bounded Borel measures into $L^1(X)$, we infer that some subsequence (ψ_{ε_ν}) of our initial subsequence converges to a limit ψ in $L^1(X)$. By the weak continuity of dd^c , we get $T = \lim(\omega + dd^c\psi_{\varepsilon_\nu}) = \omega + dd^c\psi$. $\square$

Let $\Omega \subset X$ be an open coordinate patch such that L is trivial on a neighborhood of $\overline{\Omega}$, and let e^{-h} be the weight representing the initial hermitian metric on $L|_{\overline{\Omega}}$. Then $dd^c h = \omega$ and $dd^c(h + \psi_\varepsilon) = \omega_\varepsilon$, so the function $\varphi_\varepsilon = h + \psi_\varepsilon$ defines a psh weight on $L|_\Omega$, as well as its limit $\varphi = h + \psi$. By the continuity of G , we also infer from the proof of Lemma 10.8 that the family (ψ_ε) is bounded in $L^1(X)$. The usual properties of subharmonic functions then show that there is a uniform constant C such that $\varphi_\varepsilon \leq C$ on $\overline{\Omega}$. We use this and equation (10.7) to prove that the limit φ has logarithmic poles at all points $x_j \in \Omega$, thanks to Bedford and Taylor's maximum principle for solutions of Monge-Ampère equations [BT76]:

(10.9) Lemma. *Let u, v be smooth (or continuous) psh functions on $\overline{\Omega}$, where Ω is a bounded open set in $\mathbb{C}^n$. If*

$$u|_{\partial\Omega} \geq v|_{\partial\Omega} \quad \text{and} \quad (dd^c u)^n \leq (dd^c v)^n \quad \text{on } \Omega,$$

then $u \geq v$ on Ω . $\square$

In the application of Lemma 10.9, we suppose that Ω is a neighborhood of x_j and take

$$u = \tau_j(\chi(\log|z_j - x_j|/\varepsilon) + \log\varepsilon) + C_1, \quad v = \varphi_\varepsilon,$$

where C_1 is a large constant. Then for $\varepsilon > 0$ small enough

$$\begin{aligned} u|_{\partial\Omega} &= \tau_j \log|z_j - x_j| + C_1, & v|_{\partial\Omega} &\leq C, \\ (dd^c v)^n &= \omega_\varepsilon^n \geq \tau_j^n \alpha_{j,\varepsilon}^n = (dd^c u)^n & \text{on } \Omega. \end{aligned}$$

For C_1 sufficiently large, we infer $u \geq v$ on Ω , hence

$$\varphi_\varepsilon \leq \tau_j \log(|z_j - x_j| + \varepsilon) + C_2 \quad \text{on } \Omega.$$

(10.10) Corollary. *The psh weight $\varphi = h + \psi$ on $L|_\Omega$ associated to the limit function $\psi = \lim \psi_{\varepsilon_\nu}$ satisfies $dd^c \varphi = T$. Moreover, φ has logarithmic poles at all points $x_j \in \Omega$ and*

$$\varphi(z) \leq \tau_j \log|z_j - x_j| + O(1) \quad \text{at } x_j. \quad \square$$

(10.11) Remark. The choice of the coefficients τ_j is made according to the order s_j of jets at x_j which sections in $H^0(X, K_X + L)$ should generate. Corollary 5.13 requires $\nu(\varphi, x_j) \geq n + s_j$, hence we need only take $\tau_j = n + s_j$. Accordingly, L must satisfy the numerical condition

$$L^n > \sigma = \sum (n + s_j)^n.$$

In particular, for the question of global generation, we need only take one point with $s_1 = 0$, thus $\sigma = n^n$, and for the question of very ampleness we need either two points with $s_1 = s_2 = 0$ or one point with $s_1 = 1$, thus $\sigma = \max(2n^n, (n+1)^n) = (n+1)^n$. In fact $\sigma = 2n^n$ is enough to get very ampleness: in the case of two infinitely near points $x_1 = x_2 = 0$ in the direction $\frac{\partial}{\partial z_n}$ (say), we can replace (10.5) and (10.7) respectively by

$$(10.5') \quad \alpha'_\varepsilon = dd^c \chi \left(\frac{1}{2} \log \frac{|z_1|^2 + \dots + |z_{n-1}|^2 + |z_n|^4}{\varepsilon^2} \right),$$

$$(10.7') \quad \omega_\varepsilon^n = n^n (\alpha'_\varepsilon)^n + \left(1 - \frac{2n^n}{L^n} \right) \omega^n \quad \text{with } \omega_\varepsilon = \omega + dd^c \psi_\varepsilon.$$

In this case, we have $\int_X (\alpha'_\varepsilon)^n = 2$. Arguments similar to those used in the proof of Corollary 10.10 show that

$$\varphi(z) \leq n \log (|z_1| + \dots + |z_{n-1}| + |z_n|^2) + O(1),$$

whence $\mathcal{I}(\varphi)_0 \subset (z_1, \dots, z_{n-1}, z_n^2)$ by the result of Exercise 5.10. Corollary 5.12 can then be used to obtain separation of infinitesimally near points. $\square$

Case of a big nef line bundle. All our arguments were developed under the assumption that L is ample, but if L is only nef and big, we can proceed in the following way. Let A be a fixed ample line bundle with smooth curvature form $\gamma = \frac{i}{2\pi} \Theta(A) > 0$. As $mL + A$ is ample for any $m \geq 1$, by Theorem 5.1 there exists a smooth hermitian metric on L depending on m , such that $\omega_m = \frac{i}{2\pi} \Theta(L)_m + \frac{1}{m} \frac{i}{2\pi} \Theta(A) > 0$ and

$$(10.12) \quad \omega_m^n = \frac{(L + \frac{1}{m}A)^n}{A^n} \gamma^n.$$

However, a priori we cannot control the asymptotic behaviour of ω_m when m tends to infinity, so we introduce the sequence of non necessarily positive $(1,1)$ -forms $\omega'_m = \frac{i}{2\pi} \Theta(L)_1 + \frac{1}{m} \frac{i}{2\pi} \Theta(A) \in \{\omega_m\}$, which is uniformly bounded in $C^\infty(X)$ and converges to $\Theta(L)_1$. Then we solve the Monge-Ampère equation

$$(10.13) \quad \omega_{m,\varepsilon}^n = \sum_{1 \leq j \leq N} \tau_j^n \alpha_{j,\varepsilon}^n + \left(1 - \frac{\sigma}{(L + \frac{1}{m}A)^n} \right) \omega_m^n$$

with $\omega_{m,\varepsilon} = \omega'_m + dd^c \psi_{m,\varepsilon}$ and some smooth function $\psi_{m,\varepsilon}$ such that $\int_X \psi_{m,\varepsilon} \gamma^n = 0$; this is again possible by Yau's theorem 5.1. The numerical condition needed on σ to solve (10.13) is obviously satisfied for all m if we suppose

$$\sigma = \sum \tau_j^n < L^n < \left(L + \frac{1}{m}A \right)^n.$$

The same arguments as before show that there exist a convergent subsequence $\lim_{\nu \rightarrow +\infty} \psi_{m_1, \varepsilon_\nu} = \psi$ in $L^1(X)$ and a closed positive $(1,1)$ -current $T = \lim \omega_{m_1, \varepsilon_\nu} = \frac{i}{2\pi} \Theta(L)_1 + dd^c \psi \in c_1(L)$ such that Corollary 10.10 is still valid; in this case, h is taken to be the weight function corresponding to $\Theta(L)_1$. Everything thus works as in the ample case.

11. Numerical Criteria for Very Ample Line Bundles

In the last section, we explained a construction of psh weights admitting singularities at all points in a given finite set. Our next goal is to develop a technique for bounding the Lelong numbers at other points. An explicit numerical criterion can then be derived from Corollary 5.13.

First suppose that L is an ample line bundle over X . The idea is to apply the self-intersection inequality 9.5 to the $(1, 1)$ -current $T = \lim \omega_{\varepsilon_\nu}$ produced by equation (10.7), and to integrate the inequality with respect to the Kähler form $\omega = \frac{1}{2\pi} \Theta(L)$. Before doing this, we need to estimate the excess of intersection in terms of T_{ac}^n .

(11.1) Proposition. *The absolutely continuous part T_{ac} of T satisfies*

$$T_{\text{ac}}^n \geq \left(1 - \frac{\sigma}{L^n}\right) \omega^n \quad \text{a.e. on } X.$$

Proof. The result is local, so we can work in an open set Ω which is relatively compact in a coordinate patch of X . Let ρ_δ be a family of smoothing kernels. By a standard lemma based on the comparison between arithmetic and geometric means (see e.g. [BT76], Proposition 5.1), the function $A \mapsto (\det A)^{1/n}$ is concave on the cone of nonnegative hermitian $n \times n$ matrices. Thanks to this concavity property we get

$$\left[(\omega_\varepsilon \star \rho_\delta(x))^n\right]^{1/n} \geq (\omega_\varepsilon^n)^{1/n} \star \rho_\delta(x) \geq \left(1 - \frac{\sigma}{L^n}\right)^{1/n} (\omega^n)^{1/n} \star \rho_\delta(x),$$

thanks to equation (6.5). As ε_ν tends to 0, $\omega_{\varepsilon_\nu} \star \rho_\delta$ converges to $T \star \rho_\delta$ in the strong topology of $C^\infty(\Omega)$, thus

$$((T \star \rho_\delta)^n)^{1/n} \geq \left(1 - \frac{\sigma}{L^n}\right)^{1/n} (\omega^n)^{1/n} \star \rho_\delta \quad \text{on } \Omega.$$

Now, take the limit as δ goes to 0. By the Lebesgue density theorem $T \star \rho_\delta(x)$ converges almost everywhere to $T_{\text{ac}}(x)$ on Ω , so we are done. $\square$

According to the notation used in § 9, we consider an arbitrary subset $\Xi \subset X$ and introduce the jumping values

$$b_p = \inf \{c > 0; \text{codim}(E_c(T), x) \geq p, \forall x \in \Xi\}.$$

By Proposition 11.1 and Inequality 10.2 a), we have

$$(11.2) \quad T_{\text{ac}}^j \wedge \omega^{n-j} \geq \left(1 - \frac{\sigma}{L^n}\right)^{j/n} \omega^n.$$

Now, suppose that the “formal vector bundle” $T_X \otimes \mathcal{O}(aL)$ is nef, i.e. that the $\mathbb{R}$ -divisor $\mathcal{O}_{T_X}(1) + a\pi^*L$ is nef for some constant $a \geq 0$. We can then apply Theorem 9.5 with $u = a\omega$ and

$$\{\Theta_p\} = (1 + b_1 a) \cdots (1 + b_p a) \{\omega^p\};$$

by taking the wedge product of Θ_p with ω^{n-p} , we get

$$(1 + b_1 a) \dots (1 + b_p a) \int_X \omega^n \geq \sum_{k \geq 1} (\nu_{p,k} - b_1) \dots (\nu_{p,k} - b_p) \int_X [Z_{p,k}] \wedge \omega^{n-p} \\ + \int_X (T_{ac} + b_1 a \omega) \wedge \dots \wedge (T_{ac} + b_p a \omega) \wedge \omega^{n-p}.$$

Combining this inequality with (11.2) for T_{ac}^{p-j} yields

$$(1 + b_1 a) \dots (1 + b_p a) L^n \geq \sum_{k \geq 1} (\nu_{p,k} - b_1) \dots (\nu_{p,k} - b_p) L^{n-p} \cdot Z_{p,k} \\ + \sum_{0 \leq j \leq p} S_j^p(b) a^j \left(1 - \frac{\sigma}{L^n}\right)^{(p-j)/n} L^n,$$

where $S_j^p(b)$, $1 \leq j \leq p$, denotes the elementary symmetric function of degree j in $b_1, \dots, b_p$ and $S_0^p(b) = 1$. As $\prod(1 + b_j a) = \sum S_j^p(b) a^j$, we get

$$(11.3) \quad \sum_{k \geq 1} (\nu_{p,k} - b_1) \dots (\nu_{p,k} - b_p) L^{n-p} \cdot Z_{p,k} \\ \leq \sum_{0 \leq j \leq p} S_j^p(b) a^j \left(1 - \left(1 - \frac{\sigma}{L^n}\right)^{(p-j)/n}\right) L^n.$$

If L is only supposed to be big and nef, we follow essentially the same arguments and replace ω in all our inequalities by $\omega_m = \frac{i}{2\pi}(\Theta(L)_m + \frac{1}{m}\Theta(A))$ with A ample (see section 6). Note that all (n, n) -forms ω_m^n were defined to be proportional to $\gamma^n = (\frac{i}{2\pi}\Theta(A))^n$, so inequality 11.1 becomes in the limit

$$T_{ac}^n \geq \left(1 - \frac{\sigma}{L^n}\right) \frac{L^n}{A^n} \gamma^n = \left(1 - \frac{\sigma}{L^n}\right) \frac{L^n}{(L + \frac{1}{m}A)^n} \omega_m^n.$$

The intersection inequality (11.3) is the expected generalization of Proposition 8.2 in arbitrary codimension. In this inequality, $\nu_{p,k}$ is the generic Lelong number of T along $Z_{p,k}$, and $Z_{p,k}$ runs over all p -codimensional components Y of $\bigcup_{c > b_p} E_c(T)$ intersecting Ξ ; by definition of b_j we have $\max_k \nu_{p,k} = b_{p+1}$. Hence we obtain:

(11.4) Proposition. *Let L be a big nef line bundle such that $\mathcal{O}_{T_X}(1) + a\pi^*L$ is nef, and let $T \in c_1(L)$ be the positive curvature current obtained by concentrating the Monge-Ampère mass L^n into a finite sum of Dirac measures with total mass σ , plus some smooth positive density spread over X (equation (10.7)). Then the jumping values b_p of the Lelong number of T over an arbitrary subset $\Xi \subset X$ satisfy the recursive inequalities*

$$(11.5) \quad (b_{p+1} - b_1) \dots (b_{p+1} - b_p) \leq \frac{1}{\min_Y L^{n-p} \cdot Y} \sum_{0 \leq j \leq p-1} S_j^p(b) a^j \sigma_{p-j},$$

where $\sigma_j = (1 - (1 - \sigma/L^n)^{j/n}) L^n$, and where Y runs over all p -codimensional subvarieties of X intersecting Ξ .

Observe that σ_j is increasing in j ; in particular $\sigma_j < \sigma_n = \sigma$ for $j \leq n-1$. Moreover, the convexity of the exponential function shows that

$$t \mapsto \frac{1}{t} (1 - (1 - \sigma/L^n)^t) L^n$$

is decreasing, thus $\sigma_j > \sigma_p j/p$ for $j < p$; in particular $\sigma_j > \sigma j/n$ for $j \leq n-1$. We are now in a position to prove the following general result.

(11.6) Main Theorem. *Let X be a projective n -fold and let L be a big nef line bundle over X . Fix points $x_j \in X$ and multiplicities $s_j \in \mathbb{N}$. Set*

$$\begin{aligned} \sigma_0 &= \sum_{1 \leq j \leq N} (n + s_j)^n \quad \text{resp. } \sigma_0 = 2n^n \text{ if } N = 1 \text{ and } s_1 = 1, \quad \text{and} \\ \sigma_p &= (1 - (1 - \sigma_0/L^n)^{p/n}) L^n, \quad 1 \leq p \leq n-1. \end{aligned}$$

Suppose that the formal vector bundle $T_X \otimes \mathcal{O}(aL)$ is nef for some $a \geq 0$, that $L^n > \sigma_0$, and that there is a sequence $0 = \beta_1 < \dots < \beta_n \leq 1$ with

$$(11.7) \quad L^{n-p} \cdot Y > (\beta_{p+1} - \beta_1)^{-1} \dots (\beta_{p+1} - \beta_p)^{-1} \sum_{0 \leq j \leq p-1} S_j^p(\beta) a^j \sigma_{p-j}$$

for every subvariety $Y \subset X$ of codimension $p = 1, 2, \dots, n-1$ passing through one of the points x_j . Then there is a surjective map

$$H^0(X, K_X + L) \twoheadrightarrow \bigoplus_{1 \leq j \leq N} \mathcal{O}(K_X + L)_{x_j} \otimes (\mathcal{O}_{X, x_j} / \mathfrak{m}_{X, x_j}^{s_j+1}),$$

i.e. $H^0(X, K_X + L)$ generates simultaneously all jets of order s_j at x_j .

As the notation is rather complicated, it is certainly worth examining the particular case of surfaces and 3-folds, for the problem of getting global generation (taking $\sigma_0 = n^n$), resp. very ampleness (taking $\sigma_0 = 2n^n$). If X is a surface, we find $\sigma_0 = 4$ (resp. $\sigma_0 = 8$), and we take $\beta_1 = 0, \beta_2 = 1$. This gives only two conditions, namely

$$(11.7^{n=2}) \quad L^2 > \sigma_0, \quad L \cdot C > \sigma_1$$

for every curve C intersecting Ξ . These bounds are not very far from those obtained with Reider's theorem, although they are not exactly as sharp. If X is a 3-fold, we have $\sigma_0 = 27$ (resp. $\sigma_0 = 64$), and we take $\beta_1 = 0 < \beta_2 = \beta < \beta_3 = 1$. Therefore our condition is that there exists $\beta \in]0, 1[$ such that

$$(11.7^{n=3}) \quad L^3 > \sigma_0, \quad L^2 \cdot S > \beta^{-1} \sigma_1, \quad L \cdot C > (1 - \beta)^{-1} (\sigma_2 + \beta \sigma_1)$$

for every curve C or surface S intersecting Ξ . If we take β to be of the order of magnitude of $a^{-2/3}$, these bounds show that the influence of a on the numerical conditions for L is at worst $a^{1/3}$ in terms of the homogeneous quantities $(L^p \cdot Y)^{1/p}$. In higher dimensions, a careful choice of the β_p 's shows that the influence of a on these quantities is always less than $O(a^{1-\delta_n})$, $\delta_n > 0$.

Proof of Theorem 11.6. Select $\tau_j > n + s_j$ so that $L^{n-p} \cdot Y$ still satisfies the above lower bound with the corresponding value $\sigma = \sum \tau_j^n > \sigma_0$. Then apply Theorem 11.4 with $\Xi = \{x_1, \dots, x_N\}$. Inequality (11.5) shows inductively that $b_p < \beta_p$ for $p \geq 2$, so $b_n < 1$ and therefore x_j is an isolated point in $E_1(T)$. On the other hand, the Monge-Ampère equation gives us a weight function φ admitting a logarithmic pole at each point x_j , in such a way that $\nu(\varphi, x_j) \geq \tau_j > n + s_j$ (Corollary 10.10). However $T = dd^c \varphi$ need not be positive definite. In order to apply our vanishing theorems, we still have to make the curvature positive definite everywhere. Since L is nef and big, it has a singular metric for which the curvature current $T_0 = dd^c \varphi_0$ satisfies $\nu(T_0, x) < 1$ everywhere and $T_0 \geq \varepsilon \gamma$ for some Kähler metric (Corollary 6.8). Then $T' = (1 - \delta)T + \delta T_0$ still has Lelong numbers $\nu(T', x_j) > n + s_j$ for $\delta > 0$ small, and the inequality $\nu(T', x) \leq (1 - \delta)\nu(T, x) + \delta$ implies that we have $\nu(T', x) < 1$ for x near x_j , $x \neq x_j$. Corollaries 5.12 and 5.13 imply the Theorem. When $N = 1$ and $s_1 = 1$, we can use Remark 10.11 to show that the value $\sigma_0 = 2n^n$ is admissible in place of $(n + 1)^n$. $\square$

(11.8) Corollary. *Let X be a smooth algebraic surface, and let L be a big nef line bundle over X . Then on a given subset $\Xi \subset X$*

$K_X + L$		<i>is spanned</i>	<i>separates points</i>			<i>generates s-jets</i>
<i>when</i>	$L^2 >$	4	8	9	12	$(2 + s)^2$
	$\forall C, \quad L \cdot C >$	2	6	5	4	$2 + 3s + s^2$

for all curves $C \subset X$ intersecting Ξ . In particular, if L is ample, $K_X + mL$ is always globally spanned for $m \geq 3$ and very ample for $m \geq 5$.

Proof. For s -jets, we have $\sigma_0 = (2 + s)^2$, so we find the condition

$$L^2 > (2 + s)^2, \quad L \cdot C > (1 - (1 - (2 + s)^2/L^2)^{1/2})L^2.$$

The last constant decreases with L^2 and is thus at most equal to the value obtained when $L^2 = (2 + s)^2 + 1$; the integral part is then precisely $2 + 3s + s^2$. $\square$

(11.9) Remark. Ein and Lazarsfeld [EL94] have recently obtained an algebraic proof of the Main Theorem, in which they can even weaken the condition that $T_X \otimes \mathcal{O}(aL)$ is nef into the condition that $-K_X + aL$ is nef (by taking determinants, $T_X \otimes \mathcal{O}(aL)$ nef $\Rightarrow -K_X + naL$ nef). However, it seems that the lower bounds they get for $L^{n-p} \cdot Y$ are substantially larger than the ones given by the analytic method. Also, the assumption on T_X is sufficient to get universal bounds for $2K_X + L$, as we will see later. $\square$

For the applications, we introduce a convenient definition of higher jet generation, following an idea of [BS93].

(11.10) Definition. *We say that L generates s -jets on a given subset $\Xi \subset X$ if $H^0(X, L) \twoheadrightarrow \bigoplus J_{x_j}^{s_j} L$ is onto for any choice of points $x_1, \dots, x_N \in \Xi$ and integers*

$s_1, \dots, s_N$ with $\sum(s_j + 1) = s + 1$. We say that L is s -jet ample if the above property holds for $\Xi = X$.

With this terminology, L is 0-jet ample if and only if L is generated by global sections and 1-jet ample if and only if L is very ample. In order that $K_X + L$ generates s -jets on Ξ , the constant to be used in Theorem 11.6 is $\sigma_0 = \max \sum (n + s_j)^n$ over all decompositions $s + 1 = \sum (s_j + 1)$. In fact, if we set $t_j = s_j + 1$, the following lemma gives $\sigma_0 = (n + s)^n$, that is, the maximum is reached when only one point occurs.

(11.11) Lemma. *Let $t_1, \dots, t_N \in [1, +\infty[$. Then*

$$\sum_{1 \leq j \leq N} (n - 1 + t_j)^n \leq \left(n - 1 + \sum_{1 \leq j \leq N} t_j \right)^n.$$

Proof. The right hand side is a polynomial with nonnegative coefficients and the coefficient of a monomial t_j^k involving exactly one variable is the same as in the left hand side (however, the constant term is smaller). Thus the difference is increasing in all variables and we need only consider the case $t_1 = \dots = t_N = 1$. This case follows from the obvious inequality

$$n^n N = n^n + \binom{n}{1} n^{n-1} (N - 1) \leq (n + N - 1)^n. \quad \square$$

(11.12) Corollary. *If all intersection numbers $L^p \cdot Y$ satisfy the inequalities in Theorem 11.6 with $\sigma_0 = (n + s)^n$, then $K_X + L$ is s -jet ample.*

In order to find universal conditions for $K_X + L$ to be very ample, our main theorem would require a universal value a depending only on $n = \dim_{\mathbb{C}} X$ such that $T_X \otimes \mathcal{O}(aL)$ is always nef. However, this is clearly impossible as the example of curves already shows: if X is a curve of genus g and L has degree 1, then $T_X \otimes \mathcal{O}(aL)$ is nef if and only if $a \geq 2g - 2$. As we will see in § 13, an explicit value a depending only on the intersection numbers L^n and $L^{n-1} \cdot K_X$ exists, but this value is very large. Here, these difficulties can be avoided by means of the following simple lemma.

(11.13) Lemma. *Let F be a very ample line bundle over X . Then the vector bundle $T_X \otimes \mathcal{O}(K_X + nF)$ is nef and generated by global sections.*

Proof. By the assumption that F is very ample, the 1-jet bundle $J^1 F$ is generated by its sections. Consider the exact sequence

$$0 \longrightarrow T_X^* \otimes F \longrightarrow J^1 F \longrightarrow F \longrightarrow 0$$

where $\text{rank}(J^1 F) = n + 1$ and $\det(J^1 F) = K_X + (n + 1)F$. The n -th exterior power $\Lambda^n(J^1 F) = (J^1 F)^* \otimes \det(J^1 F)$ is also generated by sections. Hence, by dualizing the sequence, we get a surjective morphism

$$\Lambda^n(J^1 F) \longrightarrow (T_X^* \otimes F)^* \otimes \det(J^1 F) = T_X \otimes \mathcal{O}(K_X + nF).$$

Therefore $T_X \otimes \mathcal{O}(K_X + nF)$ is generated by sections and, in particular, it is nef. $\square$

Before going further, it is convenient to introduce the following quantitative measurement of ampleness:

(11.14) Definition. Let F be a nef line bundle and let $\Xi \subset X$ be an arbitrary subset. We define

$$\mu(F, \Xi) = \min_{1 \leq p \leq n} \min_{\dim Y = p, Y \cap \Xi \neq \emptyset} (F^p \cdot Y)^{1/p}$$

where Y runs over all p -dimensional subvarieties intersecting Ξ . The main properties of this invariant are:

- *Linearity with respect to F :* $\forall k \geq 0, \mu(kF, \Xi) = k \mu(F, \Xi)$;
- *Nakai-Moishezon criterion:* F is ample if and only if $\mu(F, X) > 0$.

The next idea consists in the following iteration trick: Lemma 11.13 suggests that a universal lower bound for the nefness of $T_X \otimes \mathcal{O}(aL')$ can be achieved with $L' = K_X + L$ if L is sufficiently ample. Then it follows from the Main Theorem 11.6 that $K_X + L' = 2K_X + L$ is very ample under suitable numerical conditions. Lemma 11.13 applied with $F = 2K_X + L$ shows that the bundle $T_X \otimes \mathcal{O}((2n+1)K_X + nL)$ is nef, and thus $T_X \otimes \mathcal{O}((2n+1)L'')$ is nef with $L'' = K_X + \frac{1}{2}L \leq L'$. Hence we see that the Main Theorem can be iterated. The special value $a = 2n+1$ will play an important role.

(11.15) Lemma. Let L' be an ample line bundle over X . Suppose that the vector bundle $T_X \otimes \mathcal{O}((2n+1)L')$ is nef. Then, for any $s \geq 1$, $K_X + L'$ is s -jet ample as soon as $\mu(L', X) \geq 3(n+s)^n$. (By Remark 10.11, this is still true with $3(n+s)^n$ replaced by $6n^n$ if $s = 1$.)

Proof. We apply the Main Theorem 11.6 with $\sigma_0 = (n+s)^n$ and $a = 2n+1$. Thanks to the inequality $\sigma_j \leq \sigma_0$ and to the identity

$$\sum_{0 \leq j \leq p-1} S_j^p(\beta) = (1 + \beta_1 a) \dots (1 + \beta_p a),$$

(recall that $\beta_1 = 0$), we find lower bounds $L^{n-p} \cdot Y > M_p$ with

$$M_p \leq \frac{(1 + \beta_1 a) \dots (1 + \beta_p a) \sigma_0}{(\beta_{p+1} - \beta_1) \dots (\beta_{p+1} - \beta_p)}.$$

If we choose an increasing sequence $0 = \beta_1 < \dots < \beta_n = 1$ such that β_p / β_{p+1} is increasing, we get $\beta_j / \beta_{p+1} < \beta_{j+n-p-1} / \beta_n = \beta_{j+n-p-1}$, hence

$$\begin{aligned} \prod_{1 \leq j \leq p} (\beta_{p+1} - \beta_j) &= \beta_{p+1}^p \prod_{1 \leq j \leq p} \left(1 - \frac{\beta_j}{\beta_{p+1}}\right) \geq \beta_{p+1}^p \prod_{1 \leq j \leq p} (1 - \beta_{j+n-p-1}), \\ M_p &\leq \prod_{1 \leq j \leq n-1} \frac{(1 + \beta_j a)}{(1 - \beta_j)} \beta_{p+1}^{-p} \sigma_0. \end{aligned}$$

A suitable choice is $\beta_p = n^{-n(n-p)/(p-1)}$. We then find

$$M_p \leq C_n n^{n(n-p-1)} (n+s)^n \leq C_n (n+s)^{n(n-p)}$$

where

$$C_n = \prod_{1 \leq p \leq n-1} \frac{(1 + (2n+1)n^{-n(n-p)/(p-1)})}{(1 - n^{-n(n-p)/(p-1)})}.$$

Numerical calculations left to the reader show that that $C_n < 3$ for all $n \geq 2$ and that $C_n = 3 - 4 \log n/n + O(1/n)$ as $n \rightarrow +\infty$. Lemma 11.15 follows. $\square$

(11.16) Lemma. *Let F be a line bundle which generates s -jets at every point. Then $F^p \cdot Y \geq s^p$ for every p -dimensional subvariety $Y \subset X$.*

Proof. Fix an arbitrary point $x \in Y$. Then consider the singular metric on F given by

$$\|\xi\|^2 = \frac{|\xi|^2}{\sum |u_j(z)|^2},$$

where $(u_1, \dots, u_N)$ is a basis of $H^0(X, F \otimes \mathfrak{m}_x^s)$. By our assumption, these sections have an isolated common zero of order s at x . Hence F possesses a singular metric such that the weight $\varphi = \frac{1}{2} \log \sum |u_j|^2$ is psh and has an isolated logarithmic pole of Lelong number s at x . By the comparison inequality (3.6) with $\psi(z) = \log |z - x|$, we get

$$F^p \cdot Y \geq \int_{B(x, \epsilon)} [Y] \wedge (dd^c \varphi)^p \geq s^p \nu([Y], \psi) = s^p \nu(Y, x) \geq s^p. \quad \square$$

(11.17) Theorem. *Let $s \geq 1$ and $m \geq 2$ be arbitrary integers and let L be an ample line bundle. If L satisfies the numerical condition*

$$(m-1)\mu(L, X) + s \geq 6(n+s)^n,$$

then $2K_X + mL$ is s -jet ample. If $s = 1$, the result still holds with $6(n+s)^n$ replaced by $12n^n$, in particular $2K_X + 12n^n L$ is always very ample.

Proof. We denote here simply $\mu(L, X) = \mu(L)$, for $\Xi = X$ everywhere in the proof. As L is ample, there exists an integer q (possibly very large) such that

$$(11.18) \quad \begin{cases} K_X + qL & \text{is ample,} \\ T_X \otimes \mathcal{O}((2n+1)(K_X + qL)) & \text{is nef,} \\ \mu(K_X + qL) \geq 3(n+s)^n. \end{cases}$$

By lemma 11.15 applied to $L' = K_X + qL$, we find that $F = K_X + L' = 2K_X + qL$ is very ample and generates s -jets. In particular $K_X + \frac{q}{2}L$ is an ample $\mathbb{Q}$ -divisor, and for any p -dimensional subvariety $Y \subset X$ we have

$$\begin{aligned} (K_X + (q-1)L)^p \cdot Y &= \left(\frac{1}{2}F + (q/2-1)L\right)^p \cdot Y \\ &= \sum_{0 \leq k \leq p} \binom{p}{k} 2^{k-p} (q/2-1)^k F^{p-k} \cdot L^k \cdot Y. \end{aligned}$$

By the convexity inequality 10.2 b) and Lemma 11.16 we get

$$F^{p-k} \cdot L^k \cdot Y \geq (F^p \cdot Y)^{1-k/p} (L^p \cdot Y)^{k/p} \geq s^{p-k} (\mu(L))^k.$$

Hence $(K_X + (q-1)L)^p \cdot Y \geq ((q/2-1)\mu(L) + s/2)^p$ and

$$\mu(K_X + (q-1)L, X) \geq \frac{1}{2}((q-2)\mu(L) + s).$$

Moreover, Lemma 11.13 applied to F shows that

$$T_X \otimes \mathcal{O}(K_X + nF) = T_X \otimes \mathcal{O}((2n+1)K_X + nqL)$$

is nef. As $nq/(2n+1) \leq q/2 \leq q-1$ for $q \geq 2$, we find that all properties (11.18) except perhaps the last one remain valid with $q-1$ in place of q :

$$(11.19) \quad \begin{cases} K_X + (q-1)L & \text{is ample,} \\ T_X \otimes \mathcal{O}((2n+1)(K_X + (q-1)L)) & \text{is nef,} \\ \mu(K_X + (q-1)L, X) \geq \frac{1}{2}((q-2)\mu(L) + s). \end{cases}$$

By induction we conclude that (11.19) is still true for the smallest integer $q-1 = m$ such that

$$\frac{1}{2}((q-2)\mu(L) + s) = \frac{1}{2}((m-1)\mu(L) + s) \geq 3(n+s)^n.$$

For this value of m , Lemma 11.15 implies that $2K_X + mL$ generates s -jets. $\square$

(11.20) Remark. The condition $(m-1)\mu(L, X) + s \geq 6(n+s)^n$ is never satisfied for $m = 1$. However, Lemma 8.6 applied with $F = K_X + L$ and $\mu = 2$ allows us to obtain also a sufficient condition in order that $2K_X + L$ generates s -jets. It is sufficient that $2(K_X + L)$ generates jets of order $s'_j = 2(n+s_j) + 1$ at any of the points x_j whenever $\sum(s_j + 1) = s + 1$. For $n \geq 2$ we get

$$\sum(n + s'_j)^n = \sum(2s_j + 3n + 1)^n \leq (3n + 3 + 2s)^n$$

after a short computation. The proof of Theorem 11.17 then yields the sufficient condition $\mu(L, X) \geq 6(3n + 3 + 2s)^n$. $\square$

(11.21) Remark. If G is a nef line bundle, the Main Theorem 11.6 is still valid for the line bundle $K_X + L + G$, with the same lower bounds in the numerical conditions for L ; indeed, the proof rests on the existence of suitable singular hermitian metrics with positive definite curvature on L , and adding G preserves all properties of these metrics. It follows that Theorem 11.17 and Remark 11.20 can be applied as well to the line bundle $2K_X + mL + G$, under the same numerical conditions. $\square$

12. Holomorphic Morse Inequalities

Let X be a compact Kähler manifold, E a holomorphic vector bundle of rank r and L a line bundle over X . If L is equipped with a smooth metric of curvature form $\Theta(L)$, we define the q -index set of L to be the open subset

$$(12.1) \quad X(q, L) = \left\{ x \in X ; i\Theta(L)_x \text{ has } \begin{array}{l} q \text{ negative eigenvalues} \\ n - q \text{ positive eigenvalues} \end{array} \right\}$$

for $0 \leq q \leq n$. Hence X admits a partition $X = \Delta \amalg \coprod_q X(q, L)$ where $\Delta = \{x \in X ; \det(\Theta(L)_x) = 0\}$ is the degeneracy set. We also introduce

$$(12.1') \quad X(\leq q, L) = \bigcup_{0 \leq j \leq q} X(j, L).$$

It is shown in [Dem85b] that the cohomology groups $H^q(X, E \otimes \mathcal{O}(kL))$ satisfy the following asymptotic *weak Morse inequalities* as $k \rightarrow +\infty$

$$(12.2) \quad h^q(X, E \otimes \mathcal{O}(kL)) \leq r \frac{k^n}{n!} \int_{X(q, L)} (-1)^q \left(\frac{i}{2\pi} \Theta(L) \right)^n + o(k^n).$$

A sharper form is given by the *strong Morse inequalities*

$$(12.2') \quad \begin{aligned} & \sum_{0 \leq j \leq q} (-1)^{q-j} h^j(X, E \otimes \mathcal{O}(kL)) \\ & \leq r \frac{k^n}{n!} \int_{X(\leq q, L)} (-1)^q \left(\frac{i}{2\pi} \Theta(L) \right)^n + o(k^n). \end{aligned}$$

These inequalities are a useful complement to the Riemann-Roch formula when information is needed about individual cohomology groups, and not just about the Euler-Poincaré characteristic.

One difficulty in the application of these inequalities is that the curvature integral is in general quite uneasy to compute, since it is neither a topological nor an algebraic invariant. However, the Morse inequalities can be reformulated in a more algebraic setting in which only algebraic invariants are involved. We give here two such reformulations.

(12.3) Theorem. *Let $L = F - G$ be a holomorphic line bundle over a compact Kähler manifold X , where F and G are numerically effective line bundles. Then for every $q = 0, 1, \dots, n = \dim X$, there is an asymptotic strong Morse inequality*

$$\sum_{0 \leq j \leq q} (-1)^{q-j} h^j(X, kL) \leq \frac{k^n}{n!} \sum_{0 \leq j \leq q} (-1)^{q-j} \binom{n}{j} F^{n-j} \cdot G^j + o(k^n).$$

Proof. By adding ε times a Kähler metric ω to the curvature forms of F and G , $\varepsilon > 0$ one can write $\frac{i}{2\pi} \Theta(L) = \theta_\varepsilon(F) - \theta_\varepsilon(G)$ where $\theta_\varepsilon(F) = \frac{i}{2\pi} \Theta(F) + \varepsilon\omega$ and $\theta_\varepsilon(G) = \frac{i}{2\pi} \Theta(G) + \varepsilon\omega$ are positive definite. Let $\lambda_1 \geq \dots \geq \lambda_n > 0$ be the eigenvalues of $\theta_\varepsilon(G)$ with respect to $\theta_\varepsilon(F)$. Then the eigenvalues of $\frac{i}{2\pi} \Theta(L)$ with respect to $\theta_\varepsilon(F)$

are the real numbers $1 - \lambda_j$ and the set $X(\leq q, L)$ is the set $\{\lambda_{q+1} < 1\}$ of points $x \in X$ such that $\lambda_{q+1}(x) < 1$. The strong Morse inequalities yield

$$\sum_{0 \leq j \leq q} (-1)^{q-j} h^j(X, kL) \leq \frac{k^n}{n!} \int_{\{\lambda_{q+1} < 1\}} (-1)^q \prod_{1 \leq j \leq n} (1 - \lambda_j) \theta_\varepsilon(F)^n + o(k^n).$$

On the other hand we have

$$\binom{n}{j} \theta_\varepsilon(F)^{n-j} \wedge \theta_\varepsilon(G)^j = \sigma_n^j(\lambda) \theta_\varepsilon(F)^n,$$

where $\sigma_n^j(\lambda)$ is the j -th elementary symmetric function in $\lambda_1, \dots, \lambda_n$, hence

$$\sum_{0 \leq j \leq q} (-1)^{q-j} \binom{n}{j} F^{n-j} \cdot G^j = \lim_{\varepsilon \rightarrow 0} \int_X \sum_{0 \leq j \leq q} (-1)^{q-j} \sigma_n^j(\lambda) \theta_\varepsilon(F)^n.$$

Thus, to prove the Lemma, we only have to check that

$$\sum_{0 \leq j \leq n} (-1)^{q-j} \sigma_n^j(\lambda) - \mathbf{1}_{\{\lambda_{q+1} < 1\}} (-1)^q \prod_{1 \leq j \leq n} (1 - \lambda_j) \geq 0$$

for all $\lambda_1 \geq \dots \geq \lambda_n \geq 0$, where $\mathbf{1}_{\{\dots\}}$ denotes the characteristic function of a set. This is easily done by induction on n (just split apart the parameter λ_n and write $\sigma_n^j(\lambda) = \sigma_{n-1}^j(\lambda) + \sigma_{n-1}^{j-1}(\lambda) \lambda_n$). $\square$

In the case $q = 1$, we get an especially interesting lower bound (this bound has been observed and used by S. Trapani [Tra95] in a similar context).

(12.4) Consequence. $h^0(X, kL) - h^1(X, kL) \geq \frac{k^n}{n!} (F^n - nF^{n-1} \cdot G) - o(k^n)$.
Therefore some multiple kL has a section as soon as $F^n - nF^{n-1} \cdot G > 0$.

(12.5) Remark. The weaker inequality

$$h^0(X, kL) \geq \frac{k^n}{n!} (F^n - nF^{n-1} \cdot G) - o(k^n)$$

is easy to prove if X is projective algebraic. Indeed, by adding a small ample $\mathbb{Q}$ -divisor to F and G , we may assume that F, G are ample. Let $m_0 G$ be very ample and let k' be the smallest integer $\geq k/m_0$. Then $h^0(X, kL) \geq h^0(X, kF - k'm_0 G)$. We select k' smooth members G_j , $1 \leq j \leq k'$ in the linear system $|m_0 G|$ and use the exact sequence

$$0 \rightarrow H^0(X, kF - \sum G_j) \rightarrow H^0(X, kF) \rightarrow \bigoplus H^0(G_j, kF|_{G_j}).$$

Kodaira's vanishing theorem yields $H^q(X, kF) = 0$ and $H^q(G_j, kF|_{G_j}) = 0$ for $q \geq 1$ and $k \geq k_0$. By the exact sequence combined with Riemann-Roch, we get

$$\begin{aligned}
h^0(X, kL) &\geq h^0(X, kF - \sum G_j) \\
&\geq \frac{k^n}{n!} F^n - O(k^{n-1}) - \sum \left(\frac{k^{n-1}}{(n-1)!} F^{n-1} \cdot G_j - O(k^{n-2}) \right) \\
&\geq \frac{k^n}{n!} \left(F^n - n \frac{k' m_0}{k} F^{n-1} \cdot G \right) - O(k^{n-1}) \\
&\geq \frac{k^n}{n!} \left(F^n - n F^{n-1} \cdot G \right) - O(k^{n-1}).
\end{aligned}$$

(This simple proof is due to F. Catanese.) $\square$

(12.6) Corollary. *Suppose that F and G are nef and that F is big. Some multiple of $mF - G$ has a section as soon as*

$$m > n \frac{F^{n-1} \cdot G}{F^n}.$$

In the last condition, the factor n is sharp: this is easily seen by taking $X = \mathbb{P}_1^n$ and $F = \mathcal{O}(a, \dots, a)$ and $G = \mathcal{O}(b_1, \dots, b_n)$ over $\mathbb{P}_1^n$; the condition of the Corollary is then $m > \sum b_j/a$, whereas $k(mF - G)$ has a section if and only if $m \geq \sup b_j/a$; this shows that we cannot replace n by $n(1 - \varepsilon)$. $\square$

We now discuss another application of Morse inequalities in the case where $c_1(L) \in N_{\text{psef}}$. Then the regularization theorem 9.1 allows us to measure the distance of L to the nef cone N_{nef} . In that case, a use of singular metrics combined with 9.1 produces smooth metrics on L for which an explicit bound of the negative part of the curvature is known. It follows that (12.2) gives an explicit upper bound of the cohomology groups of $E \otimes \mathcal{O}(kL)$ in terms of a polynomial in the first Chern class $c_1(L)$ (related techniques have already been used in [Sug87] in a slightly different context).

(12.7) Theorem. *Suppose that there is a nef cohomology class $\{u\}$ in $H^{1,1}(X)$ such that $c_1(\mathcal{O}_{T_X}(1)) + \pi^*\{u\}$ is nef over the hyperplane bundle $P(T_X^*)$. Suppose moreover that L is equipped with a singular metric such that $T = \frac{1}{2\pi} \Theta(L) \geq 0$. For $p = 1, 2, \dots, n, n+1$ set*

$$b_p = \inf \{c > 0; \text{codim } E_c(T) \geq p\},$$

with $b_{n+1} = \max_{x \in X} \nu(T, x)$. Then for any holomorphic vector bundle E of rank r over X we have

$$h^q(X, E \otimes \mathcal{O}(kL)) \leq A_q r k^n + o(k^n)$$

where A_q is the cup product

$$A_q = \frac{1}{q!(n-q)!} (b_{n-q+1}\{u\})^q \cdot (c_1(L) + b_{n-q+1}\{u\})^{n-q}$$

in $H^{2n}(X, \mathbb{R})$, identified with a positive number.

(12.8) Remark. When X is projective algebraic and $\kappa(L) = n$, the proof of 6.6 f) shows that $mL \simeq \mathcal{O}(A + D)$ with A ample and D effective, for some $m \geq 1$.

Then we can choose a singular metric on L such that $T = \frac{i}{2\pi}\Theta(L) = \omega + m^{-1}[D]$, where $\omega = m^{-1}\frac{i}{2\pi}\Theta(A)$ is a Kähler metric. As $\nu(T, x) = m^{-1}\nu(D, x)$ at each point, the constants b_j of theorem 12.7 are obtained by counting the multiplicities of the singular points of D ; for example, if D only has isolated singularities, then $b_1 = 0$, $b_2 = \dots = b_n = 1/m$. Observe moreover that the nefness assumption on $\mathcal{O}_{T_X}(1)$ is satisfied with $\{u\} = c_1(G)$ if G is a nef $\mathbb{Q}$ -divisor such that $\mathcal{O}(T_X) \otimes \mathcal{O}(G)$ is nef, e.g. if $\mathcal{O}(S^m T_X) \otimes \mathcal{O}(mG)$ is spanned by sections for some $m \geq 1$. $\square$

Proof of theorem 12.7. By definition, we have $0 = b_1 \leq b_2 \leq \dots \leq b_n \leq b_{n+1}$, and for $c \in]b_p, b_{p+1}]$, $E_c(T)$ has codimension $\geq p$ with some component(s) of codimension p exactly. Let ω be a fixed Kähler metric on X . By adding $\varepsilon\omega$ to u if necessary, we may assume that $u \geq 0$ and that $\mathcal{O}_{T_X}(1)$ has a smooth hermitian metric such that $c(\mathcal{O}_{T_X}(1)) + \pi^*u \geq 0$.

Under this assumption, the approximation theorem 9.1 shows that the metric of L can be approximated by a sequence of smooth metrics such that the associated curvature forms T_j satisfy the uniform lower bound

$$(12.9) \quad T_j \geq -\lambda_j(x)u(x) - \varepsilon_j\omega(x)$$

where $\lim_{j \rightarrow +\infty} \varepsilon_j = 0$ and $(\lambda_j)_{j>0}$ is a decreasing sequence of continuous functions on X such that $\lim_{j \rightarrow +\infty} \lambda_j(x) = \nu(T, x)$ at each point.

The estimate (12.2) cannot be used directly with $T = \frac{i}{2\pi}\Theta(L)$ because wedge products of currents do not make sense in general. Therefore, we replace $\frac{i}{2\pi}\Theta(L)$ by its approximations T_j and try to find an upper bound for the limit.

(12.10) Lemma. *Let $U_j = X(q, T_j)$ be the q -index set associated to T_j and let c be a positive number. On the open set $\Omega_{c,j} = \{x \in X; \lambda_j(x) < c\}$ we have*

$$(-1)^q \mathbf{1}_{U_j} T_j^n \leq \frac{n!}{q!(n-q)!} (cu + \varepsilon_j\omega)^q \wedge (T_j + cu + \varepsilon_j\omega)^{n-q}.$$

Proof. Write $v = cu + \varepsilon_j\omega > 0$ and let $\alpha_{1,j} \leq \dots \leq \alpha_{n,j}$ be the eigenvalues of T_j with respect to v at each point. Then $T_j^n = \alpha_{1,j} \dots \alpha_{n,j} v^n$ and

$$v^q \wedge (T_j + v)^{n-q} = \frac{q!(n-q)!}{n!} \sum_{1 \leq i_1 < \dots < i_{n-q} \leq n} (1 + \alpha_{i_1,j}) \dots (1 + \alpha_{i_{n-q},j}) v^n.$$

On $\Omega_{c,j}$ we get $T_j \geq -v$ by inequality (12.9), thus $\alpha_{i,j} \geq -1$; moreover, we have $\alpha_1 \leq \dots \leq \alpha_q < 0$ and $0 < \alpha_{q+1} \leq \dots \leq \alpha_n$ on U_j . On $\Omega_{c,j}$ we thus find

$$0 \leq (-1)^q \mathbf{1}_{U_j} \alpha_{1,j} \dots \alpha_{n,j} \leq \mathbf{1}_{U_j} \alpha_{q+1,j} \dots \alpha_{n,j} \leq (1 + \alpha_{q+1,j}) \dots (1 + \alpha_{n,j}),$$

therefore $(-1)^q \mathbf{1}_{U_j} T_j^n \leq (n!/q!(n-q)!) v^q \wedge (T_j + v)^{n-q}$. $\square$

Proof of theorem 12.7 (end). Set $\Lambda = \max_X \lambda_1(x)$. By Lemma 12.10 applied with an arbitrary $c > \Lambda$ we have

$$(-1)^q \mathbf{1}_{U_j} T_j^n \leq \frac{n!}{q!(n-q)!} (\Lambda u + \varepsilon_1\omega)^q \wedge (T_j + \Lambda u + \varepsilon_1\omega)^{n-q} \quad \text{on } X.$$

Then estimate (12.2) and Lemma 12.10 again imply

$$\begin{aligned}
 h^q(X, E \otimes \mathcal{O}(kL)) &\leq r \frac{k^n}{n!} \int_X (-1)^q \mathbf{1}_{U_j} T_j^n + o(k^n) \\
 &\leq \frac{r k^n}{q!(n-q)!} \left\{ \int_{\Omega_{c,j}} (cu + \varepsilon_j \omega)^q \wedge (T_j + cu + \varepsilon_j \omega)^{n-q} \right. \\
 (12.11) \quad &\quad \left. + \int_{X \setminus \Omega_{c,j}} (\Lambda u + \varepsilon_1 \omega)^q \wedge (T_j + \Lambda u + \varepsilon_1 \omega)^{n-q} \right\} + o(k^n).
 \end{aligned}$$

Since $\lambda_j(x)$ decreases to $\nu(T, x)$ as $j \rightarrow +\infty$, the set $X \setminus \Omega_{c,j}$ decreases to $E_c(T)$. Now, $T_j + \Lambda u + \varepsilon_1 \omega$ is a closed positive $(1, 1)$ -form belonging to a fixed cohomology class, so the mass of any wedge power $(T_j + \Lambda u + \varepsilon_1 \omega)^p$ with respect to ω is constant. By weak compactness, there is a subsequence (j_ν) such that $(T_{j_\nu} + \Lambda u + \varepsilon_1 \omega)^p$ converges weakly to a closed positive current Θ_p of bidegree (p, p) , for each $p = 1, \dots, n$. For $c > b_{p+1}$, we have $\text{codim } E_c(T) \geq p+1$, hence $\mathbf{1}_{E_c(T)} \Theta_p = 0$. It follows that the integral over $X \setminus \Omega_{c,j}$ in (12.11) converges to 0 when $c > b_{n-q+1}$. For the same reason the integral over $\Omega_{c,j}$ converges to the same limit as its value over X : observe that $(T_j + cu + \varepsilon_j \omega)^{n-q}$ can be expressed in terms of powers of u, ω and of the positive forms $(T_j + \Lambda u + \varepsilon_1 \omega)^p$ with $p \leq n-q$; thus the limit is a linear combination with smooth coefficients of the currents Θ_p , which carry no mass on $E_c(T)$. In the limit, we obtain

$$h^q(X, E \otimes \mathcal{O}(kL)) \leq \frac{r k^n}{q!(n-q)!} (c\{u\})^q \cdot (c_1(L) + c\{u\})^{n-q} + o(k^n),$$

and since this is true for every $c > b_{n-q+1}$, Theorem 12.7 follows. $\square$

13. Effective Version of Matsusaka's Big Theorem

An important problem of algebraic geometry is to find effective bounds m_0 such that multiples mL of an ample line bundle become very ample for $m \geq m_0$. From a theoretical point of view, this problem has been solved by Matsusaka [Mat72] and Kollár-Matsusaka [KoM83]. Their result is that there is a bound $m_0 = m_0(n, L^n, L^{n-1} \cdot K_X)$ depending only on the dimension and on the first two coefficients L^n and $L^{n-1} \cdot K_X$ in the Hilbert polynomial of L . Unfortunately, the original proof does not tell much on the actual dependence of m_0 in terms of these coefficients. The goal of this section is to find effective bounds for such an integer m_0 , along the lines of [Siu93]. However, one of the technical lemmas used in [Siu93] to deal with dualizing sheaves can be sharpened. Using this sharpening of the lemma, Siu's bound will be here substantially improved. We first start with the simpler problem of obtaining merely a nontrivial section in mL . The idea, more generally, is to obtain a criterion for the ampleness of $mL - B$ when B is nef. In this way, one is able to subtract from mL any multiple of K_X which happens to get added by the application of Nadel's vanishing theorem (for this, replace B by B plus a multiple of $K_X + (n+1)L$).

(13.1) Proposition. *Let L be an ample line bundle over a projective n -fold X and let B be a nef line bundle over X . Then $K_X + mL - B$ has a nonzero section for some integer m such that*

$$m \leq n \frac{L^{n-1} \cdot B}{L^n} + n + 1.$$

Proof. Let m_0 be the smallest integer $> n \frac{L^{n-1} \cdot B}{L^n}$. Then $m_0 L - B$ can be equipped with a singular hermitian metric of positive definite curvature. Let φ be the weight of this metric. By Nadel's vanishing theorem, we have

$$H^q(X, \mathcal{O}(K_X + mL - B) \otimes \mathcal{I}(\varphi)) = 0 \quad \text{for } q \geq 1,$$

thus $P(m) = h^0(X, \mathcal{O}(K_X + mL - B) \otimes \mathcal{I}(\varphi))$ is a polynomial for $m \geq m_0$. Since P is a polynomial of degree n and is not identically zero, there must be an integer $m \in [m_0, m_0 + n]$ which is not a root. Hence there is a nontrivial section in

$$H^0(X, \mathcal{O}(K_X + mL - B)) \supset H^0(X, \mathcal{O}(K_X + mL - B) \otimes \mathcal{I}(\varphi))$$

for some $m \in [m_0, m_0 + n]$, as desired. $\square$

(13.2) Corollary. *If L is ample and B is nef, then $mL - B$ has a nonzero section for some integer*

$$m \leq n \left(\frac{L^{n-1} \cdot B + L^{n-1} \cdot K_X}{L^n} + n + 1 \right).$$

Proof. By Fujita's result 8.3 a), $K_X + (n+1)L$ is nef. We can thus replace B by $B + K_X + (n+1)L$ in the result of Prop. 13.1. Corollary 13.2 follows. $\square$

(13.3) Remark. We do not know if the above Corollary is sharp, but it is certainly not far from being so. Indeed, for $B = 0$, the initial constant n cannot be replaced by anything smaller than $n/2$: take X to be a product of curves C_j of large genus g_j and $B = 0$; our bound for $L = \mathcal{O}(a_1[p_1]) \otimes \dots \otimes \mathcal{O}(a_n[p_n])$ to have $|mL| \neq \emptyset$ becomes $m \leq \sum (2g_j - 2)/a_j + n(n+1)$, which fails to be sharp only by a factor 2 when $a_1 = \dots = a_n = 1$ and $g_1 \gg g_2 \gg \dots \gg g_n \rightarrow +\infty$. On the other hand, the additive constant $n+1$ is already best possible when $B = 0$ and $X = \mathbb{P}^n$. $\square$

So far, the method is not really sensitive to singularities (the Morse inequalities are indeed still true in the singular case as is easily seen by using desingularizations of the ambient variety). The same is true with Nadel's vanishing theorem, provided that K_X is replaced by the L^2 dualizing sheaf ω_X (according to the notation introduced in Remark 5.17, $\omega_X = K_X(0)$ is the sheaf of holomorphic n -forms u on X_{reg} such that $i^{n^2} u \wedge \bar{u}$ is integrable in a neighborhood of the singular set). Then Prop. 13.1 can be generalized as

(13.4) Proposition. *Let L be an ample line bundle over a projective n -fold X and let B be a nef line bundle over X . For every p -dimensional (reduced) algebraic subvariety Y of X , there is an integer*

$$m \leq p \frac{L^{p-1} \cdot B \cdot Y}{L^p \cdot Y} + p + 1$$

such that the sheaf $\omega_Y \otimes \mathcal{O}_Y(mL - B)$ has a nonzero section. $\square$

To proceed further, we need the following useful “upper estimate” about L^2 dualizing sheaves (this is one of the crucial steps in Siu’s approach; unfortunately, it has the effect of producing rather large final bounds when the dimension increases).

(13.5) Proposition. *Let H be a very ample line bundle on a projective algebraic manifold X , and let $Y \subset X$ be a p -dimensional irreducible algebraic subvariety. If $\delta = H^p \cdot Y$ is the degree of Y with respect to H , the sheaf*

$$\mathcal{H}om(\omega_Y, \mathcal{O}_Y((\delta - p - 2)H))$$

has a nontrivial section.

Observe that if Y is a smooth hypersurface of degree δ in $(X, H) = (\mathbb{P}^{p+1}, \mathcal{O}(1))$, then $\omega_Y = \mathcal{O}_Y(\delta - p - 2)$ and the estimate is optimal. On the other hand, if Y is a smooth complete intersection of multidegree $(\delta_1, \dots, \delta_r)$ in $\mathbb{P}^{p+r}$, then $\delta = \delta_1 \dots \delta_r$ whilst $\omega_Y = \mathcal{O}_Y(\delta_1 + \dots + \delta_r - p - r - 1)$; in this case, Prop. (13.5) is thus very far from being sharp.

Proof. Let $X \subset \mathbb{P}^N$ be the embedding given by H , so that $H = \mathcal{O}_X(1)$. There is a linear projection $\mathbb{P}^N \dashrightarrow \mathbb{P}^{p+1}$ whose restriction $\pi : Y \rightarrow \mathbb{P}^{p+1}$ to Y is a finite and regular birational map of Y onto an algebraic hypersurface Y' of degree δ in $\mathbb{P}^{p+1}$. Let $s \in H^0(\mathbb{P}^{p+1}, \mathcal{O}(\delta))$ be the polynomial of degree δ defining Y' . We claim that for any small Stein open set $W \subset \mathbb{P}^{p+1}$ and any L^2 holomorphic p -form u on $Y' \cap W$, there is a L^2 holomorphic $(p+1)$ -form $\tilde{u}$ on W with values in $\mathcal{O}(\delta)$ such that $\tilde{u}|_{Y' \cap W} = u \wedge ds$. In fact, this is precisely the conclusion of the Ohsawa-Takegoshi extension theorem [OT87], [Ohs88] (see also [Man93] for a more general version); one can also invoke more standard local algebra arguments (see Hartshorne [Har77], Th. III-7.11). As $K_{\mathbb{P}^{p+1}} = \mathcal{O}(-p-2)$, the form $\tilde{u}$ can be seen as a section of $\mathcal{O}(\delta - p - 2)$ on W , thus the sheaf morphism $u \mapsto u \wedge ds$ extends into a global section of $\mathcal{H}om(\omega_{Y'}, \mathcal{O}_{Y'}(\delta - p - 2))$. The pull-back by π^* yields a section of $\mathcal{H}om(\pi^*\omega_{Y'}, \mathcal{O}_Y((\delta - p - 2)H))$. Since π is finite and generically $1:1$, it is easy to see that $\pi^*\omega_{Y'} = \omega_Y$. The Proposition follows. $\square$

By an appropriate induction process based on the above results, we can now improve Siu’s effective version of the Big Matsusaka Theorem [Siu93]. Our version depends on a constant λ_n such that $m(K_X + (n+2)L) + G$ is very ample for $m \geq \lambda_n$ and every nef line bundle G . Corollary (8.5) shows that $\lambda_n \leq \binom{3n+1}{n} - 2n$, and a similar argument involving the recent results of Angehrn-Siu [AS94] implies $\lambda_n \leq n^3 - n^2 - n - 1$ for $n \geq 2$. Of course, it is expected that $\lambda_n = 1$ in view of the Fujita conjecture.

(13.6) Effective version of the Big Matsusaka Theorem. *Let L and B be nef line bundles on a projective n -fold X . Assume that L is ample and let H be the very ample line bundle $H = \lambda_n(K_X + (n+2)L)$. Then $mL - B$ is very ample for*

$$m \geq (2n)^{(3^{n-1}-1)/2} \frac{(L^{n-1} \cdot (B+H))^{(3^{n-1}+1)/2} (L^{n-1} \cdot H)^{3^{n-2}(n/2-3/4)-1/4}}{(L^n)^{3^{n-2}(n/2-1/4)+1/4}}.$$

In particular mL is very ample for

$$m \geq C_n (L^n)^{3^{n-2}} \left(n + 2 + \frac{L^{n-1} \cdot K_X}{L^n} \right)^{3^{n-2}(n/2+3/4)+1/4}$$

with $C_n = (2n)^{(3^{n-1}-1)/2}(\lambda_n)^{3^{n-2}(n/2+3/4)+1/4}$.

Proof. We use Prop. (13.4) and Prop. (13.5) to construct inductively a sequence of (non necessarily irreducible) algebraic subvarieties $X = Y_n \supset Y_{n-1} \supset \dots \supset Y_2 \supset Y_1$ such that $Y_p = \bigcup_j Y_{p,j}$ is p -dimensional, and Y_{p-1} is obtained for each $p \geq 2$ as the union of zero sets of sections

$$\sigma_{p,j} \in H^0(Y_{p,j}, \mathcal{O}_{Y_{p,j}}(m_{p,j}L - B))$$

with suitable integers $m_{p,j} \geq 1$. We proceed by induction on decreasing values of the dimension p , and find inductively upper bounds m_p for the integers $m_{p,j}$.

By Cor. (13.2), an integer m_n for $m_n L - B$ to have a section σ_n can be found with

$$m_n \leq n \frac{L^{n-1} \cdot (B + K_X + (n+1)L)}{L^n} \leq n \frac{L^{n-1} \cdot (B + H)}{L^n}.$$

Now suppose that the sections $\sigma_n, \dots, \sigma_{p+1,j}$ have been constructed. Then we get inductively a p -cycle $\tilde{Y}_p = \sum \mu_{p,j} Y_{p,j}$ defined by $\tilde{Y}_p =$ sum of zero divisors of sections $\sigma_{p+1,j}$ in $\tilde{Y}_{p+1,j}$, where the multiplicity $\mu_{p,j}$ on $Y_{p,j} \subset Y_{p+1,k}$ is obtained by multiplying the corresponding multiplicity $\mu_{p+1,k}$ with the vanishing order of $\sigma_{p+1,k}$ along $Y_{p,j}$. As cohomology classes, we find

$$\tilde{Y}_p \equiv \sum (m_{p+1,k} L - B) \cdot (\mu_{p+1,k} Y_{p+1,k}) \leq m_{p+1} L \cdot \tilde{Y}_{p+1}.$$

Inductively, we thus have the numerical inequality

$$\tilde{Y}_p \leq m_{p+1} \dots m_n L^{n-p}.$$

Now, for each component $Y_{p,j}$, Prop. (13.4) shows that there exists a section of $\omega_{Y_{p,j}} \otimes \mathcal{O}_{Y_{p,j}}(m_{p,j}L - B)$ for some integer

$$m_{p,j} \leq p \frac{L^{p-1} \cdot B \cdot Y_{p,j}}{L^p \cdot Y_{p,j}} + p + 1 \leq p m_{p+1} \dots m_n L^{n-1} \cdot B + p + 1.$$

Here, we have used the obvious lower bound $L^{p-1} \cdot Y_{p,j} \geq 1$ (this is of course a rather weak point in the argument). The degree of $Y_{p,j}$ with respect to H admits the upper bound

$$\delta_{p,j} := H^p \cdot Y_{p,j} \leq m_{p+1} \dots m_n H^p \cdot L^{n-p}.$$

We use the Hovanski-Teissier concavity inequality (10.2 b)

$$(L^{n-p} \cdot H^p)^{\frac{1}{p}} (L^n)^{1-\frac{1}{p}} \leq L^{n-1} \cdot H$$

to express our bounds in terms of the intersection numbers L^n and $L^{n-1} \cdot H$ only. We then get

$$\delta_{p,j} \leq m_{p+1} \dots m_n \frac{(L^{n-1} \cdot H)^p}{(L^n)^{p-1}}.$$

By Prop. (13.5), there is a nontrivial section in

$$\mathcal{H}om(\omega_{Y_{p,j}}, \mathcal{O}_{Y_{p,j}}((\delta_{p,j} - p - 2)H)).$$

Combining this section with the section in $\omega_{Y_{p,j}} \otimes \mathcal{O}_{Y_{p,j}}(m_{p,j}L - B)$ already constructed, we get a section of $\mathcal{O}_{Y_{p,j}}(m_{p,j}L - B + (\delta_{p,j} - p - 2)H)$ on $Y_{p,j}$. Since we do not want H to appear at this point, we replace B with $B + (\delta_{p,j} - p - 2)H$ and thus get a section $\sigma_{p,j}$ of $\mathcal{O}_{Y_{p,j}}(m_{p,j}L - B)$ with some integer $m_{p,j}$ such that

$$\begin{aligned} m_{p,j} &\leq pm_{p+1} \dots m_n L^{n-1} \cdot (B + (\delta_{p,j} - p - 2)H) + p + 1 \\ &\leq pm_{p+1} \dots m_n \delta_{p,j} L^{n-1} \cdot (B + H) \\ &\leq p(m_{p+1} \dots m_n)^2 \frac{(L^{n-1} \cdot H)^p}{(L^n)^{p-1}} L^{n-1} \cdot (B + H). \end{aligned}$$

Therefore, by putting $M = n L^{n-1} \cdot (B + H)$, we get the recursion relation

$$m_p \leq M \frac{(L^{n-1} \cdot H)^p}{(L^n)^{p-1}} (m_{p+1} \dots m_n)^2 \quad \text{for } 2 \leq p \leq n-1,$$

with initial value $m_n \leq M/L^n$. If we let $(\overline{m}_p)$ be the sequence obtained by the same recursion formula with equalities instead of inequalities, we get $m_p \leq \overline{m}_p$ with $\overline{m}_{n-1} = M^3(L^{n-1} \cdot H)^{n-1}/(L^n)^n$ and

$$\overline{m}_p = \frac{L^n}{L^{n-1} \cdot H} \overline{m}_{p+1}^2 \overline{m}_{p+1}$$

for $2 \leq p \leq n-2$. We then find inductively

$$m_p \leq \overline{m}_p = M^{3^{n-p}} \frac{(L^{n-1} \cdot H)^{3^{n-p-1}(n-3/2)+1/2}}{(L^n)^{3^{n-p-1}(n-1/2)+1/2}}.$$

We next show that $m_0L - B$ is nef for

$$m_0 = \max(m_2, m_3, \dots, m_n, m_2 \dots m_n L^{n-1} \cdot B).$$

In fact, let $C \subset X$ be an arbitrary irreducible curve. Either $C = Y_{1,j}$ for some j or there exists an integer $p = 2, \dots, n$ such that C is contained in Y_p but not in Y_{p-1} . If $C \subset Y_{p,j} \setminus Y_{p-1}$, then $\sigma_{p,j}$ does not vanish identically on C . Hence $(m_{p,j}L - B)|_C$ has nonnegative degree and

$$(m_0L - B) \cdot C \geq (m_{p,j}L - B) \cdot C \geq 0.$$

On the other hand, if $C = Y_{1,j}$, then

$$(m_0L - B) \cdot C \geq m_0 - B \cdot \tilde{Y}_1 \geq m_0 - m_2 \dots m_n L^{n-1} \cdot B \geq 0.$$

By the definition of λ_n (and by Cor. (8.5) showing that such a constant exists), $H + G$ is very ample for every nef line bundle G , in particular $H + m_0L - B$ is very ample. We thus replace again B with $B + H$. This has the effect of replacing M with $M = n(L^{n-1} \cdot (B + 2H))$ and m_0 with

$$m_0 = \max(m_n, m_{n-1}, \dots, m_2, m_2 \dots m_n L^{n-1} \cdot (B + H)).$$

The last term is the largest one, and from the estimate on $\overline{m}_p$, we get

$$\begin{aligned} m_0 &\leq M^{(3^{n-1}-1)/2} \frac{(L^{n-1} \cdot H)^{(3^{n-2}-1)(n-3/2)/2+(n-2)/2} (L^{n-1} \cdot (B + H))}{(L^n)^{(3^{n-2}-1)(n-1/2)/2+(n-2)/2+1}} \\ &\leq (2n)^{(3^{n-1}-1)/2} \frac{(L^{n-1} \cdot (B + H))^{(3^{n-1}+1)/2} (L^{n-1} \cdot H)^{3^{n-2}(n/2-3/4)-1/4}}{(L^n)^{3^{n-2}(n/2-1/4)+1/4}} \end{aligned}$$

□

(13.7) Remark. In the surface case $n = 2$, one can take $\lambda_n = 1$ and our bound yields mL very ample for

$$m \geq 4 \frac{(L \cdot (K_X + 4L))^2}{L^2}.$$

If one looks more carefully at the proof, the initial constant 4 can be replaced by 2. In fact, it has been shown recently by Fernández del Busto that mL is very ample for

$$m > \frac{1}{2} \left[\frac{(L \cdot (K_X + 4L) + 1)^2}{L^2} + 3 \right],$$

and an example of G. Xiao shows that this bound is essentially optimal (see [FdB94]).

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Jean-Pierre Demailly
 Université de Grenoble I,
 Institut Fourier, URA 188 du CNRS, BP 74,
 F-38402 Saint-Martin d'Hères, France
e-mail address: demailly@fourier.ujf-grenoble.fr

Manifolds of semi-positive curvature

Thomas Peternell

Introduction

The uniformisation theorem in one complex variable determines the universal cover $\tilde{X}$ of compact Riemann surfaces X ; $\tilde{X}$ is the projective line resp. $\mathbf{C}$ resp. the unit disc according to whether X itself is the projective line (genus 0), a torus (genus 1) or of genus ≥ 2 . Since a long time the question was studied what should be "the" generalisation to higher dimensions. This question is a main theme of these notes. Already in dimensions 1 and 2 it is completely clear that the behaviour of varieties of general type (i.e. genus ≥ 2 for curves) is quite different from that one of curves of genus ≤ 1 resp. surfaces of Kodaira dimension at most 1. Having this in mind and due to the fact that certainly there will not be a convincing uniformisation theorem for manifolds of general type we will only consider the uniformisation problem for the higher dimensional analogues of the projective line and tori. So we ask : what are natural generalisations of Riemann surfaces of genus 0 or 1? Maybe one is tempted to take manifolds of Kodaira dimension $-\infty$ resp. 0. However n -folds X with $\kappa(X) = -\infty$ are at least as general as arbitrary $(n-1)$ -folds Y : just observe that $Y \times \mathbf{P}_1$ has $\kappa = -\infty$. Moreover we do not want to study varieties from a birational point of view but want to have a biregular theory; so for example the class of n -folds with $\kappa = 0$ is too large.

Let us next take a differential geometric point of view : the projective line is the only curve having a metric of positive curvature while tori are characterised by the existence of flat metrics. In higher dimensions basically two concepts of curvature (in the complex situation) are available : holomorphic bisectional curvature and Ricci curvature. The first one means that we consider the full tangent bundle of X , while considering Ricci curvature says that we are only interested in the determinant of the tangent bundle, i.e. the anti-canonical bundle. So the classification problem asks: what are the projective (or compact Kähler) manifolds (admitting an hermitian metric) with semi-positive holomorphic or semi-positive Ricci curvature ?

Since metrics are often very difficult to construct one is interested in algebraic settings of the uniformisation problem. It turns out that the correct algebraic reformulation of the existence of a metric of semi-positive holomorphic bisectional curvature is to say that the tangent bundle is "nef", although these two notions are not completely equivalent. Nefness can (in the projective case) be tested just by restricting the tangent bundle to curves (however all curves); so it is to some extent easier to check because not dependent on the construction of a global object (a metric). Intuitively speaking, a vector bundle on a curve is nef, if the tensor product with any small positive rational multiple of any ample line bundle is an

ample " $\mathbf{Q}$ -vector bundle". The same of course holds for the anti-canonical bundle. Hence we ask : what are the manifolds whose tangent bundles resp. anti-canonical bundles are nef ?

We do not want to state precise results in this introduction but merely give some rough description how the classes look like. We consider only the algebraic, i.e. "nef" case and begin with the tangent bundle T_X . Mori's famous theorem says that if T_X is ample, then X is the projective space; on the other hand, if T_X is trivial, then it is a well-known and easy fact, that X must be a torus. A general manifold with T_X nef is somehow between these two extremes. Many examples are provided by homogeneous manifolds. Rational homogeneous manifolds (projective spaces, quadrics, Grassmannians, etc.) and tori should be the building blocks of manifolds with nef tangent bundles. What this precisely means and to what extent this is already proved, will be discussed at length in sect. 4.

While one can expect a very precise classification of manifolds with nef tangent bundles, the class of manifolds with nef anti-canonical bundles $-K_X$ is much broader. If $-K_X$ is ample, one obtains by definition the Fano manifolds, including all hypersurfaces in projective n -space of degree less than $n+1$. The other extreme are manifolds with trivial (or numerically trivial) (anti-) canonical bundle, i.e. tori, Calabi-Yau manifolds and symplectic manifolds. Again these four types of varieties should be thought of as the building blocks of manifolds with nef anti-canonical bundles.

The structure of these notes is as follows. We begin in sect. 1 by describing the theory of nef vector bundles and geometric consequences of the structure of the ample cone on a projective variety : base point free theorem, contraction theorem etc. Section 2 provides a new proof of Mori's theorem that every projective manifold with ample tangent bundle is projective space, without using characteristic p . So this proof is completely inside complex algebraic geometry or complex analysis. In sect. 3 we discuss the metric analogue of Mori's theorem due to Siu and Yau as well as the classification of compact manifolds admitting a Kähler metric of semi-positive holomorphic bisectional curvature by Mok. The next two sections form the heart of these notes. In sect. 4 we describe the structure of projective or compact Kähler manifolds whose tangent bundles are nef, while the last section is concerned with manifolds with nef anti-canonical bundles.

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Notations

If X is a complex manifold, we denote by K_X the canonical divisor (line bundle) and by $\kappa(X)$ the Kodaira dimension of X (X compact) . T_X denotes the tangent

bundle of X . We let Ω_X^q be the sheaf of q -forms. As usual we often do not distinguish between divisors and line bundles and between locally free sheaves and vector bundles. If E is a vector bundle, $\mathbf{P}(E)$ denotes the associated projectivised bundle in the sense of Grothendieck, i.e. taking hyperplanes in the fibers.

As usual, $\text{Pic}(X)$ is the Picard group of X . If D is an effective divisor in X , we let $\mathcal{N}_{D|X}$ be its normal bundle. If X is compact Kähler, we set $q(X) = \dim H^1(X, \mathcal{O}_X) = \dim H^0(X, \Omega_X^1)$. Furthermore we set

$$\tilde{q}(X) = \sup\{q(\tilde{X}) | \tilde{X} \longrightarrow X \text{ finite étale}\}.$$

Of course $\tilde{q}(X)$ need not be finite, however in the situations we are interested in, it will be.

1. Nef Vector Bundles

This section consists of two parts : first we discuss nef line bundles on projective varieties and compact manifolds and the geometry of the ample cone, i.e. some basic facts of what is often called Mori theory. Second we study nef vector bundles on compact complex manifolds in general and investigate their main properties. We assume basic knowledge of the theory of ample line bundles, as presented e.g. in the first chapter of [Ha70]. General references for this section are [DPS94a],[De91],[KMM87],[FJ83].

1.1 Definition Let X be a projective variety. A line bundle L on X is said to be nef, if $\deg(L|C) \geq 0$ for every irreducible curve $C \subset X$. We usually write $(L.C)$ or $(c_1(L).C)$ instead of $\deg(L|C)$.

1.2 Examples and Basic Properties

- (1) Every ample line bundle is nef. More generally every line bundle for which some tensor power is generated by global sections, is nef. Those bundles are often called semi-ample. However not every nef line bundle is semi-ample (see (1.3) below).
- (2) A line bundle L is nef if and only if for some (hence every) ample line bundle A the line bundles $L^k \otimes A$ are ample for all $k \geq 1$. (This follows easily e.g. from Seshadri's ampleness criterion that A is ample if and only if there is a positive constant K such that $(A.C) \geq Km(C)$, where $m(C)$ is the maximum of all multiplicities $m_x(C)$, $x \in C$).
- (3) If L is nef, then $(c_1(L)^k.Y) \geq 0$ for all irreducible subvarieties $Y \subset X$ of dimension k .
- (4) Let L be a line bundle admitting a hermitian metric such that the curvature Θ of the associated canonical ("Chern") connection is semi-positive. This means that if the metric is locally given by the positive function h , then $-\partial\bar{\partial}\log h$ is a semi-positive $(1,1)$ -form, i.e. $-\log h$ is plurisubharmonic. (see (1.13) for the general definition). Then $(c_1(L).C) = \int_C \Theta \geq 0$ for every curve $C \subset X$, hence L is nef.

We have seen that line bundles with semi-positive curvature are nef. The converse however is not true :

1.3 Example

Let $\Gamma = \mathbf{C}/(\mathbf{Z} + \mathbf{Z}\tau)$ be an elliptic curve. We construct a vector bundle E of rank 2 on Γ which will fit into an exact sequence

$$0 \longrightarrow \mathcal{O}_\Gamma \longrightarrow E \longrightarrow \mathcal{O}_\Gamma \longrightarrow 0.$$

The construction of E is as follows. Let $G = \mathbf{Z} + \mathbf{Z}\tau$, operating on $\mathbf{C} \times \mathbf{C}^2$ by virtue of

$$\begin{aligned} g(x, z_1, z_2) &= (x + 1, z_1, z_2), \\ g_\tau(x, z_1, z_2) &= (x + \tau, z_1 + z_2, z_2). \end{aligned}$$

Let $E = \mathbf{C} \times \mathbf{C}^2/G$, the projection to Γ being given by the projection to x . Then the trivial subbundle of E is defined by the quotient of $z_2 = 0$ and the quotient bundle by $z_1 = 0$.

Now let $X = \mathbf{P}(E)$ and let $L = \mathcal{O}_X(1)$. Then one can show by explicit calculation that L does not carry a hermitian metric of semi-positive curvature, see [DPS94a], whereas L is nef which can easily be checked, see [Ha77, chap.5.2] for the relevant fact about curves on ruled surfaces.

If X is merely a compact manifold, not necessarily algebraic, then the above definition of nefness is not suitable since X need not have any curve. So we give now an alternative definition which in case X is algebraic turns out to be equivalent to the old one.

1.4 Definition

Let X be a compact complex manifold. Let ω be a positive (1,1)-form on X . A line bundle L on X is called nef if for every $\epsilon > 0$, there exists a hermitian metric h_ϵ on L whose curvature satisfies the condition

$$\Theta_{h_\epsilon} \geq -\epsilon\omega.$$

A few remarks are in order here. First, it is an obvious consequence from compactness of X that the definition is independent of the choice of ω . Second, a line bundle L which is nef in the sense of (1.4) fulfills clearly $(L.C) \geq 0$ for all curves $C \subset X$. The converse follows from (1.2(2)) and the classical fact that every ample line bundle has a metric of positive curvature.

One reason why nef line bundles are important is that they behave better than ample line bundles with respect to functoriality, e.g. we have :

if $f : X \longrightarrow Y$ is a holomorphic map of compact manifolds, then the pull-back $f^*(L)$ of a nef line bundle L on Y is nef.

Next we discuss nef and ample line bundles from a numerical point of view : we introduce the ample cone of an algebraic variety and consider its structure.

1.5 Definition

We let X always denote a normal projective variety.

(1) Two line bundles L_i on X are said to be numerically equivalent, in signs $L_1 \equiv L_2$, if $(L_1, C) = (L_2, C)$ for all curves $C \subset X$. The vector space $N^1(X) = (Pic(X)/\equiv) \otimes \mathbf{R}$ is a finite-dimensional vector space, the Néron-Severi space of X ; its dimension is called the Picard number of X and is denoted by $\rho(X)$. Inside $N^1(X)$ sits the closed cone K generated by the classes of nef line bundles which is also the smallest closed cone containing all classes of ample line bundles. K is called the ample cone of X .

(2) In an analogous way we say that two curves $C_i \subset X$ are numerically equivalent, in signs $C_1 \equiv C_2$, if $(L, C_1) = (L, C_2)$ for all line bundles on X . The corresponding real vector space is denoted $N_1(X)$. Inside $N_1(X)$ we have the closed cone generated by the classes of effective curves, denoted $\overline{NE}(X)$. It is a well-known fact that $\dim N^1(X) = \dim N_1(X)$; moreover K and $\overline{NE}(X)$ are dual cones.

(3) Often we will identify line bundles and Cartier divisors on projective varieties, in particular in connection with the ample cone. Denoting the group of divisors by $Div(X)$, an element $D \in Div(X) \otimes \mathbf{Q}$ is called a $\mathbf{Q}$ -divisor.

(4) A divisor D is said to be big if its Iitaka dimension $\kappa(D) = \dim X$. In case D is nef, this is equivalent to saying that $D^n > 0$, where $n = \dim X$.

A basic tool in the theory is the Kawamata-Viehweg vanishing theorem which is a generalisation of Kodaira's vanishing theorem :

1.6 Theorem

Let X be a projective manifold, D a big and nef divisor . Then $H^q(X, D + K_X) = 0, q \geq 1$.

For a proof and the following very important generalisation see e.g. [KMM87]. Let D be a $\mathbf{Q}$ -divisor, which is big and nef. Let D' be the round up, D'' the round down so that D', D'' are Cartier divisors. Assume that the fractional part $D' - D''$ has support with only normal crossings. Then $H^q(X, D' + K_X) = 0, q \geq 1$. There is also a relative version.

A very important application is the

1.7 Base Point Free Theorem

Let X be a projective manifold , D a nef divisor on X . If $aD - K_X$ is big and nef for some positive a , then D is semi-ample : there exists m_0 , such that mD is generated by global sections for $m \geq m_0$. In particular this holds if D is big and nef and if $aD - K_X$ is nef.

The base point free theorem is due to Kawamata and Shokurov. For a proof see [KMM87]. Again there are more general versions, it is of great importance that certain singularities ("terminal" singularities) can be allowed on X .

(1.8) Now consider a projective manifold X such that K_X is not nef. Let H be an ample divisor on X . Then there exists a unique positive real number t such that

$K_X + tH$ is nef but not ample, i.e. $K_X + tH$ is on the boundary of the ample cone. The rationality theorem of Shokurov says that in fact $t \in \mathbf{Q}$. Take a positive integer m , such that $D = m(K_X + tH)$ is Cartier. Then the base point free theorem can be applied to D to the effect that D is semi-ample. This and much more is the content of the following two theorems due to Mori, Kawamata, Kollár, Reid and others, describing the geometry of the ample cone and its consequences for the variety itself.

1.9 Cone Theorem

Let X be a projective manifold. Let $\overline{NE}_+(X) = \{D \in \overline{NE}(X) | (D.K_X) \geq 0\}$. Then

$$\overline{NE}(X) = \overline{NE}_+(X) + \sum R_i,$$

where the R_i are extremal rays of the cone $\overline{NE}(X)$ which are discrete in

$$\{D \in \overline{NE}(X) | (D.K_X) < 0\}.$$

1.10 Contraction Theorem

Let X be a projective manifold and R an extremal ray on X , i.e. a (geometrically) extremal ray in $\overline{NE}(X)$, with $(R.K_X) < 0$. Then there exists a morphism

$\phi : X \longrightarrow Y$ to a normal projective variety Y with the following properties.

- (1) $\phi_*(\mathcal{O}_X) = \mathcal{O}_Y$
- (2) if $C \subset X$ is an irreducible curve, then $\dim \phi(C) = 0$ if and only if $[C] \in R$.
- (3) $-K_X$ is ϕ -ample.

Proofs can again be found in [KMM87]. The map ϕ in (1.10) will be referred to as a Mori contraction or contraction of an extremal ray. An immediate consequence of (1.10(2)) is $\rho(Y) = \rho(X) - 1$. We should note that in general Y might be singular but often the singularities can be controlled.

We now introduce a very important class of manifolds which play a crucial role in all what follows.

1.11 Definition

- (1) A projective manifold whose anticanonical bundle $-K_X$ is ample, is called a Fano manifold.
- (2) The index of X is the largest positive integer such that there is a line bundle L on X with $L^r = K_X$.

Fano manifolds of dimension at most three are completely classified, we refer to [Is77,90], [Mu81], [Mu89], [MM81]. Most basic examples are projective spaces, quadrics, Grassmannians, hypersurfaces of degree at most n in $\mathbf{P}_n$, etc. Fano manifolds are simply connected which is a consequence of a far stronger property : rational connectedness ([Ca92] and [KoMiMo92]). This means that any two points

in the variety can be joined by a chain of rational curves. Note that the cone theorem gets quite easy in the Fano case because then $\overline{NE}_+(X) = 0$. Note further that the general smooth fiber of a Mori contraction is a Fano manifold by (1.10(3)).

We now turn to nef vector bundles.

1.12 Definition

Let E be a vector bundle on a compact complex manifold (or on a projective variety). E is nef iff the line bundle $\mathcal{O}_{\mathbf{P}(E)}(1)$ is nef.

The relevant differential-geometric analogue is given in

1.13 Definition

Let (E, h) be a hermitian vector bundle, i.e. a vector bundle equipped with a hermitian metric h . We say that (E, h) is semi-positive (in the sense of Griffiths), if the curvature tensor $\Theta_h(E)$ fulfills the following condition.

$$(-i)\Theta_h(E)(\zeta \otimes v) \geq 0$$

for all $x \in X$, $\zeta \in T_x X$ and $v \in E_x$. In case of strict inequality for non-zero vectors ζ, v we say that E is Griffiths-positive.

We recall that, if D is the canonical "Chern" connection on E , i.e. compatible with both the hermitian and holomorphic structure, then $\Theta_h(E)$ is a hermitian $(1,1)$ -form with values in $\text{Hom}(E, E)$ given by

$$\Theta_h(E) = \frac{i}{2\pi} D^2 = i \sum_{1 \leq j, k \leq n; 1 \leq \lambda, \mu \leq r} a_{j,k,\lambda,\mu} dz_j \wedge d\bar{z}_k \otimes e_\lambda^* \otimes e_\mu.$$

Here $z_1, \dots, z_n$ are local coordinates on X and $e_1, \dots, e_r$ are a local orthogonal frame of E . For details on hermitian differential geometry we refer to [Gr69].

1.14 Remark

It is an easy exercise to prove that a Griffiths-positive vector bundle is ample. Whether the converse holds is unknown and certainly a deep problem. The bundle E from example 1.3 shows that there are nef vector bundles which are not Griffiths semi-positive; this can also be easily seen as follows : if E would admit a metric with curvature $\Theta \geq 0$, then $\det E = \mathcal{O}$ forces the trace $\text{tr} \Theta = 0$, hence E would be hermitian flat. Then however the exact sequence

$$0 \longrightarrow \mathcal{O} \longrightarrow E \longrightarrow \mathcal{O} \longrightarrow 0$$

would split. On the other hand we have

1.15 Theorem

Let X be a compact complex manifold and ω a positive "reference" $(1,1)$ -form on X . Let E be a vector bundle on X . Then E is nef if and only if the following holds.

There exists a sequence (h_m) of metrics on $S^m E$ such that for every $\epsilon > 0$ and $m \geq m_0(\epsilon)$ the following inequality holds

$$\Theta_{h_m}(S^m E) \geq -m\epsilon\omega \otimes id_{S^m E}.$$

Of course the last inequality has to be understood in the "sense of Griffiths," i.e. one has to insert vectors ζ, v as in the definition of semi-positivity. A proof of (1.15) can be found in [DPS94a]. The next proposition collects basic properties of nef bundles.

1.16 Proposition

Let E and F be vector bundles on the compact manifold X .

- (1) If E and F are nef, then also $E \otimes F$ is nef.
- (2) If E is nef, so also all tensor powers of E .
- (3) If some symmetric power $S^m E$ is nef, $m \geq 1$, then E is nef.
- (4) If $f: Y \rightarrow X$ is a holomorphic map of a compact manifold Y to X , then f^*E is nef if E is nef. The converse holds if f is onto with equidimensional fibers.
- (5) Given an exact sequence of vector bundles $0 \rightarrow F \rightarrow E \rightarrow Q \rightarrow 0$, then
 - (a) E nef implies Q nef,
 - (b) F, Q nef imply E nef,
 - (c) E nef and $(\det Q)^{-1}$ nef imply F nef.
- (6) If E is nef and $s \in H^0(X, E^*)$ is a non-zero section, then s has no zeroes at all.

We finish this section by giving basic Chern class inequalities for nef vector bundles. For ample bundles these are due to Fulton and Lazarsfeld [FL83], the general case for nef bundles on Kähler manifolds is done in [DPS94a]. We denote by $\lambda(k, r)$ the set of partitions of k by non-negative integers $\leq r$; i.e an element a of $\lambda(k, r)$ has the form $(a_1, \dots, a_r)$ with $a_i \leq r, a_i \geq a_{i+1}$ and $\sum a_i = k$. Now define the Schur polynomial $P_a \in \mathbb{Z}[c_1, \dots, c_r]$ associated with a by the formula

$$P_a(c) = \det(c_{a_i - i + j})_{1 \leq i, j \leq r}.$$

We say that $\deg P_a = 2k$. Here c_i is a variable with $\deg c_i = 2i$. We moreover set $c_0 = 1$ and $c_i = 0$ if $i < 0$ or if $i > r$. In our situation the c_i are of course the Chern classes of a vector bundle E .

1.17 Theorem

Let (X, ω) be a compact Kähler manifold and E a nef vector bundle on X . Let P_a be a Schur polynomial of degree $2k$. Then for any compact analytic set $Y \subset X$ of dimension $d \geq k$ the inequality

$$\int_Y P_a(c(E)) \wedge \omega^{d-k} \geq 0$$

holds.

Important examples of Schur polynomials of degree $2k$ are c_k , the k -th Chern class, s_k , the k -th Segre class and $c_1 c_{k-1} - c_k$.

A polynomial of the type

$$P = \sum \lambda_a P_a$$

where all P_a are Schur polynomials of a fixed degree $2k$ and where $\lambda_a > 0$, are called numerically positive. These are studied in [FL83]. In particular a product of numerically positive polynomials is again numerically positive. Thus we see that any Chern monomial $c_I = c_1^{i_1} \cdots c_r^{i_r}$ of degree $2k$ satisfies the condition

$$0 \leq c_I \leq c_1^k.$$

Thus we obtain

1.18 Corollary

If E is nef, then any Chern monomial c_I of degree $2k$ satisfies the inequality

$$0 \leq \int_X c_I(E) \wedge \omega^{n-k} \leq \int_X c_1(E)^k \wedge \omega^{n-k}.$$

In particular, if $c_1(E)^n = 0$, then all Chern polynomials of degree $2n$ vanish.

2. Manifolds with ample tangent bundles

In this section we are beginning to study projective manifolds whose tangent bundles have positivity properties. More specifically we want to give a new proof of Mori's famous

2.1 Theorem

Let X be a projective manifold with ample tangent bundle. Then $X \simeq \mathbf{P}_n$.

Mori's approach is to detect the lines through a given point in X . His methods work even over any algebraically closed field. Since it is not at all clear whether the Fano manifold X carries even one single rational curve, a major point in his proof is the construction of rational curves (as already mentioned after (1.11)). This is done by his famous characteristic p -trick : first he creates rational curves in the analogous situation in characteristic p (with essential help of the Frobenius morphism), then he shows that these curves survive in characteristic 0. Instead of discussing this technique (which is very important in many circumstances) we want to give here another proof which works completely in complex algebraic geometry. The idea is to consider $\mathbf{P}(T_X)$. An important tool will be the classical and basic

2.2 Theorem of Kobayashi-Ochiai

Let X be a projective manifold of dimension n and index r . Then

- (1) $r \leq n + 1$
- (2) *If $r = n + 1$, then $X \simeq \mathbf{P}_n$*
- (3) *If $r = n$, then $X \simeq Q_n$, the n -dimensional quadric.*

The proof, whose essential ingredients are Kodaira vanishing and Riemann-Roch, can be found in [KO73].

We begin with the following

2.3 Observation

$\mathbf{P}(T_{\mathbf{P}_n})$ carries two different $\mathbf{P}_{n-1}$ -bundle structures.

In fact, one bundle structure is obvious and for the other consider the Euler sequence on $\mathbf{P}_n$:

$$0 \longrightarrow \mathcal{O}(-1) \longrightarrow \mathcal{O}^{n+1} \longrightarrow T(-1) \longrightarrow 0.$$

Taking $\mathbf{P}$ it follows that $\mathbf{P}(T_{\mathbf{P}_n})$ can be embedded into $\mathbf{P}(\mathcal{O}^{n+1}) \simeq \mathbf{P}_n \times \mathbf{P}_n$. Now the restriction of the two projections to $\mathbf{P}(T)$ give the two bundle structures we are looking for.

Our aim will be to reconstruct this situation.

From now on let X be a projective manifold of dimension n with ample tangent bundle T_X and let $\mathbf{P} = \mathbf{P}(T_X)$. Let $\pi : \mathbf{P} \longrightarrow X$ be the projection. Then $\mathbf{P}$ has a nice structure, described in the following

2.4 Observation

$\mathbf{P}$ is a Fano manifold with $\dim \mathbf{P} = 2n - 1$ and index n .

In fact, we have

$$-K_{\mathbf{P}} = \mathcal{O}_{\mathbf{P}}(n).$$

The ampleness of T_X is equivalent to that of $\mathcal{O}_{\mathbf{P}}(1)$, by definition.

The following remark makes life less complicated :

2.5 Proposition

$$b_2(X) = \rho(X) = 1.$$

Proof. Assume that $b_2(X) \geq 2$. Since X is Fano, there exists a Mori contraction $\phi : X \longrightarrow Y$ on X with $\dim Y \geq 1$. In case $\dim Y < n$, take a general fiber F . Then its normal bundle $\mathcal{N}$ is trivial; on the other hand it is a quotient of $T_X|_F$, contradiction. In case ϕ is birational, things are a little bit more complicated. Take a non-trivial fiber F . If F is given the appropriate fiber structure, then clearly the conormal sheaf N_F^* is generated by global sections. On the other hand $N_{red F}^*$ is a subsheaf of $\Omega_X^1|_{red F}$, hence a subsheaf of a negative vector bundle. Now it is an exercise to prove that these two statements are incompatible. Another argument, not using different cases, is as follows. In (4.4) below we show that if T_X is nef, then every effective divisor on X is nef. But the same arguments show that in our case even every effective divisor $\neq 0$ is ample. Now the cone theorem and the fact that X is Fano imply $\rho(X) = 1$. Compare [MS77].

Since X is a Fano manifold, we conclude

$$\text{Pic}(X) \simeq \mathbf{Z}.$$

We now start the proof of (2.1). Since $\mathbf{P}$ is Fano with $\rho = 2$ we find another extremal ray different from that one defining π by the cone theorem. Let $\rho : \mathbf{P} \rightarrow Z$ be the associated contraction. Note that by the contraction theorem $\pi|_F$ is finite for every fiber F of ρ and that $\rho|_G$ is finite for every fiber G of π . This fact, referred to as (*), will be used several times.

(2.6) First we assume $\dim Z < 2n - 1$. Then by (*) we have $\dim F \leq n$. Now specify F to be the general smooth fiber ρ . By adjunction we have

$$K_F = \mathcal{O}_{\mathbf{P}}(-n)|_F,$$

hence it follows from (2.2) that either

$$(2.6.1) \quad \dim F = n - 1, \text{ index}(F) = n, \text{ thus } F \simeq \mathbf{P}_{n-1}, \text{ or}$$

$$(2.6.2) \quad \dim F = n, \text{ index}(F) = n, \text{ thus } F \simeq Q_n.$$

Of course, (2.6.1) is the expected case while (2.6.2) has to be excluded. In both cases however, it is a priori clear that we have lots of rational curves in X !

First we assume (2.6.1). Let l be a line in F . We are going to show

$$(\pi(F) \cdot \pi(l)) = 1. \quad (+)$$

Granted (+) we conclude as follows. We obtain $\mathcal{O}_X(1) \simeq \mathcal{O}_X(\pi(F))$. On the other hand we have $-K_X = \mathcal{O}_X(r)$, hence

$$r = (-K_X \cdot \pi(l)) \geq n + 1,$$

the last inequality coming from the ampleness of T_X on $\pi(l)$ and Grothendieck's splitting theorem applied to the normalisation of $\pi(l)$. Thus we get $r \geq n + 1$ and $X \simeq \mathbf{P}_n$. Since the quadric does not have ample tangent bundle (compute the restriction to lines), we conclude. Let us sketch the proof of (+), for details we refer to [Pe91, p.316/17]. Assume that (+) does not hold. Let $F_z = \rho^{-1}(z)$, $z \in Z$. For a smooth point in $p \in \pi(l)$ set

$$T_p = \{z \in Z \mid F_z \cap \pi^{-1}(p) \neq \emptyset\} = \rho(\pi^{-1}(p)),$$

and

$$T_{pq} = T_p \cap T_q.$$

Note that T_p has codimension 1 in Z and T_{pq} has codimension 1 in T_p . Moreover we may assume $n \geq 3$, since the surface case is anyway clear by Kodaira-Enriques classification. Thus $\dim T_{p,q} \geq 1$. Fixing p we now choose $t_0 \in T_p$ general such that

$$(a) \quad \pi(l) \not\subset \pi(F_{t_0})$$

$$(b) \quad \pi(l) \cap \pi(F_{t_0}) \text{ contains a smooth point } q \text{ on } \pi(l) \text{ different from } p.$$

It is easy to verify the possibility of doing so, (a) follows immediately from the fact that $\rho|_{\mathbf{P}}(E|_{\pi(l)})$ is surjective, hence generically finite which in turn is a consequence of $\text{Pic}(X) = \mathbf{Z}$. We see that T_p is the closure of the union of the T_{pq} , where q varies over the smooth points in $\pi(l)$. Now choose a general curve C contained in some

T_{pq} . We let $C_p = G_p \cap \rho^{-1}(C)$, analogously C_q , and let $\tilde{C}$ be the fiber product of these two curves over C . Let $\tilde{L}$ be the closure of

$$\{(p', q', l_{p'q'})\},$$

where $l_{p'q'}$ is the line connecting p' and q' in $F_{\rho(p')} = F_{\rho(q')}$, $(p', q') \in \tilde{C}$. Then $\tilde{L}$ can be viewed as deformation (l_t) of our original line $l = l_0 \subset F$ with parameter space $\tilde{C}$. Hence $(\pi(l_t))$ is a deformation of $\pi(l)$ fixing the points p and q . By Mori's breaking lemma [Mo79], there exists a point $\infty \in \tilde{C}$ such that $\pi(l_\infty)$ splits into several (possibly not distinct) components. So does l_∞ . But this contradicts $(-K_{\mathbf{P}} \cdot l_\infty) = n$ and the divisibility of $-K_{\mathbf{P}}$. One word to Mori's breaking lemma. Consider a 1-dimensional family (C_t) through two points p, q . Then pass to the normalisation of the graph, which is a $\mathbf{P}_1$ -bundle if the family does not split. But then this ruled surface would have two exceptional curves which is not possible (see [Ha77, chap5, sect.2]). So (C_t) has to split.

This settles (2.6.1).

In order to exclude (2.6.2) look at rational curves $C \subset X$ with small $(-K_X \cdot C)$. By the breaking lemma, we can achieve $(-K_X \cdot C) \leq n + 1$, see [Mo79]. Then consider $Y = \mathbf{P}(T_X|C)$, which after normalisation has to split as $\mathcal{O}(2) \oplus \mathcal{O}(1)^{n-1}$. Now $\rho|Y$ is surjective and the comparison with the known contraction of the Fano manifold Y (namely projection to $\mathbf{P}_1$ and contraction of the exceptional divisor $\mathbf{P}(\mathcal{O}(1)^n)$) gives a contradiction.

It remains to treat the case

(2.7) ρ is birational.

T_X being ample, there is a uniquely determined real positive number ϵ such that $T_X(-\epsilon)$ is nef but not ample. In other words

$$\mathcal{O}_{\mathbf{P}(T_X(-\epsilon))}(1) = \mathcal{O}_{\mathbf{P}}(1) \otimes \pi^*(\mathcal{O}_X(\epsilon))$$

is nef but not ample. This shows that ϵ is in fact rational since the real divisor in question is dual to the extremal ray defining ρ . Now choose a large positive integer λ such that $\epsilon\lambda \in \mathbf{Z}$ and such that n divides both λ and $\epsilon\lambda$. Put

$$\mu = \frac{-\epsilon\lambda}{n}, \quad \nu = \frac{\lambda}{n} + 1.$$

Let

$$L = \mathcal{O}_{\mathbf{P}}(\nu) \otimes \pi^*(\mathcal{O}_X(\mu)).$$

Then L is a line bundle on $\mathbf{P}$ and we have

$$K_{\mathbf{P}} \otimes L^{\otimes n} = \mathcal{O}_{\mathbf{P}(T_X(-\epsilon))}(\lambda).$$

Thus

$$K_{\mathbf{P}} \otimes L^{\otimes n} \in \rho^*(\text{Pic}(Z)),$$

i.e. is a "good supporting divisor of ρ ". Obviously L is ample. Now we will make use of the following theorem due to Fujita [Fj87,2.5], [Fj91,1.5].

(2.8) Proposition

Let (X, L) be a polarised manifold, i.e. X is a projective manifold and L an ample line bundle. Let R be an extremal ray on X and $\phi : X \rightarrow Z$ be the associated contraction and assume that ϕ is given by the global sections of $(K_X \otimes L^{\otimes k})^{\otimes m}$ for some positive k . Assume that ϕ is birational. If $z \in Z$ is a point with $r = \dim \phi^{-1}(z) > 0$, then $((K_X \otimes L^{\otimes r}).R) \geq 0$. If equality holds and if F_0 denotes a fiber component of dimension r , then the normalisation of F_0 is $\mathbf{P}_r$ and the pull-back of L is $\mathcal{O}(1)$.

Some remarks to the proof of (2.8). Let $f : Y \rightarrow F_0$ be a desingularisation. Assuming that $((K_X \otimes L^r).R) < 0$, it follows that $-(K_X \otimes L^s)$ is ample on F_0 for all $0 \leq s \leq r$. Now Kawamata-Viehweg vanishing gives

$$H^i(f^*(K_X \otimes L^r)) = 0, i < r.$$

On the other hand Fujita proves by an inductive argument the vanishing of H^r , too. Hence

$$\chi(g^*(K_X \otimes L^s)) = 0 \text{ for } 0 \leq s \leq r.$$

Since $\chi(g^*(K_X \otimes L^s))$ is a polynomial in s of degree $\leq r$, it follows

$$\chi(g^*(K_X \otimes L^s)) = 0$$

for all s which contradicts $(c_1(L)^r.Y) > 0$ by the Riemann-Roch theorem (which gives that χ grows like s^r as s approaches ∞). In case of equality one has

$$H^r(Y, f^*(L^{-t})) = 0 \tag{+}$$

for $1 \leq t \leq r$. Thus $Y \simeq \mathbf{P}_r$ by [Fj89,2.2]. The way to see this last statement is as follows. Let $\pi : \hat{Y} \rightarrow Y$ be a desingularisation and put $M = \pi^* f^*(L)$. Letting $\chi(t) = \chi(M^{\otimes t})$ we get from Kawamata-Viehweg vanishing and (+) that

$$\chi(t) = 0 \text{ for } -1 \geq t \geq -r,$$

thus

$$\chi(t) = \frac{d}{t!}(t+1) \cdots (t+r).$$

Writing down Riemann-Roch and comparing the highest coefficients we conclude

$$d = M^r; K.M^{r-1} = -d(r+1),$$

where K is the canonical divisor of $\hat{Y}$. Now observe $h^0(K \otimes M^{\otimes(r+1)}) = d$. Taking $D \in |K + (r+1)M|$, it follows $M^{r-1}.D = 0$ which easily yields $d = 1$ and moreover $h^0(M) = h^0(M+D) = d$. Then one can compute Fujita's Δ -invariant :

$$\Delta(\hat{Y}, M) = r + M^r - h^0(M) = 0.$$

A basic result of Fujita [Fj89,1.1] tells us that the vanishing of Δ implies that M is the pull-back of a very ample line bundle by a birational map. Hence $H^0(M)$ gives a birational surjective map $\hat{Y} \rightarrow \mathbf{P}_r$ and it follows $Y \simeq \mathbf{P}_r$.

Now back to our situation (2.7). We can conclude by (2.8) that every non-trivial fiber F of ρ contains a component of dimension n , since we a priori know that $\dim F \leq n$. Pick a component F_0 of dimension n and let again $g : Y \rightarrow F_0$ be the normalisation. We have $Y \simeq \mathbf{P}_n$. Since $K_{\mathbf{P}} \otimes L^n|_{F_0}$ is trivial, we obtain

$$g^*(\mathcal{O}_{\mathbf{P}}(1)) \simeq \mathcal{O}(1).$$

Putting this into the definition of L and letting d be the degree of $\pi|_{F_0}$ it follows $ed = 1$. Now a dimension count shows that the exceptional locus A of ρ has codimension 1 (either use [Wi91] or consider directly $T_X|\pi(l), l \subset F_0$ a line). Note that we do not use char p if we apply Wisniewski's result; in fact char p is used in [Wi91] only to show that there is a rational curve in F_0 through the general point of F_0 (in our notations), which is obvious here. Now let $C \subset X$ be a rational curve of minimal degree in X and denote by $F : \tilde{C} \rightarrow C$ its normalisation. Then

$$f^*(T_X) = \mathcal{O}(2) \oplus \mathcal{O}(1)^{n-1}.$$

Let $A_C := A \cap \pi^{-1}(C)$ and $\mathbf{P}_C = \pi^{-1}(C)$. Then $\psi = \rho|_{\mathbf{P}_C}$ is a non-finite birational map. Let $\tilde{\mathbf{P}}_C \rightarrow \mathbf{P}_C$ be the normalisation and $\tilde{\psi} : \tilde{\mathbf{P}}_C \rightarrow \psi(\tilde{\mathbf{P}}_C)$ be the induced map. Let $\hat{\psi} : \tilde{\mathbf{P}}_C \rightarrow W$ be the Stein factorisation. A general lemma on modifications of projective bundles over $\mathbf{P}_1$ (lemma 6 in [Pe91]) shows easily that $\hat{\psi}$ has to contract exactly $\mathbf{P}(\mathcal{O}(1)^{n-1}) \subset \tilde{\mathbf{P}}_C$ and thus

$$\tilde{A}_C = \mathbf{P}(\mathcal{O}(1)^{n-1}).$$

So $A \cap \pi^{-1}(x) = A_C \cap \pi^{-1}(x) \simeq \mathbf{P}_{n-2}$ for general $x \in C$. By the existence of lines in Fano 3-folds with $\rho = 1$, the case $n = 3$ is obvious, so we may assume $n \geq 4$. Hence $A \cap \pi^{-1}(x)$ is a linear hypersurface in $\mathbf{P}_{n-1}, (*)$. This holds now for every $x \in X$ (note that A does not contain $\pi^{-1}(x)$). Hence A is a $\mathbf{P}_{n-1}$ -bundle over X , in particular A is smooth. Concerning the other projection, the general fiber of F of $\rho|_A$ is now clearly $\mathbf{P}_n$. From the adjunction formula we deduce $N_A|_F = \mathcal{O}(-1)$, so generically ρ is the blow-up of Z along $\rho(A)$ (and Z is smooth generically along $\rho(A)$). The same computation gives

$$(\mathcal{O}_{\mathbf{P}}(1).l) = 1 \tag{**}$$

for a line $l \subset F$. Now write

$$\mathcal{O}_{\mathbf{P}}(A) = \mathcal{O}_{\mathbf{P}}(\nu) \otimes \pi^*(\mathcal{O}_X)(\mu).$$

Then $(*)$ gives $\nu = 1$ and $(**)$ yields $\mu = -2d$, where $d = \deg(\pi|_l)$. Taking π_* we have $H^0(T_X(-2d)) \neq 0$, so there is a non-zero section $s \in H^0(T_X)$ which is tangential to all minimal rational curves, which is clearly impossible.

Thus the proof of (2.1) is completed .

We should remark here that we could have also used generalisations of Fujita's results, due to Andreatta-Wisniewski [AW93], to conclude (see Theorem 4.1 in [AW93].) Theorem 2.1 can be generalised in the following way

(2.9) Theorem

Let X be a compact n -dimensional manifold and E an ample vector bundle on X with $\text{rank} \geq n + 1$ and $\det E = -K_X$. Then $X \simeq \mathbf{P}_n$ and $E \simeq \mathcal{O}(1)^{\oplus n+1}$.

If E has rank n , then either $X \simeq \mathbf{P}_n$ and $E = T_X$ resp. $E = \mathcal{O}(2) \oplus \mathcal{O}(1)^{\oplus n-1}$ or $X \simeq Q_n$ and $E = \mathcal{O}(1)^{\oplus n}$.

This was conjectured by Mukai and proved independently in [Fj90], [Pe90,91] and [YZ90]. One can still go a step further and consider the case of rank $n - 1$, this was solved in [PSW92]. (2.9) can be viewed as a kind of "vector bundle adjunction", but it is not the right place to go into details here.

I want to finish this section with some related theorems and problems. First we have

(2.10) Theorem

Let X be a projective n -dimensional manifold such that $\bigwedge^2 T_X$ is ample. Then either $X \simeq \mathbf{P}_n$ or $X \simeq Q_n$.

This is proved in [CS93].

Using (a refined version of) Mori's theorem (2.1) Lazarsfeld [La84] was able to solve an old conjecture of Remmert and van de Ven :

(2.11) Theorem

Let X be a compact n -dimensional manifold and $f : \mathbf{P}_n \rightarrow X$ be a surjective holomorphic map. Then $X \simeq \mathbf{P}_n$.

If we have instead a surjective map $f : Q_n \rightarrow X$, then X is either the projective space or the quadric. This was proved by Cho-Sato [CS89] and Paranjapé-Srinivas [PS89].

(2.12) Conjectures and Problems

(1) Let X be a projective manifold of dimension n such that $\bigwedge^q T_X$ is ample for some $3 \leq q \leq n - 1$. Give an estimate for the index r in terms of n and q . Of course the cases $q = 1, 2$ are settled by (2.1) and (2.10) and $q = n$ is just to say that X is Fano.

(2) Let X and Y be Fano manifolds of index r resp. r' , assume $b_2(Y) = 1$. Let $f : Y \rightarrow X$ be a surjective holomorphic map. Is then $r' \leq r$? This would generalise Lazarsfeld's theorem and its quadric analogue.

(3) Let X be a projective manifold of dimension n with $\text{Pic}(X) = \mathbf{Z}$ and E be an ample vector bundle on X of rank m with $\det E = -K_X$. Is then $\text{index}(X) \geq m$?

3. Kähler manifolds with semi-positive curvature

In this section we discuss Siu-Yau's analogue to Mori's theorem and the "semi-positive" generalisation due to Mok.

3.1 Theorem (Siu-Yau [SY80])

Let X be a compact Kähler manifold with positive holomorphic bisectional curvature. Then $X \simeq \mathbf{P}_n$.

To say that a Kähler manifold X has positive holomorphic bisectional curvature means that there exists a **Kähler** metric whose curvature is positive in Griffiths' sense (1.13). This is a priori stronger than to say that there exists a Hermitian metric on X whose curvature is positive which in turn is a stronger condition than ampleness of T_X . Of course, a posteriori all three conditions are equivalent. We will however see that in the semi-positive case the third condition (T_X is nef) is much weaker than the first. In summary Siu-Yau's theorem is - apart from the fact that it works only over $\mathbf{C}$ - weaker than Mori's theorem.

(3.2) We would like to discuss the methods used by Siu and Yau. By a theorem of Goldberg-Kobayashi [GK67] one knows that $b_2(X) = 1$ (in fact one proves that every harmonic $(1,1)$ -form on (X, ω) is a multiple of the Kähler form ω , using the curvature condition). Moreover X is a Fano manifold, hence simply connected. Then $\pi_2(X)$ is canonically isomorphic to $H_2(X, \mathbf{Z}) \simeq \mathbf{Z}$. Now fix a generator h of $H_2(X, \mathbf{Z})$ with the property that

$$(c_1(\mathcal{L}).h) = 1$$

where $\mathcal{L}$ is the ample generator of $\text{Pic}(X) \simeq H^2(X, \mathbf{Z}) \simeq \mathbf{Z}$. By definition, h , considered as an element of $\pi_2(X)$, is represented by a C^∞ -map $f : S^2 \rightarrow X$. The aim is to show that f is homotopic to a holomorphic map $f_0 : \mathbf{P}_1 \rightarrow X$. Given this it follows from the ampleness of T_X on the rational curve C defined by f_0 and Grothendieck's splitting theorem that $(-K_X.C) \geq n + 1$ (with $n = \dim X$). On the other hand we have $-K_X = L^r$ with r the index of X . Since $(c_1(L).C) = 1$, Kobayashi-Ochiai's theorem gives first $r = n + 1$ and thus $X \simeq \mathbf{P}_n$.

Of course the difficulty is to prove the existence of f_0 . For this purpose the notion of the $(\bar{\partial})$ -energy of a C^∞ -map $f : X \rightarrow Y$ of compact Kähler manifolds is important. If $(h_{\alpha\beta})$ is the Kähler metric on X resp. (g_{ij}) that one on Y (in local coordinates), then we define the pointwise energy by

$$|\bar{\partial}f|^2 = \sum_{i,j} g^{ij} f_i^\alpha \bar{f}_j^\beta h_{\alpha\beta},$$

where $(g^{ij}) = (g_{ij})^{-1}$, $f_i^\alpha = \frac{\partial f^\alpha}{\partial w_i}$. Moreover we set

$$E(f) = \int_X |\bar{\partial}f|^2.$$

Finally one defines $E([f])$ as the infimum of sums of energies of maps f_i with $\sum f_i$ homotopic to f . Now coming back to the C^∞ -map $f : S^2 = \mathbf{P}_1 \rightarrow X$ representing h , Siu and Yau first prove that f is homotopic to $\sum f_i$ with $f_i : S^2 \rightarrow X$ energy-minimizing such that $E([f]) = \sum_{i=1}^k E(f_i)$. This is a rather general statement for compact Riemannian manifolds X . But now, thanks to the positivity of the bisectional curvature of the Kähler metric, each f_i is either holomorphic or antiholomorphic. This is the first main point and proves in particular the existence of rational curves in X . Of course we want $k = 0$ in the above sum, so the case $k > 0$ has to be excluded which is the second main point. If $k > 0$, then say f_0 is holomorphic and f_1 is antiholomorphic. Then Siu and Yau construct a sum $\sum g_j$ homotopic to f with $\sum E(g_j) < \sum E(f_i)$ which is a contradiction. Here the positivity of T_X comes a second time into the game : namely it is shown (this is algebraic geometry) that for a general choice of (x, v) with $x \in X$ and $v \in T_x X$ there exists a rational curve $C \subset X$ whose normalisation map is homotopic to f_0 such that x is a smooth point of C and $v \in T_x C$. To prove this it is actually sufficient to know the ampleness of T_X on C .

Concerning the semi-positive case Mok proved in [Mk88]

3.3 Theorem (Mok)

Let X be a compact Kähler manifold of semi-positive holomorphic bisectional curvature. Then up to finite étale cover X is of the form

$$X_1 \times \cdots \times X_r$$

with X_i either a torus or a compact irreducible hermitian symmetric manifold.

The relevant definition is

3.4 Definition A hermitian manifold (X, h) is called a hermitian symmetric manifold if for every $x \in X$ there exists a holomorphic isometry σ of (X, h) with $\sigma^2 = id$ such that x is an isolated fix point of σ . (X, h) is said to be irreducible if it is not a product of hermitian manifolds (X_i, h_i) .

Most basic example are projective spaces, quadrics and Grassmannians. As general references we recommend [Bo53,54], [He78],[Mk89]. Note that all irreducible hermitian symmetric manifolds have $b_2 = 1$. They are special rational homogeneous manifolds, cp. 4.3; see [He78] for a complete classification.

We give a very rough sketch of the proof. First, using a splitting theorem of Howard-Smyth-Wu - a much more general situation will be discussed in sect. 4 and 5 - one can assume that $b_2(X) = 1$ and that the Ricci curvature is positive at one point. Then X is a Fano manifold and it has to be proved that X is hermitian symmetric. Using the evolution equation of Hamilton, Mok shows that one may even assume the Ricci curvature of the given metric to be positive everywhere. Now there is a large algebraic part in the proof. Namely, one considers rational curves of minimal degree in X . These exist in every Fano manifold by Mori [Mo79], however the proof requires characteristic p-methods as explained in sect.2. So Mok's proof is not

purely complex-analytic. Since T_X is nef on any curve, the rational curves have lots of deformations, e.g. the family fills up X . Now Mok considers the collection of all tangent vectors at smooth points to any minimal rational curve. This gives a variety in $\mathbf{P}(\Omega_X^1)$ whose closure is called S . The algebraic property of S which is important can be described as follows. Take a general minimal rational curve C and let f be its normalisation. Then $f^*(T_X) = \mathcal{O}(a_1) \oplus \dots \oplus \mathcal{O}(a_{n-l}) \oplus \mathcal{O}^l$ where $a_i > 0$ for $1 \leq i \leq n-l$. Assume now $X \neq \mathbf{P}^n$. Then S has codimension l except possibly for the (a posteriori non-existing) case $l = 0$; in this case S has codimension 1. Using this information Mok now proves that S is invariant under the holonomy action. To explain what this means, fix a point $x \in X$. Let M be the set of closed paths through x . By parallel transport along paths in M , we obtain a map $M \rightarrow O(T_x X)$ into the orthogonal group of the tangent space $T_x X$ at x (with respect to h). The image of this map is called the holonomy group of (X, h) , denoted H ; it does not depend on the choice of x . Now, h being a Kähler metric, H lies in fact in the unitary group $U(n)$. As a general reference on holonomy stuff we recommend [KN63] and [Be83]. The invariance of S implies in particular that the holonomy action is not transitive on the unit sphere of $T_x X$. A theorem of Berger [Be55] says that then X has to be hermitian symmetric:

3.5 Theorem

Let X be an irreducible simply connected compact Kähler manifold. Then either the holonomy group acts transitively on every real tangent space $T_x X$, $x \in X$ or X is an hermitian symmetric manifold of rank ≥ 2 , i.e. a hermitian symmetric manifold different from projective space.

4. Manifolds with nef tangent bundles

Mok's theorem 3.3 is the semi-positive generalisation of Siu-Yau's result. We are now aiming for the algebraic case : the nef generalisation of Mori's theorem. Thus we want to investigate the structure of compact Kähler or projective manifolds whose tangent bundles are nef. The results discussed here are due to [CP91], [DPS93,94a,94b], in case of dimension 3 see also [Zh90].

4.1 Examples

- (1) Every homogeneous compact Kähler manifold has nef tangent bundle. In fact, the homogeneity condition is equivalent to saying that T_X is generated by global sections. In particular, all hermitian symmetric manifolds, flag manifolds and tori have nef tangent bundles.
- (2) Let $f : \tilde{X} \rightarrow X$ be a finite étale cover. Then $T_{\tilde{X}}$ is nef if and only if T_X is nef. This follows from (1.16) since $T_{\tilde{X}} = f^*(T_X)$.
- (3) Let $X = \mathbf{P}(T_{\mathbf{P}^n})$. It is an easy exercise to show that T_X is generated by global sections, so T_X is nef. On the other hand there is no Kähler metric on X whose holomorphic bisectional curvature is semi-positive. In fact, otherwise X would be a product by Mok's theorem which is clearly false. Thus the class of manifolds

with nef tangent bundles is strictly larger than the class of semi-positively curved compact Kähler manifolds.

(4) Let $X = \mathbf{P}(E) \rightarrow Y$ be the projectivisation of a vector bundle E of rank r over a torus Y . Then T_X is nef if and only if $E \otimes (\det E)^{\frac{1}{r}}$ is numerically flat. This needs some explanation, since $\det E$ is not necessarily divisible by r in $\text{Pic}(X)$. However if E is a given vector bundle on X and L a line bundle and t a real number, then we say that $E \otimes L^t$ is nef, if the numerical equivalence class $[\mathcal{O}_{\mathbf{P}(E)}(1)] + t[\pi^*(L)]$ is contained in the closure of the ample cone of $\mathbf{P}(E)$. Thus only the sentence " $E \otimes L^t$ is nef " is defined without giving a concrete meaning to $E \otimes L^t$.

Now, coming back to our specific example, T_X is nef if and only if $T_{X/Y}$ is nef by virtue of the exact sequence

$$0 \rightarrow T_{X/Y} \rightarrow T_X \rightarrow \pi^*(T_Y) \rightarrow 0$$

and the triviality of T_Y . The relative Euler sequence

$$0 \rightarrow \mathcal{O}_X \rightarrow \pi^*(E^*) \otimes \mathcal{O}_{\mathbf{P}(E)}(1) \rightarrow T_{X/Y} \rightarrow 0$$

proves that $T_{X/Y}$ is nef if and only if $\pi^*(E^*) \otimes \mathcal{O}_{\mathbf{P}(E)}(1)$ is nef. Now it is not difficult to translate this into the numerical flatness of $E \otimes (\det E)^{\frac{1}{r}}$. Observe that $c_1(E \otimes (\det E)^{\frac{1}{r}}) = 0$.

4.2 The structure of compact homogeneous Kähler manifolds

A compact homogeneous Kähler manifold is a compact homogeneous manifold which is at the same time Kähler, so there is a priori no connection between the group action and the metric. It can however be shown that the above notion is the same as to say that the group of isometries (with respect to the Kähler metric) acts transitively on X . A theorem of Borel and Remmert [BR62] states that any compact homogeneous Kähler manifold X is a product $T \times Y$ with T a torus and Y a simply connected homogeneous manifold (cp. 4.16, 4.17). A result of Goto [Go54] says that such a simply connected homogeneous Kähler manifold is in fact rational homogeneous, i.e. Y is a rational manifold. Now the structure of those varieties is completely described by the following facts also due to Goto.

(1) Let G be a connected semi-simple complex Lie group and $H \subset G$ a parabolic subgroup. Then G/H is rational homogeneous.

(2) Let X be rational homogeneous. Then $\text{Aut}(X)$ is semi-simple and if G is any connected semi-simple complex Lie group acting transitively on X , then the isotropy group $\text{Iso}_G(x)$ is a parabolic subgroup of G for any x , with $X \simeq G/\text{Iso}_G(x)$.

Actually if X is a rational homogeneous manifold, then X is irreducible, i.e. not a product, if and only if $\text{Aut}(X)$ is simple.

4.3 Remarks

(1) Not every Kähler manifold with nef tangent bundle is homogeneous. In fact, in [DPS94a] the following example is given. Let $\Gamma = \mathbf{C}/(\mathbf{Z} + \mathbf{Z}\tau)$, with $\text{Im}\tau > 0$, be an elliptic curve. Let $G = \{1, g_1, g_2, g_1g_2\}$ be the group given by

$$g_1(z_1, z_2, z_3) = (z_1 + \frac{1}{2}, -z_1, -z_2),$$

$$g_2(z_1, z_2, z_3) = (-z_1, z_2 + \frac{1}{2}, -z_3 + \frac{1}{2}),$$

$$g_1 g_2(-z_1 + \frac{1}{2}, -z_2 + \frac{1}{2}, z_3 + \frac{1}{2}).$$

Then G acts freely on $\Gamma \times \Gamma \times \Gamma$, hence the quotient X is smooth. It is easy to check that no vector field on Γ^3 is G -invariant, hence T_X has no section. Moreover we have $q(X) = 0$.

(2) We conclude that the classification of manifolds with nef tangent bundles should be only up to finite étale cover. Therefore $\tilde{q}(X)$ (defined in the notations) plays a more important role than the irregularity $q(X)$ itself; we note that by (4.15) below we always have $\tilde{q}(X) \leq \dim X$.

(3) Of course one can also look at compact manifolds with nef tangent bundles which are not Kähler. We ignore this aspect completely and refer to [DPS94a].

In view of (4.3.1) one might suspect that every Kähler manifold with T_X nef has a finite étale cover which is homogeneous. But even this is false: let Γ and E be as in (1.3). Let $X = \mathbf{P}(E)$. It is easily checked that the relative tangent bundle of $X \rightarrow \Gamma$ is nef, hence T_X is nef. If there would be a finite étale cover $\tilde{X}$ of X which is homogeneous, then in particular $-K_{\tilde{X}}$ would have a metric of semi-positive curvature. Thus $-K_X$ would have the same property contradicting (1.3). Note that by virtue of the action

$$(x, z_1, z_2) \longrightarrow (x + a, z_1 + b, z_2)$$

the automorphism group $\text{Aut}(X)$ acts with an open orbit (one says that X is almost homogeneous) and there is one closed orbit, namely the unique curve C with $C^2 = 0$, given by $\{z_2 = 0\}$.

We now study general properties of manifolds with nef tangent bundle. We begin with the following easy

4.4 Proposition

Let X be a compact Kähler manifold with T_X nef. Then every effective divisor on X is nef.

Proof. We give a proof in case X is projective. Let D be an effective divisor, we may assume that D is irreducible and reduced. Consider the exact sequence

$$0 \longrightarrow \mathcal{N}_{D|X}^* \longrightarrow \Omega_X^1|D \longrightarrow \Omega_D^1 \longrightarrow 0.$$

If $C \not\subset \text{Sing}(D)$, then $\mathcal{N}_D^*|C \rightarrow \Omega_X^1|C$ is still non-zero (since the derivatives of the local equation f of D do not all vanish along C), hence $\mathcal{N}^*|C$ is a subsheaf of a vector bundle whose dual is nef, thus $\mathcal{N}_D|C$ is nef. This proves $D \cdot C \geq 0$ for all $C \not\subset \text{Sing}(D)$. In case $C \subset \text{Sing}(D)$ one has to take into account the vanishing order of f along C and obtains at least a morphism $\mathcal{N}_D^* \rightarrow S^m(\Omega_X^1)$ whose restriction to C is not zero for a suitable m . Then the conclusion is as above.

4.5 Proposition

Let X be compact Kähler, $\dim X = n$, with T_X nef. Let $\mathcal{L}$ be a nef line bundle on X . If $c_1(\mathcal{L})^n \neq 0$, then $\mathcal{L}$ is ample. In particular, if $c_1(X)^n \neq 0$, then X is Fano.

Proof. Fix a Kähler metric ω on X . For any positive ϵ choose a metric h_ϵ on $\mathcal{L}$ with curvature $\Theta_\epsilon \geq -\epsilon\omega$. Let $V = \bigcup_{k \geq 2} V_k$, where V_k is the set of points where Θ_ϵ has exactly k negative and $n - k$ positive eigenvalues. By elementary calculation in the eigenvalues of the curvature one has the inequality on V_q

$$0 \leq \frac{1}{n!}(-1)^q \Theta_\epsilon^n \leq \frac{1}{q!(n-q)!} \epsilon^q \omega^q \wedge (\Theta_\epsilon + \epsilon\omega)^{n-q}.$$

Integrating we see that

$$\| \int_{V_q} \Theta_\epsilon^n \| = O(\epsilon^2), q \geq 2.$$

By virtue of Demailly's holomorphic Morse inequalities [De85]

$$h^q(X, \mathcal{L}^k) \leq \frac{k^n}{n!} \int_{V_q} (-1)^q \Theta_\epsilon^n + o(k^n).$$

We conclude from Riemann-Roch and our assumption $c_1(\mathcal{L})^n > 0$ that $h^0(\mathcal{L}^k) - h^1(\mathcal{L}^k)$ grows as k^n , so does $h^0(\mathcal{L}^k)$. It follows that $\kappa(\mathcal{L}) = n$, in particular X is Moishezon. Another way to conclude the Moishezon property is as follows. By Kawamata-Viehweg's vanishing theorem resp. its generalisation due to Enoki [En93] in the non-algebraic case, we obtain

$$H^q(\mathcal{L}^k \otimes K_X) = 0$$

for all $k > 0, q > 0$. Then by Riemann-Roch we obtain

$$h^0(\mathcal{L}^k \otimes K_X) \sim k^n.$$

X being Moishezon we conclude either by Moishezon's general theorem (Moishezon plus Kähler is projective) or by (4.6) that X is actually projective. In order to show that $\mathcal{L}$ is ample we fix a very ample line bundle $\mathcal{H}$ on X and a general smooth member $H \in |\mathcal{H}|$. Taking cohomology of the exact sequence

$$0 \longrightarrow \mathcal{L}^k \otimes \mathcal{H}^* \longrightarrow \mathcal{L}^k \longrightarrow \mathcal{L}^k|_H \longrightarrow 0$$

we see from $\kappa(\mathcal{L}) = n$ and $\kappa(\mathcal{L}|_H) \leq n - 1$ that

$$H^0(\mathcal{L}^k \otimes \mathcal{H}^*) \neq 0$$

for k sufficiently large. Hence we can write $\mathcal{L}^k = \mathcal{H} \otimes \mathcal{O}(D)$ with D effective. Since D is nef by (4.4), $\mathcal{L}$ is ample.

4.6 Proposition

Every Moishezon manifold with nef tangent bundle is projective.

We give a very rough idea of the proof and refer to [DPS94a] for details. X being Moishezon we can find a line bundle $\mathcal{L}$ on X with $\kappa(\mathcal{L}) = n = \dim X$. By (4.4) $\mathcal{L}$ is nef, hence also big. Since $-K_X$ is nef we can apply the base point free theorem to $\mathcal{L}$ and hence obtain a birational map to a projective variety. Let E be its exceptional set. Its conormal sheaf has the tendency to be positive along the fibers; on the other it has a non-zero map to some $S^m(\Omega_X^1|_E)$ by power series expansion of the equations of E . Then we obtain a contradiction, T_X being nef. This argument proves also

4.7 Proposition

A projective manifold X with T_X nef does not admit a birational holomorphic not biholomorphic map to any compact complex space.

Certainly this is true also in the Kähler case.

4.8 Proposition

Let X be compact Kähler, $\dim X = n$ with T_X nef. If $c_1(X)^n = 0$, then there exists a holomorphic p -form for some odd integer p . Moreover $\hat{q}(X) > 0$, i.e. X has a holomorphic 1-form after finite étale cover.

We indicate two proofs, an analytic one and an algebraic one. The analytic proof is based on

4.9 Theorem

Let E be a holomorphic vector bundle on the compact Kähler manifold X . Then E is numerically flat (i.e. E and E^ are nef) if and only if there exists a filtration*

$$0 = E_0 \subset E_1 \subset \dots \subset E_p = E$$

by subbundles such that the quotients E_i/E_{i+1} are hermitian flat. In particular all Chern classes $c_i(E)$ vanish for a numerically flat vector bundle.

4.10 Digression on hermitian flat bundles

We say that a vector bundle E is hermitian flat if it is defined by a representation $\rho : \pi_1(X) \rightarrow U(r)$, where r is the rank of E . The bundle E is constructed from ρ as follows. Let $\tilde{X}$ be the universal cover of X . Then E is the quotient of $\tilde{X} \times \mathbb{C}^r$ by the following action of π_1 :

$$(x, v) \longrightarrow (\gamma(x), \rho(\gamma)(v)),$$

where $\gamma \in \pi_1$ and $\gamma(x)$ indicates the action of π_1 as group of decktransformations on $\tilde{X}$.

The following three statements are equivalent.

- (1) E is hermitian flat
- (2) E carries a metric whose canonical connection D is flat : $\Theta = 0$
- (3) E is given by constant unitary matrices as transition functions.

A proof of this elementary fact can be found in [Ko87]. Note that given a bundle with flat connection we get the associated representation by parallel displacement.

Proof of 4.9. Of course only one direction is difficult, namely the existence of a filtration for a numerically flat bundle. There is a general philosophy behind the proof : we will show that E is ω -semi-stable but possibly not stable. The defect of E being stable will give the first part of the filtration . First note that, since $\det E$ is numerically flat, $c_1(E) = 0$. Of course we want to construct the filtration by induction. Fix a Kähler form ω on X . In a first step we consider an arbitrary subsheaf $\mathcal{F} \subset \mathcal{O}(E)$ of rank p . Consider $\det \mathcal{F} = (\bigwedge^p \mathcal{F})^{**} \subset \bigwedge^p E$. This gives a bundle morphism

$$f : \bigwedge^p E^* \longrightarrow (\det \mathcal{F})^*.$$

From this one concludes that

$$c_1(\mathcal{F}).\omega^{n-1} \leq 0 \quad (*)$$

This would be clear if f is surjective which is not necessarily the case. The technical difficulty raised by this fact is omitted (we do not claim that $\det \mathcal{F}^*$ is nef!). Note that $(*)$ is just the same as to say that E is ω -semi-stable (by definition). In case of equality in $(*)$ one can show that $c_1(\mathcal{F}) = 0$ and consequently $\det \mathcal{F}$ is hermitian flat. In this case the map $\det \mathcal{F} \longrightarrow \bigwedge^p E$ has no zeros, because otherwise $\bigwedge^p E \otimes \det \mathcal{F}^*$ would have a section with zeros while its dual is nef, contradicting (1.16).

In the second step let $\mathcal{F}$ be a subsheaf of $\mathcal{O}(E)$ of minimal positive rank such that $c_1(\mathcal{F}).\omega^{n-1} = 0$. Of course, such a sheaf exists since we do not require $rk \mathcal{F} < rk E$. We may assume that $\mathcal{F}$ is reflexive by possibly substituting $\mathcal{F}$ by $\mathcal{F}^{**}$. Hence $\det \mathcal{F}$ is a subbundle by the first step. A simple algebraic argument shows that then $\mathcal{F}$ itself is locally free and actually a subbundle of E . It follows that $\mathcal{F}^*$ is nef as a quotient of E^* . Since $\det \mathcal{F}$ is nef by (a), we deduce that $\mathcal{F}$ is numerically flat.

Hence $\mathcal{F}$ is numerically flat. From the Chern class inequality (1.17) it follows that

$$c_2(\mathcal{F}).\omega^{n-2} = 0 \quad (**).$$

By the minimality assumption $\mathcal{F}$ cannot have a proper subsheaf $\mathcal{H}$ with

$$c_1(\mathcal{H}).\omega^{n-1} = 0,$$

which is to say (by virtue of (a)) that $\mathcal{F}$ is ω -stable. By a deep theorem of Uhlenbeck-Yau [UY86], $\mathcal{F}$ carries a Hermite-Einstein metric; see also [Ko87]. A flatness criterion of Kobayashi and Lübke [Ko87, 4.11] says that Hermite-Einstein bundles satisfying $(**)$ are hermitian flat. So $\mathcal{F}$ is hermitian flat. Then let $E_1 = \mathcal{F}$ and proceed inductively.

4.11 We indicate the analytic proof of (4.8). Our condition $c_1(X)^n = 0$ forces by (1.18) all Chern polynomials $P(c_1, \dots, c_n)$ of degree n to vanish. Since the Riemann-Roch theorem

$$\chi(X, \mathcal{O}_X) = (ch(T_X).Td(X))_{2n}$$

expresses the holomorphic Euler characteristic as a sum of such Chern polynomials, we get

$$\chi(X, \mathcal{O}_X) = 0.$$

Since $h^0(X, \mathcal{O}_X) = 1$, we find an odd number p such that $h^0(X, \Omega_X^p) = h^p(X, \mathcal{O}_X) > 0$. It remains to show, and this is the hard part, that after finite étale cover we can achieve $p = 1$.

In order to construct a 1-form we fix a non-zero p -form u which gives rise to a contraction morphism

$$C : \bigwedge^{p-1} T_X \longrightarrow \Omega_X^1, C(v) = v \lrcorner u.$$

By definition, $C(v_1 \wedge \dots \wedge v_{p-1})(v_p) = u(v_1 \wedge \dots \wedge v_p)$ for vector fields v_i . If r denotes the generic rank of C , we can form $\bigwedge^r C$, which can be considered as a section of

$$\left(\bigwedge^r \left(\bigwedge^{p-1} T_X \right) \otimes \bigwedge^r T_X \right)^*.$$

The dual of the last bundle being nef, we conclude by (1.16) that $\bigwedge^r C$ has no zeros which is to say that C is a vector bundle homomorphism. Thus $E = \text{Im} C$ is a subbundle of Ω_X^1 , and a quotient bundle of $\bigwedge^{p-1} T_X$. Thus E is numerically flat. By Theorem 4.9 (applied to E^*) we find a subbundle $F \subset E$ which is hermitian flat, thus defined by a representation $\rho : \pi_1(X) \longrightarrow U(q)$, with $q = rk F$. We would like to show that F becomes trivial after a finite étale cover, which would be the case if $\Im \rho$ is finite. Granted this we obtain the 1-form we have been looking for. In order to find the étale cover, we need some information on the fundamental group $\pi_1(X)$. The relevant fact is that π_1 is of subexponential growth. This will be proved in (5.3) for all manifolds whose anticanonical bundle $-K_X$ is nef, a far more general situation than we are studying here. Now let $G = \rho(\pi_1)$, G is then also of subexponential growth. On the other hand G is a subgroup of a linear algebraic group, namely $U(q)$. Thus a purely group-theoretic theorem of Tits [Ti72] can be applied which asserts that either G contains a solvable subgroup of finite index or G contains a non-abelian free subgroup. Since non-abelian free groups are not of subexponential growth, G must contain a solvable subgroup of finite index. After a finite étale cover we may assume that G itself is solvable. Hence there exists a series

$$G = G_0 \supset G_1 \supset \dots \supset G_N = \{e\},$$

with $G_{i+1} = [G_i, G_i]$. Of course we may assume that G is infinite. Then let i be the smallest number such that G_i/G_{i+1} is infinite. Since G_i has finite index in G , $H_i = \rho^{-1}(G_i)$ has finite index in π_1 . Thus we can find a finite étale cover $f : \tilde{X} \longrightarrow X$ with $\pi_1(\tilde{X}) = H_i$. Now consider the induced surjective map

$$\bar{\rho} : H_i/[H_i, H_i] = H_1(\tilde{X}, \mathbf{Z}) \longrightarrow G_i/[G_i, G_i] = G_i/G_{i+1}.$$

By our choice of i and the surjectivity of $\bar{\rho}$ we conclude that $H_1(\tilde{X}, \mathbf{Z})$ is infinite. Thus $H^1(\tilde{X}, \mathbf{C}) \neq 0$ and from Hodge decomposition we get $q(\tilde{X}) \neq 0$.

4.12 Corollary

Let X be Kähler with T_X nef. Then $\tilde{q}(X) > 0$ if and only if $\pi_1(X)$ is infinite.

In fact, only one direction is not obvious. By (1.2(3)) we have $c_1(X)^n \geq 0$, where $n = \dim X$. If $c_1(X)^n > 0$ we have already seen that X is Fano, hence simply connected, hence we must have $c_1(X)^n = 0$, so (4.8) can be applied to conclude.

4.13 We shall next discuss an algebraic approach to prove (4.8). Of course we then have to assume X to be projective. We start with

4.13.1 Proposition

Let X be a complex manifold, Y a reduced locally irreducible complex space and $\varphi : X \rightarrow Y$ a surjective holomorphic map. Assume $\dim Y < \dim X$ and that T_X is nef on every compact curve contracted by φ . Assume for a fixed point $y_0 \in Y$ that every component of $\varphi^{-1}(y_0)$ of maximal dimension is compact. Let F be such a component. Then the following assertions hold.

- (1) $\text{red}F$ is smooth
- (2) $N_{\text{red}F|X}^*$ is topologically trivial up to a torsion line bundle
- (3) $N_{\text{red}F|X}^*$ is numerically flat.

Proof. We treat only a special case in order to see what is going on. For the technical details of the general case see [DPS94a, 5.1]. Let $n = \dim X$ and assume $\dim F = n - 1$. Let $F' = \varphi^{-1}(y_0)$. Then F' carries a natural complex structure as follows. Let I be the maximal ideal sheaf of the reduced point $\{y_0\}$ in Y . Let

$$J = \text{Im}(\varphi^*(I) \rightarrow \mathcal{O}_X).$$

Then J is the defining ideal for F' . Of course F' can have nilpotent elements. Let k be the maximal integer such that the infinitesimal neighborhood F_k of F is contained in F' . For simplicity assume $k = 1$. Thus the canonical map

$$h : N_{F'|X}^*|_F \rightarrow N_{F|X}^*$$

is non-zero. Since $N_{F'|X}^*$ is globally generated by definition of the fiber structure, we obtain many sections $h(s), s \in H^0(N_{F'|X}^*|_F)$.

Now in the canonical exact sequence

$$N_F^* \xrightarrow{g} \Omega_X^1|_F \rightarrow \Omega_F^1 \rightarrow 0$$

the map g is injective. In fact, g is generically injective, F being generically smooth, hence $\text{Ker } g$ is torsion, thus 0 since N_F^* is locally free. Consider $g(h(s))$. Since $T_X|_F$ is nef by assumption, $g(h(s))$ does not have any zero, in particular $h(s)$ has no zeros for all s . It follows first that g is a bundle morphism, so F is a connected component of F' , thus N_F^* is globally generated, hence trivial, and second that g is a bundle morphism, so F is smooth.

Of course, the general situation is more complicated. Philosophically speaking, singularities of F give a negative part in the normal bundle which contradicts the nefness of T_X .

4.13.2 Proposition

Let X be compact Kähler with T_X nef, let $\varphi : X \longrightarrow Y$ be a surjective holomorphic map with Y a normal compact complex space which is a subspace of a Kähler manifold (or a so-called Kähler space). Assume $\varphi_*(\mathcal{O}_X) = \mathcal{O}_Y$ and that $-K_X$ is ample on every fiber (e.g. X is projective and φ the contraction of an extremal ray). Then both Y and φ are smooth.

Proof. First we exclude the case $\dim X = \dim Y$. In fact, since $-K_X$ is ample on the fibers of φ , it follows from Mori's characteristic p- method that one can find rational curves in the fibers. Now we conclude by the following observation

4.13.3 Observation

Let X be a compact manifold with T_X nef and let $C \subset X$ be a rational curve. Then the deformations of C fill up X .

In order to see (4.13.3) let us assume that C is smooth for simplicity (see [CP91] for the general case). Since $T_X|_C$ is nef, it follows that the normal bundle N_C is nef, so $N_C = \bigoplus \mathcal{O}(a_i)$ with $a_i \geq 0$, so N_C is globally generated and $H^1(N_C) = 0$, thus the deformations of C fill up X by a theorem of Griffiths [Gr66].

Returning to the proof of (4.13.2) we have to prove that every fiber F' of φ is smooth. Let F be an irreducible component of F' with reduced structure. Then by (4.13.1) F is smooth with numerically flat normal bundle. From the adjunction formula and the ampleness of $-K_X|_F$ we conclude that F is Fano. From (4.9) and the simply connectedness of F we deduce that N_F is trivial. Applying Griffiths' theorem a second time, we easily conclude that $\dim F = \dim X - \dim Y$ and that $F = F'$ as complex spaces (the fibers of φ are connected by $\varphi_*(\mathcal{O}_X) = \mathcal{O}_Y$).

Now we finish the algebraic proof of (4.8). We proceed by induction on n ; the case $n = 1$ being obvious. If K_X is nef, then $K_X \equiv 0$, hence T_X is numerically flat, and X is covered by a torus. So we may assume that K_X is not nef. So there exists a contraction $\varphi : X \longrightarrow Y$ of an extremal ray. By (4.13.2) Y and φ are smooth. Since X is not Fano, Y is not a point. Now $\phi^*(T_Y)$ is a quotient of T_X , so T_Y is nef. Let $m = \dim Y$. If $c_1(Y)^m \neq 0$, it is strictly positive, hence Y is Fano by (4.5). But then $\chi(\mathcal{O}_X) = \chi(\mathcal{O}_Y) = 1$, contradicting $\chi(\mathcal{O}_X) = 0$. So $c_1(Y)^m = 0$ and it follows by induction that $\tilde{q}(Y) > 0$, whence our claim.

This ends the algebraic proof of (4.8).

4.14 Remark

(4.13.2) can be generalised to the following. If X is a projective manifold with T_X nef, then X does not admit a non-trivial birational map to a normal compact complex space. In fact, one can construct rational curves in the fibers, cp. [Kw91]. This should certainly also be true in the Kähler case.

We next study the Albanese map of manifolds with nef tangent bundles. Fixing a base $u_1, \dots, u_q$ of holomorphic 1-forms on a compact Kähler manifold X , the Albanese

map α is a holomorphic map $X \longrightarrow A$ to a torus of dimension q , with the property that every u_i is induced by a 1-form on A , i.e. α induces a bijective map

$$\alpha^* : H^0(A, \Omega_A^1) \longrightarrow H^0(X, \Omega_X^1).$$

For details on the Albanese map we refer to [Ue75]. Note that in general α is neither equidimensional nor surjective and has disconnected fibers. However we have

4.15 Proposition

Let X be compact Kähler with T_X nef. Then the Albanese map $\alpha : X \longrightarrow A$ is surjective, smooth and has connected fibers. If F is a fiber of α , then T_F is nef. The relative tangent bundle $T_{X|A}$ is nef, too.

Proof. In fact, if α would not be smooth or surjective, then by the exact sequence

$$\alpha^*(\Omega_A^1) \longrightarrow \Omega_X^1 \longrightarrow \Omega_{X|A}^1 \longrightarrow 0$$

we would obtain 1-forms on X which have zeros. This contradicts (1.16). Dualising the sequence (which is now also left exact and a vector bundle sequence) we see that the relative tangent bundle is nef, since T_A is trivial (1.16). Hence T_F is nef (just restrict). The connectedness of the fibers follows from Stein factorisation and the universal property of the Albanese.

In section 5 we will see that the smoothness and surjectivity of the Albanese hold under far more general assumptions. Note that (4.15) yields $\tilde{q}(X) \leq n = \dim X$ with equality holding if and only if X is covered by a torus.

Now that we know that the Albanese $\alpha : X \longrightarrow A$ is a surjective submersion with T_F nef for every fiber, we want to have a closer look to the structures of the fibers. One might hope that $q(F) = 0$. This is however not the case; there are "hyperelliptic" surfaces, i.e. elliptic bundles over an elliptic curve admitting a torus as finite étale cover and the fibration is nothing than the Albanese map. However we will see that the above claim holds after finite étale cover; in the example the Albanese map is then (after passing to the torus) the identity. The computation of $q(F)$ and $\tilde{q}(F)$ is done in

4.16 Proposition

Let X and Y be compact Kähler manifolds and $g : X \longrightarrow Y$ be a smooth surjective map with connected fibers. Let F be a fiber of g . Then the following assertions hold.

(1) $q(X) \leq q(Y) + q(F)$

(2) $\tilde{q}(X) \leq \tilde{q}(Y) + \tilde{q}(F)$

(3) *We have equality in (2) if the following additional conditions hold:*

(a) Y is a torus

(b) $\pi_1(F)$ is almost abelian, i.e. contains an abelian subgroup of finite index

(c) for every finite étale cover $\tilde{F} \longrightarrow F$ the Albanese of $\tilde{F}$ has constant rank.

All conditions in (3) will hold for the Albanese map of a manifold with nef tangent bundles. Without additional assumptions we cannot expect the equality in (2) to be true. There are monodromy problems as seen in the proof. Although (1) and (2)

are clear in our situation of the Albanese, we nevertheless have to examine the case of equality and therefore the general case is done by no extra cost. Anyway we will use (4.16) also in section 5 in case only $-K_X$ is nef.

Proof. (1) By the theorem of Ehresmann g is differentiably locally trivial. Thus the direct image sheaf $R^1 g_*(\mathbf{R})$ is a locally constant real bundle of rank $2q$ (Hodge decomposition) containing the locally constant system $R^1 g_*(\mathbf{Z})$. Thus :

$$\dim H^0(Y, R^1 g_*(\mathbf{R})) \leq 2q,$$

with equality holding if and only if $R^1 g_*(\mathbf{Z}) = \mathbf{Z}^{2q}$, which is the same as to say that the monodromy action of $\pi_1(Y)$ on $H^1(F, \mathbf{Z})$ is trivial. Now an application of the Leray spectral sequence

$$E_2^{pq} = H^p(Y, R^q g_*(\mathbf{R}))$$

together with Blanchard's classical theorem that this spectral sequence degenerates for Kähler fibrations [Bl56] at E_2 , i.e. $E_2 = E_\infty$ gives the inequality (1), with equality if and only if the monodromy action is trivial.

In order to prove (2) take a finite étale cover $\tilde{X}$ of X . We want to see that $\tilde{X}$ induces a fibration $\tilde{X} \rightarrow \tilde{Y}$ with fiber $\tilde{F}$, where $\tilde{Y}$ is an étale cover of Y and $\tilde{F}$ an étale cover of F . Since finite étale covers are determined by subgroups of finite index in the fundamental group, we proceed as follows. Since g is topologically locally trivial, we can use the homotopy exact sequence

$$\dots \rightarrow \pi_2(Y) \rightarrow \pi_1(F) \rightarrow \pi_1(X) \rightarrow \pi_1(Y) \rightarrow 1.$$

Then the preimage of $\pi_1(\tilde{X}) \subset \pi_1(X)$ in $\pi_1(F)$ has finite index, so defines a finite étale cover $\tilde{F} \rightarrow F$. In the same spirit we can also take the image in $\pi_1(Y)$ inducing a finite étale cover $\tilde{Y} \rightarrow Y$. The problem is to put things together to obtain a fibration $\tilde{X} \rightarrow \tilde{Y}$ with fiber $\tilde{F}$. By topology this is possible if the induced sequence

$$\pi_2(\tilde{Y}) \rightarrow \pi_1(\tilde{F}) \rightarrow \pi_1(\tilde{X}) \rightarrow \pi_1(\tilde{Y}) \rightarrow 1$$

is exact. Since $\pi_2(Y) = \pi_2(\tilde{Y})$ this is easily checked. Now taking the supremum over all fibrations $\tilde{X} \rightarrow \tilde{Y}$ we obtain claim (2).

We discuss the most difficult part (3). By our assumption $\pi_1(F)$ contains a finitely generated free abelian subgroup G , so $G \simeq \mathbf{Z}^{2m}$ for some integer m . Now pass to a finite étale cover $\tilde{F} \rightarrow F$ such that $\pi_1(\tilde{F}) = G$. Thus $q(\tilde{F}) = \tilde{q}(\tilde{F}) = m$ by Hodge decomposition and the fact that the abelianised π_1 is H_1 . Since G may not be invariant under the action of $\text{Aut}(\pi_1(F))$, we introduce

$$G' = \bigcap_{\varphi \in \text{Aut}(\pi_1(F))} \varphi(G),$$

which is easily checked to be of finite index in G . Since G' is invariant under the monodromy action, we can form the semi-direct product $H = G' \rtimes \pi_1(Y)$. We obtain an exact sequence

$$0 \rightarrow G' \rightarrow H \rightarrow \pi_1(Y) \rightarrow 0$$

which is " of finite index " in the sequence

$$0 \longrightarrow \pi_1(F) \longrightarrow \pi_1(X) \longrightarrow \pi_1(Y) \longrightarrow 0.$$

This gives rise to a smooth fibration $g' : X' \longrightarrow Y$ with fiber F' such that $\pi_1(X') = H$ and $\pi_1(F') = G'$, in particular F' comes up as a finite étale cover of $\tilde{F}$.

As we have seen earlier, to prove our equality comes down to show that the monodromy action is trivial. Thus our central claim is

$$\text{the image of } \pi_1(Y) \longrightarrow \text{Aut}(H^1(F', \mathbf{Z})) \text{ is finite.} \quad (*)$$

In fact, having proved this, the monodromy action will be trivial after passing to a finite étale cover of Y .

For simplicity of notations we assume $X = X'$ and $F = F'$. The fibration g has still the "disadvantage" that the fibers might have positive irregularity. In order to deal with this we take simultaneously the Albanese map of every fiber. This is possible under quite general circumstances - we refer to [Ca85] and [Fu83] - and rather elementary in our situation. We obtain a relative Albanese fibration

$$X \longrightarrow A(X|Y) \longrightarrow Y.$$

$A(X|Y) \longrightarrow Y$ is a smooth fibration with every fiber a torus, moreover the total space $A(X|Y)$ turns out to be Kähler. We claim that the fibration is even locally trivial. Y being a torus, we easily reduce to $Y = \mathbf{C}$. Fixing a Kähler form on $A(X|Y)$ we obtain a holomorphic map $h : \mathbf{C} \longrightarrow D$ into the moduli space of polarised tori. Now it is a well-known theorem of Griffiths and Schmid [GS69] that D does not admit non constant maps $\mathbf{C} \longrightarrow D$, thus our fibration is locally trivial. Actually in our situation this can rather easily be checked by hand and it comes down to an application of Liouville's theorem.

In order now to prove the finiteness statement for the monodromy action, we introduce an euclidean structure on the real bundle $R^1 g_*(\mathbf{R})$ by representing local sections as real parts of holomorphic 1-forms, integrating the forms over the fibers with respect to a given Kähler metric and taking again real part. Thus for every $y \in Y$ we obtain a euclidean structure $(\ , \)_y$ on $H^1(F_y, \mathbf{R})$. Since the relative Albanese fibration is locally trivial, we conclude that $(\ , \)_y$ is independent of y . Thus the euclidean structure is locally flat, so the monodromy group, i.e. the image of $\pi_1(Y)$, is contained in the orthogonal group $O(2q)$. Since it is anyway contained in $Sl(2q, \mathbf{Z})$, it must be finite.

This ends the proof (4.16).

We apply (4.16) to obtain the main structure theorem of the Albanese map of compact Kähler manifolds with nef tangent bundles.

4.17 Theorem

Let X be a compact Kähler manifold with nef tangent bundle. Let $\tilde{X}$ be a finite étale cover such that $q(\tilde{X}) = \tilde{q}(X)$. Then

(1) $\pi_1(\tilde{X}) = \mathbf{Z}^{2q}$, so $\pi_1(X)$ is a finite extension of $\mathbf{Z}^{2q}$.

(2) *The Albanese map $\alpha : \tilde{X} \longrightarrow A(\tilde{X})$ is a smooth surjective map with nef relative tangent bundle and the fibers are Fano manifolds with nef tangent bundles.*

Proof. We proceed by induction on $n = \dim X$. So assume that (4.16) holds in dimension $< n$. Consider a fiber F of the Albanese map α . Then $\dim F < n$ and T_F is nef, thus $\pi_1(F)$ contains an abelian subgroup of finite index. Thus the assumptions of (4.14.5(3)) are satisfied and hence we conclude $\tilde{q}(F) = 0$. It follows that F is Fano. Thus F is simply connected and the homotopy exact sequence for α shows $\pi_1(\tilde{X}) = \mathbf{Z}^{2q}$. Now it follows easily that $\pi_1(X)$ is an extension of a finite group and $\mathbf{Z}^{2q}$.

Having in mind example (4.3.3) we might hope that the way in which the fibers of the Albanese map vary is very much restricted. In fact, we have :

4.18 Theorem

Let X be a projective manifold with T_X nef. Let $\alpha : X \longrightarrow A$ be the Albanese map. Then the vector bundles $\alpha_(K_X^{-m})$ are numerically flat for every positive integer.*

For a proof we refer to [DPS94a]. It is also interesting to note that $\tilde{q}(X)$ can be measured in terms of $-K_X$:

$\dim X - \tilde{q}(X) = \nu(-K_X)$, where $\nu(L)$ is the numerical Kodaira dimension of a line bundle L , i.e. the maximal integer k with $c_1(L)^k \neq 0$.

Continuing the study of manifolds X with nef tangent bundles we want now to have a closer look at the fibers of the Albanese map in case $q(X) = \tilde{q}(X)$, which is achieved after possibly passing to a finite étale cover. These fibers then have $\tilde{q}(X) = 0$, hence we are interested in Fano manifolds with nef tangent bundles.

4.19 Conjecture

Let X be a Fano manifold with T_X nef. Then X is homogeneous.

4.20 Comments

(1) Here are some evidences for the conjecture. First of all, it holds in dimension ≤ 3 . In fact, if $n = \dim X = 2$, then X is a del Pezzo surface without exceptional curves, hence $X = \mathbf{P}_2$ or $X = \mathbf{P}_1 \times \mathbf{P}_1$. In dimension 3 one can use the Iskovskih classification of Fano 3-folds ([Is77,90], [Mu89]) to obtain the following list : $\mathbf{P}_3, Q_3, \mathbf{P}(T_{\mathbf{P}_2})$, see [CP91]. Even for $n = 4$ the conjecture can be proved [CP93] except that we cannot rule out Fano 4-folds of index 1 without knowing the existence of lines, a problem which is still unsolved (in this context a line would be a rational curve C with $(-K_X.C) = 1$). Second, X behaves in various contexts as if it is homogeneous : there are no exceptional sets, every rational has many deformations which fill up X , ... In (4.21) we shall try to give a scheme of a proof; it comes down to show that T_X is generated by global sections.

(2) Assume that Conjecture 4.19 holds. Then Mori's theorem as well as (2.10) follow easily. The reason is - assuming already $b_2(X) = 1$ - that $H_2(X, \mathbf{Z})$ is generated by the class of a rational curve C , hence picking such a rational curve C we immediately

conclude that the index of X is $n + 1$ resp. n . It should also be not too difficult to deduce Mok's theorem from 4.19.

(3) Since rational-homogeneous manifolds are rigid, the positive solution of (4.19) would also strengthen the general structure theorem on compact Kähler manifolds X with nef tangent bundles. In fact, let $\alpha : X \rightarrow A$ be its Albanese map. Then α is a (flat) locally trivial fiber bundle with typical fiber a rational-homogeneous manifold.

4.21 Scheme of a proof of 4.19

(1) The first step would be a reduction to $b_2(X) = 1$. In fact, if $b_2(X) \geq 2$, then let $\varphi : X \rightarrow Y$ be the contraction of an extremal ray. By (4.13.2) both Y and φ are smooth. The fibers have again nef tangent bundle and are Fano, since $-K_X$ is φ -ample. By induction on $n = \dim X$, φ is a fiber bundle over a rational-homogeneous manifold with typical fiber again a rational-homogeneous manifold. It remains to construct "enough" vector fields on X from the vector fields on the fibers and the base.

(2) Now suppose $b_2(X) = 1$. Then we consider $\mathbf{P} = \mathbf{P}(T_X)$. Since T_X is nef, the anti-canonical bundle $-\mathbf{K}_{\mathbf{P}}$ is nef. The plan is to try to construct sections in $\mathcal{O}_{\mathbf{P}}(1)$, an equivalent problem to the constructions of vector fields on X . A weak solution to the problem would be to construct sections of $\mathcal{O}_{\mathbf{P}}(m)$ for large m , i.e. of sections of $S^m(T_X)$. By the base point free theorem this would be achieved if the self-intersection

$$c_1(\mathcal{O}_{\mathbf{P}}(1))^{2n-1} > 0,$$

which is equivalent to saying that the highest Segre class $s_n(T_X)$ is positive. Since $\mathcal{O}_{\mathbf{P}}(1)$ is nef, $c_1(\mathcal{O}_{\mathbf{P}}(1))^{2n-1} \geq 0$, thus only the case $c_1(\mathcal{O}_{\mathbf{P}}(1))^{2n-1} = 0$ has to be excluded. Assuming this is achieved, $\mathcal{O}_{\mathbf{P}}(m)$ is generated by global sections for all large m . Let $\varphi : \mathbf{P} \rightarrow Z$ be the associated morphism (independent of m) with Z a normal projective of dimension $2n - 1$. Now Z is easily seen to be a Gorenstein variety, i.e. Cohen-Macaulay and the canonical Weil divisor K_Z is Cartier. In fact we have

$$-\mathbf{K}_{\mathbf{P}} = \varphi^*(-K_Z). \quad (+)$$

Note that a priori we only have $\mathcal{O}_{\mathbf{P}}(m) = \varphi^*(L_m)$ for some very ample line bundle L_m , m large, but necessarily $L_m^n = -mK_Z$ and since the same also holds for $m + 1$, we conclude the above equality (+). So Z is a singular Gorenstein Fano variety with only canonical singularities because of (+) and $\rho(Z) = 1$. Its index is n , since the above argument works also for $\mathcal{O}_{\mathbf{P}}(1)$, namely $\mathcal{O}_{\mathbf{P}}(1) = \varphi^*(H)$ for some ample line bundle H on Z . Now it comes down to prove the generatedness of H , which is automatically the ample generator on Z . We have extracted this as problem (4.22) below. Note that by Fujita's conjecture (in the singular case for normal Gorenstein spaces with rational singularities; see Demailly's article in this volume), we would get the global generatedness of $-K_Z = H^n$. A first step would be to prove that the linear system $|H|$ is not empty, which is very likely and fulfilled in all known examples of Fano varieties (however if the index is small compared to the dimension, e.g. $= 1$, then the linear system may have base points). Then, in a next step, one

might hope to prove global generatedness, if one knows already the existence of sufficiently many vector fields. A problem in this direction is formulated in (4.23).

4.22 Problem

Let X be a normal projective Fano Gorenstein variety with at most canonical singularities and $\rho(X) = 1$. Assume that $\dim X = 2n - 1$ and that X has index n . Let H be the ample generator of X . Prove that H has sections. Is H even generated by sections or at least "large" ?

Of course there should be a more general version of (4.22) : if $\dim X$ is "large" compared to the index, say at least around one half of the index, then H is effective. In small dimensions this is known. By Kodaira vanishing this comes down to prove $\chi(H) \neq 0$. Of course the second part will be much more difficult.

A more modest problem is

4.23 Problem

Let X be an almost homogeneous Fano manifold with T_X nef. Is X homogeneous ?

It is interesting to note that for almost homogeneous compact Kähler manifolds, Barth and Oeljeklaus proved the structure theorem (4.16) [BO74].

Already ruled surfaces over elliptic curves coming from flat non-trivial vector bundles show that one cannot expect that a manifold with T_X nef admits a finite étale cover which is homogeneous. However these examples are almost homogeneous. So we may ask

4.24 Problem

Let X be compact Kähler with T_X nef. Does there exist a finite étale cover $\tilde{X}$ which is almost homogeneous ?

Even if (4.19) holds, this is not completely clear. Choose $\tilde{X}$ such that $q(\tilde{X}) = \tilde{q}(X)$. Then consider the Albanese map $\alpha : \tilde{X} \rightarrow A$. Then one has to prove two things

- (a) every vector fields on A lifts to $\tilde{X}$.
 - (b) the relative tangent bundle is generically generated by global sections.
- (a) holds at least if $-K_X$ is hermitian semi-positive, see (5.6) below. For (b) one would like to extend some sections of T_F from a fiber to a section of the relative tangent bundle.

A final remark. Compact Kähler n -folds with $n \leq 3$ are completely classified in [DPS94a], in the surface case even without the Kähler assumption.

5. Manifolds with semi-positive anti-canonical bundles

Instead of looking at the tangent bundle we now consider merely the determinant of the tangent bundle, i.e. the anticanonical bundle $-K_X$ of a compact Kähler manifold X . We say that $-K_X$ is hermitian semi-positive if there is a metric on $-K_X$, whose curvature is semi-positive. It is not required that the metric comes from a metric on X . However a fundamental theorem of Aubin [Au76] and Yau [Y77,78] says that given a semi-positive $(1,1)$ -form ω representing $c_1(X) = c_1(-K_X)$, then there exists a Kähler metric on X whose Ricci curvature is ω . Thus the three conditions

- (1) $c_1(X)$ is represented by a semi-positive form
- (2) $-K_X$ has a metric of semi-positive curvature, and
- (3) X carries a Kähler metric with semi-positive Ricci curvature

are equivalent. More generally we will also consider compact Kähler manifolds with nef anti-canonical bundle, but the results will not be as good as in the semi-positive case.

5.1 Basic Examples

There are two extreme types of examples :

- (a) Fano manifolds
- (b) manifolds with $c_1(X) = 0$.

Fano manifolds are defined by the condition that $-K_X$ is ample, or, equivalently by a Kähler metric of positive Ricci curvature. They are simply connected and have lots of rational curves ; in fact they are rationally connected, i.e. two general points can be joined by (a chain of) rational curves ([Ca92],[KoMiMo92]). In sharp contrast, we have a look at manifolds with $c_1(X) = 0$. By the decomposition theorem due to Bogomolov, Beauville and Kobayashi (see [Be83]) X admits a finite étale cover of the form

$$T \times X_1 \times \cdots \times X_m \times Y_1 \times \cdots \times Y_n,$$

where T is a torus, X_i are Calabi-Yau manifolds and Y_j are symplectic manifolds. The relevant definitions are given below.

- (1) A compact Kähler manifold is a Calabi-Yau manifold if $K_X \simeq \mathcal{O}_X$ and if $H^q(\tilde{X}, \mathcal{O}_{\tilde{X}}) = 0$ for all finite étale covers $\tilde{X} \rightarrow X$.
- (2) A compact Kähler manifold X is symplectic if $K_X \simeq \mathcal{O}_X$ and if there exists a nowhere degenerate holomorphic 2-form ω on X such that every holomorphic p -form on X is up to a scalar of the form $\omega^{\frac{p}{2}}$ (so necessarily p and $\dim X$ are even).

The proof of the decomposition theorem heavily depends on the existence of a Kähler-Einstein metric on X , which is by definition a Kähler metric which vanishing Ricci curvature. The existence of those metric has been proved by Aubin and Yau (see above.)

In this section we want to investigate to what extent a general compact Kähler manifold with $-K_X$ hermitian semi-positive (or nef) is built up by Fano manifolds and manifolds with $c_1(X) = 0$. Let us note that sometimes a Fano manifold can be blown up along a subspace without loosing the Fano property (or the property that

$-K_X$ is nef). The most basic examples are del Pezzo surface : the projective plane is blown up in up to eight points in general position; if one blows up nine points in general position then still $-K_X$ is nef (but not ample). See [Dm77].

We begin with the investigation of the fundamental group of a semi-positively curved manifold.

5.2 Definition

Let G be a finitely generated group with generators $g_1, \dots, g_p$.

- (1) We let $N(k)$ be the number of elements $x \in G$ which can be written as "words" of length at most k in the "letters" $g_1, \dots, g_p$:

$$x = g_{i_1}^{\epsilon_1} \cdots g_{i_k}^{\epsilon_k},$$

where $\epsilon \in \{0, -1, 1\}$.

- (2) G has subexponential growth if for every $\epsilon > 0$ there exists $C(\epsilon) = C > 0$ such that

$$N(k) \leq C e^{\epsilon k}, \text{ for } k \geq 0.$$

- (3) G is said to be of polynomial growth of order $\leq m$, if $N(k)$ can be bounded by a polynomial of degree at most m .

It should be remarked that the above definitions (2) and (3) do not depend on the choice of the generators $g_1, \dots, g_p$. For later use we note that non abelian free groups never have subexponential growth. The following theorem has been proved in [DPS93] .

5.3 Theorem

Let X be a compact Kähler manifold with $-K_X$ nef. Then $\pi_1(X)$ has subexponential growth. If $-K_X$ is even hermitian semi-positive, then $\pi_1(X)$ has polynomial growth of order at most $2 \dim X$ (in particular $q(X) \leq \dim X$.)

We will more or less skip the proof and only mention two things. First it is very important to use special metrics a la Aubin-Yau. Second we want to explain how the function $N(k)$ can be computed by (real) differential geometry. So let (X, g) be a compact m -dimensional Riemannian manifold and let $\tilde{X} \rightarrow X$ be the universal cover. Let $E \subset \tilde{X}$ be a fundamental domain for the action of $\pi_1(X)$ on $\tilde{X}$. Let

$$E_k = \bigcup_{\delta \in \pi_1(X), \text{length}(\delta) \leq k} \delta(E).$$

Observing that $\pi_1(X)$ acts isometrically on $\tilde{X}$ with respect to the pulled back metric of g , we obtain that $\text{vol}(E_k) = N(k) \text{vol}(X)$ and moreover that $E_k \subset B = B(y, \alpha k + \beta)$, where B is the geodesic ball with center $y \in E$, where β is the diameter of E with respect to g and where α is the maximum over the length of all loops representing the generators $g_1, \dots, g_p$. Thus one obtains the inequality

$$N(k) \leq \frac{\text{vol}(B)}{\text{vol}(X)}.$$

Now the task is to estimate the volume of a geodesic ball in $\tilde{X}$. For details we refer to [DPS93].

As an immediate consequence we obtain

5.4 Corollary

Let X be compact Kähler with $-K_X$ nef. Let $f : X \rightarrow C$ be a holomorphic map onto a compact Riemann surface C of genus g . Then $g \leq 1$.

Proof. Assume to the contrary that $g \geq 2$. Then $\pi_1(C)$ is well-known to contain a non-abelian free group of $2g - 1$ generators, in particular $\pi_1(C)$ has exponential growth. From the exact sequence

$$\pi_1(X) \rightarrow \pi_1(C) \rightarrow \pi_0(F),$$

where F is a general fiber, we conclude that $f_*(\pi_1(X))$ has finite index in $\pi_1(C)$. This contradicts Theorem 5.3 since $\pi_1(C)$ has exponential growth.

(5.4) will be considerably generalised later on. We proceed by investigating the Albanese map. Unfortunately the methods we are discussing work - at least at the moment - only in the hermitian semi-positive case. The following proposition is important.

5.5 Proposition

Let (X, ω) be a compact Kähler manifold. Let $\sharp : T_X \rightarrow \Omega_X^1$ denote the conjugate linear C^∞ -isomorphism given by $\sharp v = i\bar{v} \lrcorner \omega$. $\sharp$ extends clearly to tensor products. Let v be a C^∞ -section of $T_X^{\otimes r}$. Then the following formula holds.

$$\int_X \|\bar{\partial}(\sharp v)\|^2 d\mu = \int_X \|\bar{\partial}v\|^2 d\mu + \int_X \langle R(v), v \rangle d\mu.$$

Here $d\mu$ means integration with respect to the volume form of the Kähler metric and R is the following operator associated to the Ricci curvature of the Kähler metric :

$$R(v) = R\left(\sum_{|I|=p} v_I \frac{\partial}{\partial z_I}\right) = \sum_{|I|=p} \left(\sum_{k \in I} \lambda_k\right) v_I \frac{\partial}{\partial z_I},$$

where the $\frac{\partial}{\partial z_j}$ are an orthonormal frame with respect to the Kähler metric and where λ_k are the eigenvalues of the Ricci curvature in this orthonormal frame.

The proof is done by pointwise calculation of various operators and then by integrating. For details see [DPS93].

(5.5) is the main ingredient to prove the following theorem due to Lichnerowicz.

5.6 Theorem

Let X be a compact Kähler manifold with $-K_X$ hermitian semi-positive.

- (a) For every $\omega \in H^0(\Omega_X^{\otimes r})$ we find $v \in H^0(T_X^{\otimes r})$ such that $\omega \perp v = \|v\|^2$, in particular ω has no zeros and we have a canonical embedding $H^0(\Omega_X^{\otimes r}) \rightarrow H^0(T_X^{\otimes r})$, where r is any positive integer.
- (b) The Albanese map $\alpha: X \rightarrow A$ is a surjective submersion.
- (c) Every ω as in (a) is parallel with respect to any Kähler metric of semi-positive Ricci curvature.
- (d) Every vector field (resp. automorphism) of A lifts to one on X .

Before we approach the proof of (5.6), we comment on (5.6). First note that the original proof of Lichnerowicz from 1971 had to avoid Calabi-Yau and therefore is more complicated. Of course (5.6) holds equally for sections in wedge powers or symmetric powers of Ω_X^1 resp. for sections in irreducible tensor representations $\Gamma^a \Omega_X^1$. To say that a form is parallel means that it is invariant under parallel transport with respect to the given metric. Therefore to know a parallel form means to know it at one point (we assume all manifolds to be connected, of course), in particular a parallel form does not have zeroes except it is identically zero. The assertion of (5.6) holds of course also for all (irreducible) tensor representations, hence also for symmetric and wedge powers and compositions of these.

Proof. Fix $\omega \in H^0(\Omega_X^{\otimes r})$. Let $v = \sharp^{-1}(\omega)$. Take a Kähler metric of semi-positive curvature. Then (5.5) gives :

$$\int_X \|\bar{\partial}\omega\|^2 d\mu \geq \int_X \|\bar{\partial}v\|^2 d\mu.$$

Hence $\bar{\partial}v = 0$ and the a priori only differentiable v is holomorphic. Now (a) is clear and (b) is an immediate consequence as well as (c).

It remains to prove the parallelity. This can be done by a similar formula as in (5.5), the Bochner formula. It reads :

$$\Delta \|\omega\|^2 \geq \|\nabla \omega\|^2 + A(\omega),$$

where A is an operator defined in terms of the Ricci curvature to the effect that $A(\omega) \geq 0$ if the Ricci curvature is semi-positive. Integrating over X , the left hand side of the inequality is 0, hence we obtain $\nabla \omega = 0$, which is equivalent to saying that ω is parallel. Of course this gives also another proof of parts of (a) and of (b).

We should remark here that one can even prove that the Albanese map is a locally trivial holomorphic fiber bundle, for the proof we refer to [DPS94b].

In the further investigations we choose a different approach from [DPS94b]. We need the following purely group-theoretical result of Gromov [Gr81].

5.7 Theorem (Gromov)

Every group of polynomial growth contains a nilpotent subgroup of finite index.

We recall the definition of nilpotency. Define inductively $C^1(G) = G$, and

$C^{n+1}(G) = [G, C^n(G)]$, where $[a, b] = aba^{-1}b^{-1}$. Then G is nilpotent, if there exists a number n such that $C^n(G) = \{e\}$. In particular any nilpotent group is solvable. By means of Gromov's theorem we can prove

5.8 Proposition

Let X be a compact Kähler manifold with $-K_X$ hermitian semi-positive (or any compact Kähler manifold whose fundamental group has polynomial growth). If $\tilde{q}(X) = 0$, then $\pi_1(X)$ is finite.

Proof. By (5.3) π_1 has polynomial growth. Hence π_1 has a nilpotent subgroup of finite index. After passing to a finite étale cover of X we may assume that π_1 itself is nilpotent, hence solvable. Thus we have a composition series

$$\pi_1 = G_0 \supset G_1 \supset \dots \supset G_m = \{e\}$$

where G_{i+1} is a normal subgroup of G_i with abelian quotient G_i/G_{i+1} . Now assume that π_1 is infinite. Then let i_0 be minimal with the property that G_{i_0}/G_{i_0+1} is infinite. Since this quotient is abelian, we conclude that $G_{i_0}/[G_{i_0}, G_{i_0}]$ is infinite. By the minimality assumption G_{i_0} has finite index in G_0 , hence there exists a finite étale cover $\tilde{X} \rightarrow X$ such that $\pi_1(\tilde{X}) = G_{i_0}$. Then

$$H_1(\tilde{X}, \mathbf{Z}) \simeq G_{i_0}/[G_{i_0}, G_{i_0}].$$

It follows that $b_1(\tilde{X}) > 0$ since $H_1(\tilde{X}, \mathbf{Z})$ is finitely generated. By Hodge decomposition we conclude $\tilde{q}(X) \neq 0$, contradiction.

Of course (5.8) holds for every compact Kähler manifold whose fundamental group has polynomial growth.

5.9 Corollary

Let X be compact Kähler with $-K_X$ hermitian semi-positive. Assume $\tilde{q}(X) = q(X)$ (which can always be achieved after finite étale cover). Let $\alpha : X \rightarrow A$ be the Albanese. Then $\alpha_ : \pi_1(X) \rightarrow \pi_1(A)$ is onto with finite kernel. In particular $\pi_1(X)$ contains an abelian subgroup of finite index, i.e. $\pi_1(X)$ is almost abelian.*

Proof. Using the exact homotopy sequence for the fibration α we see that it is sufficient to prove finiteness of $\pi_1(F)$, where F is a fiber. Hence by (5.8) we need only to know $\tilde{q}(F) = 0$. This follows at once by induction and (4.16).

For further investigation we make use of a refinement of the Kodaira dimension (see [Ca93]), taking into account not only sections in some multiple of K_X but also of tensor powers of the cotangent bundle.

5.10 Definition

Let X be a compact (Kähler) manifold. Then we set

$$\kappa_+(X) = \max\{\kappa(\det \mathcal{F}) \mid \mathcal{F} \text{ is a subsheaf of } \Omega_X^p \text{ for some } p > 0\}.$$

Here $\det \mathcal{F} = (\bigwedge^r \mathcal{F})^{**}$, where r is the (generic) rank of $\mathcal{F}$.

(5.11) Here we collect some basic properties of κ_+ , for the proofs see [Ca93].

- (1) $\kappa_+(X) \geq \kappa(X)$. This is obvious.
- (2) $\kappa_+(X)$ is invariant under finite étale covers.

- (3) If $f : X \rightarrow Y$ is a finite map, then $\kappa_+(X) \geq \kappa_+(Y)$.
 (4) $\kappa_+(X)$ is a birational invariant of compact manifolds.

5.12 Proposition

Let X be a compact Kähler manifold with $-K_X$ hermitian semi-positive. Then

- (1) $\kappa_+(X) \leq 0$
 (2) If $f : X \rightarrow Y$ is a dominant meromorphic map to a normal compact Kähler space Y (e.g. a projective variety) then $\kappa(Y) \leq 0$. Here we define $\kappa(Y) = \kappa(\hat{Y})$ for a desingularisation $\hat{Y}$ of Y .

Proof. (1) follows easily from (5.6), because $\kappa_+(X) > 0$ gives a section with zeroes in some bundle $(\bigwedge^r \Omega_X^p)^{\otimes m}$. (2) is then a consequence of (5.11).

The structure of simply connected compact Kähler manifolds with $-K_X$ hermitian semi-positive is studied by the following

5.13 Theorem

Let X be a compact Kähler manifold with $-K_X$ hermitian semi-positive and finite fundamental group. Then the universal cover has a decomposition

$$\tilde{X} \simeq X_1 \times \dots \times X_r,$$

where every X_i is either a Calabi-Yau manifold or symplectic or X_i satisfies

$$\kappa_+(X_i) = -\infty.$$

Proof. Equip X with a Kähler metric g of semi-positive Ricci curvature. Let $(X, g) \simeq \Pi(X_i, g_i)$ the de Rham decomposition of (X, g) . This means that every (X_i, g_i) has irreducible holonomy. For the de Rham decomposition we refer to [KN63]. Although the de Rham decomposition is in the first instance a real theorem dealing with Riemannian manifolds, in our situation the (X_i, g_i) are Kähler manifolds. The holonomy group H_i of (X_i, g_i) is therefore contained in the unitary group $U(n_i)$, $n_i = \dim X_i$. Now it remains to show (by virtue of (5.12)) that $\kappa_+(X_i) = 0$ implies that X_i is Calabi-Yau or symplectic. The plan is to identify X_i by its holonomy group. Let $X = X_i$ for simplicity of notations. If $\kappa_+(X) = 0$, then we find $p > 0$ and a subsheaf $\mathcal{F} \subset \Omega^p$ with $\kappa(\det \mathcal{F}) = 0$. We obtain a non-zero element $\omega \in H^0(X, S^m \bigwedge^r \Omega^p)$ for suitable m, r. Actually we may assume that $\omega \in H^0(\Gamma^a \Omega^p)$ for an irreducible tensor representation Γ^a of highest weight a . We refer to e.g. [Bt57] and [De88]. By (5.6) ω is invariant under holonomy. Now we deduce from Berger's fundamental result [Be55] that there are only three possibilities for H ; namely $U(n)$, $SU(n)$ or the symplectic group $Sp(\frac{n}{2})$, in case n is even. Since Calabi-Yau manifolds are characterised by $H = SU(n)$ and symplectic manifolds by $H = Sp(\frac{n}{2})$ (see e.g. [Be83]), we only have to exclude that $H = U(n)$. This is however obvious since there is no $U(n)$ -invariant linear subspace.

5.13.a Reformulation

Using a result of Cheeger-Gromoll it follows by the same arguments that the universal cover $\tilde{X}$ of a compact Kähler manifold X with $-K_X$ hermitian semi-positive is of the form

$$\tilde{X} \simeq \mathbf{C}^q \times X_1 \times \dots \times X_r,$$

where the X_i are as in (5.13).

5.14 Corollary

Let X be compact Kähler with $-K_X$ hermitian semi-positive and finite π_1 . Then

- (1) $\kappa_+(X) = 0$ if and only if X admits a Ricci-flat Kähler metric.
- (2) $\kappa_+(X) = -\infty$ if and only if $H^0(\tilde{X}, \Omega_{\tilde{X}}^p) = 0$ for all finite étale covers $\tilde{X} \rightarrow X$ and all $p \geq 1$.

(5.15)

It remains to determine the structure of compact Kähler manifolds with $-K_X$ hermitian semi-positive and $\kappa_+(X) = -\infty$. In some sense those X should be near to being Fano. So we ask : does it follow that X is rationally connected? Campana conjectures this in [Ca93] even without the semi-positivity assumption on X . For $n = \dim X = 2$ the conjecture is clear by classification, for $n = 3$ it follows from the fact that every threefold with $\kappa(X) = -\infty$ is uniruled (for this one needs Mori's deep theorem on minimal models, [Mo88].) See [Ca93]. The converse statement : " X rationally connected implies $\kappa_+(X) = -\infty$ " is by the way rather easy.

So far for the hermitian semi-positive case. Of course we expect everything to hold also if $-K_X$ is merely nef. Since it seems difficult to prove polynomial growth of $\pi_1(X)$ we discuss another approach which studies $\pi_1(X)$ from a more geometric point of view. Let X at the moment be an arbitrary compact Kähler manifold and $\tilde{X} \rightarrow X$ its universal cover. Then $\tilde{X}$ admits a dominant meromorphic map $\tilde{X} \rightarrow \Gamma$, sometimes called Shafarevitch map, contracting generically the maximal compact connected analytic subspace through a point. We refer to [Ca92] and [Ko92] for the construction. Note that $\tilde{X} \rightarrow \Gamma$ yields a map (by dividing through the action of π_1) $X \rightarrow \Gamma/\pi_1(X)$. This is also called Shafarevitch map. The background is the following conjecture of Shafarevitch :

The universal cover of a compact Kähler manifold is holomorphically convex and therefore admits a proper holomorphic surjective map to a Stein space (the so-called Remmert reduction) which identifies two points if and only if they both lie in a connected compact complex subspace.

The meromorphic Shafarevitch map is a weak substitute for the conjecture. Now Campana proved in [Ca93] the following theorem, generalising a result of Gromov [Gr91].

5.16 Theorem

Let X be a compact Kähler manifold with $\chi(X, \mathcal{O}_X) \neq 0$. Then the inequality $\kappa_+(X) \geq \dim \Gamma$ holds except if $\kappa_+(X) = -\infty$ and $\dim \Gamma = 0$. If $\kappa_+(X) = -\infty$, then X is even simply connected.

Note that the condition $\dim \Gamma = 0$ is equivalent to the finiteness of π_1 .

5.17 Proposition

Let X be a smooth projective threefold with $-K_X$ nef. If $\tilde{q}(X) = 0$, then $\pi_1(X)$ is finite.

Proof. Due to the decomposition theorem of Bogomolov-Beauville-Kobayashi (see 5.1), we may assume $\kappa(X) = -\infty$.

(a) We are going to apply (5.16); therefore we have to compute $\kappa_+(X)$, i.e. we want to show

$$\kappa_+(X) \leq 0 \tag{1}.$$

Now every smooth projective 3-fold with $\kappa(X) = -\infty$ is uniruled by [Mo88], hence Campana [Ca93] shows that $\kappa_+(X)$ is the largest number d such that there exists a dominant meromorphic map $f : X \rightarrow Y$ to a projective variety Y with $\kappa(Y) = d$. Assume that $d > 0$. In case $\dim Y = d = 1$, f must be holomorphic, which case is excluded by (5.4). So $\dim Y = 2$. Now this case is excluded by (5.20) below, since by [Mo88] we may assume that f factors into a sequence of divisorial contractions, flips, and, in the last step, a fibration over Y with general fiber $\mathbf{P}_1$. This proves (1).

(b) In order to be able to apply (5.16) we still have to note that $\chi(X, \mathcal{O}_X) \neq 0$, which is clear from $q(X) = 0$ and $\kappa(X) = -\infty$. This ends the proof of (5.17).

Of course there are serious difficulties to make the proof work in larger dimensions. The first most serious is that the Mori programme works up to now only in dimension 3 and thus it is e.g. not known and in fact one of the most important open problems in birational geometry that manifolds of negative Kodaira dimension are uniruled. The second difficulty is that we have to know $\chi(\mathcal{O}_X) \neq 0$ in order to apply Campana's result.

We want to have a look at the Albanese map $\alpha : X \rightarrow A$ in case $-K_X$ is nef. Let $\text{alb}(X)$ denote the Albanese dimension of X , that is the dimension of the image of the Albanese map. We have already seen in (5.4) that α is surjective in case $\text{alb}(X) = 1$. Now we have

5.18 Proposition

Let X be a compact Kähler manifold with $-K_X$ nef. Then there is no holomorphic surjective map $\varphi : X \rightarrow Y$ to a projective variety Y with $\kappa(Y) > 0$, such that K_F is big for the general F of φ .

Proof. By considering the Stein factorisation we may assume Y normal and the fibers of φ connected. Now take H very ample on Y and take general members of $|H|$ whose intersection is a smooth curve C and such that $X_C = \varphi^{-1}(C)$ is smooth. Then one sees easily that the dual of the relative dualising sheaf $\omega_{X_C|C}^{-1}$ is big and nef. From Kawamata-Viehweg vanishing we get

$$H^1(X_C, \omega_{X_C} \otimes \omega_{X_C|C}^{-1}) = 0.$$

Since $\omega_{X_C} \otimes \omega_{X_C|C}^{-1} = \varphi^*(\omega_C)$, this is absurd.

An easy consequence is

5.19 Corollary

Let X be a n -dimensional projective manifold with $-K_X$ nef. If $\text{alb}(X) = n - 1$, then the Albanese map is surjective.

"Of course" it should be true that a projective or a compact Kähler manifold does not admit any dominant meromorphic map to a variety of positive Kodaira dimension. We prove in this direction

5.20 Proposition

Let X be a smooth projective 3-fold with $-K_X$ nef. Let $f : X \rightarrow Y$ be a dominant rational map to a normal surface S which factors into a sequence of Mori contractions and flips (i.e. certain birational transformations which are biholomorphic in codimension 1, see [KMM87]). Then $\kappa(S) \leq 0$.

Proof. Let $C \subset S$ be a general hyperplane section. If one could prove that f is holomorphic over $f^{-1}(C)$ then the method of the proof of (5.18) would apply also here. However this need not be the case (it happens of course, if no flip is involved). Instead, consider the first Mori contraction $\psi : X \rightarrow W$. We may assume that ψ is birational; let E be the contracted prime divisor. If $\dim \psi(E) = 0$, then $K_X^3 < 0$, so $-K_X$ is big and nef ([DPS93]), and X is rationally connected ([KoMiMo92]), so our claim follows. So let $\psi(E)$ be a curve C . Then by [DPS93] the manifold W has again $-K_W$ nef with two exceptions; then we proceed by induction on the Picard number. The two exceptions are described as follows: C is a smooth rational curve with normal bundle of type $(-1, -2)$ or $(-2, -2)$. In the first case it is proved in [DPS93] that X is birational to some Y with $-K_Y$ big and nef, so X is rationally connected. The other case is more complicated and needs a special analysis; this will be carried out in [PP95]. In [DPS93] it is already proved that in the case one has $K_X^2 = 0$.

(5.21)

Let X be a smooth projective threefold with $-K_X$ nef. By the previous results the Albanese map α of X is surjective. So we now ask whether it is smooth. The strategy is as follows. Since we may assume $\kappa(X) = -\infty$, the Albanese factors into a sequence of Mori contractions and flips and eventually an additional elliptic fibration. In order to make an induction over $\rho(X)$ we would like to prove (as in 5.20) that, after performing a Mori contraction on X the result is still smooth with nef anticanonical bundle. This works as already mentioned above except for one unlucky case which might occur: there is a blow-up $f : X \rightarrow W$ along a smooth rational curve $C \subset W$ with normal bundle $\mathcal{O}(-2) \oplus \mathcal{O}(-2)$ with $-K_X$ nef and $q(X) > 0$. Necessarily we then have $K_X^2 = 0$ ([DPS93]). This case will be excluded in [PP95], mainly by carefully examining preimages of curves in the Albanese torus in case of a 2-dimensional Albanese torus resp. by studying the fibers of the Albanese map in the other case.

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Thomas Peternell
Math. Institut
95440 Bayreuth, Germany
(11.7.95)

Kähler-Einstein Metrics on Algebraic Manifolds

Gang Tian *

Courant Institute of Mathematical Sciences
New York University, New York 10012

0. Introduction

The study of Kähler-Einstein metrics was pioneered by E. Calabi more than forty years ago. In the mid-70's, Yau's celebrated solution of the Calabi conjecture opened a new chapter and provided the first existence theorem on Kähler-Einstein metrics for Kähler manifolds with nonpositive first Chern class. Also for the first time, Yau found many important applications of those metrics to algebraic geometry. For instance, he proved the Chern number inequality in any dimension and gave a characterization of compact complex hyperbolic spaces. He also showed the uniqueness of the algebraic structure on CP^n . Since then, more applications have been found.

In these lecture notes, we will discuss basic results on Kähler-Einstein metrics and their applications. These notes are based on a course I gave at the C.I.M.E. session "Transcendental Methods in Algebraic Geometry", during July, 1994, at Cetraro, Italy. I would like to thank the organizers, Fabrizio Catanese and Ciro Ciliberto, for giving me the opportunity to speak at that very stimulating meeting.

Section 1 contains basic definitions and results in Kähler geometry. We also state the main existence theorems for Kähler-Einstein metrics. In Section 2, we give applications of Kähler-Einstein metrics, in particular, deriving the Chern number inequalities. Using the Chern number inequalities for singular surfaces, we derive an explicit bound on the degree on curves in complex surfaces of general type. Sections 3 and 4 are devoted to the proof of the main existence theorem stated in Section 1. In particular, we will prove Yau's theorem on the Calabi conjecture. We will discuss the existence of Kähler-Einstein metrics on Fano manifolds. We will give the solution for Kähler-Einstein metrics on Del-Pezzo surfaces. In Section 5, we study the connection between Kähler-Einstein metrics and the stability of algebraic manifolds. In particular, we show a general theorem on stability of algebraic manifolds which admit Kähler-Einstein metrics through determinant line bundles. I believe that this connection will have more profound implications. In Section 6, we give a residue formula involving the fixed-point set of holomorphic vector fields. This formula was already known to the author more than three years ago. I first lectured on it in the summer school held by "Graduiertenkolleg Geometrie und Mathematische Physik" at Ruhr-Universität-Bochum, 1993. Ms. Habermann took notes for

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me during the lecture. In the preparation of this section and Section 3, I benefitted from her hand-written notes. I would like to thank her for her generosity.

1. Kähler-Einstein Metrics

Let M be a complex manifold. A hermitian metric g on M is a Riemannian metric such that $g(JX, JY) = g(X, Y)$ for all tangent vectors X, Y . Its Kahler form ω_g is the (1,1)-form defined by

$$\omega_g(X, Y) = \sqrt{-1} g(X, JY), \quad X, Y \in TM \quad (1.1)$$

The hermitian metric g is Kähler if its Kähler form ω_g is closed, i.e., $d\omega_g = 0$. Then g is called a Kähler metric. We will denote by $[\omega_g]$ the cohomology class in $H^2(M, \mathbb{R}) \cap H^{1,1}(M, \mathbb{C})$. It is called the Kähler class of g . A complex manifold which admits a Kähler metric is a Kähler manifold. All algebraic manifolds are Kähler manifolds. The simplest example of these manifolds is the complex projective space with the standard Fubini-Study metric (cf. (1.8)).

Let M be a Kähler manifold with a Kähler metric g . In local coordinates $(z_1, \dots, z_n)$, where n is the complex dimension of M , the metric g is represented by the positive definite hermitian matrix $\{g_{i\bar{j}}\}_{1 \leq i, j \leq n}$, where $g_{i\bar{j}} = g(\frac{\partial}{\partial z_i}, \frac{\partial}{\partial \bar{z}_j})$. Then the Kähler form is given by

$$\omega_g = \sum_{i,j=1}^n g_{i\bar{j}} dz_i \wedge d\bar{z}_j \quad (1.2)$$

That ω_g is closed is equivalent to

$$\frac{\partial}{\partial z_k} g_{i\bar{j}} = \frac{\partial}{\partial z_i} g_{k\bar{j}} \quad (1.3)$$

It follows that the Levi-Civita connection ∇ of g is given by

$$\nabla_{\frac{\partial}{\partial z_i}} \frac{\partial}{\partial z_j} = \Gamma_{ij}^k \frac{\partial}{\partial z^k}, \quad \text{where } \Gamma_{ij}^l = g^{k\bar{l}} \frac{\partial}{\partial z_i} g_{j\bar{k}} \quad (1.4)$$

where $\{g^{k\bar{l}}\}$ is the inverse matrix of $\{g_{k\bar{l}}\}$. The bisectional curvature tensor $\mathcal{R}(g)$ of g is given by

$$\mathcal{R}(g)_{i\bar{j}k\bar{l}} = -\frac{\partial^2 g_{i\bar{j}}}{\partial z^k \partial \bar{z}^l} + g^{p\bar{q}} \frac{\partial g_{p\bar{j}}}{\partial \bar{z}^l} \frac{\partial g_{i\bar{q}}}{\partial z^k} \quad (1.5)$$

Consequently, the Ricci curvature tensor of g is

$$\mathcal{R}(g)_{i\bar{j}} = g^{k\bar{l}} \mathcal{R}_{i\bar{j}k\bar{l}} = -\frac{\partial^2}{\partial z^i \partial \bar{z}^j} \log \det(g_{k\bar{l}}) \quad (1.6)$$

Let $\text{Ric}(g)$ be the Ricci form

$$\text{Ric}(g) = \mathcal{R}(g)_{i\bar{j}} dz^i \wedge d\bar{z}^j \quad (1.7)$$

Clearly, the Ricci form is real and closed. It is well-known that it represents the real part of the first Chern class $c_1(M)$ of M .

Definition 1.1. We say that M is a compact Kähler manifold with $c_1(M) > 0$ (resp. $c_1(M) < 0$) if the real part $c_1(M)_R$ of the first Chern class can be represented by a positive (resp. negative) form ϕ , that is, a d -closed form ϕ which is given in local coordinates by

$$\phi = \sum_{i,j=1}^n \phi_{i\bar{j}} dz^i \wedge d\bar{z}^j$$

where $\{\phi_{i\bar{j}}\}$ is a positive (resp. negative) definite Hermitian matrix. We say that M is a compact manifold with vanishing first Chern class if $c_1(M)_R$ is cohomologically zero.

In the case that M is an algebraic manifold, $-c_1(M)$ is just the first Chern class of the canonical bundle K_M of M , so the conditions in Definition 1.1 are equivalent to the ampleness of the anti-canonical bundle, the canonical bundle, etc..

Definition 1.2. A Kähler metric g is Kähler-Einstein if $\text{Ric}(g) = \lambda \omega_g$ for some real number λ .

Obviously, the existence of a Kähler-Einstein metric implies that the first Chern class $c_1(M)$ of M is either positive, zero, or negative.

Examples:

- (1) The complex projective space CP^n . Let $[z^0 : z^1 : \cdots : z^n]$ be the homogeneous coordinates of CP^n . We define a d -closed $(1,1)$ -form

$$\omega_{FS} = \partial\bar{\partial} \log(|z^0|^2 + \cdots + |z^n|^2) \quad (1.8)$$

It is not hard to show that it is a positive form. Therefore, it is a Kähler form of a Kähler metric g_{FS} , called Fubini-Study metric of CP^n . The Fubini-Study metric is Kähler-Einstein, since $\text{Ric}(g_{FS}) = (n+1)\omega_{FS}$. In fact, its bisectional curvature $\mathcal{R}(g_{FS})$ is given in local coordinates

$$\mathcal{R}(g_{FS})_{i\bar{j}k\bar{l}} = g_{i\bar{j}}g_{k\bar{l}} + g_{i\bar{l}}g_{k\bar{j}}$$

where $\{g_{i\bar{j}}\}$ is the metric tensor of g_{FS} in the local coordinates.

- (2) The complex torus $T_C^n = C^n/\Gamma$. The flat metric on C^n descends to a flat Kähler metric on T_C^n . In particular, such a metric is Kähler-Einstein.
- (3) Consider $\mathcal{B}^n = \{z \in C^n \mid |z| < 1\} \subset C^n$ with the Kähler metric g given by

$$\omega_g = \sqrt{-1}\partial\bar{\partial} \log(1 - |z|^2).$$

This gives a complete Kähler metric with $\text{Ric}(g) = -(n+1)\omega_g$. Moreover, in local coordinates,

$$\mathcal{R}(g)_{i\bar{j}k\bar{l}} = -g_{i\bar{j}}g_{k\bar{l}} - g_{i\bar{l}}g_{k\bar{j}}.$$

If Γ is a lattice in $SU(n, 1)$, then $\mathcal{B}^n/\Gamma$ is a compact Kähler-Einstein manifold with negative first Chern class.

Definition 1.3. A Kähler metric g is of constant bisectonal curvature, if in any local coordinate system, $\mathcal{R}_{\bar{i}\bar{j}k\bar{l}} = \lambda(g_{\bar{i}\bar{j}}g_{k\bar{l}} + g_{\bar{i}l}g_{k\bar{j}})$ for some constant λ .

All manifolds in above example have constant bisectonal curvature. The following result is classical (cf. [Be]).

Proposition 1.4. Every simply-connected Kähler manifold of constant bisectonal curvature is one of the manifolds in above examples.

Theorem 1.5. Let M be a compact Kähler manifold. Then

- (1) ([Y1]) If $c_1(M) = 0$, then M has a unique Kähler-Einstein metric g in each given Kähler class. This metric g has vanishing Ricci curvature;
- (2) ([Y1], [Au]) If $c_1(M) < 0$, then M has a unique Kähler-Einstein metric g . This metric has negative Ricci curvature;
- (3) ([BM]) If $c_1(M) > 0$, then there is at most one Kähler-Einstein metric on M ;
- (4) ([T2]) If $c_1(M) > 0$ and $n = 2$, then M has a Kähler-Einstein metric if and only if $\eta(M)$ is reductive, where $\eta(M)$ is the Lie algebra of holomorphic vector fields. This metric has positive Ricci curvature.

The proof of Theorem 1.5, (1), (2) will be outlined in section 3, while the proof of (3), (4) will be outlined in section 4.

Remarks:

- 1) The uniqueness part of (1) and (2) in the above theorem was shown by Calabi in the 50's ([Ca]). It follows from the maximum principle.
- 2) It is unknown when there is a Kähler-Einstein metric on a compact Kähler manifold with $c_1(M) > 0$ and dimension $n > 2$. It is a very hard problem. There is a folklore conjecture that M has a Kähler-Einstein metric if M has no holomorphic vector fields. However, I do not believe that this conjecture is true. The existence of Kähler-Einstein metrics should be equivalent to certain stability of the underlying manifold with respect to the anti-canonical polarization. We refer the readers to Section 5 for more details.

There are some generalizations of this theorem to singular Kähler-Einstein metrics. For technical reasons, we have to confine ourselves to the cases corresponding to the first two cases in the above theorem, i.e., Kähler-Einstein metrics of non-positive scalar curvature. In fact, Yau proved a very general existence theorem on the existence of singular Kähler-Einstein metrics in his celebrated paper [Y1]. This theorem of Yau has many applications. For instance, by a standard argument of using the Gauss equation for subbundles, one can apply those metrics to proving the stability of the tangent sheaves of algebraic manifolds of general type (cf. [Si], [Ts1]). Similarly, we can deduce the stability of a certain extension of the tangent sheaves of minimal varieties of general type. However, the theorem does not give enough information on the singularities of those metrics and how the singularities are related to the geometry of the underlying manifolds. This limits their applications. This problem was later addressed by many people, including Cheng and Yau, Bando, R. Kobayashi, Tsuji and myself (cf. [CY], [Ko1], [Ko2], [Ts2], [TY1], [TY2],

[TY3], [Ba], [BK]). I am not going to list all these generalizations. One of them is of particular importance to the study of quasi-projective manifolds of logarithmic general type.

Theorem 1.6 ([TY1]). *Let M be an n -dimensional compact Kähler manifold and D be a divisor in M with normal crossings. Assume that 1) $K_M + D$ is ample outside D , where K_M denotes the canonical bundle of M ; 2) $K_M + D$ is numerically positive; 3) $K_M + D$ is big, i.e., $(K_M + D)^n > 0$. Then there is a unique complete Kähler-Einstein metric on $M \setminus D$.*

We will refer the readers to [TY1] (also see [Ba]) for its proof.

Remarks:

- 1) In [TY1], we proved that there is a Kähler-Einstein metric and that this metric is almost complete, which is slightly weaker than completeness but good enough for all known applications. The completeness was eventually proved by Yau in [Y3].
- 2) Theorem 1.6 in the case of $K_M + D$ being ample had been proved by Cheng and Yau, R. Kobayashi, independently ([CY], [Ko1]). In [Ts2], Tsuji also proved a special case of this theorem.

Another interesting generalization is singular Kähler-Einstein metrics which are conic along a divisor with normal crossings. Their applications will be given in next section. We consider a compact Kähler manifold M and a divisor D with normal crossings. Let $\{D_i\}_{1 \leq i \leq m}$ be the irreducible components of D and $\{\alpha_i\}_{1 \leq i \leq m}$ be a family of positive numbers less than one. We assume that $K_M + \sum_{i=1}^m \alpha_i D_i$ is ample over M . A conic Kähler metric g with cone angles $2\pi(1 - \alpha_i)$ along D_i is a smooth Kähler metric g on $M \setminus D$ satisfying: let x_0 be in $\bigcap_{j=1}^k D_{i_j} \setminus \bigcup_{i \neq i_j} D_i$, where $k \leq n$, and $z_1, \dots, z_n$ be local coordinates such that z_j locally defines D_{i_j} , then ω_g is equivalent to the following Kähler form near x_0 ,

$$\sum_{j=1}^k |z_j|^{-2\alpha_j} dz_j \wedge d\bar{z}_j + \sum_{j=k+1}^n dz_j \wedge d\bar{z}_j \quad (1.8)$$

We call the restriction of g to $M \setminus D$ the regular part of the conic metric g . We will also define the Kähler class of g to be the cohomology class represented by ω_g as a d-closed, positive (1,1)-current on M . A Kähler orbifold metric is a particular conic Kähler metric with cone angles of the form $\frac{2\pi}{m}$. We are interested in conic Kähler-Einstein metrics, i.e., those conic Kähler metrics whose regular part has constant Ricci curvature. The Kähler-Einstein orbifold metrics are among these metrics (cf. [CY], [Ko1]).

Theorem 1.7. *Let M , $D = \{D_i\}$, $\{\alpha_i\}$ be as above such that $K_M + \sum_{i=1}^m \alpha_i D_i$ is ample over M . Then there is a unique conic Kähler-Einstein metric on M with the Kähler class representing the Chern class of $K_M + \sum_{i=1}^m \alpha_i D_i$ and cone angles $2\pi(1 - \alpha_i)$ along D_i .*

The proof of this theorem involves two steps. In the first step, one constructs a conic Kähler metric with the Kähler class $K_M + \sum_{i=1}^m \alpha_i D_i$ and the cone angles $2\pi(1 - \alpha_i)$ along D_i , such that its regular part has bounded curvature. Then one can set up a complex Monge-Ampère equation whose solutions give rise to conic Kähler-Einstein metrics. In the second step, one solves this Monge-Ampère equation. The arguments in this step are similar to those in Yau's proof of Theorem 1.5, (1). The details will be carried out in the thesis of my student T. Jeffres.

The conic Kähler-Einstein metrics in Theorem 1.7 have negative scalar curvature. Therefore, this theorem can be considered as a generalization of Theorem 1.5, (2).

2. Applications of Kähler-Einstein Metrics

In this section, we give some applications to algebraic geometry. We will only consider the cases that the first Chern class is non-positive. Other cases will be discussed in later sections. First we list a few well-known theorems which are classical applications of Kähler-Einstein metrics. I will omit their proof. The following is simply a corollary of the existence of Kähler-Einstein metrics and the standard Bochner identities (cf. [Kob]).

Theorem 2.1. *Suppose that M is an irreducible algebraic manifold. We have*

- 1) *If the first Chern class $c_1(M)$ is negative, then $H^0(M, S^k TM \otimes K_M^l) = 0$ for either $k > nl$ or $k = nl$ and M is not biholomorphic to a locally Hermitian symmetric manifold;*
- 2) *If the first Chern class $c_1(M)$ is zero, then $H^0(M, S^k TM) = 0$ unless M is flat;*
- 3) *If the first Chern class $c_1(M)$ is positive and M admits a Kähler-Einstein metric, then $H^0(M, S^k TM \otimes K_M^l) = 0$ for either $k < nl$ or $k = nl$ and M is not biholomorphic to a locally Hermitian symmetric manifold.*

Theorem 2.2 (Structure Theorem). *Let M be a compact Kähler manifold with vanishing first Chern class. Then a finite covering of M is of the form $T_C^k \times N$, where T_C^k is a complex torus of dimension k , which is the half of the first Betti number of M , and N is a simply-connected Kähler manifold with vanishing first Chern class. In particular, any Calabi-Yau manifold has finite fundamental group.*

The proof of this can be found in [Bea].

Theorem 2.3. ([Y2]) *Let M be a projective algebraic manifold. (1) If the canonical bundle K_M is ample, then*

$$\left(\frac{2(n+1)}{n} c_2(M) - c_1(M)^2 \right) \cdot (-c_1(M))^{n-2} \geq 0, \quad (2.1)$$

the equality in (2.1) holds if and only if M is a compact quotient of the complex ball by a discrete subgroup in $SU(n, 1)$; (2) If K_M is trivial, then $c_2(M)[\omega]^{n-2} \geq 0$

for any Kähler class $[\omega]$ on M of dimension 2, and if the equality holds for some $[\omega]$, then M is flat; (3) If $c_1(M) > 0$ and M has a Kähler-Einstein metric, then (2.1) holds with $-c_1(M)$ replaced by $c_1(M)$, and the equality holds if and only if $M = \mathbb{C}P^n$.

Proof: We only sketch a proof for (1) here. The proof for other cases is identical.

Let g be the unique Kähler-Einstein metric on M with $\text{Ric}(g) = -\omega_g$ (cf. Theorem 1.5). Recall that the traceless part of the curvature $R(g)$ is defined in local coordinates by

$$R^\circ(g)_{i\bar{j}k\bar{l}} = R(g)_{i\bar{j}k\bar{l}} + \frac{1}{n+1} (g_{i\bar{j}}g_{k\bar{l}} + g_{i\bar{l}}g_{k\bar{j}}) \quad (2.2)$$

This traceless curvature measures the deviation of g away from the metric with constant bisectional curvature. For any tensor T , we denote by $|T|$ the norm of T with respect to g . It follows from straightforward computations (cf. [Be], p80) that

$$\frac{1}{n(n-1)} |R^\circ|^2 \omega_g^n = -\text{Tr}(R(g) \wedge R(g)) \wedge \omega_g^{n-2} + \frac{1}{n+1} \omega_g^n \quad (2.3)$$

Integrating (2.3) over M , we obtain

$$-\frac{1}{n(n-1)} \int_M |R^\circ|^2 \omega_g^n = (c_1(M)^2 - 2c_2(M))[\omega_g]^{n-1} - \frac{1}{n+1} [\omega_g]^n.$$

Clearly, (2.1) follows from this. If the equality holds, the traceless curvature R° is identically zero, i.e., g is of constant negative bisectional curvature, and consequently, M is a quotient of the complex hyperbolic space.

Remark: In case $n = 2$, (2.1) is the Miyaoka-Yau inequality: $c_1(M)^2 \leq 3c_2(M)$, which was also proved by Miyaoka for surfaces of general type. By using conic Kähler-Einstein metrics, one can also show that (2.1) still holds even if K_M is only numerically positive (Corollary 2.6, also see [Ko1], [CY]).

More generally, let $\pi : M \mapsto M_0$ be a smooth resolution. Assume that M_0 is a minimal variety of general type, namely, M_0 has only canonical singularities and K_{M_0} is big and numerically positive. Then one expects

$$\left(\frac{2(n+1)}{n} c_2(M) - c_1(M)^2 \right) \cdot \pi^* K_{M_0}^{n-2} \geq 0$$

This can be proved by establishing the stability of the Hicks bundle $\Omega^1 M \oplus \mathcal{O}_M$ with the canonical Hicks field $\text{id} \in H^0(M, T_M \otimes \Omega^1 M)$ (cf. [LT]).

Using Theorem 1.6 in place of Theorem 1.5, we can have an analog of (2.1) for quasi-projective manifolds. More precisely, if M is a compact Kähler-manifold and D is a divisor with normal crossings such that all assumptions in Theorem 1.6 are satisfied, then

$$\left(\frac{2(n+1)}{n} C_2(T^*M(\log D)) - C_1(T^*M(\log D))^2 \right) \cdot C_1(T^*M(\log D))^{n-2} \geq 0, \quad (2.4)$$

where $T^*M(\log D)$ is the logarithmic cotangent bundle of M , D . Moreover, if the equality holds, then $M \setminus D$ is the quotient of the complex hyperbolic ball. We refer the readers to [TY1] (also see [Ba]) for details.

In the following, we would like to derive the Chern number inequalities for conic Kähler metrics.

Let $M, D, \{\alpha_i\}$ be as in Theorem 1.7 and g be the conic Kähler-Einstein metric in Theorem 1.7. We denote by $c_i(M, g)$ the i^{th} Chern form of g . Since the curvature of g is bounded, $c_i(M, g)$ defines a cohomology class $c_i(M, g)$.

Lemma 2.4. *We have*

$$c_1(M, g) = c_1(M) - \sum_{i=1}^m \alpha_i [D_i] \quad (2.5)$$

$$\begin{aligned} c_2(M, g) = c_2(M) + \sum_{i=1}^m \alpha_i (K_M \cdot D_i + D_i^2) \\ + \frac{1}{2} \sum_{i \neq j} \alpha_i \alpha_j D_i \cdot D_j - \sum_{i=1}^m \alpha_i (2 - \alpha_i) \text{Sing}(D_i) \end{aligned} \quad (2.6)$$

where $\text{Sing}(D_i)$ denotes the singular set of D_i , which is a codimension 2 smooth cycle in M . In particular, in the case $n = 2$,

$$c_2(M, g) = c_2(M) + \sum_{i=1}^m \alpha_i \mathcal{N}(\tilde{D}_i) + \frac{1}{2} \sum_{i \neq j} \alpha_i \alpha_j D_i \cdot D_j + \sum_{i=1}^m \alpha_i^2 \text{Sing}(D_i) \quad (2.7)$$

where $\tilde{D}_i$ is the normalization of D_i .

Proof: The equality (2.5) is trivial, since g is a conic Kähler-Einstein metric with Kähler class $K_M + \sum_{i=1}^m \alpha_i D_i$. Therefore, we only need to prove (2.6). We first reduce the general case to the case of complex surfaces. Let F be any complex surface in M , which intersects D transversally. Then we have

$$c_1(M)|_F = c_1(N_{F|M}) + c_1(F) \quad (2.8)$$

$$c_2(M)|_F = c_2(N_{F|M}) + c_2(F) + c_1(F) \cdot c_1(N_{F|M}) \quad (2.9)$$

where $N_{F|M}$ is the normal bundle of F in M . The metric g restricts to a conic Kähler metric g_F on F and a continuous metric on $N_{F|M}$, therefore, by (2.5) and (2.8),

$$\begin{aligned} C_1(M, g)|_F &= c_1(N_{F|M}) + c_1(F, g_F) \\ &= c_1(F) - \sum_{i=1}^m \alpha_i D_i \cdot F + c_1(N_{F|M}) \\ &= c_1(M)|_F - \sum_{i=1}^m \alpha_i D_i \cdot F \end{aligned} \quad (2.10)$$

$$c_2(M, g)|_F = c_2(N_{F|M}) + c_2(F, g_F) + c_1(F, g_F) \cdot c_1(N_{F|M}) \quad (2.11)$$

Suppose that we have proved (2.6) for complex surfaces. Then

$$\begin{aligned}
c_2(F, g_F) &= c_2(F) + \sum_{i=1}^m \alpha_i (K_F \cdot (D_i \cap F) + (D_i \cap F)^2) \\
&\quad + \frac{1}{2} \sum_{i \neq j} \alpha_i \alpha_j (D_i \cap F) \cdot (D_j \cap F) - \sum_{i=1}^m \alpha_i (2 - \alpha_i) \text{Sing}(D_i \cap F) \\
&= c_2(F) + \sum_{i=1}^m \alpha_i (K_F \cdot (D_i \cap F) + D_i^2 \cap F) \\
&\quad + \frac{1}{2} \sum_{i \neq j} \alpha_i \alpha_j (D_i \cdot D_j) \cap F - \sum_{i=1}^m \alpha_i (2 - \alpha_i) \text{Sing}(D_i \cap F) \quad (2.12)
\end{aligned}$$

It follows from (2.10), (2.11) that

$$\begin{aligned}
c_2(M, F)|_F &= c_2(F) + c_2(N_{F|M}) + \sum_{i=1}^m \alpha_i (K_F \cdot (D_i \cap F) + D_i^2 \cap F) \\
&\quad + \frac{1}{2} \sum_{i \neq j} \alpha_i \alpha_j (D_i \cdot D_j \cap F) - \sum_{i=1}^m \alpha_i (2 - \alpha_i) \text{Sing}(D_i \cap F) \\
&\quad + (c_1(F) - \sum_{i=1}^m \alpha_i (D_i \cap F)) \cdot c_1(N_{F|X}) \\
&= \left\{ c_2(M) + \sum_{i=1}^m \alpha_i (K_M \cdot D_i + D_i^2) \right. \\
&\quad \left. + \frac{1}{2} \sum_{i \neq j} \alpha_i \alpha_j D_i \cdot D_j - \sum_{i=1}^m \alpha_i (2 - \alpha_i) \text{sing}(D_i) \right\} |_F \quad (2.13)
\end{aligned}$$

Since F is arbitrary, we have shown (2.6) for all dimensions.

It remains to prove (2.6) for complex surfaces. To do it, we just pick up a vector field and use the transgression as in the proof of the Gauss-Bonnet-Chern formula for the Euler number. We will not go through the details here. We just remark that if $\alpha_i = \frac{k_i-1}{k_i}$ and M' is the branched covering of M along $D_1, \dots, D_m$ with multiplicities $k_1, \dots, k_m$, then g pulls back to be a smooth metric on M' and both sides of (2.6) are equal to $\frac{1}{k_1 \dots k_m} c_2(M')$, so the lemma follows easily.

Theorem 2.5. *Let $M, D, \alpha_1, \dots, \alpha_m$ be as in Theorem 1.3. Then we have*

$$\begin{aligned}
&(K_M + \sum_{i=1}^m \alpha_i D_i)^n \\
&\leq \frac{2(n+1)}{n} \left(c_2(M) + \sum_{i=1}^m \alpha_i (K_M \cdot D_i + D_i^2) \right. \\
&\quad \left. + \frac{1}{2} \sum_{i \neq j} \alpha_i \alpha_j D_i \cdot D_j - \sum_{i=1}^m \alpha_i (2 - \alpha_i) \text{Sing}(D_i) \right) \left(K_M + \sum_{i=1}^m \alpha_i D_i \right)^{n-2} \quad (2.14)
\end{aligned}$$

Moreover, the equality in (2.14) holds if and only if the conic Kähler-Einstein metric is of constant bisectional curvature.

It follows from Lemma 2.4 and the same arguments as in the proof of Theorem 2.3.

Corollary 2.6. *Let M be a smooth manifold with numerically positive canonical bundle. Then the Chern number inequality (2.1) holds.*

Proof: If D is an ample smooth divisor, then $K_M + \alpha D$ is ample for any positive α . Then (2.1) follows from (2.14) by letting α go to zero.

Theorem 2.7. *Let M be a minimal complex surface of general type, and $D = D_1 + \cdots + D_m$ a divisor with only nodes as singularities. Assume that $D_i^2 \geq 0$ for all i . Then for any $0 \leq \alpha_i \leq 1$ ($i = 1, \dots, m$), we have*

$$K_M^2 + 2 \sum_{i=1}^m \alpha_i \left(1 - \frac{3\alpha_i}{4}\right) K_M \cdot D_i \leq 3c_2(M) - \frac{3}{2} \sum_{i=1}^m \alpha_i (2 - \alpha_i) \mathcal{X}(\tilde{D}_i) + \frac{1}{2} \left(\sum_{i=1}^m \alpha_i D_i \right)^2 \quad (2.15)$$

where $\mathcal{X}(\tilde{D}_i)$ is the Euler number of the normalization $\tilde{D}_i$ of D_i . In particular, if D is irreducible, we have

$$K_M^2 + 2\alpha \left(1 - \frac{3\alpha}{4}\right) K_M \cdot D \leq 3c_2(M) - \frac{3}{2} \alpha (2 - \alpha) \mathcal{X}(\tilde{D}) + \frac{\alpha^2}{2} D^2, \quad (2.16)$$

where $0 \leq \alpha \leq 1$.

Proof: Let r_i be the number of nodes in D_i . By the Adjunction Formula, we have

$$r_i = \frac{1}{2} \left(\mathcal{X}(\tilde{D}) + K_M \cdot D_i + D_i^2 \right)$$

Then (2.15) follows from this, (2.14), and simple computations.

Remark: We conjecture that (2.15) is true for any reduced curves D with $D_i^2 \geq 0$, where D_i are irreducible components of D .

Let us give a simple application of Theorem 2.7 to complex hyperbolic geometry. It has been a very important problem to bound the degree of the rational curves in a complex surface M of general type. In fact, it is a special case of the Green and Griffiths conjecture. The first substantial progress on this problem was made by Bogomolov. Lifting the rational curves to the projectified tangent spaces, he was able to prove that the degree of the rational curves is bounded from above in case the complex surface M satisfies $c_1(M)^2 > c_2(M)$. This last condition means that the cotangent bundle is big. But his proof does not give an effective bound. Recently, Miyaoka and Lu announced new results in general cases. In particular, they studied the finiteness of the rational curves on a surface of general type under certain conditions. Here we will apply (2.14) to give a new proof of Bogomolov's result, moreover, our proof gives the first effective bound on the degree in the case

of $c_1(M)^2 > c_2(M)$. Our proof can be also modified to prove that the degree of the rational curves in M is bounded even if $c_1(M)^2 = c_2(M)$, but we do not have effective bound in this case. I will only consider rational curves with nodes, which is presumably the most generic case. Let C be a curve in M . It is known that C^2 is bounded from below in terms of its genus, so we may assume that C^2 is positive. For simplicity, we assume that C has only nodes as singularities. The general case will appear elsewhere.

Since $C^2 > 0$, $K_M + \alpha C$ is ample for any positive α . Then by Theorem 2.7,

$$3c_2(M) - K_M^2 \geq \frac{3}{2}\alpha(2 - \alpha)\mathcal{X}(\tilde{C}) + 2\alpha\left(1 - \frac{3\alpha}{4}\right)K_M \cdot C - \frac{\alpha^2}{2}C^2$$

The maximum of the right side in the above is attained at

$$\alpha = \frac{2K_M \cdot C + 3\mathcal{X}(\tilde{C})}{3K_M \cdot C + C^2 + 3\mathcal{X}(\tilde{C})},$$

and it follows that

$$3c_2(M) - c_1(M)^2 \geq \frac{(2K_M \cdot C + 3\mathcal{X}(\tilde{C}))^2}{6K_M \cdot C + 2C^2 + 6\mathcal{X}(\tilde{C})}. \quad (2.17)$$

We may assume that $3K_M \cdot C + C^2 > -3\mathcal{X}(\tilde{C})$, for otherwise $0 \leq K_M \cdot C \leq -\mathcal{X}(\tilde{C})$, which is actually impossible if C is rational.

By the Index Theorem (cf. [BPV]), $C^2 K_M^2 \leq (K_M \cdot C)^2$. Therefore,

$$\frac{2}{K_M^2}(K_M^2 - c_2(M))(K_M \cdot C)^2 + (4\mathcal{X}(\tilde{C}) - 2\Delta)K_M \cdot C + (3\mathcal{X}(\tilde{C}) - 2\Delta)\mathcal{X}(\tilde{C}) \leq 0 \quad (2.18)$$

where $\Delta = 3c_2(M) - c_1(M)^2$. Then one can easily deduce the following theorem by straightforward computations.

Theorem 2.8. *Let M be a surface of general type with $K_M^2 > c_2(M)$, and C be a curve in M with only nodes as singularities. Then $\Delta - 2\mathcal{X}(\tilde{C}) \geq 0$ and*

$$K_M \cdot C \leq \frac{3c_2(M) - K_M^2 + \max\{0, -2\mathcal{X}(\tilde{C})\}}{K_M^2 - c_2(M)} K_M^2 \quad (2.19)$$

In particular, if $\Delta \leq 3$, there are no rational curves in M , and if M has nonnegative signature, i.e., $K_M^2 \geq 2c_2(M)$, then $C^2 < K_M^2$ for any curve C of genus ≤ 1 .

If $c_2(M) \geq K_M^2$, we can still bound the number of rational curves in M . Let $D_1, \dots, D_m$ be m rational curves with normal crossings and having positive self-intersection number. Then Theorem 2.7 implies that for any $\alpha > 0$,

$$(K_M + \alpha D)^2 \leq 3c_2(M) - 3m\alpha(2 - \alpha) + \frac{3\alpha^2}{2}(K_M \cdot D + D^2).$$

One can easily deduce from this

Theorem 2.9. *Let M be a minimal surface of general type. Assume that $K_M^2 \geq c_2(M)$. Then there are at most $\frac{1}{2}K_M^2 - 1$ rational curves D_i with only normal crossing singularities.*

Remark: It should be possible to remove the assumption on singularities of C in both Theorem 2.8 and 2.9. For this purpose, one needs to prove (2.15) in the general case.

3. Existence of Kähler-Einstein Metrics, I

In this section, we will prove the first two parts of Theorem 1.5. We will also reduce the proof of Theorem, (4) to an *a priori* C^0 -estimate for certain complex Monge-Ampère equations. First we derive the complex Monge-Ampère equations corresponding to the Kähler-Einstein metrics.

Assume that $c_1(M) = \mu[\omega]$, where $\mu = 0, -1, 1$ according to $c_1(M) = 0, < 0$, or > 0 . Fix a Kähler metric g with the Kähler form $\omega_g \in [\omega]$. Let g' denote a Kähler-Einstein metric. Then there is some smooth function ϕ such that

$$\omega_{g'} = \omega_g + \partial\bar{\partial}\phi$$

It follows that

$$\begin{aligned} \text{Ric}(g') &= -\partial\bar{\partial} \log \det(g'_{k\bar{l}}) \\ &= \text{Ric}(g) - \partial\bar{\partial} \log \frac{\det(g'_{k\bar{l}})}{\det(g_{k\bar{l}})}. \end{aligned}$$

There is a unique function f such that $\text{Ric}(g) - \mu\omega_g = \partial\bar{\partial}f$ and $\int_M (e^f - 1)\omega_g^n = 0$, since both $\mu\omega_g$ and $\text{Ric}(g)$ represent the same cohomology class. Furthermore, $\mu\omega_{g'} = \text{Ric}(g')$, since g' is a Kähler-Einstein metric. Hence,

$$\begin{aligned} \mu\omega_{g'} &= \text{Ric}(g) - \partial\bar{\partial} \log \left(\frac{\omega_{g'}^n}{\omega_g^n} \right) \\ &= \mu\omega_g + \partial\bar{\partial}f - \partial\bar{\partial} \log \left(\frac{\omega_{g'}^n}{\omega_g^n} \right) \end{aligned}$$

On the other hand, since $\omega_{g'} = \omega_g + \partial\bar{\partial}\phi$, we have

$$\mu\partial\bar{\partial}\phi = \partial\bar{\partial}f - \partial\bar{\partial} \log \left(\frac{\omega_{g'}^n}{\omega_g^n} \right).$$

This implies

$$\omega_{g'}^n = e^{f - \mu\phi + \text{const.}} \omega_g^n$$

If $\mu = \pm 1$, we redefine ϕ such that the constant is zero. If $\mu = 0$, the constant has to be zero, since $\int_M (e^f - 1)\omega_g^n = 0$ and $\int_M \omega_{g'}^n = \int_M \omega_g^n$. Therefore, we derive the

following complex Monge-Ampère equation

$$\begin{cases} (\omega_g + \partial\bar{\partial}\phi)^n = e^{f-\mu\phi} \omega_g^n \\ \omega_g + \partial\bar{\partial}\phi > 0 \end{cases} \quad (3.1)$$

where $\mu = 0, -1, 1$. To prove Theorem 1.5, i.e., the existence of Kähler-Einstein metrics, we only need to solve (3.1).

We will use the continuity method. Consider the following equations

$$\begin{cases} (\omega_g + \partial\bar{\partial}\phi)^n = e^{f_s - \mu t \phi} \omega_g^n \\ \omega_g + \partial\bar{\partial}\phi > 0 \end{cases} \quad (3.2)_{t,s}$$

where $0 \leq s \leq 1$, $0 \leq t \leq 1$, $\mu = 0, -1, 1$, and $f_s = sf + c_s$ for a constant c_s such that $\int_M (e^{f_s} - 1) \omega_g^n = 0$.

Define $E = \{0\} \times [0, 1] \cup [0, 1] \times \{1\}$ and I the set of $(t, s) \in E$ such that (3.2) $_{t',s'}$ is solvable for all $(t', s') \leq (t, s)$. Here $(t', s') < (t, s)$ means that either $t' < t$ or $t' = t$ and $s' < s$.

Observation: To solve (3.1), we only need to prove

- (1) $I \neq \emptyset$;
- (2) I is open in E ;
- (3) I is closed in E .

Since $(0, 0) \in I$, (1) is clear.

Proof of (2): Assume $(t, s) \in I$ and ϕ is a solution of (3.2) $_{t,s}$. We write $\omega_{g'} = \omega_g + \partial\bar{\partial}\phi$. Then we can denote a metric near $\omega_{g'}$ by

$$\omega_{g'} + \partial\bar{\partial}\psi = \omega_g + \partial\bar{\partial}(\phi + \psi).$$

Now (3.2) $_{\tilde{t},\tilde{s}}$ becomes

$$\begin{aligned} (\omega_{g'} + \partial\bar{\partial}\psi)^n &= e^{f_{\tilde{s}} - \mu \tilde{t}(\phi + \psi)} \omega_g^n \\ &= e^{f_{\tilde{s}} - \mu \tilde{t}(\phi + \psi)} e^{-f_s + \mu \phi} \omega_{g'}^n. \end{aligned}$$

Put $h = f_{\tilde{s}} - f_s - \mu(\tilde{t} - t)\phi$ and define an operator

$$P(\psi) = \log \left(\frac{(\omega_{g'} + \partial\bar{\partial}\psi)^n}{\omega_{g'}^n} \right) - h + \mu \tilde{t} \psi$$

where ψ is in $C^{2,\frac{1}{2}}(M, \mathbb{R})$ and $\|\psi\|_{2,\frac{1}{2}}$ is small. Note that $P(\psi)$ is in $C^{0,\frac{1}{2}}(M, \mathbb{R})$. The linearization of P is given by

$$D_\psi P = \Delta_{g'} + \mu \tilde{t} : C^{2,\frac{1}{2}}(M, \mathbb{R}) \rightarrow \tilde{C}^{0,\frac{1}{2}}(M, \mathbb{R}),$$

where $\tilde{C}^{0,\frac{1}{2}}(M, \mathbb{R})$ is just $C^{0,\frac{1}{2}}(M, \mathbb{R})$ for $\mu t \neq 0$, and if $\mu t = 0$, $\tilde{C}^{0,\frac{1}{2}}(M, \mathbb{R})$ consists of all $\psi \in C^{0,\frac{1}{2}}(M, \mathbb{R})$ such that $\int_M \psi \omega_{g'}^n = 0$.

To show that $P(\psi) = 0$ has a solution for small h and for $\tilde{t}$ sufficiently close to t , we need to show that the map $\Delta_{g'} + \mu t$ is surjective. This is automatically true for the cases where $\mu t \leq 0$, which correspond to the cases $c_1(M) \leq 0$ or $t = 0$. It remains to consider the cases where $0 < \mu t < 1$ and $s = \tilde{s} = 1$. In these cases, we have

$$\begin{aligned} \text{Ric}(g') &= -\partial\bar{\partial}\log\det(g') \\ &= -\partial\bar{\partial}(f - \mu\phi) + \text{Ric}(g) \\ &= \omega_g + \mu t \partial\bar{\partial}\phi \\ &= (1 - \mu t)\omega_g + \mu t(\omega_g + \partial\bar{\partial}\phi) \\ &= (1 - \mu t)\omega_g + \mu t\omega_g > \mu t\omega_{g'}. \end{aligned}$$

The following lemma is well-known and a corollary of the standard Bochner identity.

Lemma 3.1. *Let λ_1 be the first non-zero eigenvalue of $\Delta_{g'}$, where g' is given as above. Then $\lambda_1 > \mu t$.*

Proof: Let ζ be the eigenfunction of λ_1 , i.e., $-\Delta_{g'}\zeta = \lambda_1\zeta$. We normalize it by $\int_M \zeta^2 = 1$. Then

$$\begin{aligned} 0 &\leq \int_M \zeta_{ij}\bar{\zeta}_{ij} = - \int_M \zeta_{ij\bar{j}}\bar{\zeta}_i = - \int_M \zeta_{j\bar{i}j}\bar{\zeta}_i \\ &= - \int_M \left(\zeta_{j\bar{j}i}\bar{\zeta}_i + \text{Ric}(g')_{i\bar{j}}\zeta_j\bar{\zeta}_i \right) \\ &= - \int_M (\Delta_{g'}\zeta)_i\bar{\zeta}_i - \int_M \text{Ric}(g')_{i\bar{j}}\zeta_j\bar{\zeta}_i \\ &= \lambda_1 \int_M \zeta_i\bar{\zeta}_i - \int_M \text{Ric}(g')_{i\bar{j}}\zeta_j\bar{\zeta}_i. \end{aligned}$$

Hence

$$\lambda_1 \int_M \zeta_i\bar{\zeta}_i = \int_M \zeta_{ij}\bar{\zeta}_{ij} + \int_M \text{Ric}(g')_{i\bar{j}}\zeta_j\bar{\zeta}_i > \mu t \int_M \zeta_i\bar{\zeta}_i.$$

It follows $\lambda_1 > \mu t$.

This lemma implies that the linearization $D_\psi P$ is surjective in all cases. Therefore, by the Implicit Function Theorem, I is open.

It remains to prove that I is closed. For this, it suffices to have an *a priori* $C^{2, \frac{1}{2}}$ -estimate of the solutions of $(3.2)_{t,s}$. Namely, if $(t_i, s_i) \rightarrow (t, s)$ and ϕ_{t_i, s_i} are solutions of $(3.2)_{t_i, s_i}$, then by taking a subsequence, we may have a limit $\phi_{t_i, s_i} \rightarrow \phi$ and ϕ solves $(3.2)_{t,s}$.

Lemma 3.2 (Yau, [Y1]). *There exist two constants c and b depending only on (M, g) such that all solutions ϕ of $(3.2)_{t,s}$ satisfy*

$$0 < n + \Delta\phi \leq ce^{b(\sup\phi - \inf\phi)}.$$

Proof: Let ϕ be a solution of $(3.2)_{t,s}$. It is clear that $n + \Delta\phi > 0$, since $\omega_g + \partial\bar{\partial}\phi > 0$.

Let Δ' be the Laplacian of the Kähler metric g' with the Kähler form $\omega_{g'} = \omega_g + \partial\bar{\partial}\phi$. One can show (cf. [Y1], p347-351)

$$\begin{aligned} \Delta'(e^{-a\phi}(n + \Delta\phi)) &\geq e^{-a\phi}(\Delta f_s - t\mu\Delta\phi - n^2|R(g)|_g) \\ &\quad - ae^{-a\phi}n(n + \Delta\phi) + (a - |R(g)|_g)e^{-a\phi - \frac{I_s - t\mu\phi}{n-1}}(n + \Delta\phi)^{\frac{n}{n-1}} \end{aligned} \quad (3.3)$$

Taking b to be $a+1$ and $a > \sup_M |R(g)|_g + 1$, the lemma then follows from a direct application of the maximum principle to the function $e^{-a\phi}(n + \Delta\phi)$.

Lemma 3.3. *Let ϕ be a solution of (3.2)_{t,s}. There exists a constant c depending on (M, g) and $\|\phi\|_{C^0}$ such that*

$$\|\phi\|_{C^2, \frac{1}{2}} \leq C. \quad (3.4)$$

Proof: This can be proved by applying the maximum principle to the function $|\nabla^3\phi|_{g'}^2$, where the covariant derivatives and the norm are taken with respect to the metric g' . For more details, see Section 3 in [Y1]. This can be also proved by using the estimates in [CKNS], [Ev], [T6].

Thus we showed that in order to solve (3.2)_{t,s}, we only need to show an *a priori* C^0 -estimate for the solutions of (3.2)_{t,s}.

Lemma 3.4 (The Sobolev Inequality) ([Li]) *Let (X^m, ds^2) be any m -dimensional compact Riemannian manifold. Then there exist two constants $C_1, C_2 > 0$ such that for all functions $f \in C^1(X, R)$*

$$C_1 \left(\int_X |f|^{\frac{2m}{m-2}} dV \right)^{\frac{m-2}{m}} - C_2 \int_X |f|^2 dV \leq \int_X |\nabla f|^2 dV$$

holds, where C_1 and C_2 depend only on (X, ds^2) . Moreover, if $\text{Ric}(ds^2) \geq r ds^2$ for some $r > 0$ and $\text{Vol}(X, ds^2) \geq v > 0$, then C_1 and C_2 can be chosen depending only on r and v .

Now we prove the required C^0 -estimate in the following cases.

Case 1. $\mu = 0$ or $t = 0$.

Let ϕ be a solution of (3.2)_{t,s}. Without loss of generality, we may assume that $\sup_M \phi = -1$. Let $\mathcal{G}$ be a positive Green's function of the Laplacian of g . Then by the Green's formula,

$$\phi(x) = \frac{1}{V} \int_M \phi \omega_g^n - \frac{1}{V} \int_M \Delta\phi(y) \mathcal{G}(x, y) \omega_g^n(y), \quad \text{where } V = \int_M \omega_g^n,$$

for all $x \in M$, where M is the total volume of (M, g) . By $\Delta\phi + n > 0$, i.e., $-\Delta\phi < n$, we see that the second integral on the right is uniformly bounded:

$$-\frac{1}{V} \int_M \Delta\phi(y) \mathcal{G}(x, y) \omega_g^n(y) < C$$

for a constant $C > 0$. Thus for all $x \in M$, we have the inequality

$$\phi(x) \leq \frac{1}{V} \int_M \phi \omega_g^n + C,$$

which implies

$$-1 = \sup_M \phi \leq \frac{1}{V} \int_M \phi \omega_g^n + C$$

This is equivalent to $\frac{1}{V} \int_M |\phi| \omega_g^n \leq C_0$, since ϕ is nonpositive.

For simplicity, we set $\phi_- = -\phi$. In the following, C always denotes a uniform constant. We have

$$\begin{aligned} & \frac{1}{V} \int_M \phi_-^p (\omega_g^n - (\omega_g + \partial \bar{\partial} \phi)^n) \\ &= \frac{1}{V} \int_M \phi_-^p \partial \bar{\partial} \phi_- \wedge (\omega_g^{n-1} + \omega_g^{n-2} \wedge \omega_{g'} + \cdots + \omega_{g'}^{n-1}) \\ &= -\frac{1}{V} \int_M \partial(\phi_-^p) \wedge \bar{\partial} \phi_- \wedge (\omega_g^{n-1} + \omega_g^{n-2} \wedge \omega_{g'} + \cdots + \omega_{g'}^{n-1}) \\ &= -\frac{p}{V} \int_M \phi_-^{p-1} \partial \phi_- \wedge \bar{\partial} \phi_- \wedge (\omega_g^{n-1} + \omega_g^{n-2} \wedge \omega_{g'} + \cdots + \omega_{g'}^{n-1}) \\ &= -\frac{p}{V} \int_M \phi_-^{\frac{p-1}{2}} \partial \phi_- \wedge \phi_-^{\frac{p-1}{2}} \bar{\partial} \phi_- \wedge (\omega_g^{n-1} + \omega_g^{n-2} \wedge \omega_{g'} + \cdots + \omega_{g'}^{n-1}) \\ &= -\frac{4p}{(p+1)^2} \frac{1}{V} \int_M \partial \left(\phi_-^{\frac{p+1}{2}} \right) \wedge \bar{\partial} \left(\phi_-^{\frac{p+1}{2}} \right) \wedge (\omega_g^{n-1} + \omega_g^{n-2} \wedge \omega_{g'} + \cdots + \omega_{g'}^{n-1}) \\ &\leq -\frac{4p}{(p+1)^2} \frac{1}{V} \int_M \partial \left(\phi_-^{\frac{p+1}{2}} \right) \wedge \bar{\partial} \left(\phi_-^{\frac{p+1}{2}} \right) \wedge \omega_g^{n-1} \\ &= -\frac{4p}{(p+1)^2} \frac{1}{V} \int_M \left| \nabla \phi_-^{\frac{p+1}{2}} \right|^2 \omega_g^n \end{aligned} \quad (3.5)$$

where $\omega_{g'} = \omega_g + \partial \bar{\partial} \phi$. Using $\sup_M (e^{f_s} - 1) \leq C$ and $1 \leq \phi_-$, we have

$$-\frac{1}{V} \int_M \phi_-^p (\omega_g^n - (\omega_g + \partial \bar{\partial} \phi)^n) = \frac{1}{V} \int_M \phi_-^p (e^{f_s} - 1) \omega_g^n \leq \frac{C}{V} \int_M \phi_-^{p+1} \omega_g^n$$

Thus,

$$\frac{4p}{(p+1)^2} \frac{1}{V} \int_M \left| \nabla \phi_-^{\frac{p+1}{2}} \right|^2 \omega_g^n \leq \frac{C}{V} \int_M \phi_-^{p+1} \omega_g^n = \frac{C}{V} \int_M |\phi_-|^{p+1} \omega_g^n.$$

It follows from this and the Sobolev inequality in Lemma 3.4 that

$$\begin{aligned} & \frac{4p}{(p+1)^2} \left\{ C_1 \left(\frac{1}{V} \int_M |\phi_-|^{\frac{n(p+1)}{n-1}} \omega_g^n \right)^{\frac{n-1}{n}} - \frac{C_2}{V} \int_M |\phi_-|^{p+1} \omega_g^n \right\} \\ & \leq \frac{C}{V} \int_M |\phi_-|^{p+1} \omega_g^n. \end{aligned} \quad (3.6)$$

Let $p_0 = 2$, and $p_i = \frac{n}{n-1}p_{i-1}$ for $i \geq 1$. It follows from (3.6) that

$$\|\phi\|_{p_i} \leq (Cp_{i-1})^{\frac{1}{p_{i-1}}} \|\phi\|_{p_{i-1}}, \quad (3.7)$$

where $\|\cdot\|_p$ denotes the L^p -norm. On the other hand, we have

$$0 \leq \frac{1}{V} \int_M |\nabla \phi_-|^2 \omega_g^n \leq \frac{C}{V} \int_M |\phi_-| \omega_g^n.$$

Thus, by the Poincaré inequality, we deduce from this that $\|\phi_-\|_2 \leq C\|\phi_-\|_{L^1}$. Therefore, by (3.7) and letting $i \rightarrow \infty$, $\|\phi\|_{C^0} \leq C$.

Therefore, we have shown that the equation (3.2)_{0,s} is solvable for any $s \in [0, 1]$. In particular, we have proved Theorem 1.5, (1).

Case 2. $\mu = -1, s = 1$ and $0 < t \leq 1$.

Suppose ϕ is a solution of (3.2)_{t,1}. Let x_0 be a point in M where ϕ attains its maximum. Thus, $\partial\bar{\partial}\phi \leq 0$, by (3.2)_{t,1}, and it follows that $e^{f_1+t\phi} \leq 1$ at x_0 , i.e., $f_1 + t\phi \leq 0$ at x_0 . Hence,

$$t \sup_M \phi = t\phi(x_0) \leq -f_1(x_0) \leq \sup_M (-f_1) \leq \|f_1\|_{C^0}.$$

Similarly, one can show $-t \inf_M \phi \leq \|f_1\|_{C^0}$. Therefore, $\|\phi\|_{C^0} \leq \frac{1}{t} \|f_1\|_{C^0}$. Because the right side is uniformly bounded by $\frac{C}{t}$ for some constant C , we have proved Theorem 1.5, (2).

It remains to consider the case $c_1(M) > 0$, i.e., $\mu = 1$. In rest of this section, we will bound the C^0 -norm of any solution ϕ of (3.2)_{t,1} in terms of the difference of the two averages of ϕ with respect to g and g' . The above discussions imply that (3.2)_{t,1} is solvable if the parameter t is small. So we may assume that $t \geq \delta(M, g) > 0$ in the following.

Let G be a maximal compact subgroup in the group $\text{Aut}(M)$ of all holomorphic automorphisms of M . Then we may assume that g is G -invariant, i.e., $\sigma^*g = g$ for any $\sigma \in G$. We define

$$P_G(M, g) = \{\phi \in C^\infty(M, \mathbb{R}) \mid \omega_g + \partial\bar{\partial}\phi > 0 \text{ and } \phi \text{ is } G\text{-invariant}\} \quad (3.8)$$

Then we can introduce a functional $I(\cdot)$ on $P_G(M, g)$ by

$$I(\phi) = \frac{1}{V} \int_M (\phi(\omega_g^n - (\omega_g + \partial\bar{\partial}\phi)^n)) \quad (3.9)$$

where $\phi \in P_G(M, g)$, and $V = \int_M \omega_g^n = \int_M \omega_{g'}^n$. The integral in (3.9) is the difference of two averages of ϕ with respect to two Kähler metrics. It is easy to see the following:

- (1) $I(\phi) \geq 0$.
- (2) We can solve (3.2)_{t,1} among G -invariant functions.

Lemma 3.5. *For $t \geq \delta > 0$, there is a constant C_δ such that*

$$\|\phi\|_{C^0} \leq C_\delta(1 + I(\phi))$$

for any solution ϕ of (3.2)_{t,1}.

Proof: We write $\omega_{g'} = \omega_g + \partial\bar{\partial}\phi > 0$. Let g' be the Kähler metric with the Kähler form $\omega_{g'}$. We have shown before that $\text{Ric}(g') \geq tg'$. Thus the Sobolev constants in Lemma 3.4 are independent of t . Put $\psi = 1 + \sup_M \phi - \phi$. Then $\psi \geq 1$ and

$$I(\phi) = \frac{1}{V} \int_M (-\psi)(\omega_g^n - \omega_{g'}^n) = -\frac{1}{V} \int_M \psi \omega_g^n + \frac{1}{V} \int_M \psi \omega_{g'}^n \quad (3.10)$$

Since $\Delta\psi \leq n$, using the Green formula, one can show as in Case 1 that the first integral on the right side is uniformly bounded. Hence,

$$I(\phi) \geq \frac{1}{V} \int_M |\psi| \omega_{g'}^n - C \quad (3.11)$$

Let Δ' be the Laplacian of the metric g' . Then

$$\Delta'\phi = \text{tr}_{g'} \partial\bar{\partial}\phi = n - \text{tr}_{g'}(g) \quad (3.12)$$

This implies $-\Delta'\psi < n$. Multiplying this inequality by ψ^p and integrating by parts, we obtain

$$\frac{4p}{(p+1)^2} \frac{1}{V} \int_M \left| \nabla' \psi^{\frac{p+1}{2}} \right|^2 \omega_{g'}^n = -\frac{1}{V} \int_M \psi^p \Delta' \psi \omega_{g'}^n \leq \frac{n}{V} \int_M \psi^p \omega_{g'}^n \quad (3.13)$$

By the Moser iteration, we obtain

$$\sup_M \psi \leq C \|\psi\|_{L^2(g')}$$

On the other hand, using (3.13) for $p = 1$ and Lemma 3.1, we deduce $\|\psi\|_{L^2(g')} \leq C \|\psi\|_{L^1(g')}$. Therefore, by (3.11),

$$\|\phi\|_{C^0} \leq \sup_M \psi + C' \leq C(1 + I(\phi))$$

The lemma is proved.

Let us summarize our discussions in Case 3.

Proposition 3.6. *Assume that $\mu = 1$. Then M admits a Kähler-Einstein metric if $I(\phi) \leq C$ for any solution ϕ of (3.2)_{t,1}, where C is a uniform constant.*

Question 3.7. *How do we bound $I(\phi)$ for a solution ϕ of (3.2)_{t,1}?*

In general, the functional $I(\cdot)$ is not bounded for the solutions of (3.2)_{t,1}. As we will see in Section 6, there are compact Kähler manifolds which have positive first Chern class and do not admit any Kähler-Einstein metrics (cf. [Mat], [Fu1]).

4. Existence of Kähler-Einstein Metrics, II

In this section, we assume that M is a compact Kähler manifold with positive first Chern class. We will discuss the existence of Kähler-Einstein metrics on M . In particular, we will outline a proof of the main theorem in [T2] (Theorem 1.5, (4)). This theorem states that a complex surface with positive first Chern class admits a Kähler-Einstein metric if and only if its Lie algebra of the holomorphic vector fields is reductive.

Recall the complex Monge-Ampere equation:

$$\begin{cases} (\omega_g + \partial\bar{\partial}\phi)^n = e^{f-t\phi}\omega_g^n \\ \omega_g + \partial\bar{\partial}\phi > 0 \end{cases} \quad (4.1)_t$$

where g is a Kähler metric with the Kähler form ω_g in $c_1(M)$, and f is determined by

$$\Delta_g f = s(g) - n, \quad \int_M (e^f - 1)\omega_g^n = 0 \quad (4.2)$$

where $s(g)$ denotes the scalar curvature of g . As we said before, the existence of Kähler-Einstein metrics is equivalent to solvability of $(4.1)_1$.

Let us first give one sufficient condition for solvability of $(4.1)_1$. We introduce

$$J(\phi) = \int_0^1 \frac{I(s\phi)}{s} ds \quad (4.3)$$

The following lemmata can be easily proved by straightforward computations (cf. [T1]).

Lemma 4.1. $\frac{1}{n+1}I(\phi) \leq I(\phi) - J(\phi) \leq \frac{n}{n+1}I(\phi)$

Lemma 4.2. Let ϕ_t be a path in $P_G(M, g)$. Then

$$\frac{d}{dt}(I(\phi_t) - J(\phi_t)) = -\frac{1}{V} \int_M \phi_t \Delta_t \dot{\phi}_t (\omega_g + \partial\bar{\partial}\phi_t)^n \quad (4.4)$$

where Δ_t denotes the Laplacian of the Kähler metric with the Kähler form $\omega_g + \partial\bar{\partial}\phi_t$.

Proposition 4.3. (cf. [T1], Proposition 2.3) If ϕ_t is a solution of $(4.1)_t$, then $I(\phi_t) - J(\phi_t)$ is an non-decreasing function on t , moreover, we have

$$J(\phi_t) - \int_M \phi_t \omega_g^n = -\frac{1}{t} \int_0^t (I(\phi_s) - J(\phi_s)) ds \quad (4.5)$$

where $\omega_t = \omega_g + \partial\bar{\partial}\phi_t$.

Proof: Differentiating $(4.1)_t$, we have

$$\Delta_t \dot{\phi}_t = -t\dot{\phi}_t - \phi_t; \quad (4.6)$$

together with (4.4), this implies

$$\frac{d}{dt}(I(\phi_t) - J(\phi_t)) = \frac{1}{V} \int_M (\Delta_t \dot{\phi}_t + t\dot{\phi}_t) \Delta_t \dot{\phi}_t (\omega_g + \partial\bar{\partial}\phi_t)^n.$$

This last integral is always non-negative, since the first eigenvalue of Δ_t is not smaller than t (cf. Lemma 3.1). Therefore, $I(\phi_t) - J(\phi_t)$ is non-decreasing.

Furthermore, using (4.4) and (4.6), we deduce

$$\begin{aligned} t \frac{d}{dt}(I(\phi_t) - J(\phi_t)) &= \frac{t}{V} \int_M \phi_t(t\dot{\phi}_t + \phi_t)\omega_t^n \\ &= -\frac{d}{dt} \left(\frac{t}{V} \int_M \phi_t \omega_t^n \right) \end{aligned} \quad (4.7)$$

Here we have used the fact that

$$\omega_t^n = e^{f-t\phi_t} \omega_g^n.$$

Integrating (4.7), we obtain

$$-J(\phi_t) + \frac{1}{V} \int_M \phi_t \omega_g^n = I(\phi_t) - J(\phi_t) + \frac{1}{V} \int_M \phi_t \omega_t^n = \frac{1}{t} \int_0^t (I(\phi_s) - J(\phi_s)) ds.$$

Proposition 4.3 is proved.

Now we can easily prove the following theorem, which was essentially proved in [T1].

Theorem 4.4. *Suppose that there is an increasing function $h(s)$ with $\lim_{s \rightarrow \infty} h(s) = \infty$, such that for any solution ϕ_t of (4.1)_t,*

$$J(\phi_t) - \frac{1}{V} \int_M \phi_t \omega_g^n \geq h(I(\phi_t)) \quad (4.8)$$

Then (4.1)₁ is solvable, moreover, there is a uniform bound on solutions of (4.1)_t.

Proof: Let ϕ_t be any solution of (4.1)_t. It follows from (4.5) and (4.8) that $h(I(\phi_t)) \leq C$ for a uniform constant C , so $I(\phi_t)$ is uniformly bounded. Then the theorem follows from Proposition 3.6.

This simple theorem has a few corollaries. We will denote by G a maximal compact subgroup, possibly trivial or finite, in the group $\text{Aut}(M)$ of holomorphic automorphisms of M . We define a functional

$$F_g(\phi_t) = I(\phi_t) - J(\phi_t) + \frac{1}{V} \int_M \phi_t \omega_t^n = \frac{1}{V} \int_M \phi_t \omega_g^n - J(\phi_t) \quad (4.9)$$

We may take the Kähler metric g to be G -invariant.

Definition 4.5. *We say that F_g is proper if there is an increasing h with $\lim_{s \rightarrow \infty} h(s) = \infty$, such that*

$$F_g(\phi_t) \leq -h(I(\phi)) \quad (4.10)$$

for any G -invariant ϕ satisfying: $\omega_g + \partial\bar{\partial}\phi > 0$ and $\int_M (e^{f-\phi} - 1)\omega_g^n = 0$. If (4.10) holds for functions in a subset E in $C^\infty(M)$, we say F_g is proper on E .

Corollary 4.6. *If the functional F_g is proper, then M admits a Kähler-Einstein metric.*

Proof: Let ϕ_t be any solution of (4.1)_t. Choose c such that $\int_M (e^{f-\phi_t-c} - 1)\omega_g^n = 0$. Using the concavity of the logarithmic function and $\int_M (e^{f-t\phi_t} - 1)\omega_g^n = 0$, we have

$$\frac{1}{V} \int_M (c + (1-t)\phi_t)\omega_t^n \geq 0$$

By the properness of F_g , we deduce

$$F_g(\phi_t) = F_g(\phi_t + c) - \frac{c}{V} \int_M \omega_g^n \leq -h(\phi_t) + \frac{(1-t)}{V} \int_M \phi_t \omega_t^n$$

In the last section, we proved

$$\int_M \phi_t \omega_t^n \leq C$$

for some uniform constant $C > 0$. Then this corollary follows from Theorem 4.4.

Remark: This corollary is contained in [DT1]. A weaker form of it is already proved in [Di]. One can also show a weaker converse of this corollary, namely, if M has a Kähler-Einstein metric, then F_g is bounded from below. One can prove an analogy of Corollary 4.6 in terms of Mabuchi's K-energy (cf. [BM]). In fact, in proving the weaker converse of Corollary 4.6, we need to use an idea from [BM].

Let us give another corollary of Theorem 4.4. We denote by $P_G(M, g)$ the set of G -invariant functions ϕ satisfying: $\omega_g + \partial\bar{\partial}\phi > 0$ and $\sup_M \phi = 0$. Following [T1], we define

$$\alpha(M) = \sup \left\{ \alpha \mid \text{For some } C_\alpha, \int_M e^{-\alpha\phi} \omega_g^n \leq C_\alpha \text{ for all } \phi \in P_G(M, g) \right\}$$

One can show that $\alpha(M)$ is positive and an analytic invariant. Clearly, $\alpha(CP^n) = \infty$.

Corollary 4.7. ([T1], Theorem 4.1) *If $\alpha(M) > \frac{n}{n+1}$, then M admits a Kähler-Einstein metric.*

Proof: Choose $\alpha > \frac{n}{n+1}$ such that for any ϕ in $P_G(M, g)$,

$$\int_M e^{-\alpha\phi} \omega_g^n \leq C$$

where C is a uniform constant. Let ϕ_t be a G -invariant solution of (4.1)_t. Then $\phi_t - \sup_M \phi_t$ is in $P_G(M, g)$, and

$$\int_M e^{-\alpha(\phi_t - \sup_M \phi_t)} \omega_g^n \leq C$$

On the other hand, $\int_M (e^{f-t\phi_t} - 1)\omega_g^n = 0$, and it follows from the concavity of the logarithmic function,

$$\begin{aligned} \log \frac{C}{V} &\geq \log \int_M e^{-f+\alpha \sup_M \phi_t + (t-\alpha)\phi_t} \frac{e^{f-t\phi_t} \omega_t^n}{V} \\ &\geq \int_M \left(-f + \alpha \sup_M \phi_t + (t-\alpha)\phi_t \right) \frac{\omega_t^n}{V}. \end{aligned}$$

Consequently,

$$\int_M \phi_t \omega_g^n \leq -\frac{t-\alpha}{\alpha} \int_M \phi_t \omega_t^n + C'$$

where C' is a constant. Hence, by Lemma 4.1, we deduce

$$\begin{aligned} -F_g(\phi_t) &\geq -\frac{1}{n+1} \int_M \phi_t \omega_t^n - \frac{n}{n+1} \int_M \phi \omega_g^n \\ &\geq -\frac{(n+1)\alpha - n}{\alpha} \int_M \phi_t \omega_t^n - \frac{n}{n+1} C' \\ &\geq \frac{(n+1)\alpha - n}{\alpha} I(\phi_t) - \frac{n}{n+1} C', \end{aligned}$$

i.e., F_g is proper. Then this corollary follows from Theorem 4.4.

This provides an effective method to check whether M has a Kähler-Einstein metric. The invariant $\alpha(M)$ has been estimated for many manifolds M . For instance, (1) $\alpha(M) > \frac{n}{n+1}$ for Fermat hypersurfaces in CP^{n+1} of degree $d = n, n+1$ in [T1] and $\frac{n+1}{2} < d < n$ in [Na]; (2) for any $3 \leq m \leq 8$, $\alpha(M) > \frac{2}{3}$ for any blow-up M of CP^2 at m points with certain configuration; (3) $\alpha(M) \geq 1$ is proved in [T5] and [Na] for manifolds with many symmetries, such as, any smooth intersections of two quadratic hypersurfaces in CP^4 . However, there is an M such that $\alpha(M) \leq \frac{n}{n+1}$, even in the case of complex surfaces. Therefore, more needs to be done to understand when M has a Kähler-Einstein metric.

Next we consider complex surfaces with positive first Chern class, i.e., Del-Pezzo surfaces. It is known (cf. [BPV]) that any Del-Pezzo surface M is biholomorphic to either CP^2 , or $CP^1 \times CP^1$, or the blow-up of CP^2 at m points in the general position ($1 \leq m \leq 8$). Clearly, the first two surfaces admit a Kähler-Einstein metric, i.e., the canonical metric. It follows from an old theorem of Matsushima ([Mat]) that the blow-up surface with $m = 1, 2$ does not have any Kähler-Einstein metrics (cf. Section 6 for an alternative proof of this). The following theorem was proved in [T2]. This solves Calabi's problem on Kähler-Einstein metrics for complex surfaces.

Theorem 4.8 ([T2]). *Any blow-up surface of CP^2 at m points in general position, where $3 \leq m \leq 8$, admits Kähler-Einstein metrics. Consequently, a Del-Pezzo surface admits Kähler-Einstein metrics if and only if its Lie Algebra of holomorphic fields is reductive.*

In the following, we discuss the main ideas in the proof of this theorem.

Lemma 4.9. ([TY4]) *For any $3 \leq m \leq 8$, there is a configuration of points in CP^2 such that the blow-up M of CP^2 at those points admits a Kähler-Einstein metric.*

This follows from Corollary 4.7 by showing $\alpha(M) > 2/3$. For example, if $m = 5$, then M is a complete intersection of two quadratics in CP^4 , say,

$$M = \left\{ [z_0, \dots, z_4] \in CP^4 \mid \sum_{i=0}^4 z_i^2 = 0, \sum_{i=0}^4 \lambda_i z_i^2 \right\} \quad (4.11)$$

Let G be a maximal compact subgroup containing the reflection $\sigma_i : [z_0, \dots, z_i, \dots, z_4] \mapsto [z_0, \dots, -z_i, \dots, z_4]$ ($i = 0, \dots, 4$). Let π_{ij} be the projection onto the hyperplane $\{z_i z_j = 0\}$ ($i \neq j$). Then for any G -invariant function ϕ , there are ϕ_{ij} such that $\pi_{ij}^* \phi_{ij} = \phi$ and $\omega_{FS} + \phi_{ij} \geq 0$, where ω_{FS} is the Kähler form of the Fubini-Study metric. This last inequality implies

$$\left(\frac{\sqrt{-1}}{2\pi} \right)^2 \int_{CP^2} e^{-\phi_{ij} + \sup_M \phi} \omega_{FS}^2 \leq C$$

for some uniform constant C . Since $\bigcap_{i \neq j} \{\det(D\pi_{ij}) = 0\} = \emptyset$, we deduce from the above

$$\left(\frac{\sqrt{-1}}{2\pi} \right)^2 \int_M e^{-\phi + \sup_M \phi} \omega^2 < C'$$

for some constant C' . Therefore, $\alpha(M) \geq 1$ and M has a Kähler-Einstein metric.

Now we fix m . The moduli space $\mathcal{C}_m$ of the blow-ups at m points is just the space of configurations of points in general position. In particular, $\mathcal{C}_m$ is connected. We define $\mathcal{C}_m'$ to be the set of those blow-ups in $\mathcal{C}_m$ which admit a Kähler-Einstein metric. We need to prove that $\mathcal{C}_m = \mathcal{C}_m'$.

It follows from Lemma 4.9 that $\mathcal{C}_m'$ is nonempty. Note that $\mathcal{C}_m$ consists of a single Del-Pezzo surface for $m = 3, 4$, and consequently, $\mathcal{C}_m = \mathcal{C}_m'$ in these cases. So we may assume $5 \leq m \leq 8$. In fact, it is not hard to show that any smooth complete intersection of two quadratics is of the form in (4.11) after a linear transformation. Hence, $\mathcal{C}_5 = \mathcal{C}_5'$. But we will give a unified proof for $m \geq 5$ in the following.

Choose a smooth family of Kähler metrics ω_M with the Kähler class $c_1(M)$, where $V \in \mathcal{C}_m$. As we showed in section 3, the blow-up M admits a Kähler-Einstein metric if and only if the following Monge-Ampère equation is solvable,

$$(\omega_M + \partial\bar{\partial}\phi)^2 = e^{f_M - \phi} \omega_M^2, \quad \omega_M + \partial\bar{\partial}\phi > 0. \quad (4.12)$$

where f_M is determined by ω_M as in (4.2). Clearly, (4.12) depends smoothly on M in $\mathcal{C}_m$. Any blow-up surface in $\mathcal{C}_m$ ($m \geq 5$) has no nontrivial holomorphic fields. Then using the Implicit Function Theorem, one can easily prove that the set of M such that (4.12) is solvable is open. This implies that $\mathcal{C}_m'$ is open.

The most difficult part is to prove that $\mathcal{C}_m'$ is closed. Following Yau's computations for higher order estimates (cf. [Y1]), one can reduce the closedness to an *a priori* C^0 -estimate for the solutions of (4.12) for M in any compact subset of $\mathcal{C}_m$.

Such an *a priori* estimate will be achieved in two steps. First we prove a uniform partial C^0 -estimate for the solutions of (4.12).

Let $\|\cdot\|_M$ (resp. $\|\cdot\|_{M,E}$) be the hermitian metric on the anticanonical bundle K_M^{-1} , such that ω_M (resp. $\omega_{M,E}$) is its curvature form, where $M \in \mathcal{C}_m'$ and $\omega_{M,E}$ is a Kähler-Einstein metric on M . Let ϕ_M be the solution of (4.12). Then

$$\omega_{M,E} = \omega_M + \partial\bar{\partial}\phi_M$$

For any basis $\{s_i\}$ of $H^0(V, K_M^{-l})$, we have

$$\phi_M = -\frac{1}{l} \log \left(\frac{\sum_i \|s_i\|_{M,E}^2}{\sum_i \|s_i\|_M^2} \right) + \text{const.} \quad (4.13)$$

Define a function $\psi_{M,l}$ as follows: choose $\{s_i\}$ to be an orthonormal basis with respect to $\|\cdot\|_{M,E}$, put $\psi_{M,l} = \frac{1}{l} \log(\sum_i \|s_i\|_{M,E}^2)$. Note that $\psi_{M,l}$ is independent of choices of $\{s_i\}$. The key observation is the following lemma.

Lemma 4.10. *If $-\psi_{M,l} \leq C$, then there are positive numbers $\lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_N = 1$, where $N = \dim H^0(M, K_M^{-l})$, and an orthonormal basis $\{s_i\}$ of $H^0(M, K_M^{-l})$ with respect to $\|\cdot\|_M$, such that*

$$\left| \phi_M - \sup_M \phi_M - \frac{1}{l} \log \left(\sum_i \lambda_i^2 \|s_i\|_M^2 \right) \right| \leq C \quad (4.14)$$

In particular, $\phi_M - \sup_M \phi_M$ is bounded away from the divisor $\{s_N = 0\}$.

Proof: By using unitary transformations on C^N , we may choose an orthonormal basis $\{s_i\}$ of $(H^0(M, K_M^{-l}), \|\cdot\|_{M,E})$, such that $\langle s_i, s_j \rangle_M = \lambda_i^2 \delta_{ij}$, where $\langle \cdot, \cdot \rangle_M$ is the induced inner product by $\|\cdot\|_M$. Then the lemma follows from (4.13).

Theorem 4.11. ([T2]) *For any $k \geq 1$, there is a uniform constant C_k , such that for any $M \in \mathcal{C}_m'$ as above, $|\psi_{M,6k}| \leq C_k$. In particular, we have (4.14) for ϕ_M with $C = C_k$.*

Idea of Proof: First, it is not very hard to reduce the theorem to the case that k is sufficiently large (cf. [T2]).

We will use $C = C_k$ to denote a uniform constant, whose actual value may vary in different places.

Obviously, $|\psi_{M,6k}| \leq C$ is equivalent to the condition that for any $x \in M$, there is a section s of $H^0(M, K_M^{-6k})$ satisfying: $\|s\|_{M,E}(x) = 1$ and $\int_M \|s\|_{M,E}^2 \omega_{M,E}^2 \leq C$. Furthermore, by using Hörmander's L^2 -estimates, one needs only to construct s locally in a geodesic ball of fixed radius r , where r depends only on the dimension. This will follow from a deeper understanding the geometric structure of geodesic balls $B_r(x)$, where r is a small number.

We first list a few facts:

- 1) if $R(\omega_{M,E})$ denotes the curvature tensor of the Kähler-Einstein metric $\omega_{M,E}$, then the L^2 -norm of $R(\omega_{M,E})$ is bounded by C . This follows from the Gauss-Bonnet-Chern theorem.

2) There is an $\epsilon > 0$, such that for any $4r > \rho > 0$, if

$$\frac{1}{V} \int_{B_{2\rho}(x)} |R(\omega_{M,E})|^2 \omega_{M,E}^2 \leq \epsilon^2,$$

where $V = \int_M \omega_M^2$, then there is a diffeomorphism $\phi : B_\rho(x) \mapsto C^2$ satisfying: $B_\rho(0) \subset \phi(B_\rho(x))$ and

$$\|\omega_{M,E} - \phi^*(\omega_0)\|_{C^3} \leq C\epsilon,$$

where ω_0 is the flat metric on C^2 .

It follows from 1) that there are $x_1, \dots, x_s$ in M with s uniformly bounded, such that for any x outside $\bigcup_i B_{4r}(x_i)$,

$$\frac{1}{V} \int_{B_{2r}(x)} |R(\omega_{M,E})|^2 \omega_{M,E}^2 \leq \epsilon^2,$$

in particular, by 2), for such a x , $(B_r(x), \omega_{M,E})$ is uniformly equivalent to an euclidean ball.

Fix a large L . We choose r and ϵ sufficiently small. Then one can show that $(B_{Lr}(x_i) \setminus B_{2r}(x_i), \omega_{M,E})$ is close to the annulus $B_{Lr}(0) \setminus B_{2r}(0)$ in C^2/Γ_i , where Γ_i is a finite group in $U(2)$ with its only fixed point at 0. By the Bishop Volume Comparison Theorem, one can easily give a uniform upper bound on the order of Γ_i .

Next, one shows that the universal covering of $B_{Lr}(x_i)$ is biholomorphic to $B_{Lr}(0) \cap \Sigma_i$ with K_{Σ_i} trivial. In fact, Σ_i is a smooth deformation of C^2/Γ_i' , where Γ_i' is the subgroup $\Gamma_i \cap SU(2)$. The singularity of the latter space is called a Klein singularity, and Σ_i is a hypersurface in C^3 . One can further show that the order of Γ_i/Γ_i' divides 6.

Therefore, K_M^{-6} is trivial on any geodesic ball $B_r(x)$ with $x \in M$. It follows that for any x , there is a holomorphic section s of K_M^{-6} satisfying:

$$\|s\|_{M,E}(x) = 1 \quad \text{and} \quad \frac{1}{V} \int_{B_r(x)} \|s\|_{M,E}^2 \omega_{M,E}^2 \leq C.$$

Then the theorem follows.

Let $l = 6k$ be given as in the last theorem. We define invariants

$$\alpha_{l,b}(M) = \sup\{\alpha \mid \exists C_\alpha > 0, \text{ s.t. } \frac{1}{V} \int_M \frac{1}{(\sum_{i=1}^b \|s_i\|_g^2)^{\frac{q}{l}}} \omega_M^2 \leq C_\alpha$$

$$\text{for any } \{s_i\}_{1 \leq i \leq b} \subset H^0(M, K_M^{-l}) \text{ with } \langle s_i, s_j \rangle_g = \delta_{ij}\} \quad (4.15)$$

where $\langle \cdot, \cdot \rangle_g$ denotes the inner product on $H^0(M, K_M^{-l})$ induced by g . It is easy to show that all $\alpha_{l,b}(M)$ are holomorphical invariants.

Proposition 4.12. *Let M, ω_M, ϕ_M be as above. Assume that*

$$\frac{1}{\alpha_{l,1}(M)} + \frac{1}{\alpha_{l,2}(M)} < 3$$

In particular, $\alpha_{l,2}(M) > \frac{2}{3}$. Then $|\phi_M| \leq C$ for some uniform constant C .

Proof: By Theorem 4.11, one can find an orthonormal basis $\{s_i\}$ of $H^0(M, K_M^{-l})$ with respect to $\|\cdot\|_M$ and positive numbers $0 < \lambda_1 \leq \dots \leq \lambda_N = 1$ such that

$$\left| \phi_M - \sup \phi_M - \frac{1}{l} \log \left(\sum_i \lambda_i \|s_i\|_M^2 \right) \right|_{C^0} \leq C \quad (4.16)$$

We may assume that λ_{N-1} is very small, for otherwise the proposition follows easily.

Using the concavity of the logarithmic function and (4.12), one has

$$\sup_M \phi_M + \frac{(1 - \alpha_{l,2}(M) - \varepsilon)}{\alpha_{l,2}(M) - \varepsilon} \left(\frac{1}{V} \int_M \phi_M \omega_{M,E}^2 \right) \leq -\frac{1}{l} \log \lambda_{N-1} + C_\varepsilon, \quad (4.17)$$

where $\varepsilon > 0$ is a small number, and C_ε is a constant depending only on ε .

Next, one can deduce from (4.5)

$$-\frac{1}{V} \int_M \phi_M \omega_{M,E}^2 \leq \frac{2}{V} \int_M \phi_M \omega_M^2 - \frac{1}{V} \int_M \partial \phi_M \wedge \bar{\partial} \phi_M \wedge \omega_M \quad (4.18)$$

Let us estimate the second integral on the right of (4.18).

Claim: For any $\varepsilon > 0$, there exists a $C'_\varepsilon > 0$, independent of ϕ_M , such that

$$\frac{1}{V} \int_M \partial \phi_M \wedge \bar{\partial} \phi_M \wedge \omega_M \geq -\frac{(1-\varepsilon)}{l} c_1(M)^2 \log \lambda_{N-1} - C'_\varepsilon \quad (4.19)$$

Proof: Write $D = \{s_N = 0\} = \sum_{j=1}^J \mu_j E_j$, where each E_j is irreducible and μ_j is the multiplicity of E_j in D . Let η be small. Choose a finite covering $\{U_\beta\}$ of $D_{red} \setminus \text{Sing}(D_{red})$ such that 1) $U_\beta \cap U_{\beta'} = \emptyset$ for $\beta \neq \beta'$; 2) for each β , there is a coordinate system $(z_{\beta_1}, \dots, z_{\beta_n})$ on an open neighborhood of $\bar{U}_\beta$ in M such that $U_\beta = \{|z_{\beta_j}| < \eta, j = 1, 2\}$, $D_{red} \cap U_\beta = \{z_{\beta_2} = 0\}$ and z_{β_1} is tangent to $D_{red} \cap U_\beta$; 3) for a sufficiently small $\delta = \delta(\varepsilon)$, we have

$$\sum_{j=1}^J \sum_{\beta} \frac{\mu_j}{V} \int_{E_j \cap U_\beta} \omega_M \geq l - \delta$$

By taking C'_ε sufficiently large, we may assume $\lambda_{N-1}^{1/2-\delta} < \eta^3$. Let f_β be the local defining holomorphic function of D in U_β with leading coefficient 1. Then by (4.16),

$$\begin{aligned} & \frac{1}{V} \int_M \partial \phi_M \wedge \bar{\partial} \phi_M \wedge \omega_M \\ & \geq -C + \frac{(1-\delta)}{l^2} \sum_{\beta} \frac{1}{V} \int_{U_\beta \cap \{\lambda^{\frac{1}{2-\delta}} \leq |f_\beta| \leq \eta\}} \frac{\partial f_\beta \wedge \bar{\partial} f_\beta \wedge \omega_M}{|f_\beta|^2} \\ & \geq -C + \frac{1}{l^2} (1-\delta)^2 (-\log \lambda_{N-1} - C) \sum_{\beta,j} \frac{\mu_j}{V} \int_{U_\beta \cap E_j} \omega_M \\ & \geq -C - \frac{(1-\delta)^2}{l} \log \lambda_{N-1} \end{aligned}$$

Then the claim follows.

Putting (4.17)-(4.19) together, we obtain

$$-\frac{1}{V} \left(\frac{2\alpha_{l,2}(M) - 1}{\alpha_{l,2}(M)} - 8\varepsilon \right) \int_V \phi_V \omega_{M,E}^2 \leq \frac{1}{V} \int_M \phi_M \omega_M^2 + C_\varepsilon$$

On the other hand, by using the concavity of logarithmic function, we have

$$\frac{1}{V} \int_M \phi_M \omega_M^2 \leq -\frac{1}{V} \frac{1 - \alpha_{l,1}(M) - \epsilon}{\alpha_{l,1}(M)} \int_M \phi_M \omega_{M,E}^2 + C_\epsilon$$

By choosing ϵ sufficiently small, one can see from the above that $I(\phi_M)$ is uniformly bounded, whenever

$$\frac{2\alpha_{l,2}(M) - 1}{\alpha_{l,2}(M)} > \frac{1 - \alpha_{l,1}(M) - \epsilon}{\alpha_{l,1}(M)}.$$

This last inequality is just our assumption. Therefore the C^0 -norm of ϕ_M is uniformly bounded. The proposition is proved.

By straightforward computations (cf. [T2]), one can show that $\alpha_{6,1}(M) \geq \frac{2}{3}$ and $\alpha_{6,2}(M) > \frac{2}{3}$ for any M in $\mathcal{C}_m$ ($5 \leq m \leq 8$). Therefore, $\mathcal{C}_m'$ is closed. Theorem 4.8 is proved.

Finally, using the above arguments we outline a proof of the uniqueness theorem of Bando and Mabuchi (Theorem 1.5, (3)) in the case there are no nontrivial holomorphic vector fields. We refer the readers to their paper for the general case. In fact, most Kähler manifolds with positive first Chern class do not have any nontrivial holomorphic fields.

For reader's convenience, we state the theorem here.

Theorem 4.13. ([BM]) *Let M be a compact Kähler manifold with positive first Chern class and without any nontrivial holomorphic vector fields. Then there is at most one Kähler-Einstein metric on M .*

Proof: Assume that g is a Kähler-Einstein metric with $\omega_g \in c_1(M)$. Then the equation (3.2) _{$t,1$} becomes

$$\begin{cases} (\omega_g + \partial\bar{\partial}\phi)^n = e^{-t\phi} \omega_g^n \\ \omega_g + \partial\bar{\partial}\phi > 0 \end{cases} \quad (4.20)_t$$

In particular, 0 is always a solution. We need to prove that 0 is the only solution. Suppose that φ is any solution of (4.20)₁. We define

$$E = \{t \mid \text{there is a family } \{\varphi_s\}_{t \leq s \leq 1} \text{ with } \varphi_1 = \varphi, \text{ such that } \varphi_s \text{ solves (4.20)}_s\}. \quad (4.21)$$

Obviously, E contains 1. Since M has no nontrivial holomorphic vector fields, the linearized operator of (4.20) _{t} at any t is invertible (cf. lemma 3.1). It follows that E is open. As we showed before, the closedness of E follows from a uniform upper bound. By Proposition 4.3 and Lemma 3.5, the solutions of (4.20) _{t} have uniformly bounded C^0 -norm as t decreases from 1 to 0. Then it follows from Lemm 3.3 that E

is closed, so $0 \in E$. However, by the Maximum Principle, there is only one solution, i.e., 0, for $(4.20)_0$, and consequently, $\varphi_t = 0$ for all t by the construction of E and the Implicit Function Theorem. The theorem is proved.

5. Chow-Mumford Stability

In this section, we study relations between the existence of Kähler-Einstein metrics and the Chow-Mumford stability on algebraic manifolds. The stability of algebraic manifolds has been studied by many people, notably, Mumford, Gieseker, Viehweg, etc., using the geometric invariant theory (cf. [Md], [Vi]).

Let $\pi : \mathcal{X} \rightarrow Z$ be a $SL(N+1, C)$ -equivariant holomorphic fibration between smooth varieties, satisfying:

- (1) $\mathcal{X} \subset Z \times CP^N$ is a subvariety and generic fibers of π are smooth and of dimension n ; moreover, the action on $\mathcal{X}$ is induced by the canonical action of $SL(N+1, C)$ on CP^N ;
- (2) Let L be the hyperplane bundle over CP^N . Then $K_{\pi^{-1}(z)}^{\pm 1} = \mu L|_{\pi^{-1}(z)}$ for some $\mu \in Q$ and each $z \in Z_0$, where Z_0 is the subvariety of Z consisting of smooth fibers. Clearly, Z_0 is $SL(N+1, C)$ -invariant;

Remark: In general, if $\mathcal{X}$ and Z are only normal varieties, one may use resolution of singularities to reduce it to the above smooth case.

Next, we consider the virtual bundle

$$\mathcal{E} = (n+1)(\mathcal{K}^{-1} - \mathcal{K}) \otimes (\pi_2^* L - \pi_2^* L^{-1})^n - n\mu(\pi_2^* L - \pi_2^* L^{-1})^{n+1}, \quad (5.1)$$

where $\mathcal{K} = K_{\mathcal{X}} \otimes K_Z^{-1}$ is the relative canonical bundle, and π_i is the restriction to $\mathcal{X}$ of the projection from $Z \times CP^N$ onto its i^{th} -factor.

We define L_Z to be the inverse of the determinant line bundle $\det(\mathcal{E}, \pi)$.

A straightforward computation shows:

$$\begin{aligned} ch_{n+1}((n+1)(\mathcal{K}^{-1} - \mathcal{K}) \otimes (\pi_2^* L - \pi_2^* L^{-1})^n - n\mu(\pi_2^* L - \pi_2^* L^{-1})^{n+1}) \\ = 2^{n+1} ((n+1)c_1(\mathcal{K}^{-1})\pi_2^* c_1(L)^n - n\mu\pi_2^* c_1(L)^{n+1}). \end{aligned} \quad (5.2)$$

Therefore, by the Grothendick-Riemann-Roch Theorem,

$$c_1(L_Z) = 2^{n+1}\pi_{1*} ((n+1)c_1(\mathcal{K})\pi_2^* c_1(L)^n + n\mu\pi_2^* c_1(L)^{n+1}). \quad (5.3)$$

If L_Z is ample, then there is an embedding of Z into $PH^0(Z, L_Z^m)^*$ for some positive m . Let G be any one-parameter algebraic subgroup in $SL(N+1, C)$. We say $X_z = \pi^{-1}(z)$ is G -stable, if the orbit $G \cdot \tilde{z}$ in $H^0(Z, L_Z^m)^*$ is closed and the stabilizer G_z of z is finite, where $\tilde{z}$ is any lifting of z in $H^0(Z, L_Z^m)^* \setminus \{0\}$. We say X_z is G -semi-stable, if 0 is not in $G \cdot \tilde{z}$.

Theorem 5.1. *Let $\pi : \mathcal{X} \rightarrow Z$ be as above, and G be any one-parameter algebraic subgroup of $SL(N+1, C)$. We define Z_h to be the subvariety of Z consisting of all*

z' such that $X_{z'}$ has dimension higher than n . Assume that L_Z is ample over Z . Then for any z in Z_0 ,

1. if $\mu \leq 0$, then X_z is either G -stable with respect to L_Z or the closure $\overline{G \cdot z}$ intersects Z_h ;
2. If $\mu > 0$ and X_z has a Kähler-Einstein metric, then X_z is G -semi-stable with respect to L_Z or the closure $\overline{G \cdot z}$ intersects Z_h .

Remarks: The assumption in last theorem is often true for families of algebraic varieties, such as Hilbert schemes of subvarieties in a projective space. In fact, I believe that it can be proved in very general cases. We will give two such examples.

Examples: (1) Let $R_{n+k,d}$ be the space of all homogeneous polynomials on $\mathcal{C}^{n+k+1}$ of degree d . Put

$$Z = Z_1 \times \cdots \times Z_k, \quad Z_i = PR_{n+k,d_i}$$

where d_i are integers greater than one. Then Z classifies all complete intersections in $\mathcal{C}P^{n+k}$ defined by k homogeneous polynomials of degree $d_1, \dots, d_k$. Some of these intersections may be of dimension higher than n . Let $\mathcal{X}$ be the universal family of those intersections. Then one can show that the determinant line bundle L_Z is isomorphic to

$$2^{n+1} d_1 \cdots d_k \bigotimes_{i=1}^k \mathcal{O}_{\mathbb{P}^1} \left(n+1 - \frac{n+k+1 - \sum_{j=1}^k d_j}{d_i} \right)$$

Since each $d_i \geq 2$, $(n+1)d_i$ is greater than $n+k+1 - \sum_{j=1}^k d_j$. It follows that L_Z is ample on Z .

(2) Recall that $W = G(k, n)$ consists of all k -subspaces in $\mathcal{C}^n$. Let Q be the universal quotient bundle over W . Then for any $l \leq n-k$, $H^0(W, \wedge^l Q) = \wedge^l \mathcal{C}^n$ and $H^i(W, \wedge^l Q) = \{0\}$.

Let $\pi_i (i = 1, 2)$ be the projection from $W \times G(m, H^0(W, \wedge^l Q))$ onto its i^{th} -factor, where m is a positive integer, and let S be the universal bundle over $G(m, H^0(W, \wedge^l Q))$. Then there is a natural endomorphism over $W \times G(m, H^0(W, \wedge^l Q))$

$$\Phi : \pi_2^* S \mapsto \pi_1^* \wedge^l Q, \quad \Phi(v)|_{(x,P)} = v_x \in \wedge^l Q. \quad (5.4)$$

Naturally, one can regard Φ as a section in $\pi_2^* S^* \otimes \pi_1^* (\wedge^l Q)$.

We define

$$\mathcal{X} = \{(x, P) \in W \times G(m, H^0(W, \wedge^l Q)) \mid \Phi(x, P) = 0\}. \quad (5.5)$$

One can show that $\mathcal{X}$ is smooth.

If $L = \det(Q)$, then $c_1(L)$ is the positive generator of $H^2(W, \mathbb{Z})$.

Consider the fibration $\pi = \pi_2|_{\mathcal{X}} : \mathcal{X} \mapsto Z$, where $Z = G(m, H^0(W, \wedge^l Q))$. Its generic fibers are smooth and of dimension $N = k(n-k) - m \binom{n-k}{l} > 0$.

Using the Adjunction Formula, one can show

$$c_1(\mathcal{K}) = (m \binom{n-k-1}{l-1} - n) \pi_1^* c_1(L) - \binom{n-k}{l} \pi_2^* c_1(S), \quad (5.6)$$

and

$$\mu = n - m \binom{n-k-1}{l-1}$$

Therefore, it follows from (5.3) that

$$c_1(L_Z) = 2^{N+1} \left((N+1) \binom{n-k}{l} \pi_2^* c_1(S^*) \pi_1^* c_1(L)^N - \mu \pi_1^* c_1(L)^{N+1} \right). \quad (5.7)$$

Then L_Z is ample if and only if

$$\begin{aligned} & \int_{G(k,n)} c_{\binom{n-k}{l}}(\wedge^l Q)^{m-1} \left((N+1) \binom{n-k}{l} c_{\binom{n-k}{l}}(\wedge^l Q) c_1(L)^N \right. \\ & \quad \left. - (n-m \binom{n-k-1}{l-1}) c_{\binom{n-k}{l}-1}(\wedge^l Q) c_1(L)^{N+1} \right) > 0. \end{aligned} \quad (5.8)$$

This last condition should be always true.

As a concrete example, we assume $n = 7, k = 4, l = 2, m = 3$. Then $\mathcal{X}$ parametrizes a family of Fano 3-folds, which the author learned from a talk by Mukai a few years ago. The integral in (5.8) becomes

$$\int_{G(4,7)} c_3(\wedge^2 Q)^2 (12c_3(\wedge^2 Q)c_1(Q)^3 - c_2(\wedge^2 Q)c_1(Q)^4). \quad (5.9)$$

A straightforward computation shows

$$c_3(\wedge^2 Q) = c_1(Q)c_2(Q) - c_3(Q), \quad c_2(\wedge^2 Q) = c_1(Q)^2 + c_2(Q) \quad (5.10)$$

It is well-known that $c_j(Q)$ is Poincare dual to the Schubert cycle $\sigma(j, 0, 0, 0)$, where $j = 1, 2, 3$. To define a Schubert cycle $\sigma(\lambda_1, \dots, \lambda_4)$, where $3 \geq \lambda_1 \geq \dots \geq \lambda_4 \geq 0$, we take any filtration $V_0 \subset V_1 \subset \dots \subset V_7 = \mathcal{O}^7$ with $\dim V_j = j$ and define

$$\sigma(\lambda_1, \dots, \lambda_4) = \{P \in G(4, 7) \mid \dim(V_{3+j-\lambda_j} \cap P) \geq j, 1 \leq j \leq 4\}$$

By using the standard Schubert calculus (cf. [GH]), one can show:

- (1) $c_3(\wedge^2 Q)$ is Poincare dual to $\sigma(2, 1, 0, 0)$;
- (2) $12c_3(\wedge^2 Q) - c_2(\wedge^2 Q)c_1(Q)$ is Poincare dual to

$$-2\sigma(3, 0, 0, 0) + 9\sigma(2, 1, 0, 0) - \sigma(1, 1, 1, 0).$$

It follows that $c_1(Q)^3(12c_3(\wedge^2 Q) - 3c_2(\wedge^2 Q)c_1(Q))$ is Poincare dual to

$$16\sigma(3, 3, 0, 0) + 48\sigma(3, 2, 1, 0) + 17\sigma(2, 2, 2, 0) + 22\sigma(3, 1, 1, 1) + 24\sigma(2, 2, 1, 1).$$

Clearly, this cycle has positive intersection number with two generic cycles of type $\sigma(2, 1, 0, 0)$. So (5.9) is proved.

We recall that X_z is Chow-Mumford stable (resp. Chow-Mumford semi-stable), if it is G -stable (resp. G -semi-stable) for any one-parameter algebraic subgroup G in $SL(N+1, \mathbb{C})$. Then it follows from Theorem 5.1,

Theorem 5.2. *Let $\pi : \mathcal{X} \mapsto Z$ be as above. Assume that L_Z is ample over Z and Z_h is empty, i.e., $\pi : \mathcal{X} \mapsto Z$ is a fibration of equidimensional varieties. Then for any z in Z_0 ,*

1. *if $\mu \leq 0$, then X_z is Chow-Mumford stable with respect to L_Z ;*
2. *If $\mu > 0$ and X_z has a Kähler-Einstein metric, then X_z is Chow-Mumford semi-stable with respect to L_Z .*

Our proof of Theorem 5.1 also yields

Theorem 5.3. *Let $\pi : \mathcal{X} \mapsto Z$ be as in Theorem 5.1. Assume that $\mu \leq 0$, L_Z is ample over Z and $\mathcal{K} = -\mu\pi_2^*L + \pi^*F$, where F is a line bundle over Z . Then any X_z ($z \in Z_0$) is Chow-Mumford stable.*

Note that the fibrations in the last two examples satisfy the assumptions on $\mathcal{K}$ in Theorem 5.3, whenever $\mu \leq 0$. In fact, I believe that the extra assumption on $\mathcal{K}$ can be dropped from Theorem 5.3.

In the following, we outline a proof of Theorem 5.1, 5.3.

We will start with an analytic criterion for stability. We will adopt the notations in Theorem 5.1, 5.3.

Let m be an integer such that L_Z^m is very ample. The universal line bundle U on $PH^0(Z, L_Z^m)^*$ restricts to L_Z^{-m} on Z . The total space of U can be identified with

$$\{(x, v) \mid x \in PH^0(Z, L_Z^m)^*, v \in x\}$$

Then any inner product $\langle \cdot, \cdot \rangle$ on $H^0(Z, L_Z^m)^*$ induces a hermitian metric on U , and consequently, a metric $\|\cdot\|_Z$ on L_Z^{-1} , i.e., $\|(x, v)\|_Z = \sqrt{\langle v, v \rangle}$.

Lemma 5.4. *Given any z in Z_0 , we define a function on G by*

$$F_0(\sigma) = \log (\|(\sigma(z), \sigma(\tilde{z}))\|_Z), \quad \sigma \in G \quad (5.10)$$

where $\tilde{z}$ is any lifting of z in $H^0(Z, L_Z^m)^*$. Then

- i) X_z is G -stable if and only if F_0 is proper on G ;
- ii) X_z is G -semistable if and only if F_0 is bounded from below on G .

The proof of this lemma is simple and is left to the readers (cf. [T4]).

Next we recall the definition of the K-energy of Mabuchi and Bando ([BM]). Let $z \in Z_0$ and $X = X_z$. Choose any fixed metric ω with $c_1(L)$ being its Kähler class. As before, we denote by $P(X, \omega)$ the set of smooth functions φ on X such that $\omega_\varphi = \omega + \partial\bar{\partial}\varphi > 0$. Mabuchi and Bando defined the K-energy D_ω as follows: for any φ in $P(X, \omega)$, choose a path φ_t from 0 to φ in $P(X, \omega)$, then

$$D_\omega(\varphi) = -n \int_0^1 \int_X \frac{\partial \varphi}{\partial t} (\text{Ric}(\omega_{\varphi_t}) - \mu \omega_t) \wedge \omega_{\varphi_t}^{n-1} \wedge dt \quad (5.11)$$

One can show (cf. Proposition 3.1 in [T4])

$$D_\omega(\varphi) = \int_X \log \left(\left(\frac{\omega_\varphi^n}{\omega^n} \right) - f_0 \right) \omega_\varphi^n + \int_X f_0 \omega_0^n - \mu(I(\varphi) - J(\varphi)) \quad (5.12)$$

where $I(\cdot)$, $J(\cdot)$ are defined in Section 4.

Lemma 5.5. *We have*

- 1) *if $\mu \leq 0$, then D_ω is proper on $P(X, \omega)$ in the sense that there are $\epsilon > 0$ and $C > 0$, such that $D_\omega(\varphi) \geq \epsilon I(\varphi) - C$ for all φ ;*
- 2) *If $\mu > 0$ and X has a Kähler-Einstein metric, then D_ω is bounded from below.*

Proof: We only give a sketched proof of 1). The result in 2) is due to Bando and Mabuchi; we refer the readers to their paper [BM].

As in (4.11), we can define a holomorphic invariant

$$\alpha_L(X) = \sup \{ \alpha \mid \text{For some } C_\alpha, \int_X e^{-\alpha(\varphi - \sup_X \varphi)} \omega^n \leq C_\alpha \text{ for all } \varphi \in P(X, \omega) \}. \quad (5.13)$$

One can show that $\alpha_L(X)$ is always positive (cf. [T1]). Choose $\epsilon = \frac{\alpha_L(X)}{2}$. It follows from (5.13) that

$$\frac{1}{V} \int_X e^{-\epsilon(\varphi - \sup_X \varphi) - \log \frac{\omega_\varphi^n}{\omega^n}} \omega_\varphi^n \leq \frac{C_\epsilon}{V}.$$

where $V = \int_X \omega^n$. Then by the concavity of the logarithmic function, we have

$$\frac{1}{V} \int_X \log \left(\frac{\omega_\varphi^n}{\omega^n} \right) \omega_\varphi^n \geq \epsilon I(\varphi) - \log \frac{C_\epsilon}{V}.$$

The properness of D_ω follows, since $\mu \leq 0$ and $I(\varphi) - J(\varphi) \geq 0$.

Remark: Recently, I am able to prove the properness of D_ω in the suitable sense in the case of Lemma 5.5, (2).

Let ω_{FS} be the Fubini-Study metric on CP^N . For any $z' \in Z_0$, we put $\omega_{z'}$ to be the restriction of ω_{FS} to $X_{z'}$. Then D_ω induces a functional $D_{\omega, m}$ on G , i.e., $D_{\omega, m}(\sigma) = D_\omega(\sigma^*(\omega_{\sigma(z)}))$. By Lemma 5.4, 5.5, it suffices to show that there are $\epsilon > 0$ and $C > 0$ satisfying:

$$F_0(\sigma) \geq \epsilon D_{\omega, m}(\sigma) - C \quad (5.14)$$

To prove (5.14), we first compute the complex Hessian of $D_{\omega, m}$.

Let h be the pull-back metric on $\pi_2^* L$ over $\mathcal{X}$ from the standard Hermitian metric on the hyperplane bundle over CP^N . Then the curvature $\pi_2^* \omega_{FS}$ of h restricts to the Kähler metric $\omega_{z'}$ on each smooth fiber $X_{z'}$, and consequently, a hermitian metric $k_{\mathcal{X}}$ on the relative canonical bundle $\mathcal{K}$ over $\pi^{-1}(Z_0)$. We denote by $R_{\mathcal{X}|Z}$ the curvature form of $k_{\mathcal{X}}$. Then we have

Lemma 5.6. ([T4]) Define GX to be the variety $\{(\sigma, x) | x \in \sigma(X)\}$ in $G \times \mathbb{C}P^N$. Then for any smooth $2(\dim_{\mathbb{C}} G - 1)$ -form ϕ with compact support in G ,

$$-\int_G D_{\omega, m}(\sigma) \partial \bar{\partial} \phi = \int_{GX} \left(-p_z^* R_{\mathcal{X}|Z} - \frac{n\mu}{n+1} \pi_2^* \omega_{FS} \right) \wedge \pi_2^* \omega_{FS}^n \wedge \pi_1^* \phi, \quad (5.15)$$

where p_z is the map: $GX \mapsto \mathcal{X}$, assigning (σ, x) to x in $\sigma(X) \subset \mathcal{X}$.

Proof: Define $\Psi : G \times X \mapsto GX$ by sending (σ, x) to $(\sigma, \sigma(x))$. We have Hermitian metrics $\tilde{h} = \Psi^* h$ on $\Psi^* \pi_2^* L$ and $\tilde{k} = \Psi^* k_{\mathcal{X}}$ on $\pi_2^* K_X$. Then (5.15) becomes

$$-\int_G D_{\omega, m}(\sigma) \partial \bar{\partial} \phi = \int_{G \times X} \left(-R(\tilde{k}) - \frac{n\mu}{n+1} R(\tilde{h}) \right) \wedge R(\tilde{h})^n \wedge \pi_1^* \phi \quad (5.16)$$

where $R(\cdot)$ denotes the curvature form.

Let h_0 be a Hermitian metric on $L|_X$ with $\omega = R(h_0)$. Define functions φ_t on $G \times X$ by

$$\varphi_t(\sigma, x) = t \log \left(\frac{\tilde{h}(\sigma, x)}{h_0(x)} \right) \quad \text{and} \quad \tilde{h}_t = e^{\varphi_t} h_0.$$

Then $\tilde{h}_1 = \tilde{h}$ and $\tilde{h} = \pi_2^* h_0$. Notice that the curvature $R(\tilde{h}_t)$ restricts to a Kähler metric on $\sigma \times X$ for each σ . Therefore, $\tilde{h}_t$ induces a Hermitian metric $\tilde{k}_t$ on $\pi_2^* K_X$ such that $\tilde{k}_1 = \tilde{k}$ and $\tilde{k}_0$ is independent of σ .

We have

$$\begin{aligned} & -\int_G D_{\omega, m}(\sigma) \partial \bar{\partial} \phi \\ &= \int_G \partial \bar{\partial} \phi(\sigma) \wedge \int_0^1 \int_X \frac{\partial}{\partial t} \varphi_t(\sigma, x) \left(-nR(\tilde{k}_t) - n\mu R(\tilde{h}_t) \right) \wedge R(\tilde{h}_t)^{n-1} \wedge dt \\ &= \int_0^1 \int_{G \times X} \frac{\partial}{\partial t} \varphi_t \left(-nR(\tilde{k}_t) - n\mu R(\tilde{h}_t) \right) \wedge R(\tilde{h}_t)^{n-1} \wedge \partial \bar{\partial} \phi(\sigma) \wedge dt \\ &= \int_0^1 \int_{G \times X} \pi_1^* \phi \wedge \partial \bar{\partial} \left(\frac{\partial}{\partial t} \varphi_t \right) \wedge \left(-nR(\tilde{k}_t) - n\mu R(\tilde{h}_t) \right) \wedge R(\tilde{h}_t)^{n-1} \wedge dt \\ &= \int_0^1 \int_{G \times X} \pi_1^* \phi \wedge \left(-R(\tilde{k}_t) \wedge \frac{\partial}{\partial t} R(\tilde{h}_t)^n - \frac{n\mu}{n+1} \frac{\partial}{\partial t} R(\tilde{h}_t)^{n+1} \right) \wedge dt \\ &= \int_{G \times X} \pi_1^* \phi \wedge \left(-R(\tilde{k}) - \frac{n\mu}{n+1} R(\tilde{h}) \right) \wedge R(\tilde{h})^n \\ &\quad + \int_0^1 \int_{G \times X} \pi_1^* \phi \wedge \frac{\partial}{\partial t} R(\tilde{k}_t) \wedge R(\tilde{h}_t)^n \wedge dt. \end{aligned} \quad (5.17)$$

However,

$$\frac{\partial}{\partial t} R(\tilde{k}_t)(\sigma, x) = \partial \bar{\partial} (\Delta_{\sigma, t} \varphi_t)(\sigma, x),$$

where $\Delta_{\sigma, t}$ denotes the Laplacian of the Kähler metric induced by $R(\tilde{h}_t)|_{\sigma \times X}$, so the last integral in (5.17) vanishes by integration by parts. Then (5.15) follows from (5.16), (5.17).

Lemma 5.7. *Let $g_{\mathcal{X}}$ and g_Z be Hermitian metrics on the canonical bundles $K_{\mathcal{X}}$ and K_Z , respectively. Then*

$$R_{\mathcal{X}|Z} = R(g_{\mathcal{X}}) - R(g_Z) + \partial\bar{\partial}\psi \quad \text{on } \mathcal{X} \setminus \{x \mid \text{Im}(d\pi(x)) \neq T_{\pi(x)}Z\} \quad (5.18)$$

where ψ is a smooth function on $\mathcal{X} \setminus \{x \mid \text{Im}(d\pi(x)) \neq T_{\pi(x)}Z\}$ satisfying $\sup \psi < \infty$.

Proof: For any x such that $\text{Im}(d\pi(x)) = T_{\pi(x)}Z$, we will define $\psi(x)$ as follows: choose $s_{\mathcal{X}}$ in $K_{\mathcal{X}}|_x$ and s_Z in $K_Z|_{\pi(x)}$, such that $g_{\mathcal{X}}(s_{\mathcal{X}}, s_{\mathcal{X}})(x) = 1$ and $g_Z(s_Z, s_Z)(\pi(x)) = 1$. Since $df|_x : T_x\mathcal{X} \rightarrow T_{\pi(x)}Z$ is surjective, there is a unique vector s of $\mathcal{K}_x$ such that $s \cdot (df)^*(s_Z) = s_{\mathcal{X}}$. We simply define $\psi(x) = -\log k_{\mathcal{X}}(s, s)(x)$. Clearly, $\sup \psi < \infty$ and $\psi(x)$ diverges to $-\infty$ as x tends to any point where df is not surjective.

We can also write

$$\psi(x) = -\log \left(\frac{k_{\mathcal{X}} g_Z}{g_{\mathcal{X}}} \right) \quad (5.19)$$

Then (5.18) follows easily.

Corollary 5.8. *For any smooth $2(\dim_{\mathbb{C}} G - 1)$ -form ϕ with compact support in G , we have*

$$\begin{aligned} & - \int_G D_{\omega, m} \partial\bar{\partial}\phi + \int_G \psi_Z \partial\bar{\partial}\phi \\ & = \int_{GX} \left(-R(g_{\mathcal{X}}) + R(g_Z) - \frac{n\mu}{n+1} \pi_2^* \omega_{FS} \right) \pi_2^* \omega_{FS}^n \wedge \pi_1^* \phi \end{aligned} \quad (5.20)$$

where ψ_Z is a smooth function on Z_0 , moreover, ψ_Z is bounded from above and for any $z \in Z_0$,

$$\psi_Z(z) = \int_{X_z} \psi \omega_{FS}^n.$$

Consequently, ψ extends to a continuous function outside the set of points where either X_z is of dimension $> n$ or X_z contains a component of multiplicity > 1 .

This follows directly from Lemma 5.6, 5.7.

Let Φ be the push-forward current

$$\pi_* \left((-R(g_{\mathcal{X}}) + R(g_Z) - \frac{n\mu}{n+1} \pi_2^* \omega_{FS}) \wedge \pi_2^* \omega_{FS}^n \right).$$

Then $(n+1)\Phi$ represents the Chern class of L_Z^{-1} . Therefore, there is a function θ_Z , which is smooth in Z_0 , such that in the weak sense,

$$\Phi = \frac{1}{n+1} R(\|\cdot\|_Z) + \partial\bar{\partial}\theta_Z, \quad \text{on } Z. \quad (5.21)$$

Lemma 5.9. *Let Z_h be the subvariety in Z consisting of points z such that X_z is of dimension $> n$. Then Z_h has codimension at least two, and θ_Z is Hölder continuous outside Z_h .*

Proof: We only sketch the main steps in the proof of this lemma. The details are left to the readers.

Given any point $z' \in Z \setminus Z_h$, using the fact that fibers $X_{z'}$ are of equal dimension n , one can show that for sufficiently small $r > 0$,

$$r^{2-2b} \int_{B_r(z')} |\mathrm{tr}_{\omega_Z} \Phi| \omega_Z^b \leq C r^\delta,$$

where $b = \dim_C Z$, $\delta, C > 0$ are some uniform constants, ω_Z is a fixed Kähler metric on Z , and $B_r(z')$ is the ball of ω_Z with radius r and center at z' .

Then the lemma can be deduced from the standard Green's formula.

By Corollary 5.8 and Lemma 5.9, we have

$$\partial \bar{\partial} \left((D_{\omega, m} - \psi_Z + \theta_Z)(\sigma) - \frac{1}{n+1} \log \left(\frac{\|\cdot\|_Z(\sigma(z))}{\|\cdot\|_Z(z)} \right) \right) = 0.$$

Namely, the function $(D_{\omega, m} - \psi_Z + \theta_Z)(\sigma) - \frac{1}{n+1} \log \left(\frac{\|\cdot\|_Z(\sigma(z))}{\|\cdot\|_Z(z)} \right)$ is of the form $\log |F|^2$ for some holomorphic function F on $SL(N+1, C)$.

Let us denote by $\{z_{ij}\}_{0 \leq i, j \leq N}$, w the homogeneous coordinates of $CP^{(N+1)^2}$. Then $SL(N+1, C)$ can be naturally identified with the affine subvariety $W \cap \{w \neq 0\}$, where

$$W = \{[\{z_{ij}\}_{0 \leq i, j \leq N}, w] \mid \det(z_{ij}) = w^{N+1}\}.$$

Then by using the definition of $D_{\omega, m}$ and straightforward computations, one can show

Lemma 5.10. *The function F has at most polynomial growth near $W \setminus SL(N+1, C)$, i.e., there are constants $\ell > 0$, $C > 0$, such that $F(\sigma) \leq C d(\sigma, W \setminus SL(N+1, C))^\ell$, where $d(\sigma, W \setminus SL(N+1, C))$ denotes the distance from σ to $W \setminus SL(N+1, C)$ with respect to the standard metric on $CP^{(N+1)^2}$.*

Therefore, F extends to be a meromorphic function on W . Notice that W is normal and $W \setminus SL(N+1, C)$ is irreducible. It follows that F has to be a nonzero constant c , and consequently, we have

$$\left(\frac{\|\cdot\|_Z(\sigma(z))}{\|\cdot\|_Z(z)} \right)^{\frac{1}{n+1}} = e^{c - \psi_Z(\sigma) + \theta_Z(\sigma)} e^{D_{\omega, m}(\sigma)} \geq C e^{\theta_Z(\sigma) + D_{\omega, m}(\sigma)}, \quad (5.22)$$

or equivalently,

$$F_0(\sigma) \geq (n+1)D_{\omega, m}(\sigma) + (n+1)\theta_Z(\sigma) - C', \quad (5.23)$$

where $C, C' > 0$ are some uniform constants. Here we have used the fact that ψ_Z is bounded from above.

Then Theorem 5.1 follows easily from (5.23) and Lemma 5.4, 5.5 and 5.9.

To prove Theorem 5.3, we only need to show that θ_Z is bounded from below. By our assumption, the current Φ can be decomposed into $\Phi_1 + \frac{\mu}{n+1} \Phi_2$, such that

Φ_1 is continuous on Z and $\Phi_2 = \pi_*(\pi_2^* \omega_{FS}^{N+1})$. Clearly, Φ_2 is a positive current. It follows that $\partial\bar{\partial}\theta_Z$ is a negative current modulo continuous terms, since $\mu \leq 0$. Therefore, θ is bounded from below. Theorem 5.3 is proved.

6. The Calabi-Futaki invariant and A Residue Formula

In [Fu1], Futaki introduced an analytic invariant for any Kähler manifold with the first Chern class positive. The vanishing of this invariant is a necessary condition for the existence of Kähler-Einstein metrics on the manifold. Later, Calabi and Futaki generalized the invariant to any compact Kähler manifolds. This generalized invariant, i.e., Calabi-Futaki invariant, is an analytic obstruction to existence of Kähler metrics with constant scalar curvature.

In this section, we give a residue formula computing the Calabi-Futaki invariant in terms of zeroes of holomorphic vector fields. The residue formula was first obtained by Futaki in [Fu2] in a special case (cf. Corollary 6.4). He proved his residue formula by imitating Atiyah-Bott's proof of the Lefschetz Theorem for fixed points. Our idea is to write the Calabi-Futaki invariant as the integration of certain Bott-Chern classes. Then the residue formula can be obtained for the Calabi-Futaki invariant by using the transgression method involving the Bott-Chern classes.

Let (M, ω) be a compact Kähler manifold of complex dimension n , and $K_+(M)$ be the cone of Kähler classes on M . We denote by $\eta(M)$ the Lie algebra of holomorphic vector fields on M .

The Calabi-Futaki invariant is a map $f_M : K_+(M) \times \eta(M) \mapsto \mathbb{C}$. Let us recall its definition: for any Kähler class Ω in $K_+(M)$, choose any Kähler metric g with the Kähler class Ω , i.e., $\frac{\sqrt{-1}}{2\pi} \omega_g$ represents Ω , and let h_g be the unique function, determined by

$$s(g) - \frac{\int_M s(g) \omega_g^n}{\int_M \omega_g^n} = \Delta_g h_g, \quad \int_M (e^{h_g} - 1) \omega_g^n = 0$$

where ω_g is the Kähler form of g , and $s(g)$ is the scalar curvature of g . Then for any $X \in \eta(M)$,

$$f_M(\Omega, X) = \left(\frac{\sqrt{-1}}{2\pi} \right)^n \int_M X(h_g) \omega_g^n. \quad (6.1)$$

It can be shown (cf. [Fu2]) that f_M is independent of choices of g in Ω . In fact, this will also follow from our residue formula. Moreover, for each Ω , $f_M(\Omega, \cdot)$ is a Lie algebra character.

We denote by $i(X)$ the interior multiplication on differential forms by X . For example, $i(X)\omega_g$ is the 1-form α_X defined by

$$\alpha_X(v) = \omega_g(X, v), \quad \text{for any tangent vector } v.$$

Since X is holomorphic, $i(X)\omega_g$ is $\bar{\partial}$ -closed. Then by the Hodge Theorem, we can write

$$i(X)\omega_g = \alpha_0 - \bar{\partial}\theta_X,$$

where α_0 is a harmonic 1-form and $\theta_X = \theta_X(g)$ is a smooth function. Such a $\theta_X(g)$ is uniquely determined by g modulo addition of constants. Since $\bar{\partial}^* \alpha_0 = 0$, one can deduce

$$f_M(\Omega, X) = - \left(\frac{\sqrt{-1}}{2\pi} \right)^n \int_M (\bar{\partial} \theta_X)(h_g) \omega_g^n,$$

and therefore, without loss of generality, we may assume that $\alpha_0 = 0$. This is actually true if M is simply-connected.

Lemma 6.1. *Let Ω, g, X be as above. Then*

$$\begin{aligned} & (n+1)2^{n+1}f_M(\Omega, X) \\ &= \sum_{j=0}^n (-1)^j \frac{1}{j!(n-j)!} \left(\frac{\sqrt{-1}}{2\pi} \right)^n \int_M ((-\Delta_g \theta_X + \text{Ric}(g) + (n-2j)(\theta_X + \omega_g))^{n+1} \\ & \quad - (\Delta_g \theta_X - \text{Ric}(g) + (n-2j)(\theta_X + \omega_g))^{n+1}) \\ & \quad - n\mu \sum_{j=0}^{n+1} (-1)^j \frac{1}{j!(n+1-j)!} (n+1-2j)^{n+1} \left(\frac{\sqrt{-1}}{2\pi} \right)^n \int_M (\theta_X + \omega_g)^{n+1} \end{aligned} \quad (6.2)$$

where

$$\mu = \frac{c_1(M) \cdot \Omega^{n-1}}{\Omega^n} = \frac{\int_M \text{Ric}(g) \wedge \omega_g^{n-1}}{\int_M \omega_g^n}.$$

Proof: There are identities:

$$\begin{aligned} \sum_{j=0}^l (-1)^j \binom{l}{j} (l-2j)^k &= 0, \quad \text{if } k < l \text{ or } l+1, \\ \sum_{j=0}^l (-1)^j \binom{l}{j} (l-2j)^l &= 2^l l! \end{aligned} \quad (6.3)$$

Then the lemma follows from straightforward computations and the fact that

$$\int_M \Delta_g \theta_X \omega_g^n = 0.$$

We say X is nondegenerate if the following conditions are satisfied:

- (1) The zero set of X consists of complex submanifolds Z_λ ($\lambda \in \Lambda$);
- (2) X is nondegenerate in the normal directions of each Z_λ , i.e., $\det(DX|_{TM/TZ_\lambda}) \neq 0$ is nowhere zero along Z_λ .

For each λ , there is a natural endomorphism $L_\lambda(X)$ of the normal bundle $N_{M|Z_\lambda} = TM/TZ_\lambda$. If g is Kähler metric on M and ∇ is the Levi-Civita connection of g . Then

$$L_\lambda(X)(Y) = (\nabla_Y X)^\perp \in N_{M|Z_\lambda} \quad \text{for any } Y \in N_{M|Z_\lambda}.$$

We should remark that $L_\lambda(X)$ is actually independent of g . Now X being nondegenerate means that $L_\lambda(X)$ is nonsingular at any point of Z_λ .

Next we will introduce a “trace” $\text{tr}_\Omega(X)$, which is only well defined up to addition of constants. Pick up any metric g with Kähler class Ω , and let θ_X be the function above. Since $\bar{\partial}(\theta_X|_{Z_\lambda}) = -i(X)\omega_g|_{Z_\lambda} = 0$, $\theta_X|_{Z_\lambda}$ is a constant. If g' is another metric in Ω and θ'_X is chosen such that $\bar{\partial}\theta'_X = -i(X)\omega_{g'}$, then $\theta'_X = \theta_X - X(\varphi) + c$, where $\omega_{g'} = \omega_g + \partial\bar{\partial}\varphi$ and c is a constant. It follows that $\theta'_X(Z_\lambda) = \theta_X(Z_\lambda) + c$ for any $\lambda \in \Lambda$. Therefore, the line $(\theta_X(Z_\lambda) + t)_{\lambda \in \Lambda}$, where $t \in \mathbb{C}$, is independent of g . We simply define $\text{tr}_\Omega(X)_{Z_\lambda} = \theta_X(Z_\lambda)$. This is only well-defined modulo addition of a common constant. If there is no confusion, we just pretend that $\text{tr}_\Omega(X)_{Z_\lambda}$ is a well-defined number.

If $\Omega = c_1(L)$ for some ample line bundle L and X lifts to a holomorphic vector field X^* on L , then

$$\text{tr}_\Omega(X)_{Z_\lambda} = \text{tr}(X^* : L_z \mapsto L_z),$$

where z is any point in Z_λ . If $\Omega = c_1(M)$, then

$$\text{tr}_\Omega(X)_{Z_\lambda} = \text{tr}(L_\lambda(X)) = \text{tr}(X_c^* : \wedge^n TM \mapsto \wedge^n TM),$$

where X_c^* is the canonical lifting of X to $\wedge^n TM$.

Now we can state our main theorem of this section.

Theorem 6.3. *For any (Ω, X) in $K_+(M) \times \eta(M)$, we have*

$$\begin{aligned} f_M(\Omega, X) &= \sum_{\lambda \in \Lambda} \int_{Z_\lambda} \frac{(\text{tr}(L_\lambda(X)) + c_1(M))(\text{tr}_\Omega(X)_{Z_\lambda} + \Omega)^n - \frac{n\mu}{n+1}(\text{tr}_\Omega(X)_{Z_\lambda} + \Omega)^{n+1}}{\det(L_\lambda(X) + \frac{\sqrt{-1}}{2\pi} K_\lambda)}, \end{aligned} \quad (6.4)$$

where K_λ is the curvature form of the induced metric on $N_{M|Z_\lambda}$ by g . In particular, if $\Omega = c_1(L)$, we can replace $\text{tr}_\Omega(X)_{Z_\lambda}$ in (6.4) by $\text{tr}(X^* : L_z \mapsto L_z)$ for any $z \in Z_\lambda$.

Proof: The proof resembles Bott’s proof of the localization theorem of the Chern numbers ([Bo]). Therefore, as one expects, the proof works for much more general cases.

Let ϕ be any symmetric GL -invariant polynomial of degree $n+1$. Let E be a vector bundle over M and h be a hermitian metric.

Assume that $\theta_X(h)$ is an $\text{End}(E)$ -valued function satisfying:

$$\bar{\partial}\theta_X(h) = -i(X)R(h), \quad (6.5)$$

where $R(h)$ denotes the curvature form of h . For simplicity, we write $\phi(\theta_X(h) + R(h))$ for

$$\underbrace{\phi(\theta_X(h) + R(h), \dots, \theta_X(h) + R(h))}_{n+1}$$

Hence,

$$\begin{aligned} \bar{\partial}\phi(\theta_X(h) + R(h)) &= (n+1)\phi(\bar{\partial}\theta_X(h), \theta_X(h) + R(h), \dots, \theta_X(h) + R(h)) \\ &= -(n+1)\phi(i(X)R(h), \theta_X(h) + R(h), \dots, \theta_X(h) + R(h)) \\ (\text{since } i(X)\theta_X(h) = 0) &= -i(X)\phi(\theta_X(h) + R(h)). \end{aligned} \quad (6.6)$$

Now we define a form η on $M \setminus \text{zero}(X)$ by $\eta(Y) = g(Y, X)/g(X, X)$. Furthermore, we define a formal series of forms by

$$\alpha = \phi(\theta_X(h) + R(h)) \wedge \frac{\eta}{1 + \bar{\partial}\eta}. \quad (6.7)$$

Since X is holomorphic, we have $i(X) \cdot \bar{\partial} = \bar{\partial} \cdot i(X)$ and thus,

$$i(X)(\bar{\partial}\eta) = \bar{\partial}(i(X)\eta) = \bar{\partial}(\eta(X)) = 0.$$

It follows that

$$\begin{aligned} i(X)\alpha &= i(X)\phi(\theta_X(h) + R(h)) \wedge \frac{\eta}{1 + \bar{\partial}\eta} + \phi(\theta_X(h) + R(h)) \wedge \frac{\eta(X)}{1 + \bar{\partial}\eta} \\ &= -\bar{\partial}\phi(\theta_X(h) + R(h)) \wedge \frac{\eta}{1 + \bar{\partial}\eta} + \phi(\theta_X(h) + R(h)) \wedge \frac{1}{1 + \bar{\partial}\eta} \\ &= -\bar{\partial}\alpha + \phi(\theta_X(h) + R(h)) \wedge \frac{\bar{\partial}\eta}{1 + \bar{\partial}\eta} + \phi(\theta_X(h) + R(h)) \wedge \frac{1}{1 + \bar{\partial}\eta} \\ &= -\bar{\partial}\alpha + \phi(\theta_X(h) + R(h)), \end{aligned} \quad (6.8)$$

and therefore, we have

$$[\phi(\theta_X(h) + R(h))]_{2n} = \bar{\partial}[\alpha]_{2n-1}, \quad (6.9)$$

where $[\beta]_k$ denotes the degree k term in β . Then,

$$\begin{aligned} &\int_M \phi(\theta_X(h) + R(h)) \\ &= \lim_{\epsilon \rightarrow 0+} \int_{M - B_\epsilon(\text{zero}(X))} [\phi(\theta_X(h) + R(h))]_{2n} \\ &= \lim_{\epsilon \rightarrow 0+} \int_{M - B_\epsilon(\text{zero}(X))} \bar{\partial}[\alpha]_{2n-1} \\ &= - \sum_{\lambda \in \Lambda} \lim_{\epsilon \rightarrow 0+} \int_{\partial B_\epsilon(Z_\lambda)} \alpha \\ &= - \sum_{\lambda \in \Lambda} \lim_{\epsilon \rightarrow 0+} \int_{\partial B_\epsilon(Z_\lambda)} \phi(\theta_X(h) + R(h)) \wedge \sum_{k=0}^{n-1} (-1)^k \underbrace{\eta \wedge (\bar{\partial}\eta)^{k-1}}_{0(\epsilon^{-2k-1})} \end{aligned}$$

Using the arguments in [Bo], one can deduce

$$\begin{aligned} &\lim_{\epsilon \rightarrow 0+} \left(\frac{\sqrt{-1}}{2\pi} \right)^n \int_{B_\epsilon(Z_\lambda)} \phi(\theta_X(h) + R(h)) \wedge \sum_{k=0}^{n-1} (-1)^k \eta \wedge (\bar{\partial}\eta)^{k-1} \\ &= \left(\frac{\sqrt{-1}}{2\pi} \right)^{\dim_{\mathbb{C}} Z_\lambda} \int_{Z_\lambda} \frac{\phi(\theta_X(h) + R(h))}{\det(L_\lambda(X) + K_\lambda)}. \end{aligned} \quad (6.10)$$

Next we observe

$$\begin{aligned}\bar{\partial}\theta_X &= -i(X)\omega_g \\ \bar{\partial}(-\Delta_g\theta_X) &= -i(X)\text{Ric}(g).\end{aligned}\tag{6.11}$$

Now we let ϕ be the polynomial ch_{n+1} , i.e.,

$$\phi(x_1, \dots, x_{n+1}) = \frac{1}{(n+1)!} \text{tr}(x_1 \cdots x_{n+1}),$$

For simplicity, assume that $\Omega = c_1(L)$. Applying (6.10) to $E = K_M^{\pm 1} \otimes L^{n-2j}$ with $\theta_X(h) = \pm \Delta_g \theta_X + (n-2j)\theta_X$ ($0 \leq j \leq n$), or $E = L^{n+1-2k}$ with $\theta_X(h) = (n+1-2k)\theta_X$ ($0 \leq k \leq n+1$), we obtain

$$\begin{aligned}(n+1)2^{n+1}f_M(\Omega, X) &= \sum_{j=0}^n \frac{(-1)^j}{j!(n-j)!} \sum_{\lambda \in \Lambda} \int_{Z_\lambda} \frac{(\text{tr}(L_\lambda(X)) + c_1(M) + (n-2j)(\text{tr}_\Omega(X)_{Z_\lambda} + c_1(L)))^{n+1}}{\det(L_\lambda(X) + \frac{\sqrt{-1}}{2\pi}K_\lambda)} \\ &\quad - \sum_{j=0}^n \frac{(-1)^j}{j!(n-j)!} \sum_{\lambda \in \Lambda} \int_{Z_\lambda} \frac{(-\text{tr}(L_\lambda(X)) - c_1(M) + (n-2j)(\text{tr}_\Omega(X)_{Z_\lambda} + c_1(L)))^{n+1}}{\det(L_\lambda(X) + \frac{\sqrt{-1}}{2\pi}K_\lambda)} \\ &\quad - n\mu \sum_{j=0}^{n+1} \frac{(-1)^j(n+1-2j)^{n+1}}{j!(n+1-j)!} \sum_{\lambda \in \Lambda} \int_{Z_\lambda} \frac{(\text{tr}_\Omega(X)_{Z_\lambda} + c_1(L))^{n+1}}{\det(L_\lambda(X) + \frac{\sqrt{-1}}{2\pi}K_\lambda)}.\end{aligned}\tag{6.12}$$

Then (6.4) follows easily.

Corollary 6.4. (Futaki, [Fu2]) *If M has positive first Chern class, then*

$$f_M(c_1(M), X) = \frac{1}{n+1} \sum_{\lambda \in \Lambda} \int_{Z_\lambda} \frac{\text{tr}(L_\lambda(X) + c_1(M))^{n+1}}{\det(L_\lambda(X) + \frac{\sqrt{-1}}{2\pi}K_\lambda)}.$$

In particular, if $\dim Z_\lambda = 0$ for all $\lambda \in \Lambda$, then

$$f_M(c_1(M), X) = \frac{1}{n+1} \sum_{\lambda \in \Lambda} \frac{\text{tr}(L_\lambda(X))^{n+1}}{\det(L_\lambda(X))}.$$

Corollary 6.5. *Let M be a complex surface, and L be an ample line bundle over M . Assume that X has a lifting X^* to L . Write $\Lambda = \Lambda_0 \cup \Lambda_1$, where Λ_i consists of λ with $\dim_{\mathbb{C}} Z_\lambda = i$ ($i = 0, 1$). Then*

$$\begin{aligned}f_M(c_1(L), X) &= \sum_{\lambda \in \Lambda_0} \frac{\text{tr}(L_\lambda(X))\text{tr}(X^*|_{L_{Z_\lambda}})^2 - \frac{2\mu}{3}\text{tr}(X^*|_{L_{Z_\lambda}})^3}{\det(L_{Z_\lambda}(X))} \\ &\quad + \sum_{\lambda \in \Lambda_1} \left(\frac{\text{tr}(X^*|_{L_{Z_\lambda}})^2}{L_\lambda(X)} \left(1 - \frac{2\mu\text{tr}(X^*|_{L_{Z_\lambda}})}{3\text{tr}(L_\lambda)} \right) (2 - 2g(Z_\lambda)) \right. \\ &\quad \left. + \frac{2\mu\text{tr}(X^*|_{L_{Z_\lambda}})^3}{3\text{tr}(L_\lambda)^2} c_1(M)([Z_\lambda]) \right) \\ &\quad + 2 \sum_{\lambda \in \Lambda_1} \text{tr}(X^*|_{L_{Z_\lambda}}) \left(1 - \frac{\mu\text{tr}(X^*|_{L_{Z_\lambda}})}{\text{tr}(L_\lambda)} \right) c_1(L)([Z_\lambda])\end{aligned}$$

where $\mu = c_1(M) \cdot c_1(L)/c_1(L)^2$. In particular, we have

$$f_M(c_1(M), X) = \frac{1}{3} \sum_{\lambda \in \Lambda_0} \frac{\text{tr}(L_\lambda(X))^3}{\det(L_{Z_\lambda}(X))} + \frac{1}{3} \sum_{\lambda \in \Lambda_1} \text{tr}(L_\lambda(X))(2c_1(M)([Z_\lambda]) + (2 - 2g(Z_\lambda))).$$

Examples: (1) Let M be the blow-up of CP^2 at one point, say $[1, 0, 0]$. Let X be the vector field generating automorphisms: $[z_0, z_1, z_2] \mapsto [z_0, e^t z_1, e^t z_2]$ of CP^2 . Then

$$f_M(c_1(M), X) = \frac{1}{3}(2 + 2 - 6 - 2) = \frac{4}{3}$$

Therefore, M has no Kähler-Einstein metric. Similarly, one can show that the blow-up of CP^2 at two points has no Kähler-Einstein metric.

(2) Let N be a compact Kähler manifold with $c_1(N) > 0$ and let L be a line bundle over N . Put $M = P(L \oplus C)$. Then $c_1(M) > 0$ if $c_1(N) \pm c_1(L) > 0$.

Now we take $N = CP^1 \times CP^2$ and $L = \pi_1^* H_1 \otimes (\pi_2^* H_2)^{-1}$, where H_1 and H_2 are hyperplane bundles over CP^1 , CP^2 . Let X be the holomorphic field whose real part generates automorphisms ϕ_t associated to bundle automorphisms $(u, v) \mapsto (u, e^t v) \in L \oplus C$, then

$$F_M(c_1(M), X) \neq 0.$$

This example is due to Futaki.

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SIX LECTURES ON FOUR MANIFOLDS

ANDREI TYURIN

Steklov Math Institute

LECTURE 1. INTRODUCTION

Every nonsingular compact algebraic surface S over $\mathbb{C}$ defines three underlying structures

$$vS, \quad dS, \quad tS, \quad (1.1)$$

where tS is the topological type of S , dS the underlying 4 manifold and vS the deformation type of S .

By Freedman [F], the topological type tS of a simply connected algebraic surface S is uniquely determined by its homotopy type, that is, by the lattice $H_2(S, \mathbb{Z})$ and its intersection form q_S . This integer unimodular form is determined uniquely by its parity and by specifying two integers: the square of the canonical class K_S^2 and the second Chern class $c_2(S)$. Instead of this pair, we can use any of the following pairs of data:

$$(b_2^+, b_2^-), \quad (I, b_2^+), \quad (I, b_2^-);$$

here I is the index or signature of S (recall that an algebraic surface S has a natural complex orientation) and b_2^+ (resp. b_2^-) is the rank of the positive (resp. negative) part of $H^2(S, \mathbb{R})$. These constants are related as follows:

$$\begin{aligned} b_2^+ &= \frac{1}{6}(K_S^2 + c_2) - 1, \\ b_2^- &= \frac{1}{6}(5c_2 - K_S^2) - 1, \\ I &= \frac{1}{3}(K_S^2 - 2c_2). \end{aligned} \quad (1.2)$$

Definition 1.1. For any topological 4 manifold Y , write $\text{diff}(Y)$ for the set of all differentiable structures on Y .

This set is discrete, and we will use the notation $\# \text{diff}(Y)$ for its cardinality or number of elements (or that of some subset), although it will sometimes be ∞ .

Recall that the Kodaira dimension $k(S)$ of S can be $-\infty, 0, 1, 2$.

Definition 1.2. For any topological 4 manifold Y , let $\text{alg}_k(Y)$ be the subset of $\text{diff}(Y)$ containing all underlying smooth structures of algebraic surfaces of Kodaira dimension k .

Now let $\text{alg}(Y)$ be the set of all underlying smooth structures of all algebraic surfaces of the topological type Y . Thus

$$\text{alg}(Y) = \bigcup_{k=-\infty, 0, 1, 2} \text{alg}_k(Y). \quad (1.3)$$

Remark. It is convenient sometimes to write $\text{alg}(q)$ or $\text{diff}(q)$, where q is an unimodular intersection form, because the intersection form determines the topological type of a simple connected 4 manifold uniquely.

Definition 1.3. For any topological 4 manifold Y , let $\text{var}_k(Y)$ be the set of deformation types of all algebraic surfaces of Kodaira dimension k and of topological type Y .

Then the disjoint union

$$\text{var}(Y) = \bigcup_{k=-\infty, 0, 1, 2} \text{var}_k(Y) \quad (1.4)$$

is the set of all deformation classes of algebraic surfaces of topological type Y .

Moreover we have the forgetful map

$$f: \text{var}(Y) \rightarrow \text{alg}(Y). \quad (1.5)$$

The main problem of the Smooth Classification of Algebraic Surfaces is to describe the fibres of this forgetful map.

It has been observed many times (beginning with [Dol]) that, in contrast to the case of algebraic curves,

$$\# \text{var}(Y) > 1.$$

More precisely,

$$\# \text{var}_1(\mathbb{CP}_9^2) = \infty, \quad (1.6)$$

where $\mathbb{CP}_k^2$ is the complex projective plane with k points blown up.

On the other hand, for surfaces of general type,

$$\# \text{var}(N) = \# \bigcup_{tS, K_S^2 \leq N} \text{var}(tS) < \infty \quad (1.7)$$

(see for example [C]).

But this number can be large. The best lower bound is due to M. Manetti [Ma]

$$\# \text{var}(N) \geq N^{\frac{1}{5} \log N}. \quad (1.8)$$

It was observed recently (beginning with S. Donaldson's investigation of (2,3)-Dolgachev surface [D2]) that $\text{alg}(Y)$ is related closely to $\text{var}(Y)$. By way of contrast, remark again that by Freedman's famous result [F], the homotopy type of S uniquely determines its topological type. As a consequence of these observations, the following conjectures were raised:

Van de Ven conjecture. $k \neq k' \implies \text{alg}_k \cap \text{alg}_{k'} = \emptyset$.

Friedman–Morgan conjecture. $\# f^{-1}(dS) < \infty$.

These conjectures are finally proved now (see [F–Q] or [P] for the first and [F–M] for the second). So the set (1.3) is also a disjoint union, the forgetful map f (1.5) doesn't mix the components and

$$f^{-1}(dS) = vS, vS_1, \dots, vS_N \quad (1.9)$$

is a finite set of deformation types of algebraic surfaces.

Remark. The beautiful smooth classification of elliptic surfaces (see [M–O'G]) suggests the following conjecture

Conjecture. *The forgetful map (1.5) is one-to-one.*

However, our experience in Smooth Classification of algebraic surfaces of general type isn't so large yet.

The following is an almost complete list of known facts:

$$\# f^{-1}(\mathbb{CP}^2) = 1 \quad (1.10)$$

[T4] or [Y].

$$\# \text{alg}(\mathbb{CP}^2) = 1 \quad (1.11)$$

[Y]. Let K be the underlying 4 manifold of the K3 surface. Then

$$\# \text{alg}_0(tK) = 1.$$

However

$$\# \text{alg}_1(tK) = \infty, \quad (1.12)$$

[F–M] and for every element K_{m_1, m_2} of $\text{alg}_1(tK)$

$$\# f^{-1}(K_{m_1, m_2}) = 1 \quad (1.13)$$

[M–O'G].

Actually, the collection of algebraic surfaces homotopy equivalent to K3 surfaces was constructed by Kodaira [Ko], who introduced the operation of log transform on elliptic surfaces. The same transformation in a cusp neighbourhood on a 4 manifold gives [G–M]

$$\#(\text{diff}(tK) - \text{alg}(tK)) = \infty. \quad (1.14)$$

Remark. Recall that the Smooth Poincaré Conjecture is the following statement:

S.P.C. $\text{diff}(t\mathbb{CP}^2) = \text{alg}(t\mathbb{CP}^2) = \mathbb{CP}^2$.

But if any surface S contains a cusp neighbourhood then the construction of log transformation gives

$$\#(\text{diff}(tS) - \text{alg}(tS)) = \infty.$$

Now for rational surfaces one has

$$\begin{aligned} f^{-1}(d\mathbb{CP}^1 \times \mathbb{CP}^1) &= \{F_{2n}\} \\ f^{-1}(d\mathbb{CP}_1^2) &= \{F_{2n+1}\} \end{aligned} \quad (1.15)$$

[P-T], [F-Q], or [P], where the F_i are the Hirzebruch surfaces.

Remark. It is natural to consider the set of all Hirzebruch surfaces of the same parity as one deformation class. Indeed there exists a smooth algebraic family of F_{2n} containing all $F_{2n'}$ with $n' < n$ (namely the versal family [Bri]).

Moreover

$$\# f^{-1}(\mathbb{CP}_{1 < k < 9}^2) = 1.$$

In the general case, to describe the fibres of the forgetful map (1.5) as a consequence of the Galois ideology for the triple of structures (1.1) it is natural to consider the triple of groups

$$\text{Mon } S \subset \text{Mod } S \subset O(q_S), \quad (1.16)$$

where $O(q_S)$ is the orthogonal group of the intersection form q_S on $H^2(S, \mathbb{Z})$, that is, the group of isotopy classes of self homeomorphisms of S , $\text{Mod } S$ is the group of pseudoisotopy classes of diffeomorphisms, that is, the image of the standard representation

$$\text{Mod } S = \text{im } \rho \quad \text{where} \quad \rho : \text{Diff}^+ \rightarrow O(q_S) \quad (1.17)$$

and the monodromy subgroup $\text{Mon } S$ is generated by all monodromy transformations of all algebraic families containing S .

Recall that an algebraic family containing S is a nonsingular 3-fold X together with a proper smooth map

$$\pi : X \rightarrow C_n, \quad (1.18)$$

where C_n is a nonsingular algebraic curve minus n points, containing a point c such that

$$\pi^{-1}(c) = S.$$

Then one has the monodromy representation

$$\rho_X : \pi_1(C_n) \rightarrow O(q_S) \quad (1.19)$$

and

$$\text{Mon } S = \left\langle \bigcup_{\text{all } X} \text{im } \rho_X \right\rangle \quad (1.20)$$

is the subgroup generated by the union of the images of the monodromy representations taken over all algebraic families.

Remark. The deformation class vS is of course defined as the equivalence class generated by the relation that S and S' are contained in one algebraic family X . So the class vS determines the conjugacy class of the subgroup $\text{Mon } S$.

A very important case is the following: a family (1.18) is called an *L-family* if X is a Lefschetz pencil without singular fibres.

Definition 1.4. A surface S is called an L -surface if $\text{Mon } S$ (1.20) is generated by monodromies of L -families.

We consider the lattice $H^2(S, \mathbb{Z})$ as a representation of our groups $\text{Mon } S \subset \text{Mod } S \subset O(q_S)$. So $\text{Mon } S$ admits a fixed vector, namely the canonical class K_S of S . Indeed for every family X (1.18),

$$K_S = K_X|_S, \quad (1.21)$$

that is, the canonical class of S is the restriction of the canonical class of X to S (1.19) and it is a monodromy invariant:

$$\text{Mon } S \subset \text{Stab}_{K_S} \subset O(q_S), \quad (1.22)$$

where Stab_{K_S} is the stabilizer of K_S .

Remark. Of course the triple of groups $\text{Mon } S \subset \text{Mod } S \subset O(q_S)$ is an analog of the triple $\text{Mon } C \subset \text{Mod } C \subset \text{Sp}(q_C)$ groups represented on $H^1(C, \mathbb{Z})$ of an algebraic curve C . In this case the equality $\text{Mod } C = \text{Sp}(q_C)$ is obvious but the equality $\text{Mon } C = \text{Mod } C$ is the consequence of nontrivial classical arguments, see for example [E–C]. In particular the group

$$\text{Mon } C = \text{Mod } C = \text{Sp}(q_C) \quad (1.23)$$

is called the modular group (or Torelli group) of C . For algebraic surfaces these groups are always distinct and $\text{Mod } S$ is called the *modular group* of S .

Examples. 1. Let S be a rational surface with $K_S^2 > 0$. Then

$$\text{Mod } S = O(q_S) \quad (1.24)$$

(C.T.C. Wall [Wa], we reproduce his arguments Lecture 2).

2. Let S be a rational surface with $K_S^2 < 0$. Then $[O(q_S) : \text{Mod } S] = \infty$, but the representation of $\text{Mod } S$ on $H^2(S, \mathbb{Z})$ is irreducible (see [F–M]).

3. Let S be the K3 surface. Then

$$[O(q_S) : \text{Mod } S] = 2. \quad (1.25)$$

4. Let S be an elliptic surface of Kodaira dimension 1. Then

$$\text{Mod } S \subset \text{Stab}_{K_S} \quad (1.26)$$

[F–M].

Now for all currently known simply connected surfaces of general type, that is, for all surfaces with $p_g > 0$ and for the Barlow surface, we have

Main Theorem. 1) Every such surface has a proper diffeomorphism invariant sublattice $sV(S) \subset H^2(S, \mathbb{Z})$; that is, for any S' in $f^{-1}(dS)$ one has

$$sV(S) = sV(S'),$$

$$2) \quad sV(S) \subset \bigcap_{S' \in f^{-1}(dS)} \text{Pic } S',$$

$$3) \quad \bigcup_{S' \in f^{-1}(dS)} K_{S'} \subset sV(S).$$

So we can see that it is natural to call the sublattice $sV(S)$ the *Picard lattice* of dS .

We will prove the existence of this “algebraic geometric sublattice” $sV(S)$ later, but in the general case we can only estimate this sublattice. Actually, the main question is:

$$\text{how close is } \mathbb{Z} \cdot K_S \text{ to } sV(S)?$$

It must be noted immediately that there is another, purely differential geometric, approach to this problem. This approach predicts a more precise description of the diffeomorphism invariant sublattice

$$L_{K-M-W} = \langle C_1, \dots, C_k \rangle \quad (1.27)$$

generated by the classes of irreducible components of a general curve of the canonical linear system $|K_S|$, the so-called Kronheimer-Mrowka-Witten lattice (see [K-M 1], [Wi]).

It will be very productive to compare these two approaches. However the purpose of these lectures is to describe the algebraic geometric approach, and to show the scope of the method of geometric approximation in the case of algebraic surfaces. Unfortunately we shall not occupy ourselves with all details, but we shall attempt to evoke in the reader his intuitive recognition of the procedure described by Bertram [Ber] and Thaddeus [Th1] in the case of algebraic curves.

Exercises to Lecture 1.

1. Prove that

$$\# \text{alg}_{-\infty}(tK3) = 0,$$

$$\# \text{alg}_0(tK3) = 1,$$

$$\# \text{alg}_1(\mathbb{CP}^2) = 0.$$

Hint: use the standard invariants of algebraic surfaces and the results of Classification Theory.

2. Let I_+ be the positive unimodular form of rank 1 and x be its generator. Let I_- be the negative unimodular form of rank 1 and y be its generator. For any unimodular lattice L with the intersection form q and any vector x with $x^2 = \pm 1$ or ± 2 , let

$$R_x(z) = z - \frac{2}{x^2} \cdot (x \cdot z) \cdot x \in O(q) \quad (1.28)$$

be the reflection w.r.t. the vector x . Prove that

$$a) \quad O(I_+) = \{\text{id}, R_x\},$$

$$b) \quad O(I_-) = \{\text{id}, R_y\},$$

$$c) \quad O(I_+ \oplus I_-) = \{\text{id}, R_x, R_y, R_x \cdot R_y\},$$

d) $O(\oplus_k I_\pm)$ is generated by permutations and automorphisms of the summands;

e) $O(I_+ \oplus I_+ \oplus I_-)$ is generated by permutations, automorphisms of the summands and reflections $R_{x_1+x_2+y}$.

3. Let U be the even unimodular form of rank 2, E_8 the unique even unimodular positive definite form of rank 8 and E_{-8} the same form with opposite sign. We write kq as an abbreviation of the notation $q \oplus q \oplus \cdots$ with k summands. Prove that

$$a) \quad \# \text{alg}(E_8) = 0,$$

$$b) \quad \# \text{alg}(E_8 \oplus E_{-8}) = 0.$$

Hint: use the Hodge Index Theorem;

$$c) \quad \# \text{alg}(k(3U \oplus 2E_{-8})) \neq \emptyset \implies k = 1.$$

LECTURE 2. REFLECTIONS IN ORTHOGONAL GROUPS

As already remarked, the topological invariants tS of any simply connected algebraic surface S such as $c_2 = b_2 + 2$, $K_S^2 = 3I + 2c_2$ or $I = \frac{1}{3}(K^2 - 2c_2)$ satisfy certain well-known inequalities. For example, if q_S is even then S is minimal and as a result of Theory of Algebraic Surfaces (see for example [G-H], p. 590) we have

$$K_S^2 = 3I + 2b_2 + 4 \geq 0 \quad (2.1)$$

(the right side of the equality is the “topological form” of Noether’s formula (see [G-H], pp. 600–626)) that is

$$2b_2 + 4 \geq -3I \quad (2.2)$$

and the Bogomolov–Miyaoaka–Yau inequality:

$$3I \leq b_2 + 2 \quad (2.3)$$

(see [T6], formula (1.19') for another “topological form” of this inequality).

Remark. It is believed that these are the only restrictions, that is, that all these numbers can occur. Actually Xiao Gang and Chen Zhijie proved this except for a very small range.

On the other hand, by the Hodge index theorem the number

$$b_2^+ = 2h^{0,2} + 1 \quad (2.4)$$

must be odd.

Remark. We can use these facts for the exercises of Lecture 1. For example, for the form $k(3U \oplus 2E_{-8})$ we have

$$2b_2 + 4 = 44k + 4; \quad -3I = 48k.$$

So if the inequality (2.2) holds then $k = 1$. This is the solution of the last exercise of Lecture 1.

Now the representation of an unimodular form as a direct product is not unique. For example

$$E_8 \oplus I_- = 8I_+ \oplus I_-.$$

But all odd indefinite forms have a special decomposition of the form

$$q = b_2^+ I_+ \oplus b_2^- I_-. \quad (2.5)$$

Any even form can be written as a direct sum of E_8 and U with coefficients. These coefficients in front of E_8 and U can be arbitrary, but for the intersection form of an algebraic surface, the summands can be collected in special blocks. Namely for any even algebraic surface there are integer k and m such that

$$q_S = k(\pm q_K) \oplus m q_U, \quad (2.6)$$

where $q_K = 3U \oplus E_{-8}$ and k and m are chosen to obtain the specified signature and rank. So if the index is nonpositive then

$$b_2^+ = 3k + m, \quad b_2 = 22k + 2m$$

and

$$k = -\frac{I}{16}; \quad m = b_2^+ + \frac{3I}{16}. \quad (2.7)$$

The calculation for the positive index case is similar :

$$k = \frac{I}{16}; \quad m = b_2^- - \frac{3I}{16}. \quad (2.7')$$

By Rokhlin's congruence theorem the numbers k and m are integers and k is non-negative. Now we have to use the inequalities (2.2) and (2.3) to prove that m is also nonnegative.

Exercise. The inequality $3I + 2b_2 + 4$ is equivalent to $5b_2^+ - b_2^- \geq 0$ or $I + 4b_2^+ \geq 0$ so $\frac{4}{16}I + b_2^+ \geq 0$ so that m in (2.7) is nonnegative. On the other hand, the inequality $3I \leq c_2$ is equivalent to $b_2^- + 1 \geq I$ so that m in (2.7') is positive.

So if the index $I \leq 0$ then

$$q_S = -\frac{I}{16} q_K \oplus (b_2^+ + \frac{3I}{16}) U \quad (2.8)$$

and if $I \geq 0$ then

$$q_S = \frac{I}{16} (-q_K) \oplus (b_2^- - \frac{3I}{16}) U. \quad (2.8')$$

The operation of direct sum of quadratic forms is parallel to that of connected sum of manifolds. Namely let M_1 and M_2 be a pair of simple connected 4-manifolds then cut out of each M_i a small 4-ball D_i and glue $M_1 \setminus D_1$ and $M_2 \setminus D_2$ together along the boundaries $\partial D_1 = S^3$, $\partial D_2 = S^3$. We get a new 4-manifold

$$M = M_1 \# M_2 \quad (2.9)$$

with the intersection form $q_M = q_{M_1} \oplus q_{M_2}$.

Remark. This operation is a particular case of the “cutting-and-pasting” technique: if 4 manifold M is the union (glueing) of two 4 manifolds with boundary M_1 and M_2 meeting along a 3 manifold

$$X = \partial M_1 = \overline{\partial M_2} \quad (2.10)$$

(here $\overline{X}$ is the manifold X with the opposite orientation) then we can try to describe the topological and geometric invariants of M in terms of M_i and X . This operation is called a grafting and is very productive in the theory of Riemann surfaces: any Riemann surface S is a connected sum of g tori:

$$S = T^2 \# T^2 \# \dots \# T^2 \quad (2.11)$$

with g = genus of S summands. For 4 manifolds the collection of “elementary bricks” (analogous to T^2) is larger.

Thus using Freedman’s theorem, we can construct a topological model (*topomodel*) [T6] of a simple connected algebraic surface S . Namely if q_S is odd then

$$\text{top } S = b_2^+ \mathbb{CP}^2 \# b_2^- \overline{\mathbb{CP}^2}. \quad (2.12)$$

This is a 4 manifold topological equivalent to S , because the intersection forms are equal.

Remark. Of course this 4 manifold is a topomodel of any 4 manifold M with odd intersection form. But for even forms we can do this only when $M = S$ is an algebraic surface.

If q_S is even and $I \leq 0$ then

$$\text{top } S = -\frac{I}{16} K \# (b_2^+ + \frac{3I}{16}) Q, \quad (2.13)$$

where $Q = \mathbb{CP}^1 \times \mathbb{CP}^1$ is the complex quadric surface and K is the K3 surface. And in the same vein, if $I > 0$ then

$$\text{top } S = \frac{I}{16} \overline{K} \# (b_2^- - \frac{3I}{16}) Q. \quad (2.13')$$

From this one sees that for an even form with nonpositive index, the topomodel is a connected sum of algebraic surfaces, and for positive index it is a connected sum of antialgebraic surfaces. This is also true for odd forms. Moreover for an algebraic surface S , the blowup of a point or σ -process is equivalent to a connected

sum operation with the antiplane $\overline{\mathbb{CP}^2}$. To see this, we remark that a standard neighbourhood of a (-1) -curve coincides with a standard neighbourhood of the zero section of its normal bundle and that the degree of this normal bundle equals -1 . So the boundary of a small neighbourhood is the Hopf bundle, that is, the ordinary 3-sphere. So for odd q_S we see that

$$\text{top } S = (b_2^+ - 1)\mathbb{CP}^2 \# \mathbb{CP}_{b_2^-}^2 \quad (2.14)$$

is a connected sum of rational algebraic surfaces.

In the setup of differential geometry, our arguments give

Lemma 2.1. *Let M be any oriented 4 manifold containing a smooth 2-sphere S^2 such that the cohomology class $[S^2] \in H^2(M, \mathbb{Z})$ has selfintersection $[S^2]^2 = -1$. Then*

$$M = M_1 \# \overline{\mathbb{CP}^2}. \quad (2.15)$$

Indeed the boundary of a small normal neighbourhood of S^2 in M is the ordinary 3-sphere, so we can delete it and glue in a ball to get M_1 .

Remark. Some time ago the following conjecture was popular: any 4 manifold is a connected sum of algebraic surfaces and antisurfaces. This conjecture was disproved in [G-M].

Now the constants in front of the summands in (2.13) and (2.13')

$$k(S) = -\frac{I}{16}; \quad m(S) = b_2^\pm \pm \frac{2I}{16} \quad (2.16)$$

are expressive enough to describe the topological type of S . For even surfaces, the question about relations and restrictions, that is, “what pairs can occur” is very natural and important in the geography of algebraic surfaces (see the Exercise after this lecture).

Recall that the comparison of the underlying smooth structure of an algebraic surface S and the topomodel $\text{top } S$ was the starting point for modern study of smooth structures of algebraic surfaces. Namely Moishezon and Mandelbaum showed that for a rather wide class of surfaces S , the connected sum of S with the plane $\mathbb{CP}^2$ is diffeomorphic to its topomodel (see [M] for details). This class of surfaces is the most “intuitive” class of surfaces and we will meet it many times below. However for even surfaces and odd surfaces with $b_2^+ > 1$, the topomodel isn’t diffeomorphic to the surface. Thus for surfaces of this type, $\# \text{diff}(\text{top } S) \geq 2$. This fact was proved by Donaldson [D1].

But there is one class of simple connected surfaces which are smoothly equivalent to their topomodels:

Lemma 2.2. *If S is a rational surface then $S = \text{top } S$ smoothly.*

You can check this fact yourself using the description of the blowup and classification of algebraic surfaces.

Remark. The idea of decomposing 4-manifolds in parallel with the decomposition of the intersection form isn’t so naive. Namely Freedman and Taylor proved that we can do it using glueing along the homology 3-spheres instead of S^3 [F-T].

Now the decomposition $q_1 \oplus q_2$ of a quadratic form defines two embeddings

$$(\mathrm{O}(q_i) \times \mathrm{id}) \rightarrow \mathrm{O}(q_1 \oplus q_2) \quad (2.17)$$

and for us it will be very interesting to describe the relations between $\mathrm{Diff} M_i$ and $\mathrm{Diff} M$ in the same style. But unfortunately, we can only do this in one particular case. Namely if M is a glueing (or grafting) of M_1 and M_2 along the boundary X , and φ is a diffeomorphism of M_i whose restriction to X is isotopic to id_X then by the Isotopy Extension Theorem, φ can be extended to a diffeomorphism $\bar{\varphi} \in \mathrm{Diff} M$ whose images is contained in the subgroup (2.17).

Using this remark we can prove the following fact:

Lemma 2.1'. *In the situation of Lemma 2.1 the reflection $R_{[S^2]}$ can be realised as a diffeomorphism of M .*

Proof. We have to construct a diffeomorphism φ on $\overline{\mathbb{CP}^2}$ such that $\rho(\varphi)([\overline{\mathbb{CP}^1}]) = -[\overline{\mathbb{CP}^1}]$ and the restriction φ to the normal boundary of antiline is isotopic to id . To do this, return to the $\mathbb{CP}^2$ with the standard real structure given by the anti involution $h: \mathbb{CP}^2 \rightarrow \mathbb{CP}^2$. Let L be a real line and S^3 the boundary of a standard normal neighbourhood of L invariant under h . To see that the orientation preserving diffeomorphism $\varphi = h$ satisfies our conditions we only need remark that $h|_{S^3}$ preserves the orientation of S^3 .

Corollary 1. *If S is a nonminimal surface then the canonical class K_S up to sign is not a diffeomorphism invariant of S .*

Indeed, let E be an exceptional curve. Then the class $[E]$ is realised as a smooth 2-sphere and the reflection $R_{[E]} \in \mathrm{O}(q_S)$ can be realised by a diffeomorphism. Let $\sigma: S \rightarrow S_0$ be the blowdown of this exceptional curve. Then the canonical class satisfies

$$K_S = \sigma^*(K_{S_0}) + E; \quad \sigma^*(K_{S_0}) \cdot E = 0; \quad K_S \cdot E = -1$$

and

$$R_{[E]}(K_S) = \sigma^*(K_{S_0}) - E \neq \pm K_S.$$

As a corollary of the last problem of Exercise 2 of Lecture 1 and this Lemma we get

Corollary 2. *Let S be a rational surface with $\mathrm{rank} \mathrm{Pic} S \leq 3$. Then $\mathrm{Mod} S = \mathrm{O}(q_S)$.*

In the same vein one can prove the following slight generalisation of Lemma 2.1':

Lemma 2.3. *Let M be 4 manifold containing a 2-sphere S^2 with $[S^2]^2 = -2$. Then $R_{[S^2]}$ can be realised as an orientation preserving diffeomorphism.*

Proof. First of all, a small neighbourhood of the normal bundle of S^2 in M is a small neighbourhood N of the diagonal δ in the direct product $\mathbb{CP}^1 \times \mathbb{CP}^1$. This N is a holomorphic D^2 -bundle over $\mathbb{CP}^1$; let $[s]$ be the cohomology class of the zero section. We need only construct some diffeomorphism φ of $\bar{N}$ such that

$$\varphi|_{\partial N} = \mathrm{id}, \quad \text{and} \quad \varphi^*([s]) = -[s]. \quad (2.18)$$

Now if we consider N as the subset of $\mathbb{R}^3 \times \mathbb{R}^3$ given by the relations

$$N = \{(v, v') \in \mathbb{R}^3 \times \mathbb{R}^3 \mid \|v\| = 1, \|v'\| \leq 1, \text{ and } v \cdot v' = 0\} \quad (2.19)$$

then one can check that the following diffeomorphism

$$\begin{aligned} \varphi(v, v') &= (\cos((1 - \|v'\|) \cdot \pi) \cdot v + \sin((1 - \|v'\|) \cdot \pi) \frac{v'}{\|v'\|}, \\ &\quad - \|v'\| \cdot \sin((1 - \|v'\|) \cdot \pi) \cdot v + \cos((1 - \|v'\|) \cdot \pi) \cdot v'), \end{aligned} \quad (2.20)$$

if $v' \neq 0$ and $\varphi(v, 0) = (-v, 0)$ holds (2.18).

Remark. This direct construction (due to [F–M], Proposition 2.4) is the construction of the monodromy diffeomorphism associated to a small loop enclosing an ordinary double point in a family of algebraic surfaces (for example quadrics). S^2 is the so-called *vanishing cycle* and the display of the reflection is called the *Picard–Lefschetz formula*.

Now let S be a rational surface with $9 > \text{rank Pic } S$. Then

$$q_S = I_+ \oplus (\text{rank Pic } S - 1)I_-$$

(here we use the notation of Lecture 1, Exercise 2 and set $k = \text{rank Pic } S - 1$).

Lemma 2.4. $O(q_S)$ is generated by permutations, automorphisms of summands and the reflections $R_{x+y_1+y_2}$ and $R_{x+y_1+y_2+y_3}$.

Proof. For $k = 1$ the statement is obvious. Let $\varphi \in O(q_S)$ be an automorphism and

$$\varphi(x) = a \cdot x + \sum_1^k b_i \cdot y_i \quad (2.21)$$

and

$$1 = a^2 - \sum_1^k b_i^2. \quad (2.21')$$

Consider the first nontrivial case $k = 2$. Using permutations and automorphisms of summands we only need to consider the case when

$$a > 0; \quad b_1 \geq b_2 \geq 0. \quad (2.22)$$

Now

$$\varphi \cdot R_{x+y_1+y_2}(x) = (3a - 2b_1 - 2b_2) \cdot x + \cdots \quad (2.23)$$

and we would like to prove that $|3a - 2b_1 - 2b_2| < a$. Then by induction on $|a|$ we get our statement. Now

$$3a - 2b_1 - 2b_2 \leq -a \implies a \leq \frac{1}{2}(b_1 + b_2) \leq b_1 \implies a^2 \leq b_1^2,$$

which contradicts (2.21'). On the other hand,

$$3a - 2b_1 - 2b_2 \geq a \implies a \geq b_1 + b_2 \implies$$

$$1 = a^2 - b_1^2 - b_2^2 \geq (b_1 + b_2)^2 - b_1^2 - b_2^2 = 2b_1 \cdot b_2 \implies b_2 = 0$$

and we are done.

Now it is enough to consider the case when in (2.21)

$$b_1 \geq b_2 \geq \cdots \geq b_k > 0; \quad \text{and} \quad b_{k+i} = 0. \quad (2.24)$$

Then

$$\varphi \cdot R_{x+y_1+y_2+y_3}(x) = (2a - b_1 - b_2 - b_3) \cdot x + \cdots \quad (2.25)$$

and we will prove that

$$|2a - b_1 - b_2 - b_3| < a.$$

Indeed

$$(2a - b_1 - b_2 - b_3) \leq -a \implies a \leq \frac{1}{3}(b_1 + b_2 + b_3) \leq b_1,$$

which contradicts (2.21'). On the other hand,

$$(2a - b_1 - b_2 - b_3) \geq a \implies a \geq b_1 + b_2 + b_3 \implies$$

$$1 \geq (b_1 + b_2 + b_3)^2 - \sum_1^k b_i^2 = (b_1 \cdot b_2 + b_2 \cdot b_3 + b_1 \cdot b_3) - \sum_4^k b_i^2 \geq 6b_2 \cdot b_3 - (k-3)b_k^2.$$

Now

$$k < 5 \implies 1 \geq 2b_2 \cdot b_3$$

thus $k = 5$ and

$$b_2 \cdot b_3 = 1 \implies a^2 - b_1^2 = 8 \implies \varphi(x) = 3x - \sum_1^8 y_i; \quad (2.26)$$

but this is impossible because $3x - \sum_1^8 y_i$ is a characteristic vector for the lattice q_S , and x isn't. We are done.

Remark. Actually, in the same vein, one can also prove this statement for $k = 9$ (see the Exercises to this Lecture).

Now using these lemmas one can prove

Theorem 2.1. *Let S be a rational surface with $K_S^2 > 0$ then*

$$\text{Mod } S = O(q_S). \quad (2.27)$$

Proof. For the quadric $\mathbb{CP}^1 \times \mathbb{CP}^1$ this is obvious. Otherwise $H^2(S, \mathbb{Z}) = \text{Pic } S = I_+ \oplus (9 - K_S^2)I_-$, and $S = \mathbb{CP}_{9-K_S^2}^2$ is realised as the plane with a number of points blown up, with the generator x realised as the inverse image of the line $\mathbb{CP}^1$, and $\{y_i\}$ the classes of the exceptional (-1) -curves. Thus permutations of the summands and automorphisms of summands can be realised by diffeomorphisms preserving the orientation. We have to realise the cohomology classes $x + y_1 + y_2$ and $x + y_1 + y_2 + y_3$ by smooth 2-spheres. But because of the above it is enough to construct such realisations for the classes $x - y_1 - y_2$ and $x - y_1 - y_2 - y_3$. From the differential geometric point of view the location of blown-up points on $\mathbb{CP}^2$ has absolutely no importance so we can blow up three points p_1, p_2, p_3 on $\mathbb{CP}^2$ on one line L . The geometric inverse image of L on S gives the realisation for $x - y_1 - y_2 - y_3$. The realisation of $x - y_1 - y_2$ is obvious.

Remark. The investigation of the automorphisms group of $I_+ \oplus (9 - K_S^2)I_-$ can be extended to the case when K_S^2 is negative. Here Friedman and Morgan [F-M] proposed the very beautiful theory of supercells. Using this new combinatorical construction one can prove that the standard representation of $\text{Mod } \overline{\mathbb{CP}_k^2}$ is irreducible. The size of the present lectures has not permitted us to include this construction fully, but we could not resist touching on it at least in the series of Exercises 2.

Exercises to Lecture 2.

1. Prove the following statements for even surfaces.

a) If $I \leq 0$ then

$$k(S) \leq m(S) + 1. \quad (2.28)$$

b) If $I > 0$ then

$$m(S) \geq 13k(S) - 1. \quad (2.29)$$

c) For even surfaces, express the results of [P-P-X] in terms of the numbers $k(S)$ and $m(S)$ (2.16).

2. Let S be a rational surface satisfying the conditions: the anticanonical system $|-K_S|$ contains a smooth irreducible curve F , and every irreducible effective curve $C \neq F$ on S is a (-1) -(smooth) rational curve. Such surface will be called *good*. Prove that

a) such surfaces exist;

b) let $\mathcal{K} \subset H^2(S, \mathbb{R})$ be the Kähler cone of S and $e \in H^2(S, \mathbb{Z})$ any class with $e^2 = -1$; if the intersection

$$W_e = e^\perp \cap \overline{\mathcal{K}} \neq \emptyset, \quad (2.30)$$

then e can be realised as a nonsingular irreducible algebraic curve (see [P]);

c) let $\{E_i\}$ be the set of all (-1) -curves on S and

$$\mathbb{S} := \bigcup_i R_{E_i}(K) \quad (2.31)$$

be the supercell in $H^2(S, \mathbb{R})$. Then according to the beautiful results of Friedman and Morgan [F-M]

$$\varphi \in \text{Mod } S \implies \varphi(\mathbb{S}) = \pm \mathbb{S} \quad (2.32)$$

and

$$\varphi(\mathbb{S}) = \mathbb{S} \implies \varphi \in \text{Mod } S. \quad (2.33)$$

Could you find an elementary proof of these facts?

d) $K_S^2 \geq 0 \implies \mathbb{S} = \text{light cone}$;

e) as a representation, $\text{Mod } S$ is irreducible.

3) Let e_1, e_2 be two classes in $H^2(S, \mathbb{Z})$ with $e_i^2 = -1$ which can be realised as smooth 2-spheres. Prove that $e_1 \cdot e_2 = 0$ or $e_1 = \pm e_2$.

Hint: apply the reflection $R_{[e_1]}$ to e_2 .

LECTURE 3. MONODROMY AND D-S-CURVES.

For most of the “intuitive” classes of algebraic surfaces, including complete intersections, elliptic surfaces, Moishezon’s surfaces, and Salvetti’s Abelian covers of quadrics, good information about the underlying smooth structure comes from an investigation of the monodromy group $\text{Mon } S$ (1.20). We will consider $\text{Mon } S$ as a representation on the lattice $H^2(S, \mathbb{Z})$. This representation has a fixed vector K_S (see (1.22)) and has many invariant sublattices. We use some of these sublattices to estimate Mod -invariant sublattice of $H^2(S, \mathbb{Z})$.

If the smooth classification of algebraic surfaces is close to the classification up to deformation equivalence then configurations of effective curves on algebraic surfaces stable under deformation should be very important in understanding the underlying smooth structures. We begin with some examples of configurations of such type.

First of all one has the $\text{Mon } S$ -invariant sublattice

$$cv \text{ Pic } S := \bigcap_{\text{all } X} \text{im}(\text{res}: \text{Pic } X \rightarrow \text{Pic } S), \quad (3.1)$$

where X is any algebraic family containing S (see (1.18–20)). Every vector in $cv \text{ Pic } S$ is $\text{Mon } S$ -invariant. This sublattice isn’t empty because it contains the canonical class K_S .

Much more interesting and important is the sublattice of $\text{Pic } S$ generated by (-1) -curves. Recall that if S is nonrational then the map

$$m: S \rightarrow S_{\min} \quad (3.2)$$

to its minimal model is unique. Let

$$K_{\min} = m^*(K_{S_{\min}}) \quad (3.3)$$

be the pullback of the canonical class of the minimal model. Then the collection of (-1) -curves

$$\{E_i \subset S \mid E = \mathbb{CP}^1, [E]^2 = -1\} \quad (3.4)$$

is uniquely determined too. Consider the sublattice

$$\mathcal{E}_{-1} = \langle E_1, \dots, E_{n-1} \rangle \quad (3.5)$$

generated by (-1) -curves. This sublattice is $\text{Mon } S$ -invariant:

$$\mathcal{E}_{-1}^{\text{Mon } S} = \mathcal{E}_{-1}. \quad (3.6)$$

Moreover this sublattice contains the class

$$e = \sum_1^{n-1} E_i = K_S - K_{\min} \quad (3.7)$$

which is $\text{Mon } S$ -invariant. Therefore $K_{\min}$ is also $\text{Mon } S$ -invariant:

$$K_{\min}^{\text{Mon } S} = K_{\min}. \quad (3.8)$$

We saw that the canonical class K_S is not $\text{Mod } S$ -invariant (see Lecture 2, Corollary 1 of Lemma 1'). But we can hope that the statement is true for the canonical class of the minimal model:

$K_{\min}$ Conjecture.

$$K_{\min}^{\text{Mod } S} = \pm K_{\min}. \quad (3.9)$$

This conjecture is closely related to the following question raised by Friedman and Morgan (see Exercise 2 after this lecture).

(-1) -curve Conjecture.

$$\mathcal{E}_{-1}^{\text{Mod } S} = \mathcal{E}_{-1}. \quad (3.10)$$

An exceptional (-1) -curve is a particular case of a curve that is stable under deformation. Recall the definition of such curves:

Definition 3.1. An irreducible curve $C \subset S$ is called a d -stable curve if for any small deformation of S

$$\pi: S \rightarrow \Delta; \quad 0 \in \Delta; \quad \pi^{-1}(0) = S \quad (3.11)$$

the threefold S contains a germ of a surface D such that

$$D \cap S = C. \quad (3.12)$$

Let $T\Delta_0$ be the tangent space to Δ at $0 \in \Delta$, $k: T\Delta_0 \rightarrow H^1(TS)$ the Kodaira-Spencer map of the family (3.11) and $s_1: H^1(T(-[C])) \rightarrow H^1(T)$ the homomorphism induced by the multiplication by the equation of C . Then

$$C \text{ is } d\text{-stable} \implies k(T\Delta_0) \subset s_1(H^1(T(-[C]))). \quad (3.13)$$

Thus if C is a (-1) -curve, then $H^1(TS|_C) = 0$ and a (-1) -curve is d -stable. If C is a smooth (-2) -rational curve and the dimension of the space of deformations of S is > 1 then $H^1(TS) \rightarrow H^1(TS|_C)$ is usually nontrivial, and C is not d -stable.

Example 1. Let $\pi: S \rightarrow \Delta$ be a local family of surfaces such that $\pi^{-1}(x)$ is smooth for $x \neq 0$, and $\pi^{-1}(0) = S_0$ has one node. Let S be the desingularisation of S_0 and

L be (-2) -rational curve blown up from the node. Then it easy to see that L isn't d-stable.

Now in all our constructions we only consider the d-stable effective curves. Using Mon S -invariant classes and d-stable curves we can construct a lot of Mon S -invariant sublattices and first of them is the sublattice $v_2(S)$ which contains the sublattice $sV(S)$ in the statement of the Main Theorem (see Lecture 1): consider the set of d-stable curves $\{C_i\}$ subject to the conditions

$$2C_i \cdot K_{\min} \leq K_{\min}^2 \quad (3.14)$$

and the sublattice

$$v_2(S) = \langle \{C_i\} \rangle \quad (3.15)$$

generated by these curves. Then

$$v_2(S)^{\text{Mon } S} = v_2(S). \quad (3.16)$$

By definition,

$$\mathcal{E}_{-1} \subset v_2(S).$$

Now consider the sublattice

$$kv_2(S) = \langle K_{\min}, v_2(S) \rangle. \quad (3.17)$$

Of course this sublattice is Mon S -invariant too:

$$kv_2(S)^{\text{Mon } S} = kv_2(S) \quad (3.18)$$

and we will use it to estimate the Mod S -invariant sublattice $sV(S)$ in the statement of the Main Theorem (see Lecture 1).

Example 2. Let S_B be the Barlow surface (see [Bar]). Recall that this surface is minimal, and has four disjoint smooth rational (-2) -curves $L_1, \dots, L_4$. Blow down these curves to four nodes (ordinary double points):

$$\sigma: S_B \rightarrow S_0; \quad \sigma(L_i) = s_i; \quad \bigcup_i s_i = \text{Sing } S_0.$$

The surface S_0 admits 2×5 - diadral cover:

$$\varphi: Q \rightarrow S_0$$

where Q is the 20-nodal quintic given in projective coordinates $(x_i, \dots, x_5)$ for $\mathbb{P}^5$ by

$$\sum_1^5 x_i = 0; \quad \sum_1^5 x_i^5 - \frac{5}{4} \left(\sum_1^5 x_i^2 \right) \left(\sum_1^5 x_i^3 \right) = 0 \quad (3.19)$$

(Q is the canonical model of a surface constructed by van der Geer and Zagier [Bar]). From this we can see that every (-2) -curve L_i on S_B isn't d-stable.

Now S_B is minimal and $K_{S_B}^2 = 1$. Thus the set $\{C_i\}$ is empty and the sublattice

$$v_2(S_B) = 0 \quad (3.20)$$

and

$$kv_2(S_B) = \mathbb{Z} \cdot K_S. \quad (3.21)$$

Let

$$\text{rank } kv_2(S) = r(S). \quad (3.22)$$

Then $\Lambda^{r(S)} kv_2(S)$ is a Mon-invariant proper subspace of this representation. On the other hand it is very useful to consider the polynomial invariants of $\text{Mon } S$, that is, some eigenvectors $\gamma_d \in S^d H^2(S, \mathbb{Z})$ of the symmetric power representations of $\text{Mod } S$.

The crucial point here is the existence of some special transformations and subgroup of transformations in $\text{Mon } S$, namely, some special reflections. A smooth (-2) -sphere S^2 in S can be realised as smooth rational curve that is an exceptional (-2) -curve L which can be realised as a vanishing cycle for some algebraic family containing S . But first we recall some beautiful results of Invariant Theory of reflection groups.

Let H, q be a lattice with a unimodular quadratic form (of course it will be our $H^2(S, \mathbb{Z})$ with the intersection form q_S) and $K: H \rightarrow \mathbb{Z}$ be a linear form on H (it will be the fundamental class of the canonical class of S). Let

$$S = \{e_i\} \quad (3.23)$$

be an infinite set of vectors of H with $e_i^2 = -2$. For this set define Γ_S as the subgroup generated by the reflections R_{e_i} , with $e_i \in S$ that is

$$\Gamma_S = \left\langle \bigcup_{e_i \in S} R_{e_i} \right\rangle. \quad (3.24)$$

Let

$$S^* = \bigoplus_d S^d H^2(S, \mathbb{Z}) \quad (3.25)$$

be the graded algebra of polynomials on H and

$$S^{*\Gamma_S} = \bigoplus_d S^{d\Gamma_S} \quad (3.26)$$

be the subalgebra of Γ_S -invariant polynomials on H .

Now the orthogonal group $O(q)$ is the subgroup of the Lie group $O(q) \otimes \mathbb{C}$. Let $\overline{\Gamma_S}$ be the Zariski closure of Γ_S in $O(q)$. Then

$$S^{*\Gamma_S} = S^{*\overline{\Gamma_S}}. \quad (3.27)$$

Thus we have proved the following Lemma in pure the complex situation.

Lemma 3.1, (see [E]). If

- 1) the subset $\mathcal{S}$ (3.23) is contained in the hyperplane $\ker K$ of the linear form K ,
- 2) $\mathcal{S}$ is a $\Gamma_{\mathcal{S}}$ -orbit,

then $\overline{\Gamma_{\mathcal{S}}} = \text{Stab}_K$ is the subgroup of $O(q) \otimes \mathbb{C}$ consisting of automorphisms preserving K and

$$S^* \overline{\Gamma_{\mathcal{S}}} = \mathbb{C}[q, K] \quad (3.28)$$

that is every $\Gamma_{\mathcal{S}}$ -invariant polynomial is a polynomial of the intersection form as a form of degree 2 and K as a linear form.

The geometric meaning of this result is following: consider the projectivisation $\mathbb{P}(H) \otimes \mathbb{C}$ of the complexification of the space H and the zero set Γ of any $\overline{\Gamma_{\mathcal{S}}}$ -invariant polynomial γ of degree d . Let Q be the zero set of the quadric q and K be the hyperplane given by the linear form K . Then if d is even,

$$\Gamma = Q_1 \cup \cdots \cup Q_{\frac{d}{2}}, \quad (3.29)$$

where Q_i are quadrics from the pencil $\langle Q, 2K \rangle$ generated by Q and the quadric of rank 1 K . If d is odd then

$$\Gamma = Q_1 \cup \cdots \cup Q_{\frac{d-1}{2}} \cup K. \quad (3.29')$$

Indeed by Lemma

$$\gamma = \sum_{i=0}^{\frac{d}{2}} a_i \cdot q^i \cdot K^{d-2i} = c \prod_{i=0}^{\frac{1}{2}d} (q - \lambda_i K^2). \quad (3.30)$$

From this the following statement is obvious:

Corollary 1. In this situation if G is any group containing $\Gamma_{\mathcal{S}}$ which admits G -invariant polynomial γ and γ isn't in $\mathbb{C}[q]$ then K is G -invariant.

Indeed the pencil of quadrics $\langle Q, 2K \rangle$ is invariant with respect to G . But this pencil contains two singular quadrics only and in particular it contains one quadric of rank 2 only. It is $2K$. In the same vein we have in $\mathbb{P}H$

$$K = \left\langle \text{Sing } \Gamma \cap Q \right\rangle_{\frac{1}{2}d}, \quad (3.31)$$

where $\text{Sing}_n \Gamma$ is the set of singular points of Γ of multiplicity more $\geq n$ and $\langle N \rangle$ is as usual the linear hull of the subset N in $\mathbb{P}H$.

Remark. In the particular case when the linear form K is zero the statement of Lemma 3.1 remains true, that is, if the linear hull of the set $\mathcal{S}$ is the whole of H then $\gamma \in \mathbb{C}[q]$.

There exists a slight generalisation of the statement of Lemma to the case when the linear hull of $\mathcal{S}$ is the intersection of kernels of k linear forms $K_1, \dots, K_k$. Then $\gamma \in \mathbb{C}[q, K_1, \dots, K_k]$. The analog of Corollary 1 in this situation is the following fact:

Lemma 3.2. *If γ is a G -invariant polynomial and $\gamma \in \mathbb{C}[q, K_1, \dots, K_k]$, with $k \leq \text{rank} H - 2$, then the subspace $\langle K_1, \dots, K_k \rangle$ is G -invariant.*

Proof. The intersection of the kernels of the linear forms $\{K_i\}$ we can define as the linear hull of the locus of singular points of Γ of multiplicity $\geq [\frac{1}{2}(d+1)]$:

$$\bigcap_{i=1}^k K_i = \left\langle \text{Sing } \Gamma_{[\frac{d+1}{2}]} \cap Q \right\rangle. \quad (3.32)$$

To see this consider the double cover

$$\varphi: \tilde{Q} \rightarrow \mathbb{P}H \quad (3.33)$$

with the ramification along Q and the full inverse image

$$\tilde{\Gamma} = \varphi^{-1}(\Gamma) \quad (3.34)$$

in $\tilde{Q}$. Then $\tilde{Q}$ is the smooth quadric in $\mathbb{P}\tilde{H}'$, with $\text{rank} H' = \text{rank} H + 1$, and there exists a projection $p: \tilde{H} \rightarrow H$. Moreover there exists the linear form L on H' such that

$$\ker L \cap \tilde{Q} = Q. \quad (3.35)$$

Let $\tilde{K}_i$ be the inverse image of K_i under p . Then for any polynomial

$$\gamma = \sum_i a_{i,j_1,\dots,j_k} q^i \cdot K_1^{j_1} \cdots K_k^{j_k}$$

we have the polynomial $\tilde{\gamma} = \sum_i a_{i,j_1,\dots,j_k} L^2 \cdot \tilde{K}_1^{j_1} \cdots \tilde{K}_k^{j_k}$ with the zero set $\tilde{\tilde{\Gamma}} \subset \mathbb{P}H'$ and

$$\tilde{\Gamma} = \tilde{\tilde{\Gamma}} \cap \tilde{Q}. \quad (3.36)$$

Now the hypersurface $\tilde{\tilde{\Gamma}}$ in $\mathbb{P}H'$ is a cone of degree d with the vertex

$$\Delta = L \bigcap_{i=1}^k \tilde{K}_i \quad (3.37)$$

that is

$$\Delta = \text{Sing}_d \tilde{\tilde{\Gamma}}. \quad (3.38)$$

Thus from (3.36) and (3.38) we have (3.32)

If S is L-surface (see Definition 1.4) the group $\text{Mon } S$ is generated by reflections and we can use these arguments to describe the polynomial invariants of $\text{Mon } S$. In this case the reflections generated by (-2) -rational curves which can be realised as the collection of vanishing cycles of L-families. So considering as the set $\mathcal{S}$ the set of vanishing cycles of all L-families of L-surface we need only to check the condition 2) of Lemma 3.1 because the condition 1) with $K = K_S$ holds automatically. It can be checked for all complete intersections (W. Ebeling), simply connected elliptic surfaces, Moishezon and Salvetti coverings and so on. Such surfaces are called *surfaces with big monodromy* or surfaces with big diffeomorphism group.

Definition 3.1. The natural number $k_m(S)$ is called Mon-rank of S if there exists k linear forms $K_1, \dots, K_k$ on $H^2(S, \mathbb{Z})$ such that any Mon-invariant polynomial γ on $H^2(S, \mathbb{Z})$ is polynomial of q_S and K_i , that is, $\gamma \in \mathbb{C}[q_S, K_1, \dots, K_k]$ and k is the smallest number with this condition.

So if $k_m(S) \leq b_2 - 2$ then the sublattice

$$m(S) = \bigcap_i K_i \subset H^2(S, \mathbb{Z}) \quad (3.39)$$

is uniquely defined and

$$m(S)^{\text{Mon } S} = m(S). \quad (3.40)$$

Finally we cannot resist touching on the differential geometry analog of the Theory of minimal models due to Friedman and Morgan and Brussee [Bru1]. Recall that an embedded 2-sphere S^2 in 4 manifold is called a (-1) -sphere if $[S^2]^2 = -1$. If 4 manifold is an algebraic surface then every (-1) -curve is (-1) -sphere. Successively contracting all (-1) -curves on a surface gives the minimal model (which is unique if Kodaira dimension $\kappa \geq 0$). So the decomposition

$$H^2(S, \mathbb{Z}) = H^2(S_{\min}) \oplus^\perp \mathcal{E}_{-1} \quad (3.41)$$

is a deformation invariant. If 4 manifold M contains a (-1) -sphere then one has the decomposition $M = M_1 \# \mathbb{CP}^2$. According to Exercise 3 of Lecture 2 two different (-1) -sphere don't intersect so we can successively contract all of them to get the decomposition

$$M = M_{\min} \# \overline{k\mathbb{CP}^2}. \quad (3.42)$$

Is this decomposition a smooth invariant?

For algebraic surfaces one has the following conjectures:

(-1) sphere Conjecture. For any algebraic surface S with $\kappa \geq 0$ every a (-1) -sphere is homologous to a (-1) -curve.

Obviously

$$(-1)\text{-sphere Conjecture} \implies (-1)\text{-curve Conjecture}. \quad (3.43)$$

More generally

Negative Decomposition Conjecture. If any algebraic surface S with $\kappa \geq 0$ is decomposed as $S = M \# N$ where N is a negative definite 4 manifold then $H^2(N, \mathbb{Z})$ is generated by (-1) -curves.

Obviously

$$\text{Negative Decomposition Conjecture} \implies (-1)\text{-sphere Conjecture}. \quad (3.44)$$

The following cycle of Exercises was suggested to me by Rogier Brussee.

Exercises to Lecture 3.

1. Prove that

$$K_{\min} \text{ Conjecture} \implies (-1)\text{-sphere Conjecture.}$$

Hint: a) Let e be the cohomology class of (-1) -sphere; using Exercise 3 of Lecture 2 prove that

$$e \neq (-1)\text{-curve} \implies e \in H^2(S_{\min}, \mathbb{Z}). \quad (3.45)$$

b)

$$R_e(K_{\min}) = \pm K_{\min} \implies e \cdot K_{\min} = 0. \quad (3.46)$$

This is a contradiction: $K_{\min} \bmod 2$ is a characteristic element of $(H^2(S, \mathbb{Z}), q_S)$.

2. Prove that

$$(-1)\text{-curve Conjecture} \implies P_2(S) = h^0(\mathcal{O}_S(2K_S)) \text{ is a smooth invariant.} \quad (3.47)$$

Hint: use the equality

$$h^1(\mathcal{O}_S(2K_S)) = \#\{E_i\}. \quad (3.48)$$

3. Prove that

$$(-1)\text{-curve Conjecture} \implies P_n(S) = h^0(\mathcal{O}_S(nK_S)) \text{ is a smooth invariant of } S. \quad (3.49)$$

LECTURE 4. DONALDSON POLYNOMIALS. THE GREAT PROGRAM

As we saw the representation $\text{Mod } S$ can be irreducible (when S is rational) but Donaldson had shown that if $p_g = \frac{b_2^+ - 1}{2} > 0$ then big symmetric power $S^d \text{Mod } S$ admits eigenvectors (or semiinvariants) if

$$d = 1 + b_2^+ \bmod 4 \quad \text{or} \quad d = 1 + b_2^+ - K_S^2 \bmod 4. \quad (4.1)$$

Of course for even d there is a lot trivial semiinvariant polynomials from $\mathbb{C}[q_S]$. But the existence of nontrivial polynomials that are Mod -invariants up to sign provides some information about the representation $\text{Mod } S$.

Example 1. Let S be a surface with big monodromy. Then the Mon-rank of S is 1 (see Corollary 1 of Lemma 3.1 and Definition 3.1). So the existence of any nontrivial $\text{Mod } S$ -semi invariant polynomial provides the $\text{Mod } S$ -invariance of the canonical class K_S because of the implication

$$\text{Mod-invariance} \implies \text{Mon-invariance} \quad (4.2)$$

and Corollary of Lemma 3.1. In the same vein, for surfaces of Mon-rank $k_m(S)$ the existence of nontrivial Mod -semi invariant polynomial γ implies the existence of the $\text{Mod } S$ -invariant nontrivial sublattice of the lattice $m(S)$ (3.39) given by formula (3.32) that is the reducibility of the representation $\text{Mod } S$.

So the monodromy group provides information on the shape of any Mod -semi invariant polynomial (as in Definition 3.1) and from the shape of the polynomial we can recognize a Mod -invariant sublattice of $H^2(S, \mathbb{Z})$ (see (3.31). Of course any information about the shape of semi invariant polynomials can be used in the first

part of this program. But the investigation of the beautiful geometry of Donaldson polynomials provides a much deeper description of $\text{Mod } S$. We will give the brief survey of this theory and for simplicity we will call the full chain of results, conjectures and constructions in the setup of Donaldson polynomials the *Great Program*.

The main subject of this theory is a 4 manifold M with a fixed conformal structure, that is, a fixed Riemannian metric up to conformal transformations and a pair (M, C) where C is an oriented smooth 2 manifold in M . A conformal structure on M induces a conformal structure on C so C is a smooth compact Riemann surface, that is, an algebraic curve over $\mathbb{C}$. So it is very useful to begin this theory with the case of Riemann surfaces, that is, with the case of an oriented 2 manifolds with a conformal structure, in other words, with the case of algebraic curves.

Remark. Later it turns out in some cases that the invariants just constructed do not depend on the choice of a conformal structure, and are invariants of the smooth structures of M or C (that is, of the natural number $g(C) = \text{genus of } C \text{ in the last case}$).

So let C be an smooth algebraic curve of genus g and

$$\mathbb{S}(C) = \text{Sym}(H_{\text{even}}(C, \mathbb{R})) \otimes \Lambda(H_{\text{odd}}(C, \mathbb{R})) \quad (4.3)$$

be the graded algebra where $\deg(\sigma) = 4 - i$ if $\sigma \in H_i(C)$. Now let M_C^{odd} be the moduli space of rank 2 stable vector bundles on C with fixed determinant of odd degree. Then on the direct product $C \times M_C^{\text{odd}}$ there is the universal bundle $\mathcal{E}$ and the class

$$-p_1(\mathcal{E}) = c_1(\mathcal{E})^2 - 4c_2(\mathcal{E}) = c_2(ad\mathcal{E}) \in H^4(C \times M_C^{\text{odd}}). \quad (4.4)$$

This class is defined by the slant product, that is, the homomorphism $\mu: H_*(C) \rightarrow H^*(M_C^{\text{odd}})$ defined by

$$\mu(\sigma) = -\frac{1}{4}p_1(\mathcal{E})/\sigma. \quad (4.5)$$

So $\deg(\alpha) = i \implies \mu(\alpha) \in H^i(M_C^{\text{odd}})$.

More precisely

$$-p_1(\mathcal{E}) = -2[C] \otimes \Theta + \text{pt} \otimes x - 4 \sum_{i=1}^g (a_i \otimes \mu(a_i) + b_i \otimes \mu(b_i)), \quad (4.6)$$

where

$$\Theta \in H^2(M_C^{\text{odd}}, \mathbb{Z}); \quad x \in H^4(M_C^{\text{odd}}, \mathbb{Z});$$

$[C]$ is the fundamental class of C ; $[\text{pt}]$ is the class of a point of C and $a_i, b_i, i = 1, \dots, g$ is a symplectic basis of $H^1(C, \mathbb{Z})$.

Now we can view the Newstead invariant as a linear function

$$N_g^{\text{odd}}: \mathbb{S}(C) \rightarrow \mathbb{Q} \quad (4.7)$$

sending a typical monomial $\alpha = \sigma_1 \cdots \sigma_r$ where $\sigma_i \in H_{n_i}(C)$ of total degree $d(\alpha) = \sum_{i=1}^r (4 - n_i)$ to 0 if $d \neq 6g - 6 = \dim M_C^{\text{odd}}$ and to the index of intersection

$$\mu(\sigma_1) \cdots \mu(\sigma_r) \quad (4.8)$$

if $d = 6g - 6 = \dim M_C^{\text{odd}}$.

Every homogeneous polynomial of degree $3(g-1)$ we will call a Newstead polynomial.

Remark. Of course we can apply this constructions to any family of vector bundles, that is, to any vector bundle $\mathcal{E}$ over $C \times B$ where B is a base of this family. Of course in this case we have one polynomial (4.7) which we will denote by $N_g(\mathcal{E})$.

Every Newstead polynomial (4.7) is a Diff C -invariant. From this we can see its shape (see Exercises). Namely let

$$\gamma = 2 \sum_{i=1}^g \mu(a_i) \cdot \mu(b_i) \in H^6(M_C^{\text{odd}}) \quad (4.9)$$

then

$$N_g^{\text{odd}} \in \mathbb{C}[[\text{pt}], [C], \gamma]. \quad (4.10)$$

Thus this polynomial is nonzero only on the monomials $[C]^m \text{pt}^n \gamma^p$; for $m+2n+3p = 3g-3$.

Since the nondegenerate pairing (4.8) only involves the classes $[C], [\text{pt}], \gamma$ and does not depend on the conformal structure on C , we can consider, instead (4.3) of the graded algebra

$$\mathbb{S}_3 = \text{Sym}(\alpha \cdot \mathbb{Z} \oplus \beta \cdot \mathbb{Z} \oplus \gamma \cdot \mathbb{Z}), \quad (4.11)$$

where $\deg \alpha = 1; \deg \beta = 2, \deg \gamma = 3$; then the collection of the polynomials N_g^{odd} becomes a linear map

$$N^{\text{odd}}: \mathbb{S}_3 \rightarrow \mathbb{Q} \quad (4.12)$$

sending a monomial of degree d not divisible by 3 to 0 and a monomial of degree $3n$ to N_{n+1}^{odd} (4.8) of this monomial. Moreover there is the following recurrence relation

$$N_g^{\text{odd}}(\gamma \cdot \alpha) = g \cdot N_{g-1}^{\text{odd}}(\alpha) \quad (4.13)$$

(see for example [Th1], Proposition 26). So it is convenient to set

$$N_{3n}^1 = \frac{1}{(n+1)!} N_{n+1}^{\text{odd}}. \quad (4.14)$$

Then the linear map

$$N^1: \mathbb{S}_3 \rightarrow \mathbb{Q} \quad (4.15)$$

specifies

$$N^1(\gamma \cdot z) = N^1(z) \quad (4.16)$$

for the generator γ and arbitrary $z \in \mathbb{S}$. So we can exclude the generator γ and consider N^1 as the linear map

$$N^1: \mathbb{S}_2 = \text{Sym}_*(\alpha \cdot \mathbb{Z} \oplus \beta \cdot \mathbb{Z}) \rightarrow \mathbb{Q}. \quad (4.17)$$

Now we know that

$$N^1(\alpha^m \cdot \beta^k) \neq 0 \implies k \leq \frac{1}{3}(m+2k) \quad (4.18)$$

(the so-called Newstead Conjecture, see [Th1]). So it is convenient to let $\alpha \cdot \beta = \omega$ and

$$\begin{aligned} N^1 : \text{Sym}_*(\alpha \cdot \mathbb{Z} \oplus \omega \cdot \mathbb{Z}) &\rightarrow \mathbb{Q}, \\ N^1(\alpha^{3k} \cdot \omega^n) &= N_{3(n+k)+1}^{\text{odd}}(\alpha^{n+3k} \cdot \beta^n). \end{aligned} \quad (4.19)$$

Then

$$N^1(\alpha^{3k} \cdot \omega^n) = \frac{(-4)^{n+k} \cdot (n+3k)!}{(n+k+1)!} \cdot (2k)! \cdot (4^k - 2) \cdot B_{2k}, \quad (4.20)$$

where B_i is the i th Bernoulli number (see [Th1], formula (29)).

Now for any linear form $L \in \text{Sym}_*(\alpha \cdot \mathbb{Z} \oplus \beta \cdot \mathbb{Z}) \oplus \gamma \cdot \mathbb{Z}$, the interior product by any homogeneous element x

$$\frac{\partial}{\partial x} L(z) = (\deg z + \deg x) L(x \cdot z) \quad (4.21)$$

gives a differential operator so that we can consider the relations as (4.16) as a differential equation. From this we can recognise the generating functions of the pairing as the standard elementary functions (see for example [Th1]).

Remark 1. We must remark only that historically the procedure was inverse: Thaddeus reconstructed the pairing formulas (4.20) from the beautiful form of the generating functions for the ranks of spaces of conformal block spaces given by the Verlinde formula (see [Th1]).

Remark 2. Even for the simplest graded algebra $\text{Sym}_*(\mathbb{Z} \oplus \cdots \oplus \mathbb{Z})$ the product $z_1 \cdot z_2 \in \text{Sym}_{d_1+d_2}$ of two homogeneous elements $z_i \in \text{Sym}_{d_i}$ is defined as

$$z_1 \cdot z_2(\sigma_1, \dots, \sigma_{d_1+d_2}) = \quad (4.22)$$

$$\frac{1}{(d_1 + d_2)!} \sum_{g \in \Sigma_{d_1+d_2}} z_1(\sigma_{g(1)}, \dots, \sigma_{g(d_1)}) \cdot z_2(\sigma_{g(d_1+1)}, \dots, \sigma_{g(d_1+d_2)}), \quad (4.23)$$

where Σ_n is the symmetric group on n letters (see Exercises).

Remark 3. There exists cousins of these constructions for the even case, that is, for the compactification M_C^{even} of the moduli space of rank 2 semistable vector bundles on C with $c_1 = 0$. We will use this case below.

Now the regular way to get the pairing formulas and the computation of the collection of Newstead polynomials is the Geometric Approximation Procedure: there is a chain of birational transformations or flips (see [R])

$$\text{MP}_{\max} \leftrightarrow \cdots \leftrightarrow \text{MP}_1 \leftrightarrow \text{MP}_0, \quad (4.24)$$

where every MP_i is the moduli space of σ -stable pairs (see [Th 2], [B-D] or [Ber]), and can be realised as a family of vector bundles over C . Thus using the same construction as in (4.4–4.15) we can define the polynomials $NP_{g,i}^{\text{odd}}$ analogs of the Newstead polynomials, and the linear forms NP_i^1 , the analogs of (4.19) and (4.20).

Recall that the moduli space on the extreme left admits a birational regular map

$$\text{MP}_{\max} \rightarrow M_C^{\text{odd}} \quad (4.25)$$

to the moduli space of stable bundle with fixed determinant of odd (or even) degree. From this it is easy to see that the form

$$NP_{\max}^1 = N^1 \quad (4.26)$$

is the Newstead form.

Now the moduli space MP_0 on the extreme right-hand end of (4.24) is projective space of dimension $3g - 3$.

$$MP_0 = \mathbb{P}^{3g-3}. \quad (4.27)$$

Using the description of every flip in the chain (4.24) we can reconstruct all polynomials step by step beginning with $\mathbb{P}^{3g-3}$ (see the Exercises).

These constructions can be generat to the 4 manifold case. Namely let M be a 4 manifold with fixed generic conformal structure given by a Riemannian metric g . For simplicity we will only consider the case that M is simply connected. Then the Donaldson invariant of (M, g) is a linear map

$$D: \mathbb{S}_2(M) = \text{Sym}_*(H_0(M) \oplus H_2(M)) \rightarrow \mathbb{R} \quad (4.28)$$

defined in the same way as the Newstead polynomials. Namely let M_g^{2d} be the compactified moduli space of instantons on M of the topological type $(2, 0, \frac{d+3p_g+3}{4})$, that is, the compactified space of anti-self-dual connections on a $SU(2)$ -bundle E with second Chern class $c_2 = \frac{d+3p_g+3}{4}$; Thus we suppose that

$$d + 3p_g + 3 \equiv 0 \pmod{4}, \quad (4.29)$$

where the integer $p_g = \frac{b_2^+ - 1}{2}$, so we suppose that b_2^+ is odd as for algebraic surfaces. Then on the direct product $M \times M_g^{2d}$ the universal connection on the universal family $\mathcal{E}$ defines the class

$$-p_1(\mathcal{E}) = c_1(\mathcal{E})^2 - 4c_2(E) = c_2(\text{ad } \mathcal{E}) \in H^4(M \times M_g^{2d}). \quad (4.30)$$

This class is defined by the slant product the homomorphism $\mu: H_*(M) \rightarrow H^*(M_g^{2d})$ defined by

$$\mu(\sigma) = -\frac{1}{4}p_1(\mathcal{E})/\sigma. \quad (4.31)$$

So $\deg(\alpha) = i \implies \mu(\alpha) \in H^i(M_g^{2d})$.

More precisely

$$-p_1(\mathcal{E}) = 4\text{pt} \otimes x + 4 \sum_{i=1}^{b_2} \sigma_i \otimes \mu(\sigma_i), \quad (4.32)$$

where $x \in H^4(M_g^{2d}, \mathbb{Z})$; $[\text{pt}]$ is the class of a point of M and σ_i , for $i = 1, \dots, b_2$ is a basis of $H^2(M, \mathbb{Z})$.

Now we can view the collection of Donaldson polynomials or the Donaldson invariant as the linear function (4.28), sending a typical monomial $\alpha = \sigma_1 \cdots \sigma_r$

where $\sigma_i \in H_{n_i}(M)$ of total degree $d(\alpha) = \sum_{i=1}^r \frac{4-n_i}{2}$ to 0 if $d \neq p_g + 1 \bmod 4$, and to the intersection number

$$\mu(\sigma_1) \cdots \mu(\sigma_r) \quad \text{on } M_g^{2d} \quad (4.33)$$

if $2d = \dim M_g^{2d}$.

Remark. Of course, we can apply these constructions to any family of connections on a vector bundle E with a compact base B given by the connection on the vector bundle $\mathcal{E}$ over $C \times B$. In this case, we will denote the collection of polynomials (4.33) by $D_g(\mathcal{E})$.

Our graded symmetric algebra $\mathbb{S}_2(M)$ (4.28) contains two special elements of degree 2. Namely the class of a point $[\text{pt}]$ and the class of the intersection form $[q_M]$ which we can define in parallel to the class γ (4.9): if $\{\sigma_i\}$ is a basis in $H_2(M, \mathbb{Z})$ and $\alpha_{ij} = \sigma_i \cdot \sigma_j$ then

$$[q_M] = \sum_{i,j} \alpha^{ij} \sigma_i \cdot \sigma_j. \quad (4.34)$$

(Please, don't forget (4.23)!). The main point is following: if the homomorphism μ preserved the multiplicative structure of the algebra $\mathbb{S}(M)$ then for any two linear elements σ_1 and σ_2 we would have

$$\mu(\sigma_1) \cdot \mu(\sigma_2) = q_M(\sigma_1, \sigma_2) \cdot \mu([\text{pt}]), \quad (4.35)$$

and hence

$$\mu([q_M]) = N\mu([\text{pt}]).$$

But this equality is false always except for the case $d = 1$ (compare with the case of curves: the classes $\alpha \cdot \beta$ and γ are proportional for $g = 2$ only). But for some algebraic surfaces (namely K3 surfaces) and for some moduli spaces (for example for Hilb, that is, the moduli spaces of torsion free sheaves of rank 1) it is very productive to describe the Donaldson invariant in terms of the declination from this case. This is the explanation of the choice of the shapes of all following expressions. On the other hand a smooth Riemann surface C in M is called *tight* if

$$2g(C) - 2 = q_M([C]). \quad (4.36)$$

In parallel to the relations (4.13) and (4.16) for all currently known 4 manifolds the following relation holds

$$D_g([\text{pt}]^2 \cdot z) = 4D_g(z) \quad (4.37)$$

for arbitrary $z \in \mathbb{S}_2$. So it is natural to declare the 4 manifolds with this property as 4 manifolds of "simple type".

Remark. The general point of view now is that the assumption of simple type is purely technical. But the collection of currently known examples is too special to give confidence. It may be the equality (4.37) is too strong even for $\mathbb{CP}^2$.

So under (4.37) without losing any information we can consider the following functions on $H_2(M, \mathbb{Z})$

$$\mathbb{D}(\sigma) = D_g((1 + \frac{[\text{pt}]}{2}) \cdot \exp(\sigma)); \quad (4.38)$$

$$\mathbb{K} = \exp(-\frac{[q_M]}{2}) \cdot \mathbb{D}.$$

We drop the lower index g because we consider 4 manifolds with the standard topological restriction

$$b_2^+(M) \geq 3. \quad (4.39)$$

Donaldson proved [D1] that then D_g and hence $\mathbb{D}$ doesn't depend on the comformal structure, that is, on g .

Now the functions (4.38) are defined as formal power series, but actually they are analytic functions on $H_2(M, \mathbb{C})$ with very simple shape:

Theorem 4.1 [K-M1]. *If M is a smooth simply connected 4 manifold of simple type subject to (4.39) then there exist k_M classes*

$$K_1, \dots, K_{k_M} \in H_2(M, \mathbb{Z}) \quad (4.39)$$

and k_M nonzero rational numbers $a_1, \dots, a_{k_M}$ such that

$$K = \sum_1^{k_M} a_i \exp(K_i). \quad (4.40)$$

In particular every classical homogeneous Donaldson polynomial D^d (4.33) is a polynomials in $K_1, \dots, K_{k_M}$ and the intersection form q_M , that is,

$$D^d \in \mathbb{C}[q_M, K_1, \dots, K_{k_M}]. \quad (4.41)$$

Corollary 1. *The set $\{K_i\}$, for $i = 1, \dots, k_M$, in $H^2(M, \mathbb{Z})$ is a diffeomorphism invariant, and the sublattice*

$$km(M) = \langle K_1, \dots, K_{k_M} \rangle \subset H^2(M, \mathbb{Z}) \quad (4.42)$$

is a diffeomorphism invariant too.

So the diffeomorphism group $\text{Diff } M$ admits a representation to the symmetric group on k_M letters. The classes (4.39) can be called the *KM-canonical classes* of the 4 manifold M because of the following properties:

Canonical properties [K-M1]. 1)

$$K_i \equiv w_2(M) \pmod{2}; \quad (4.43)$$

2) if C is any smoothly embedded oriented 2 manifold in M with $[C]^2 \geq 0$ then the genus of C satisfies the lower bound

$$2g(C) - 2 \geq [C]^2 + |K_i \cdot [C]| \quad (4.44)$$

for every $1 \leq i \leq k_M$;

3) K is a KM-canonical class $\implies -K$ is a KM-canonical class .

Now let $M = dS$, that is, let M be the underlying 4 manifold of an algebraic surface S . Then $H_2(dS, \mathbb{C})$ admits a Hodge structure and the full algebra $\mathbb{S}_2$ (4.28) admits a mixed Hodge structure such that every homogeneous Donaldson polynomial D^d of degree d is of pure Hodge type (d, d) . Thus following from the interpretation of gauge orbits of anti-self-dual connections as stable holomorphic bundle, of the same topological type (see [D-K]) and from algebraic geometric interpretation of Donaldson polynomial (see [M-O'G] and [Mo]). We will discuss this point in Lecture 5 in parallel to the same question for the Spin polynomial invariants. From this, the elegant trick of Brussee [Bru 2] (see the Exercises) provides

Proposition 4.1. *If $M = dS$ then the KM-canonical classes are algebraic, that is, all K_i are of Hodge type $(1,1)$.*

Corollary. *Under the condition (4.39) the sublattice $km(dS)$ (4.42) is a proper sublattice of $H^2(S, \mathbb{Z})$ and the representation $\text{Mod } S$ is reducible.*

Now for the canonical class K_S and any irreducible smooth algebraic curve C on S we have by the adjunction formula:

$$2g(C) - 2 = [C]^2 + K_S \cdot [C]. \quad (4.45)$$

Recall the following very exciting problem in Differential Topology, Problem 4.36 from Kirby's list [Ki]:

Conjecture. *For any smoothly embedded 2 manifold Σ homologous to C we have $g(\Sigma) \geq g(C)$.*

Hence it is a very important question:

Problem 4.1. *Is the canonical class K_S of an algebraic surface S a KM-canonical class of dS ?*

In this direction one has the following result of Brussee [Bru2]:

Proposition 4.2. *If S is a minimal surface of general type (with $p_g > 0$ as usual) then for every KM-canonical class K_i we have*

$$K_S^2 \geq K_i^2 \quad \text{and} \quad K_S^2 = K_i^2 \implies K_S = K_i. \quad (4.46)$$

Using this fact Kronheimer proved recently ([K-M2], Proposition 9.6)

Proposition 4.3. *If p_g is odd and the canonical linear system $|K_S|$ contains a smooth irreducible curve then the canonical class K_S is a KM-canonical class.*

(Actually to prove this statement Kronheimer used only the “inequality Riemann relation” (see [T6], (5.7)):

$$D^d(\omega + \bar{\omega}) > 0; d \gg 0 \quad \text{for } \omega \in H^{2,0}. \quad (4.47)$$

Continuation of Example 1.. Return to the surfaces with big monodromy. We saw that any homogeneous component D_{2n} of D (4.38) is a polynomial in the intersection form and its canonical class and has form (3.30). Then

$$D_{2n} = \frac{1}{(2d)!} D^n = a_0 \frac{q_S^n}{2^n d!} + a_2 \frac{q_S^{n-1} K^2}{2^{n-1} (n-1)! 2!} + \cdots + a_{2n} \frac{k^{2n}}{(2n)!} \quad (4.48)$$

and the crucial observation of Kronheimer is that the coefficients a_i don't depend on the homogeneous degree $2n$ (for a simple proof see [F-S2]). So we have

$$\mathbb{K} = \sum a_{2n} \frac{k^{2n}}{(2n)!}. \quad (4.49)$$

Now if $\sigma: \tilde{S} \rightarrow S$ is the blowup of a point and $E \subset \tilde{S}$ is the exceptional (-1) -curve then the Donaldson invariant of $\tilde{S}$

$$\mathbb{D}_{\tilde{S}} = \mathbb{D}_S \cdot \exp\left(-\frac{[E]^2}{2}\right) \cosh(E). \quad (4.50)$$

Thus the set $\{K; \pm E\}$ is the set of KM-canonical classes for $\tilde{S}$.

Now returning to our general problem of describing the set (1.9)

$$f^{-1}(dS) = vS, vS_1, \dots, vS_N,$$

that is, the set of deformation types of diffeomorphic surfaces, we can guess that the set (4.39) of KM-canonical classes of dS is the set of the potential canonical classes up to sign of algebraic surfaces S_i up to the subset $\{\pm E_i\}$ of the (-1) -curves (see (3.10)). Recall that in all currently known cases we have one canonical class only and for minimal surfaces we have Proposition 4.2.

Now a lot of close (and not so close) cousins of the Donaldson invariant (4.28) of 4 manifold M can be defined. The closest ones are the Donaldson $\text{SO}(3)$ -invariants

$$\mathbb{D}^w: \mathbb{S}_2 \rightarrow \mathbb{R}; \quad w \in H^2(M, \mathbb{Z}_2) \quad (4.50)$$

given by the the compactified moduli spaces $M_g^{2d,w}$ of $\text{SO}(3)$ -instantons with the Stiefel–Whitney class $w \in H^2(M, \mathbb{Z}_2)$. Not so close but very important cousins are the invariants

$$D^p: \mathbb{S}_2^{\perp[\Sigma]} \rightarrow \mathbb{R}; \quad w \in H^2(M, \mathbb{Z}_2) \quad (4.51)$$

coming from the moduli spaces $M_g^{2d,p}$ of instantons with “parabolic structures” along an oriented smooth 2 manifold $C \subset M$. The parabolic structure is given by the holomorphic line subbundle L of degree $\deg L = -\frac{1}{2}(g(S) - p)$ of the restriction

of the instanton to C with the conformal structure given by the restriction of the Riemannian metric g to C . (The size of the present lecture does not permitted us to describe this moduli space in details. So we avoid the description of the holonomy parameter α along C and some other important intermedial details of the construction of [K-M2]). So we can consider this moduli space as a family of instantons and a family of holomorphic vector bundles on C . This gives the correspondence between the arguments of the Donaldson invariant (4.28) and the Newstead invariant (4.12) (for even case). Namely the generator γ (4.14) defines an element

$$\Gamma \in \mathbb{S}_2; \quad \deg \Gamma = 3 \quad (4.52)$$

and there exists the following expression of the invariant of the pair (M, C) in terms of the Donaldson invariant.

Proposition 4.4 ([K-M2], Theorem 5.10). *There exists an inhomogeneous “universal” polynomial $\omega_p \in \mathbb{Q}[[C], [\text{pt}], \Gamma]$ of degree $2p$ depending on $[C]^2 = n$ and $g(C)$ only such that for every $z \in \mathbb{S}_2^{\perp[C]}$ we have*

$$D^p(\omega_p \cdot z) = 2^{g(C)} D(z) \quad (4.53)$$

(under some natural numerical constraint on n and p).

Remark. For $p < 4$ the universal polynomial ω_p is homogeneous and

$$\begin{aligned} \omega_0 &= 1; \quad \omega_1 = [C], \\ \omega_2 &= \frac{1}{2}([C]^2 - (g-1) \cdot [\text{pt}]), \\ \omega_3 &= \frac{1}{6}([C]^3 - (3g-4)[C] \cdot [\text{pt}] - \Gamma). \end{aligned} \quad (4.54)$$

Finally the elementary transformation of a parabolic bundle along parabolic line subbundle gives the relation between $M_g^{2d,p}$ and $M_g^{2d,2g-2-[C]^2-p}$ and the equality

$$D^p = D^{2g-2-[C]^2-p} \quad (4.55)$$

if $[C] \equiv 0 \pmod{2}$. This equality and its close cousins give the recurrence relations which imply the shape theorem (4.40) and the canonical properties of the KM-canonical classes.

Example. If $C \in M$ is tight (see (4.36)) and the genus $g(C) > 6$ then for all $z \in \mathbb{S}_2^{\perp[C]}$ we have ([K-M], Lemma 7.6)

$$D([S]^2 \cdot z) = (g-1)D([\text{pt}] \cdot z) \quad (4.56)$$

(compare with (4.35)), and so on.

So we can see that the structure of the Donaldson invariants of algebraic surfaces of simple type is as perfect as the structure of the Newstead invariants of Riemann surfaces. What have we lost? We have lost only the Geometric Approximation Procedure (4.24). To repair this loss we can generalise the Donaldson invariant to the Spin canonical invariant which we describe in the following Lectures. This

invariant previously was used (in the same way as Donaldson invariant) to distinguish the smooth structure of 4 manifolds of the same topological type (see [T4] and [T5]). Secondly, it was used to prove the Main Theorem of Lecture 1. The natural next step is the realisation of the Great Program for this close cousin of the Donaldson invariant.

Remark. To realise the Great Program, the main tool is the method of “cutting-and-pasting”, which for 4 manifolds is called the “glueing technique” or “neck-stretching” arguments. This involves splitting any 4 manifold along a 3 manifold and describing how the moduli spaces of instantons decompose into pieces. The complexity of the situation depends on the complexity of the 3 manifold along which our 4 manifold is split. The most elementary 3 manifold, the ordinary sphere S^3 , was used by Donaldson to prove the Vanishing Theorem for the Donaldson invariant [D1]. The lens spaces $L(p, 1)$ were used by Fintushel and Stern to get the beautiful blowup formula [F-S1] and to realise the Great Program for 4 manifold containing smooth 2-spheres [F-S2]. Donaldson used the direct products $C \times S^1$ to classify elliptic surfaces and to get an algebraic geometric interpretation of Floer–Fukaya homology. At last Kronheimer and Mrowka used S^1 -bundle over a Riemann surface, that is, the boundary of the normal neighbourhoods of smooth Riemann surfaces C in 4 manifolds with positive square [K–M2] and reduced the problem of computing Floer–Fukaya homology to the computation with the Newstead polynomial of C [K–M2]. Remark that first constructions and results of gauge-theoretic “neck-stretching” methods for spin canonical invariants were obtained by Pidstrigach in [P].

Exercises to Lectur 3.

1. Let C be a smooth Riemann surface of genus g and $\{a_i, b_i\}$ a symplectic basis of $H^1(C, \mathbb{Z})$. Prove that

$$a) \quad N_g^{\text{odd}}(\mu([C])^m \cdot \mu([\text{pt}])^n \cdot \mu(a_{i_{\min}}) \cdot \mu(b_{j_{\min}}) \cdots) \neq 0 \implies i_{\min} = j_{\min}.$$

Hint: use the commutation relations, the diffeomorphism invariance of the Newstead polynomials and the existence of an orientation preserving diffeomorphism $\varphi: C \rightarrow C$ satisfying

$$\varphi^*(a_i) = -a_i; \varphi^*(b_i) = -b_i; \varphi^*(a_j) = a_j, \varphi^*(b_j) = b_j, i \neq j$$

(the Dehn twist);

b) $N_g^{\text{odd}}(\mu([C])^m \cdot \mu([\text{pt}])^n \cdot \mu(a_{i_1}) \cdot \mu(b_{i_1}) \cdots)$ is independent of the choice of the handles i_j .

Hint: use the existence of an orientation preserving diffeomorphism interchanging any pair of handles;

c) the expression $\gamma = 2 \sum_{i=1}^g \mu(a_i) \cdot \mu(b_i)$ is independent of the choice of symplectic basis $\{a_i, b_i\}$.

2. Using the rule (4.23) compute the element $[q_M]^3 \in \mathbb{S}_2(M)$ (4.28).

3. a) Let $\xi \in \text{Pic}_{2g-1} C$ be a fixed divisors class of degree $2g-1$ and

$$\mathbb{P}^{3g-3} = \mathbb{P}H^1(\mathcal{O}_C(-\xi)) = |K_C + \xi|^* \quad (4.57)$$

be the complete linear system. On the direct product, $C \times \mathbb{P}^{3g-3}$ consider the universal extension

$$0 \rightarrow \mathcal{O}_{\mathbb{P}^{3g-3}}(1) \rightarrow \mathbb{E} \rightarrow \mathcal{O}_C(\xi) \rightarrow 0 \quad (4.58)$$

given by the first isomorphism of (4.57). Compute the Newstead polynomial $N_g(\mathbb{E})$ (see Remark after (4.8)) and the Newstead invariant as $g \rightarrow \infty$.

b) For a point $p \in \mathbb{P}^{3g-3}$ let E_p be the extension (4.59) given by this class. Let $\varphi_{K_C+\xi}: C \rightarrow \mathbb{P}^{3g-3}$ be the embedding given by the complete linear system (4.57). Prove that

$$p = \varphi_{K_C+\xi}(c) \implies 0 \rightarrow \mathcal{O}_C(\xi - c) \rightarrow E_p \rightarrow \mathcal{O}_C(c) \rightarrow 0, \quad (4.59)$$

and on $C \times \mathbb{P}^{3g-3}$ we have the epimorphism $\varphi: \mathbb{E} \rightarrow \mathcal{O}_{C \times \varphi_{K_C+\xi}(C)}(\Delta)$, where Δ is the diagonal in $C \times \varphi_{K_C+\xi}(C)$.

c) Let

$$\sigma_1: MP_1 \rightarrow \mathbb{P}^{3g-3} \quad (4.60)$$

be the blowup of the curve $\varphi_{K_C+\xi}(C)$ in $\mathbb{P}^{3g-3}$, $\tilde{C}$ be the exceptional divisor in MP_1 and $\tilde{\Delta}$ be the inverse image of Δ in $C \times \tilde{C}$. Then on $C \times MP_1$ we have the exact sequence

$$0 \rightarrow \mathbb{E}_1 \rightarrow (\text{id} \times \sigma_1)^* \mathbb{E} \rightarrow \mathcal{O}_{C \times \tilde{C}}(\tilde{\Delta}) \rightarrow 0. \quad (4.61)$$

Then $\mathbb{E}_1$ is a new family of vector bundles on C . Compute the Newstead polynomial for $\mathbb{E}_1$ and the Newstead invariant for $g \rightarrow \infty$.

4. Prove Proposition 4.1. Hint: for any (1.1) class $z \in H^2$ consider for the function K (4.38), (4.40) the directional derivatives $\frac{\partial^n K}{\partial z^n}$ and the its restrictions to $H^{2,0}$.

LECTURE 5. INSTANTON SPACES. JUMPING FILTRATIONS. SPIN POLYNOMIALS.

For any 4 manifold M , a Riemannian metric g or its conformal class, even considered only locally, is a very serious object of algebraic geometry. For a point $p \in M$, the tangent space $T_p = \mathbb{R}^4$ and its complexification $T^{\mathbb{C}} = \mathbb{C}^4$ admits an isomorphism $g_{\mathbb{C}}: T^{\mathbb{C}} \rightarrow T^{\mathbb{C}*}$, and a complex quadric $Q \subset \mathbb{P}^3 = \mathbb{P}T_p^{\mathbb{C}}$. As usual this nonsingular complex two dimensional quadric is a direct product

$$Q_p = q_p^+ \times q_p^-, \quad q_p^{\pm} = \mathbb{C}P^1, \quad q_p^{\pm} = \mathbb{P}W_p^{\pm} = \mathbb{C}^2. \quad (5.1)$$

If $\pi_1(M) = 0$ then the projective lines $q_p^{\pm}$ are the fibres of $\mathbb{P}^1$ -bundles $q^{\pm}$ and the rank 2 spaces $W_p^{\pm}$ for $p \in M$ are the fibres of the $\mathbb{C}^2$ -bundles $W^{\pm}$, called the spinor bundles. Of course these bundles are defined up to multiplying by a line bundle. So we can (because of $\pi_1 = 0$) describe the lifting of the projective bundles $q^{\pm}$ to vector bundles by fixing the first Chern class $c_1(W^{\pm})$ with one constraint

$$c_1(W^{\pm}) \equiv w_2(M) \pmod{2}, \quad (5.2)$$

where $w_2(M)$ is the second Stiefel-Whitney class of M . A choice of $c_1(W^{\pm})$ is called a choice of $Spin^{\mathbb{C}}$ -structure. So

$$T_{\mathbb{C}} = \text{Hom}(W^-, W^+) = W^{-*} \otimes W^+. \quad (5.3)$$

Our metric g determines a pair of Levi-Civita connections on the pair of the spinor bundles $W^\pm$. One of these connections induces the homomorphism $\Gamma(W^+) \rightarrow \Gamma(W^+ \otimes T_{\mathbb{C}}^*)$ of the space of sections and the composite

$$\Gamma(W^+) \rightarrow \Gamma(W^+ \otimes T_{\mathbb{C}}^*) \rightarrow \Gamma(W^-), \quad (5.4)$$

where the last homomorphism is given by the convolution (see (5.3)) is called the Dirac operator.

Now over a point $p \in M$ the projective space $\mathbb{P}_p^5 = \mathbb{P}\Lambda^2 T_p^{\mathbb{C}}$ contains two quadrics: $\Lambda^2 g_p^{\mathbb{C}}$ and the Grassmanian G of lines in $\mathbb{P}T_p^{\mathbb{C}}$ or Klein quadric. Traditionally $\Lambda^2 g$ is considered as an identification $\Lambda^2 T = \Lambda^2 T^*$. Then G can be interpreted as a self adjoint operator

$$*: \Lambda^2 T \rightarrow \Lambda^2 T; \quad *^2 = \text{id}. \quad (5.5)$$

It is the $*$ -operator which gives the eigenspace decomposition

$$\Lambda^2 T = \Lambda^+ \oplus \Lambda^-; \quad \text{with } *|_{\Lambda^\pm} = \pm \cdot \text{id}. \quad (5.6)$$

Geometrically, the intersections

$$G \cap \mathbb{P}\Lambda^\pm = \mathbb{P}\Lambda^2 g_p = q^\pm \quad (5.7)$$

are families of line-generators of the quadric Q_p (5.1). The intersection

$$G_p \cap \mathbb{P}\Lambda_p^g = K_p = \bigcup_{L^\pm \in q^\pm} \langle L^+, L^- \rangle \quad (5.8)$$

$$\text{Sing } K_p = q^+ \cup q^-$$

is very special quadratic complex.

Remark. The third quadric in $\mathbb{C}\mathbb{P}_p^5$ is R , the full Riemannian curvature. The intersection

$$G_p \cap \mathbb{P}\Lambda^2 g_p \cap R = S_p \quad (5.9)$$

is a special K3 surface (see [T7]). If instead of R we consider the Weyl tensor R_W then the intersection (5.9) is a special Kummer surface of a direct product of elliptic curves given by the components of the Weyl tensor.

Example. Kähler structures. In this case we need to add an almost complex structure given as the operator

$$J_p: T_p^{\mathbb{C}} \rightarrow T_p^{\mathbb{C}*}; \quad J_p^2 = -\text{id} \quad (5.10)$$

which provides the eigenspaces decomposition

$$T_p^{\mathbb{C}} = T^{0,1} \oplus T^{1,0}; \quad J|_{T^{1,0}} = i \cdot \text{id}; \quad J|_{T^{0,1}} = -i \cdot \text{id}. \quad (5.11)$$

The projectivisation of the eigenspaces (5.11) is a pair of lines in $\mathbb{P}T_p^{\mathbb{C}}$ which gives the pair of points $\mathbb{P}H^{2,0}$ and $\mathbb{P}H^{0,2}$ on the conic q^+ in $\mathbb{P}\Lambda^+$ and the intersection of the tangent lines at these points is the Kähler form of the Kähler metric ω_p . Moreover

the local Hodge component $\mathbb{P}H^{1,1} = \langle \omega, \Lambda^- \rangle$. (Recall that a complex surface has a standard orientation.)

Now these constructions can be globalised. Namely, let Δ_g be the Laplace operator on the space Ω^2 of 2-forms on M and $H_g = \ker \Delta_g$ be the space of harmonic forms. Then according to classical Hodge theory we have the isomorphism

$$H^2(M, \mathbb{R}) = H_g. \quad (5.12)$$

Now the star operator $*$ (5.6) gives the globalisation of the decomposition (5.6):

$$\Omega^2 = \Lambda_g^+ \oplus \Lambda_g^-; \quad \Lambda_g^\pm = \Gamma(\text{Re ad } W_g^\pm). \quad (5.13)$$

This operator $*$ commutes with Δ_g and defines a q_M -orthogonal decomposition

$$H_g = H_g^+ \oplus H_g^-, \quad (5.13')$$

where H_g^+ is the subspace of selfdual forms and H_g^- that of antiselfdual forms. Fixing an orientation of each H_g^+ , which is called a *homology orientation* of M , we get the period map:

$$\Pi: \mathcal{S}(M) \rightarrow \Omega \subset \text{Grass}_{\text{orient}}(b_2^+, H^2(M, \mathbb{R})) \quad \text{defined by} \quad \Pi(g) = H_g^+; \quad (5.14)$$

here $\mathcal{S}(M)$ is the space of Riemannian metrics on M and Ω the subdomain of the Grassmannian of oriented subspaces $H^2(M, \mathbb{R})$ of rank b_2^+ defined by:

$$\Omega = \{H \in \text{Grass}_{\text{orient}}(b_2^+, H^2(M, \mathbb{R})) \mid q_M|_H > 0\}, \quad (5.15)$$

where q_M is the intersection form of M . The tangent space to Ω at $H \in \Omega$ is given as

$$T\Omega_H = \text{Hom}(H^\perp, H). \quad (5.16)$$

Now the tangent space to the space of all Riemannian metrics $\mathcal{S}(M)$ at a point g is given by

$$T\mathcal{S}(M)_g = \text{Hom}(\Lambda_g^-, \Lambda_g^+) \quad (5.17)$$

(see (5.13) and the differential of the period map (5.14))

$$d\Pi_g = T\mathcal{S}(M)_g \rightarrow T\Omega_{\Pi(g)}$$

is the composite of the chain of homomorphisms

$$d\Pi_g(\Lambda_g^- \xrightarrow{\varphi} \Lambda_g^+) = H_g^- \xrightarrow{i} \Gamma(\Lambda_g^-) \xrightarrow{\Gamma\varphi} \Gamma(\Lambda_g^+) \xrightarrow{pr} H_g^+ \quad (5.18)$$

where pr is the harmonic projection.

Problem. Is the period map onto? That is

$$\Pi(S(M)) = \Omega? \quad (5.19)$$

Remark. It is very natural to begin with the study of the infinitesimal structure of the period map (5.18) (see the Exercise).

Now let E be a $U(2)$ -bundle on M , that is, a $\mathbb{C}^2$ -bundle on M with fixed Hermitian form h , and with Chern classes c_1 and c_2 . Let $\mathcal{A}(E)_{\nabla_0}$ be the affine space of all Hermitian connections on E inducing the same connection ∇_0 on

$$\det E = \Lambda^2 E \quad (5.20)$$

and $\mathcal{G}$ the gauge group of determinant 1, that is, the group of unitary automorphisms of E preserving the connection ∇_0 on $\det E$.

Remark. Recall that the space $\mathcal{A}(L)$ of all $U(1)$ -connections on a line bundle L admits a “moment map”:

$$F: \mathcal{A}(L) \rightarrow \Omega^2 \quad (5.21)$$

sending a connection a to its curvature form $F(a)$. This map is equivariant with respect to the gauge group action, and the fibres of F are the gauge orbits (because of $H^1(M) = 0$, see [A–B]). So any gauge orbit of connections on L can be defined as a 2-form on M , namely the curvature form of connections, and the cohomology class $[F(a)] = c_1(L)$.

Now if we fix a conformal structure on M specified as usual by a choice of Riemannian metric g then for any Hermitian connection $a \in \mathcal{A}_{\nabla}$ on E , putting any Hermitian connection ∇_0 on $\Lambda^2 W^{\pm} = L$, with $c_1(L) = C \in H^2(M, \mathbb{Z})$, gives a coupled Dirac operator

$$D_a^{g,C,\nabla_0}: \Gamma^{\infty}(E \otimes W^+) \rightarrow \Gamma^{\infty}(E \otimes W^-). \quad (5.22)$$

determined by the connection a on E and the Dirac operator (5.4) on the spinor bundles.

So $\mathcal{A}(E)_{\nabla_0}$ is the base space of a family of operators. To make this continuous family and all our spaces and orbit spaces modelled on Banach spaces which admit partitions of unity we require the connection spaces to be L_k^p -spaces and the gauge groups to be L_{k+1}^p -gauge transformations. Then in particular the orbit space of connections modulo the gauge group

$$\mathcal{B}(E) = \mathcal{A}(E)_{\nabla_0} / \mathcal{G} \quad (5.23)$$

is modelled on a Banach space (see [D–K]). The slant product (see (4.30) and (4.31)) defines homomorphisms

$$\mu: H_i(M) \rightarrow H^{4-i}(\mathcal{B}(E)). \quad (5.24)$$

Now the space $\mathcal{B}(E)$ contains the subspace

$$\mathcal{M}^g(E) = \{a \mid *F(a) = -F(a)\} \subset \mathcal{B}(E) \quad (5.25)$$

of anti-self-dual connections with respect to the Riemannian metric g and the filtration given by interpreting $\mathcal{A}$ as the base space of a family of operators. Namely for any positive integer r , we can consider the subspace of jumping connections:

$$\mathcal{M}_r^{g,C}(E) = \{(a) \in \mathcal{M}^g(E) \mid \text{rank ker } D_a^{g,C,\nabla_0} \geq r\} \subseteq \mathcal{M}^g(E). \quad (5.26)$$

The number r is called the *jumping level* of E . The collection of these subspaces defines a filtration:

$$\mathcal{M}^g(E) \supseteq \mathcal{M}_1^{g,C}(E) \supseteq \cdots \supseteq \mathcal{M}_r^{g,C}(E) \supseteq \cdots \quad (5.27)$$

The virtual (expected) codimension of $\mathcal{M}_r^{g,C}(E)$ is given by

$$\text{v. codim } \mathcal{M}_r^{g,C}(E) = 2r^2 - 2r\chi_C(E), \quad (5.28)$$

where $\chi_C(E)$ is the index of the coupled Dirac operator (5.22), depending only on the Chern classes of E and the Spin^C structure $C = c_1(\Lambda^2 W^\pm)$.

The analog of the Freed–Uhlenbeck theorem, that for generic metric $g \in \mathcal{S}(M)$ the moduli space $\mathcal{M}^g(E)$ (5.25) is a smooth manifold of the expected dimension

$$\text{v. dim } \mathcal{M}^g(E) = 8c_2(E) - 3(b_2^+ + 1) \quad (5.29)$$

(see [D–K]) with regular ends (see [F–U], Theorem 3.13), was proved in [P–T] Chap. 2, §3 for the first step $\mathcal{M}_1^{g,C}(E)$ of the filtration (5.27):

Transversality properties. For general pair $(g, \nabla_0) \in \mathcal{S}(M) \times \Omega^1$ one has

- 1) the $\mathcal{M}_1^{g,C}(E) \setminus \mathcal{M}_2^{g,C}(E)$ is smooth and of the right dimension;
- 2) the fibre of the normal bundle of $\mathcal{M}_1^{g,C}(E) \subset \mathcal{M}^g(E)$ at (a) is given by

$$(N_{\mathcal{M}_1 \subset \mathcal{M}_2})_a = \text{Hom}_{\mathbb{C}}(\ker D_a^{g,C,\nabla_0}, \text{coker } D_a^{g,C,\nabla_0}). \quad (5.30)$$

Moreover, the homology orientation (see (5.14)) defines a natural orientation of $\mathcal{M}^g(E)$ (see [D–K]), and the natural complex orientation of the fibres of the normal bundle (5.30) provides the orientation of $\mathcal{M}_1^{g,C}(E)$. This orientation is described in details in [P–T], Chap. 1, §5.

Now under the usual restriction on the topology of M : we suppose that $b_2^+(M) = 2p_g(M) + 1$ is odd, so both

$$\text{v. dim } \mathcal{M}^g(E) = 2d \quad \text{and} \quad \text{v. dim } \mathcal{M}_r^{g,C}(E) = 2d_r$$

must be even.

The Uhlenbeck compactification of $\mathcal{M}^g(E)$ [F–U] admits the filtration

$$\overline{\mathcal{M}^g(E)} \supseteq \overline{\mathcal{M}_1^{g,C}(E)} \supseteq \cdots \supseteq \overline{\mathcal{M}_r^{g,C}(E)} \supseteq \cdots \quad (5.31)$$

Remark. In the special case when $c_1(E) = 0$, the topological type of E can be distinguished by the nonnegative number c_2 , so our moduli spaces $\overline{\mathcal{M}^g(E)}$ can be labelled by their dimensions $2d = 8c_2(E) - 3(b_2^+ + 1)$, and

$$\overline{\mathcal{M}^g(E)} = M_g^{2d} \quad (5.32)$$

in the constructions (4.29–30). Moreover since

$$\overline{\mathcal{M}^g(E \otimes L)} = \overline{\mathcal{M}^g(E)} \quad \text{for any line bundle } L, \quad (5.33)$$

it follows that $\overline{\mathcal{M}^g(E)}$ depends only on $w_2(E) = c_1 \bmod 2$ and the first Pontryagin number $p_1 = c_1^2 - 4c_2$ of E . So in general case we can label these moduli spaces by the dimension $2d = -2p_1 - 3(p_g + 1)$ and $w = w_2(E) \in H^2(M, \mathbb{Z}_2)$:

$$\overline{\mathcal{M}^g(E)} = M_g^{2d, w}. \quad (5.33')$$

Moreover, the shape of the Dirac operator (5.22) shows that if we twist E by any line bundle L and untwist the spinor bundles $W^\pm$ by L then this operator remains the same. This means that under the identification (5.33) we have

$$\overline{\mathcal{M}_r^{g, C-2c_1(L)}(E \otimes L)} = \overline{\mathcal{M}_r^{g, C}(E)} \quad \text{for any line bundle } L. \quad (5.33')$$

Hence $\mathcal{M}_r^{g, C}(E)$ depends on the class $(C + c_1(E)) = C_1 \in H^2(M, \mathbb{Z})$ and the first Pontryagin number $p_1 = c_1^2 - 4c_2$:

$$\overline{\mathcal{M}_r^{g, C}(E)} = M_g^{r, C_1, p_1}. \quad (5.34)$$

The very important case is when $C_1 = 0$. The virtual dimension of the moduli space $M_g^{r, 0, p_1}$ is given by the formula (we use (5.28), (5.30) and the Atiyah–Singer formula)

$$\frac{1}{2} \text{v. dim } M_g^{r, C_1, p_1} = 2d_r = (4 - r)c_2 - (3 - 2r)(p_g + 1) - r^2 \quad (5.35)$$

(here it is convenient to return to the second Chern class of E as a parameter). So below we will use the moduli spaces

$$M_g^{r, 0, p_1} = M_g^{r, 2d_r}, \quad (5.35)$$

which are called the moduli space of canonical jumping instantons of jumping level r .

Now for any element of our filtration and for a general metric g , the slant product (see (4.30)) defines a cohomological correspondence

$$\mu: H_i(M, \mathbb{Z}) \rightarrow H^{4-i}(\overline{M_g^{g, C_1, p_1}}, \mathbb{Z}), \quad (5.36)$$

and the collection of the Donaldson polynomials (4.28) and their Spin analogs

$$sD_g^{r, C_1, p_1}: \mathbb{S}_2(M) \rightarrow \mathbb{Q}, \quad (5.37)$$

which sends a typical monomial $\alpha = \sigma_1 \cdots \sigma_m$ where $\sigma_i \in H_{n_i}(M)$ of total degree $d(\alpha) = \sum_{i=1}^m \frac{4-n_i}{2}$ to 0 if $d(\alpha) \neq \frac{1}{2} \text{v. dim } \mathcal{M}_r^{g, C}(E)$ and to the intersection number

$$\mu(\sigma_1) \cdots \mu(\sigma_r) \text{ on } M_g^{2d} \text{ or } M_g^{2, C_1, p_1} \quad (5.38)$$

if $2d = \text{v. dim } M_g^{2d}$ or $2d = \text{v. dim } M_g^{2, C_1, p_1}$.

Our first task is to bring together all the homogeneous polynomials (5.34) into one linear function as for the Donaldson polynomials (4.28). Of course we can do it pure formally but the main observation is the following: if $r \geq 4$ we get a polynomial only (see (5.35)).

Remark. The formulas for the virtual dimensions of moduli spaces of jumping instantons are following:

$$\begin{aligned} d &= 4c_2 - 3(p_g + 1), \\ d_1 &= 3c_2 - (p_g + 1) - 1, \\ d_2 &= 2c_2 + (p_g + 1) - 4, \\ d_3 &= c_2 + 3(p_g + 1) - 9, \\ d_4 &= 5(p_g + 1) - 16. \end{aligned} \tag{5.39}$$

Below we consider the case $r = 1$ only. The reason is that the degrees of the homogeneous polynomials in this case are constant mod 3 just as for the Newstead invariant of a Riemann surface (see (4.17)).

Definition 5.1. The homogeneous polynomial $D_g^{2d_1}$ coming from M_g^{1,d_1} (5.35) is called the *Spin canonical polynomial*. The linear function sD_g^1 (5.37) is called the *spin canonical invariant*.

Our last task is to describe the dependence of the spin canonical polynomials and spin canonical invariant on the metric g . For this, first of all we have to describe the set $\text{Sing } \mathcal{M}^g(E)$ of singular points of the moduli space $\mathcal{M}^g(E)$ (5.25). Every singular point $(a) \in \mathcal{M}^g(E)$ is given by

- 1) a topological splitting

$$E = L_\delta \oplus L_{c_1 - \delta} \tag{5.40}$$

here L_δ is a line bundle on M with $c_1 = \delta \in H^2(M, \mathbb{Z})$;

- 2) a $U(1)$ -connection a_1 on L_δ such that the connection

$$\nabla_0 \otimes a^{-2} \text{ on } L_{c_1 - 2\delta} \text{ is } g\text{-antiselfdual}, \tag{5.41}$$

$$(a) = a_1 \oplus \nabla_0 \otimes a^{-1}.$$

In terms of the period map (5.14) this means that

$$e = c_1 - 2\delta \in \Pi(g)^\perp \tag{5.42}$$

(we consider $\Pi(g)$ here as a subspace of H_g (5.12)), recall that ∇_0 is fixed connection on $\det E$ (5.20) and

$$e^2 = p_1(E)). \tag{5.43}$$

Any class $e \in H^2(M, \mathbb{Z})$ with $e^2 < 0$ defines a subset in the period domain Ω (5.15):

$$W_e = \{H \in \Omega \mid e \in H^\perp\} \tag{5.44}$$

and the corresponding subset of metrics

$$\mathcal{S}_e = \{g \in \mathcal{S} \mid \Pi(g) \in W_e\} \subset \mathcal{S}(M) \tag{5.45}$$

is called a *hurdle*; a hurdle $\mathcal{S}_e$ is called a *wall* if its complement

$$\mathcal{S} \setminus \mathcal{S}_e \quad (5.46)$$

is disconnected.

Moreover from (5.16), it is easy to see that W_e has codimension

$$\text{codim } W_e = b_2^+$$

in the domain Ω (5.15). From this and the transversality conditions of the period map (5.14) along W_e , one can prove that if $b_2^+ > 1$ then a hurdle is never a wall, that is, the complement

$$\mathcal{S} \setminus \bigcup_{e^2 < 0} \mathcal{S}_e \quad (5.47)$$

is connected. We define the absolut invariant as the invariant (5.37) for a general metric in $\mathcal{S} \setminus \bigcup_{e^2 < 0} \mathcal{S}_e$. In this case if the labels of the invariant (5.37) are diffeomorphism invariants then all homogeneous polynomials are invariant with respect to the diffeomorphism group action on $\mathbb{S}_2(M)$ and its shape provide new invariants of M like the collection of KM-canonical classes and the sublattice $m(M)$ (see (4.40–42)), and so on.

Remark immediately that the labels of the spin canonical polynomials are diffeomorphism invariants. Actually it is the reason for the condition $C_1 = C + c_1 = 0$. On the other hand, the subsets of the filtration (5.31) can avoid $\text{Sing } \mathcal{M}^g(E)$. Namely a reducible connection $(a) = a_1 \oplus \nabla_0 \otimes a^{-1}$ (5.41) defines a jumping instanton (of the jumping level r) if the operators

$$\begin{aligned} D_{\alpha}^{g,C} : \Gamma^{\infty}(L_{\delta} \otimes W^+) &\rightarrow \Gamma^{\infty}(L_{\delta} \otimes W^-), \\ D_{\alpha'}^{g,C} : \Gamma^{\infty}(L_{c_1-\delta} \otimes W^+) &\rightarrow \Gamma^{\infty}(L_{c_1-\delta} \otimes W^-), \end{aligned} \quad (5.48)$$

have kernels satisfying

$$\text{rank ker } D_{\alpha}^{g,C} + \text{rank ker } D_{\alpha'}^{g,C} \geq r. \quad (5.49)$$

Now let a_1 be a fixed $U(1)$ -connection on a line bundle L , then a metric $g \in \mathcal{S}(M)$ is called (a_1) -Dirac regular, if

$$\text{rank ker } D_{a_1}^{g,C} \geq r \implies \chi_C(L) \geq r. \quad (5.50)$$

It is proved that a generic metric $g \in \mathcal{S}(M)$ is Dirac regular (see for example [P–T], Chap. 1, §1).

It is convenient to define a $\text{Spin}^{\mathbb{C}}$ hurdle in terms of the decomposition (5.41):

$$\mathcal{S}_{\delta} = \{g \in \mathcal{S}_{c_1-2\delta} \mid g \text{ is } (a_1)\text{-Dirac irregular}\}. \quad (5.51)$$

The main observation is the following

$$\text{a } \text{Spin}^{\mathbb{C}} \text{ hurdle } \mathcal{S}_{\delta} \text{ is a wall if and only if } \chi_C(L_{\delta}) \geq r.$$

Informally, this means that the $(c_1 - 2\delta)$ -irregularity condition (5.45) and the Dirac irregularity condition are codimension 1 conditions and are independent. Remark that the condition (5.41) defines a_1 uniquely.

To carry out these arguments rigorously, we consider a new space of parameters for our family of operators, in addition to the big space of all Riemannian metrics $\mathcal{S}(M)$, namely the direct product

$$\mathcal{S}(M) \times \Omega^2; \quad (5.52)$$

that is, the set of pairs (g, ∇_0) consisting of any metric g and Hermitian connection ∇_0 on the determinant spin bundle, which we can view as a 2-form (for a fixed metric g). Now we can use the new parameter ∇_0 to regularise the Dirac irregularity. Namely on twisting the spinor bundles $W^\pm$ by the line bundle L_δ^* , and changing the $\text{Spin}^{\mathbb{C}}$ structure from C to $C' = C - 2\delta$, the coupled Dirac operator $D_{a_1}^{g, C}$ becomes the ordinary Dirac operator (5.4) of the metric g (and a new $\text{Spin}^{\mathbb{C}}$ structure). So we must study how the jumping behaviour of the kernel of the ordinary Dirac operator of any $\text{Spin}^{\mathbb{C}}$ structure changes on deforming the metric g . But this was done in [P-T], Formulas (1.3.3) and (1.3.4), Proposition 3.1.1 and Corollary.

So a class $e = c_1 - 2\delta \in H^2(M, \mathbb{Z})$ defines a wall for the spin polynomials if and only if

$$\begin{aligned} 1) \quad & 0 \geq (c_1 - 2\delta)^2 \geq p_1, \\ 2) \quad & \chi_C(L_\delta) \geq r. \end{aligned} \quad (5.53)$$

These conditions imply the following two inequalities

$$\begin{aligned} 1) \quad & \frac{1}{4}c_1^2 \leq c_1 \cdot \delta - \delta^2, \\ 2) \quad & \chi_C(L_\delta) = \frac{1}{2}\delta(\delta + C) + \frac{1}{8}(C^2 - I) \geq r. \end{aligned} \quad (5.54)$$

where I is the index of M . The second inequality is the Atiyah–Singer formula and is equivalent to the following:

$$-C \cdot \delta - \delta^2 \leq \frac{1}{4}C^2 - \frac{1}{4}I - 2r.$$

Hence in the spin canonical case $c_1 = -C$. From this we have $8r \leq -I$. Hence

$$8r > -I \implies \text{the system of walls is empty} \quad (5.55)$$

and spin canonical polynomials are diffeomorphism invariants. This means that some aspects of the shape of these polynomials (and their coefficients, of course) are invariants of the smooth structure of 4-manifolds.

To explain why the spin polynomials are “canonical” and how we can compute them we consider the main example of special metrics.

Example. Algebraic surfaces. If $M = dS$ is the underlying 4 manifold of an algebraic surface S , then the anticanonical class $-K_S$ gives a canonical $\text{Spin}^{\mathbb{C}}$ structure, and the inequality $C + c_1 = 0$ predicts the equality

$$c_1 = K_S. \quad (5.56)$$

Then the standard numerical invariants of S , the topological Euler characteristic $c_2(S)$ and K_S^2 , satisfy

$$\frac{1}{3}(2c_2(S) - K_S^2) = -I.$$

Moreover, by the Noether formula,

$$-I + K_S^2 = 8\left(\frac{1}{12}(K_S^2 + c_2(S))\right) = 8(p_g + 1),$$

where p_g is the geometric genus of S .

Hence the inequality $8r > -I$ is equivalent to the inequality

$$-K_S^2 < 8(r - 1) - 8p_g.$$

If we are interested in the case $p_g = 0$

$$-K_S^2 < 8(r - 1). \quad (5.57)$$

Therefore, for a minimal surface of general type, jumping level 1 is enough to guarantee the invariance of the spin canonical polynomials.

Now consider any Hodge metric g_H on S , where H is the cohomology class of its Kähler form so H is a polarisation on S . Then by the Donaldson-Uhlenbeck identification theorem (see [D-K]), we have

$$\mathcal{M}^{g_H}(E) = M^H(2, c_1, c_2), \quad (5.58)$$

where the right-hand side is the moduli space of holomorphic H -slope stable rank 2 bundles on S with Chern classes c_1, c_2 .

The complex structure on S defines a canonical orientation and, with this orientation, the positive spinor bundle is the following direct sum:

$$W^+ = L_0 \oplus L_{K_S}; \quad (5.59)$$

the original Dirac operator is

$$D: \Omega^{0,0} \oplus \Omega^{0,2} \xrightarrow{d'' \oplus (d'')^*} \Omega^{0,1}, \quad (5.60)$$

and the coupled Dirac operator

$$D_{a=E}^{g_H, -K_S, \nabla_0}: \Omega^{0,0}(E) \oplus \Omega^{0,2}(E) \xrightarrow{d'' \oplus (d'')^*} \Omega^{0,1}(E) \quad (5.61)$$

is the convoluted Dolbeault complex of E . Making this identification $(a) = E$, we have identifications

$$\ker D_a^{g_H, -K} = H^0(E) \oplus H^2(E) \quad \text{and} \quad \text{coker } D_a^{g_H, -K} = H^1(E), \quad (5.62)$$

where $H^i(E)$ denote coherent cohomology groups. Thus the index of the coupled Dirac operator for this special Spin^C structure is

$$\chi_{-K_S} D_a^{g_H, -K} = h^0(E) - h^1(E) + h^2(E).$$

So the subspace $\mathcal{M}_r^{g_H, -K_S}(E)$ is the *Brill-Noether locus*

$$\mathcal{M}_r^{g_H, -K}(2, c_1, c_2) = \{E \in M^H(r, c_1, c_2) \mid h^1(E) \geq -\chi(E) + r\}. \quad (5.63)$$

But for surfaces, the last inequality can be rewritten

$$h^1(E) \geq -\chi(E) + r \iff h^0(E) + h^2(E) \geq r.$$

Hence we have a decomposition

$$\mathcal{M}_r^{g_H, -K}(2, c_1, c_2) = \bigcup_{i+j=r} M_{i,j}^H(2, c_1, c_2),$$

where the components on the right-hand side are the algebraic subvarieties of $M^H(2, c_1, c_2)$ defined by

$$M_{i,j}^H(2, c_1, c_2) = \{E \in M^H(2, c_1, c_2) \mid h^0(E) \geq i, h^2(E) \geq j\}. \quad (5.64)$$

On the other hand, sending $E \rightsquigarrow E^*(K_S)$ identifies

$$M^H(2, c_1, c_2) = M^H(2, 2K - c_1, c_2 - c_1 \cdot K - K^2),$$

and Serre duality gives

$$M_{i,j}^H(2, c_1, c_2) = M_{j,i}^H(2, rK - c_1, c_2 - c_1 \cdot K - K^2). \quad (5.65)$$

Now the Gieseker compactification $\overline{M^H(2, c_1, c_2)}$ (see [G]) gives the compactifications $\overline{M_{i,j}^H(r, c_1, c_2)}$.

The standard definition of μ -homomorphism in the algebraic geometric context (see [T6] or [O'G]) gives the collection of polynomials

$$asD_{H,i,j}^{r,c_1-K_S,c_2} : \mathbb{S}_2(S) \rightarrow \mathbb{Q}. \quad (5.66)$$

Now to compute the algebraic geometric version of the spin polynomial of jumping level r (1.6), we must sum the individual polynomials (5.66)

$$asD_H^{r,c_1-K_S,c_2} = \sum_{i+j=r} asD_{i,j}^{r,c_1-K_S,c_2}. \quad (5.67)$$

But here care is necessary, because the natural orientations of the components (5.66) can be different (see [P-T], Chap. 1, §5).

One can prove that the algebraic geometric polynomials (5.67) and spin polynomials (5.37) are equal using the same arguments as for the original Donaldson polynomials (see Morgan [Mo]). More precisely, for a Hodge metric g_H , if all the moduli spaces (5.54) have the expected dimension and avoid the reducible connections then

$$asD_H^{r,c_1-K_S,c_2} = sD_{g_H}^{r,c_1-K_S,c_1^2-4c_2} \quad (5.68)$$

see [T3], §6.

So we can use algebraic geometric constructions to get the moduli spaces, the cohomological correspondence μ and the homogeneous polynomials. But to provide some information on $\text{Mod } S$ we have to restrict ourself the case $C_1 = 0$, that is, to the case $c_1 = K_S$. A small difficulty is here that this case is singular for the ad hoc reason that. It means that a Hodge metric is never regular in the sense of Freed-Uhlenbeck. Namely by Serre duality

$$M_{i,j}^H(2, K_S, c_2) = M_{j,i}^H(2, K_S, c_2). \quad (5.69)$$

This means that even if the space $M_{i,j}^H(2, K_S, c_2)$ has the right dimension, this locus has nontrivial multiple structure; that is, as a subscheme, it is not reduced, it has nilpotents. Fortunately, we can describe the scheme-theoretic structure precisely. We do this in the case $r = 1$, that is, for the minimal jumping level.

First of all, recall that in the regular case the fibre of the normal cone of the sublocus $M_{1,0}^H(2, c_1, c_2)$ in $M^H(2, c_1, c_2)$ at a point $E \in M_{1,0}^H(2, c_1, c_2)$ is given by

$$(N_{M_{1,0} \subset M})_E = \text{Hom}(H^0(E), H^1(E)). \quad (5.70)$$

So the codimension of $M_{1,0}^H(2, c_1, c_2)$ in $M^H(2, c_1, c_2)$ is

$$\text{v. codim } M_{1,0}^H(2, c_1, c_2) = h^0(E) \cdot (h^0(E) - \chi(E)). \quad (5.71)$$

Moreover if $E \in M_{1,0}^H(2, c_1, c_2)$ is a singular point then in the space (5.70), we have the fibre of the *normal cone* of $M_{1,0}^H(2, c_1, c_2)$ in $M^H(2, c_1, c_2)$:

$$(C_{M_{1,0} \subset M})_E \subset \text{Hom}(H^0(E), H^1(E)), \quad (5.72)$$

defined as in [Fu].

The spaces $H^i(E)$ involved in (5.72) admit additional structures. Namely, by Serre duality

$$H^0(E) = H^2(E)^* \quad \text{and} \quad H^1(E) = H^1(E)^*.$$

This means that the vector space $H^1(E)$ has a nondegenerate quadratic form and a *light cone* of isotropic vectors,

$$Q_E \subset H^1(E). \quad (5.73)$$

In the simplest case $\text{rank } H^0(E) = h^0(E) = 1$, a formal normal vector n in the fibre of the formal bundle $\text{Hom}(H^0(E), H^1(E))$ is given as a nontrivial homomorphism $n: H^0(E) \rightarrow H^1(E)$. Then n is contained in the fibre of the normal cone (5.72) if and only if the image of this homomorphism is contained in Q_E :

$$n(H^0(E)) \subset Q_E; \quad (5.74)$$

that is, the image is isotropic with respect to the quadratic form q_E . So

$$(C_{M_{1,0} \subset M})_E = Q_E \quad (5.75)$$

(see [T3]), §3). Hence for the case $c_1 = K_S$ we have

$$2asD_H^{1,0,c_2} = sD_{gH}^{1,0,c_1-4c_2}. \quad (5.75)$$

Now we are ready to apply the Geometric Approximation Procedure to compute the shape of the spin canonical polynomials.

Exercises to Lecture 5.

1. Let (S, H, α) be a polarised marked K3 surface and $\mathcal{M}_H$ the moduli spaces of polarised marked K3 surfaces. Recompute the classical period map for $\mathcal{M}_H$ in terms of the period map Π (5.14). Using this interpretation and the Kulikov result prove that $\Pi(\mathcal{S}(S)) = \Omega$ (5.19).

2. We can consider the Hilbert scheme Hilb^d of a surface S as the moduli space $M^H(1, K_S, d)$ of the torsion free sheaves on S of rank 1, $c_1 = K_S$ and $c_2 = d$. Obviously this moduli space doesn't depend on the polarisation H . Define and compute the Donaldson and spin canonical polynomials for Hilb^d and the Donaldson invariant and spin canonical invariant for Hilb^d , as $d \rightarrow \infty$.

3. Let C be a smooth compact algebraic curve of genus g with fixed Spin^c structure θ , that is, $\theta \in \text{Pic}_{g-1}$, with $\theta^2 = K_C$. Let M^c be the moduli space of rank 2 stable vector bundles with $c_1 = K_S$. Define and compute the spin canonical polynomials and spin canonical invariant for this geometric situation.

LECTURE 6. GEOMETRIC APPROXIMATION PROCEDURE FOR SPIN CANONICAL INVARIANTS.

The aim of this final lecture is to show how we can get some information about the shape of Spin canonical Mod-semi invariant polynomials to prove the Main Theorem of Lecture 1 without using any information about the monodromy group. This information is enough to prove the reducibility of $\text{Mod } S$ by the same arguments as at the beginning of Lecture 4 (see (4.2)). But beforehand we would like to propose some system of definitions to emphasize the deep parallelism of the theories of algebraic surfaces and algebraic curves.

Return to a nonsingular compact nonruled algebraic surface S , to a map $m: S \rightarrow S_{\min}$ (3.2) and to the pullback $K_{\min}$ of the canonical class of the minimal model. Then the positive real ray $\mathbb{R}^+K_{\min}$ is contained in the closure of the Kähler cone $K(S) \subset \text{Pic } S \otimes \mathbb{R}$, and we can define the set of almost canonical polarisations (ac-polarisations for short) which are close to the ray $\mathbb{R}^+K_{\min}$ up to finite set of spin chambers (see [T2], Cap. 1, §2 for details). Our computations don't depend on a choice of a spin chamber.

Definition 6.1. 1. The Gieseker closure $\overline{M^{ac}(2, K_S, c_2(S) + k)} = J_k(S)$ is called the k -th Jacobian of S .

2. The subspace

$$\Theta_k = \{F \in J_k(S) \mid h^0(F) \geq 1\} \quad (6.1)$$

is called the theta locus of $J_k(S)$.

(We denote this subspace by theta on the analogy of Riemann Theorem in the case of algebraic curves).

Remark. Actually, the jumping conditions define the Brill-Noether filtration:

$$J_k(S) \supseteq \Theta_k(S) \supseteq W_k^1(S) \supseteq \cdots \supseteq W_k^r \cdots \quad (6.2)$$

(see [T5] and [T2]).

The virtual (expected) dimension of $J_k(S)$ is given by

$$\text{v. dim } J_k(S) = 5\mu(S) + 4k, \quad (6.3)$$

where $\mu(S) = \frac{3c_2(S) - K_S^2}{4}$ is the Miyaoka number, and the virtual dimension of the theta locus by

$$\text{v. dim } \Theta_k = \frac{35c_2(S) - 13K_S^2}{12} + 3k - 1 = 4\mu(S) - (p_g + 1) - 1. \quad (6.4)$$

The Jacobians $J_k(S)$ and the theta loci Θ_k are families of torsion free sheaves, hence they define Donaldson polynomials and spin canonical polynomials:

$$\gamma_k = 2asD_{ac}^{0,0,c_2(S)+k} = D_{gac}^{0,0,-4\mu(S)-c_2(S)-4k}, \quad (6.5)$$

$$s\gamma_k = 2asD_{ac}^{1,0,c_2(S)+k} = sD_{gH}^{1,0,-4\mu(S)-c_2(S)-4k},$$

(see (5.75)).

Remark. Of course this is nothing other than a new notation for the objects of Lecture 5. We are using them because we have reduced the problem of computing the Donaldson and spin invariants of smooth structure of an algebraic surface to a purely algebraic geometric problem.

Not much is known at present on general properties of Jacobians and theta loci: if k is large enough then

$$1. \quad \dim J_k(S) = \text{v. dim } J_k(S)$$

[D1], [Z],

$$2. \quad J_k(S) \text{ is irreducible}$$

[G-L],

$$3. \quad S \text{ is of general type and minimal} \implies J_k(S) \text{ is of general type}$$

[Jun Li] and

$$4. \quad \omega \in H^{2,0} \text{ generic} \implies \gamma_k(\omega + \bar{\omega}) > 0,$$

$$5. \quad S \text{ isn't minimal} \implies J_k(S) \text{ is ruled and}$$

$$\gamma_k(\omega + \bar{\omega}) = 0 \text{ for every } \omega \in H^{2,0}. \quad (6.6)$$

About theta loci, we will see that Θ_k is rationally fibred over the Hilbert scheme

$$\Theta_k \xrightarrow{\pi} Hilb^{c_2(S)+k}; \quad \text{with} \quad \pi^{-1}(\xi_{gen}) = \mathbb{P}^{c_2(S)+k-K_S^2-p_g-2}, \quad (6.7)$$

so even for minimal surfaces of general type

$$s\gamma_k(\omega + \bar{\omega}) = 0 \text{ for any } \omega \in H^{2,0}.$$

To compute the shape of the spin canonical polynomials we construct a chain of flips of intermediate moduli spaces MP_i^k like (4.24), and describe the shape of the increments $\gamma P_i^k - \gamma P_{i+1}^k$ of the polynomials which obtained by the standard slant product construction from all moduli spaces.

Remark. In the case of curves the moduli spaces MP_i^g are moduli spaces of τ -vortices (see [B-D]) which are moduli spaces of σ -stable pairs by Bradlow's Identification Theorem. The last interpretation reduces the problem to an algebraic geometric one, which can be solved using the standard algebraic geometric constructions as done by Bertram in [Ber].

On a polarised algebraic surface (S, H) , by a *pair* we always mean a pair $s: \mathcal{O}_S \rightarrow F$ consisting of a sheaf F (as usual without torsion) and a nonzero section $s \in H^0(F)$. Let $\sigma = \sigma_1 z + \sigma_2$ be a linear polynomial with rational coefficients and $\sigma_1 > 0$, and let $\chi_H(F)$ be the Hilbert polynomial of F .

Definition 6.1. A pair $(s: \mathcal{O}_S \rightarrow F)$ is called σ -(semi)stable if for every nontrivial subsheaf $F_1 \subset F$

- 1) $\text{rank } F \cdot \chi_H(F_1)(\leq) < \text{rank } F_1 \cdot (\chi_H(F) - \sigma)$ if $F_1 \subset s(\mathcal{O}_S)$,
 - 2) $\text{rank } F \cdot \chi_H(F_1)(\leq) < \text{rank } F_1 \cdot (\chi_H(F) - \sigma) + \text{rank } F \cdot \sigma$ otherwise.
- (6.8)

Remark 1. Of course we will only apply this construction to the case $\text{rank } F = 2$, $\text{rank } F_1 = 1$.

Remark 2. The "classical" definition of slope stability of a pair is

- 1) $\text{rank } F \cdot \deg F_1(\leq) < \text{rank } F_1 \cdot (\deg F - \sigma_1)$ if $F_1 \subset s(\mathcal{O}_S)$
- 2) $\text{rank } F \cdot \deg F_1(\leq) < \text{rank } F_1 \cdot (\deg F - \sigma_1) + \text{rank } F \cdot \sigma_1$ otherwise

(see [B-D] and [H-L]). It is easy to see that

$$\text{slope-stable} \implies \text{stable} \implies \text{semistable} \implies \text{slope-semistable}$$

and 0-stable means Gieseker stable.

Using Geometric Invariant Theory, Huybrechts and Lehn proved (see [H-L], 2.2) that for every polynomial σ there exists a moduli space $MP_\sigma^H(2, c_1, c_2)$ of σ -semistable pairs $(s: \mathcal{O}_S \rightarrow F)$, where F is of topological type $(2, c_1, c_2)$. (possibly empty). Moreover every such moduli space is a family of torsion free sheaves on S , and hence slant product defines a cohomology correspondence μ and a collection of polynomials

$$\gamma P_\sigma^H(2, c_1, c - 2). \quad (6.9)$$

However, under lexicographic order of the polynomials $\{\sigma\}$, there exists a sequence of polynomials

$$0 = \sigma^{\max} < \dots < \sigma^i < \dots < \sigma^0 \quad (6.10)$$

such that for obvious numerical reason the σ -stability conditions (6.8) remain the same for any polynomial σ in the interval $\sigma^{i+1} > \sigma > \sigma^i$ (see [H-L], 2.2). So for σ in this interval we get a fixed moduli space $MP_i^H(2, c_1, c_2)$ and as σ varies we have a chain of birational (up to components) transformations or flips

$$MP_{\max}^H(2, c_1, c_2) \leftrightarrow \cdots \leftrightarrow MP_1^H(2, c_1, c_2) \leftrightarrow MP_0^H(2, c_1, c_2). \quad (6.11)$$

We will specify the polarisation H as an almost canonical polarisation, so we can use the Hilbert polynomials χ_{K_S} in all our inequalities in spite of the fact that K_S may be nef only. Thus we can omit almost all indexes of moduli spaces of pairs:

$$MP_i^{ac}(2, K_S, c_1(S) + k) = MP_i^k. \quad (6.12)$$

Now the moduli space on the extreme left admits a regular map of degree one

$$MP_{\max}^k \rightarrow \Theta_k \quad (6.13)$$

hence the polynomial (6.9)

$$\gamma P_{\max}^{ac}(2, K_S, c_2(S) + k) = s\gamma_k. \quad (6.14)$$

Now we would like to compute the moduli space MP_0^k on the right-hand end of the chain (6.11) and the polynomial γP_0^k .

First of all from the definitions,

$$MP_0^k = GA\Theta_k = \{0 \xrightarrow{s} \mathcal{O}_S \rightarrow F \rightarrow J_\xi(K_S) \rightarrow 0\} \quad (6.15)$$

is the set of all nontrivial extensions of all ideal sheaves for all zero dimensional subschemes $\xi \in \text{Hilb}^d S$ of degree $d = c_2(S) + k$ by the trivial line bundle $\mathcal{O}_S$. So there is a natural map of moduli space

$$\pi: GA\Theta^k \rightarrow \text{Hilb}^d S \quad (6.16)$$

sending a pair $(\mathcal{O}_S \xrightarrow{s} F)$ to the zero set of the section s . It is easy to see (see Exercises) that if k is large enough then this map is onto. So we have to describe the fibres of this map, which are the projective spaces $\mathbb{P} \text{Ext}^1(J_\xi(K_S), \mathcal{O}_S)$.

Now let

$$Z_d \subset S \times \text{Hilb}^d S, \quad \text{with} \quad Z_d \cap (S \times \xi) = \xi \quad (6.17)$$

be the universal subscheme and

$$\begin{array}{ccc} & Z_d & \\ \bar{p}_S \swarrow & & \searrow \bar{p}_H \\ S & & \text{Hilb}^d S \end{array} \quad (6.18)$$

its two projections, induced by the projections p_H and p_S of the direct product $S \times \text{Hilb}^d S$.

For any divisor class $D \in \text{Pic } S$ consider the vector bundle

$$\mathcal{E}_D = R^0 \bar{p}_H(\bar{p}_S^* \mathcal{O}_S(D)). \quad (6.19)$$

These sheaves are locally free because the canonical homomorphism is surjective.

Let H be the divisor class of the Grothendieck sheaf $\mathcal{O}_{\mathbb{P}\mathcal{E}_D^*}(1)$. Each section s of $\mathcal{E}_D^d$ corresponds to a section $\bar{s}$ of the Grothendieck sheaf $\mathcal{O}_{\mathbb{P}\mathcal{E}_D^*}(1)$. So any section $s \in H^0(\mathcal{O}_S(D))$ defines a section $\text{res}(s)$ of the vector bundle $\mathcal{E}_D$ and a section $\overline{\text{res}(s)}$ of $\mathcal{O}_{\mathbb{P}\mathcal{E}_D^*}(1)$.

If $s_1, \dots, s_{h^0(\mathcal{O}(D))}$ is a basis of the space of sections $H^0(\mathcal{O}_S(D))$ then the intersection of divisors

$$\bigcap_{i=1}^{h^0(D)} H_{\overline{\text{res}(s_i)}} \subset \mathbb{P}\mathcal{E}_D^* = E_{d,D} \quad (6.20)$$

is the space of all nontrivial extensions of type

$$0 \rightarrow \mathcal{O}_S \rightarrow E \rightarrow J_\xi(D - K_S) \rightarrow 0 \quad (6.21)$$

up to $\mathbb{C}^*$ for all $\xi \in \text{Hilb}^d S$ (see [T1]). So if $S = S_{\min}$ then

$$D = 2K_S \implies E_{d,D} = GA\Theta_{d-c_2(S)} \quad (6.22)$$

is a first geometric approximation of the theta locus.

Remark. If a surface S isn't minimal then $h^1(\mathcal{O}_S(2K_S)) = \text{number of } (-1)\text{-curves}$ and instead of $\mathbb{P}\mathcal{E}_{2K_S}^*$ we have to use the projectivisation

$$\mathbb{P}(H^1(\mathcal{O}_S(2K_S)) \otimes \mathcal{O}_{\mathbb{P}(\mathcal{E}_{2K_S}^*)} \oplus \mathcal{O}_{\mathbb{P}(\mathcal{E}_{2K_S}^*)}(1)) \quad (6.23)$$

and its restriction to the complete intersection (6.20).

Now we can construct the universal family of extensions of type (6.21) as the extension

$$0 \rightarrow p_1^* \mathcal{O}(H) \rightarrow \mathbb{E} \rightarrow (\text{id} \times) \pi^* \mathcal{I}_Z \otimes p_5^* \mathcal{O}_S(K_S) \rightarrow 0 \quad (6.24)$$

(for the description of the cocycle of this extension see [T1], the diagram (4.25) and (4.28)).

As does any family of sheaves, $\mathbb{E}$ defines a cohomological correspondence

$$\mu_{\mathbb{E}}: H^2(S, \mathbb{Z}) \rightarrow H^2(GA\Theta, \mathbb{Z})$$

by the slant product construction (see (4.30), (4.31)), and from (6.24) we have

$$\mu_{\mathbb{E}}(\sigma) = \pi^*(\tilde{\sigma}) + \frac{1}{2}(\sigma \cdot (D - K_S))H, \quad \text{for any } \sigma \in H^2(S, \mathbb{Z}) \quad (6.25)$$

where $\tilde{\sigma} = \mu_{\text{Hilb}}(\sigma)$, and

$$\mu_{\text{Hilb}}: H^2(S, \mathbb{Z}) \rightarrow H^2(\text{Hilb}, \mathbb{Z}) \quad (6.26)$$

is given by the slant product of the family of torsion free sheaves of rank 1 over Hilb (see Exercise 2 of Lecture 5). Roughly speaking, if the fundamental 2-cycle of σ is given as a smooth oriented 2 manifold Σ then

$$\tilde{\sigma} = \mu_{\text{Hilb}}(\sigma) = \{\xi \in \text{Hilb} \mid \text{Supp} \xi \cap \Sigma \neq \emptyset\} \quad (6.27)$$

(that is, $\tilde{\sigma}$ contains clusters $\xi = p_1 + \cdots + p_d$ such that at least one point p_i is contained in Σ).

On the other hand by the same reason for any point $p \in S$ we have

$$\mu_{\text{Hilb}}([\text{pt}]) = \text{Hilb}_p = \{\xi \in \text{Hilb}^d S \mid p \in \text{Supp} \xi\} = \text{Hilb}_p = \bar{p}_H(\bar{p}_S^{-1}(p)) \quad (6.28)$$

(in terms of the diagram (6.18)). Obviously Hilb_p is birationally equivalent to Hilb^{d-1} . More precisely let $\sigma: \widetilde{\text{Hilb}^{d-1}} \rightarrow \text{Hilb}^{d-1}$ be the blowup of Hilb_p in Hilb^{d-1} ; then Ellingsrud's construction gives the birational morphism

$$e: \widetilde{\text{Hilb}^{d-1}} \rightarrow \text{Hilb}_p \subset \text{Hilb}^d S \quad (6.29)$$

(see [P-T], the diagram (II, 3.7)). Now

$$\mu_{\mathbb{E}}([pt]) = \pi^*(\text{Hilb}_p) - \frac{1}{4}H^2. \quad (6.30)$$

Now the value of the first approximation $\gamma P_0(2, K_S, c_2 + k)$ of the spin canonical polynomial on a typical monomial $\alpha = [\text{pt}]^n \cdot \sigma_1 \cdots \sigma_m$ is the intersection number

$$\begin{aligned} & \gamma P_0(2, K_S, d)(\alpha) = \\ & (\pi^*(\text{Hilb}_p) - \frac{1}{4}H^2)^n (\pi^*(\widetilde{\sigma_1}) + \frac{1}{2}\sigma_1 \cdot K_S \cdot H) \cdots (\pi^*\widetilde{\sigma_m} + \frac{1}{2}\sigma_m \cdot K_S \cdot H) \cdot H^{h^0(\mathcal{O}_S(2K_S))} \end{aligned} \quad (6.31)$$

of 2-cocycles on $\mathbb{P}\mathcal{E}_{2K_S}^{d*}$ if S is minimal and otherwise we have a formula of the same type (see (6.23)).

Thus our polynomials (6.31) are the linear combinations of polynomials of the special type:

$$\pi^*(\sigma_1 \cdots \sigma_n) \cdot H^{3d-1-n}$$

on the projectivisation $\mathbb{P}\mathcal{E}_{2K_S}^{d*}$. By the projection formula these polynomials are equal to the following standard polynomials:

$$p_d^n(\sigma) = s_{2d-n}(\mathcal{E}_{2K_S}) \cdot \sigma^n. \quad (6.32)$$

Proposition 6.1(Ellingsrud).

$$p_d^n \in S^*[q_S, K_S] \quad (6.33)$$

that is, these polynomials are polynomials of the intersection form q_S as a quadratic form and K_S as a linear form.

Proof. Let $i: Z_d \rightarrow S \times \text{Hilb}^d$ be the embedding of the universal subscheme (6.17) and $s_k(Z_d, S \times \text{Hilb}^d S)$ the relative Segre classes of the subscheme Z_d in $S \times \text{Hilb}^d S$ (see [Fu], 4.2). We consider any polynomial on $H^2(S, \mathbb{Z})$ of the form

$$p(\sigma) = i_*(s_{i_1} \cdots s_{i_p})^*_H(s_n(\mathcal{E}_D^d)\tilde{\sigma}^m) \cdot p_S^*([\text{pt}]) \quad (6.34)$$

and prove that it is in $S^m(q_S, D, K_S)$. We use induction on d , and label our objects by this number. Let

$$\sigma: \widetilde{S \times \text{Hilb}^d} \rightarrow S \times \text{Hilb}^d \quad (6.35)$$

be the blowup of $S \times \text{Hilb}^d$ along the universal subscheme Z_d and $\widetilde{Z}_d$ the exceptional divisor on $S \times \text{Hilb}^d$. Ellingsrud constructs the morphism

$$e: \widetilde{S \times \text{Hilb}^d} \rightarrow \text{Hilb}^{d+1} \quad (6.36)$$

extending the natural map (for details see [P-T], (II, 3 8) and (II, 3, 9)). Then it is easy to see that

$$e^*(\tilde{\sigma}_{d+1}) = (\sigma \cdot p_H)^*(\tilde{\sigma}_d) + (\sigma \cdot p_S)^*(\sigma), \quad (6.37)$$

and there is exact sequence:

$$0 \rightarrow (\sigma \cdot p_S)^* \mathcal{O}_S(D)(-\widetilde{Z}_d) \rightarrow e^* \mathcal{E}_D^{d+1} \rightarrow (\sigma \cdot p_H)^* \mathcal{E}_D^d \rightarrow 0. \quad (6.38)$$

Recall that $\sigma_*(\widetilde{Z}_d)^i = s_{i-2}(Z_d, S \times \text{Hilb}^d)$. So to finish the proof we have to recompute the Segre classes of Z_{d+1} in the terms of the objects of the label d . For this consider the map of direct products:

$$\widetilde{S \times \text{Hilb}^d} \times S \xrightarrow{(e \times \text{id})} \text{Hilb}^{d+1} \times S. \quad (6.39)$$

Then

$$(e \times \text{id})^{-1}(Z_{d+1}) = ((\sigma \cdot p_H) \times p_S)^{-1}(Z_d) \cup ((\sigma \cdot p_S) \times p_S)^{-1}(\Delta), \quad (6.40)$$

where Δ is the diagonal of $S \times S$. From this for the total classes we have

$$s((e \times \text{id})^{-1}(Z_{d+1})) = s(Z_d) + (1 - s(Z_d)) \cdot i_* c(N_{((\sigma \cdot p_S) \times p_S)^{-1}(\Delta)}(\widetilde{Z}_d))^{-1}, \quad (6.41)$$

where $c(N)$ is the total Chern class of the normal bundle, and we are done.

The last task is to see that if $d \gg 0$ then the polynomials (6.31) are nonvanishing. To see this it is enough to observe the following result

Proposition 6.2. *Let S be a simply connected surface of general type. Then*

$$d \geq 2h^0(\mathcal{O}_S(2K_S)) \implies \dim GA\Theta_{d-c_2(S)} = \text{v. dim } \Theta_k, \quad (6.42)$$

and $GA\Theta_{d-c_2(S)}$ is irreducible.

We propose to prove it in Exercise 1 or to see [T1], proof of Lemma 4.1.

So if $d \geq h^0(\mathcal{O}_S(2K_S))$ then the nonpolarised polynomial (6.31) has the form

$$\gamma P_0(2, K_S, d)(\sigma) = 2^d (K \cdot \sigma)^{(d-K^2-p_g-2)} \sum_{j=0}^{2d} 2^j \binom{2d}{j} (K \cdot \sigma)^{2d-j} s_{2d-j}(\mathcal{E}_{2K}) \tilde{\sigma}^j \quad (6.43)$$

and we would like to prove that this polynomial is nonzero.

Proposition 6.3. *Let $D \in \text{Pic } S$ be a polarisation of S , and $d \gg 0$ then*

$$\gamma P_0(2, K_S, d)(D) \neq 0.$$

Proof. We can use the same arguments as for the standard proof of the non-degeneracy condition of the Donaldson polynomials. Namely, replacing D if necessary by a large multiple we can assume that there exists a smooth curve $C \subset S$ in $|D|$ such that the restriction map of the (noncompactified) moduli space $M^D(2, K_S, d)$ to the moduli space $M_C^{C \cdot K_S}$ of stable bundles on C with $c_1 = C \cdot K_S$ is an embedding (see [T8] or [Bog]). Let $\Theta \in \text{Pic } M_C^{C \cdot K_S}$ be the positive generator or theta-divisor and $[M_{1,0}^D(2, K_S, d)]$ the fundamental class of the subvariety of jumping instantons (5.64). Then

$$\gamma P_0(2, K_S, d)(D) = [M_{1,0}^D(2, K_S, d)] \cdot \Theta^{\text{codim}} \neq 0, \quad (6.44)$$

and we are done.

Remark. A more productive way is to use [H-L], Theorem 2.17 which gives the restriction embedding for the moduli spaces of stable pairs directly.

Now if some extension of type

$$0 \rightarrow \mathcal{O}_S \rightarrow F \rightarrow J_\xi(K_S) \rightarrow 0$$

is (ac)-slope nonstable and $J_\eta(D) \rightarrow F$ is the destabilising subsheaf then we have a nontrivial homomorphism

$$\varphi: J_\eta(D) \rightarrow J_\xi(K_S), \quad \text{with } (\varphi)_0 = C + \varepsilon, \quad (6.45)$$

where C is the 1-dimensional component of the zero-set of φ and $\varepsilon \in \text{Hilb } S$ is the 0-dimensional component. So

$$D = K_S - C \quad (6.46)$$

and

$$D \cdot K_{\min} \geq \frac{1}{2} K_{\min}^2 \implies C \cdot K_{\min} \leq \frac{1}{2} K_{\min}^2 \quad (6.47)$$

hence the class of C is contained in the collection of curves (3.14). Because of the uniqueness of the destabilising subsheaf the components of the nonstable subloci of $GA\Theta$ can be labeled by the classes of the curves in the collection (3.14). On the other hand every curve C (3.14) gives a new family of extensions, of the type

$$0 \rightarrow \mathcal{O}_S(C) \rightarrow F \rightarrow J_\xi(K_S - C) \rightarrow 0. \quad (6.48)$$

Thus to get the spin canonical polynomial we have to reconstruct our family (6.24) by a chain of elementary transformations. Before describing the centres of elementary transformations we would like to observe that we have already got some important results about the invariance of the canonical classes for algebraic surfaces of special type:

Theorem 6.1. *Let S be a surface of general type such that S doesn't contain d -stable curves C (see Definition 3.1) such that $C \cdot K_S \leq \frac{1}{2} K_{\min}^2$ (for example a surface S with $\text{Pic } S = \mathbb{Z} \cdot K_S$ or the Barlow surface (3.19)). Then the canonical class K_S is invariant under diffeomorphism.*

Proof. Indeed such surfaces are minimal, hence $K_S^2 > 0$ and $s\gamma_k$ is diffeomorphism invariant. On the other hand, in this case

$$\Theta_k = GA\Theta(6.22)$$

and $s\gamma_k \in S[q_S, K_S]$ with (6.7). We are done (see Corollary 1 of Lemma 3.1).

Now suppose that there exists a curve C with $C \cdot K_{\min} \leq \frac{1}{2} K_{\min}^2$, and for simplicity assume that C is rigid. Let CR_C^0 be the subset of $GA\Theta_k$ containing nonstable sheaves destabilised by $\mathcal{O}_S(K_S - C)$, that is, CR_C^0 is the set of all diagrams of type

$$\begin{array}{ccccccc}
 & & & 0 & & & \\
 & & & \uparrow & & & \\
 & & & J_\eta(C) & & & \\
 & & \nearrow \varphi' & \uparrow & & & \\
 0 \rightarrow \mathcal{O}_S & \rightarrow & F & \rightarrow & J_\xi(K_S) & \rightarrow 0 & (6.49) \\
 & & \uparrow & \nearrow \varphi & & & \\
 & & \mathcal{O}_S(K_S - C) & & & & \\
 & & \uparrow & & & & \\
 & & 0 & & & &
 \end{array}$$

(which is called a *cross* for short), where $\xi_i \in \text{Hilb}^d$ and $\eta \in \text{Hilb}^{d+K_S \cdot C + C^2}$. From this diagram we can see that $\xi \subset C$, and we have the fibration

$$\pi: CR_C^0 \rightarrow \text{Hilb } C \subset \text{Hilb}^d S. \quad (6.50)$$

To describe the fibre

$$CR_{\xi \subset C}^0 \subset \mathbb{P} \text{Ext}^1(J_\xi(K_S), \mathcal{O}_S) \quad (6.51)$$

we can remark that the homomorphism $\varphi: \mathcal{O}_S(K_S - C) \rightarrow J_\xi(K_S)$ can be obtained from a homomorphism $\mathcal{O}_S(K_S - C) \rightarrow F$ if and only if the induced homomorphisms

$$\text{Ext}^1(J_\xi(K_S), \mathcal{O}_S) \xrightarrow{\bar{\varphi}} \text{Ext}^1(\mathcal{O}_S(K_S - C), \mathcal{O}_S) \quad (6.52)$$

satisfies

$$\bar{\varphi}(e_F) = 0, \quad (6.53)$$

where e_F is the cocycle which defines the horizontal extension in (6.49). Moreover the homomorphism (6.52) is a part of the long exact sequence $\text{Ext}(\cdot, \mathcal{O}_S)$ of the short sequence

$$0 \rightarrow \mathcal{O}_S(K_S - C) \rightarrow J_\xi(K_S) \rightarrow \mathcal{O}_C(K_S \cdot C - \xi) \rightarrow 0, \quad (6.54)$$

from which we have

$$\ker \bar{\varphi} = \text{Ext}^1(\mathcal{O}_C(K_S \cdot C - \xi), \mathcal{O}_S) = H^1(\mathcal{O}_C(2K_S \cdot C - \xi))^* = H^0(\mathcal{O}_C(C^2 - K_S \cdot C + \xi)), \quad (6.55)$$

(we used Serre duality on S and on C). So

$$CR_{\xi \subset C}^0 = |C^2 - K_S \cdot C + \xi|$$

Now it is easy to see that for the cocycle defining the horizontal extension in (6.49) we have

$$\mathbb{P}e_F = \eta \subset C. \quad (6.56)$$

(Please, prove this as an exercise!). Geometrically, this means that the map

$$(\pi \times \pi'): CR_C^0 \rightarrow \text{Hilb}^d C \times_{\text{Pic } C} \text{Hilb}^{d-K_S \cdot C + C^2} \quad (6.57)$$

sending a cross (6.49) to (ξ, η) is an isomorphism.

Now from the same long cohomological sequence of (6.54) we have

$$\text{coker } \bar{\varphi} = H^1(\mathcal{O}_S(C - K_S)) = H^1(\mathcal{O}(2K_S - C))^*. \quad (6.58)$$

So in the same vein as (6.20), we have

$$CR_C^0 = \bigcap_{i=1}^{h^0(2K_S) + h^1(\mathcal{O}_S(2K_S - C))} H_i \subset \mathbb{P}\mathcal{E}_{2K_S}^*|_{\text{Hilb } C}. \quad (6.59)$$

Thus if $h^1(\mathcal{O}_S(2K_S - C)) = 0$ then for $\pi: GA\Theta \rightarrow \text{Hilb } S$ we have

$$CR_C^0 = \pi^{-1}(\text{Hilb } C). \quad (6.60)$$

For simplicity consider this case only. Then to blowup CR_C^0 in $GA\Theta$ it is enough to blowup $\text{Hilb } C$ in $\text{Hilb } S$ and to describe the normal cone of $\text{Hilb } C$ and the relative Segre classes $s(\text{Hilb } C, \text{Hilb } S)$. For this the following observation is very productive:

Let $s \in H^0(\mathcal{O}_S(C))$ be the section with our curve C as zero set (that is, $(s)_0 = C$) and $\text{res}(s)$ be the section of the standard vector bundle $\mathcal{E}_C$ (6.19) on $\text{Hilb } S$ defined by s (see the sentence between (6.19) and (6.20)). Then for the zero set of this section we have

$$(\text{res}(s))_0 = \text{Hilb } C \quad (6.61)$$

and hence the projectivisation of the normal cone is given as

$$\mathbb{P}C_{\text{Hilb } C \subset \text{Hilb } S} = \mathbb{P}\mathcal{E}_C|_{\text{Hilb } C} \quad (6.62)$$

and from this for the total relative Segre class we have

$$s(\text{Hilb } C, \text{Hilb } S) = s(\mathcal{E}_C) \cap \text{Hilb } C. \quad (6.63)$$

Finally, every point $p \in C$ define on $\text{Hilb } C$ the Schwarzenberger divisor

$$D_p = \{\eta \in \text{Hilb } C \mid p \in \text{Supp } \eta\} \quad (6.63')$$

(see [ACGH], Ch.8, 2.5). So for the restriction of the cycles on $\text{Hilb } S$ to $\text{Hilb } C$ we have

$$\tilde{\sigma} \cdot \text{Hilb } C = (\sigma \cdot C) \cdot D_p. \quad (6.63'')$$

The partial modification of the component $GA\Theta$ along the component CR_C^0 is the blowup of this subvariety in $GA\Theta$, an “elementary transformation” of the lift of the universal family $\mathbb{E}$ (6.24), a recomputation of the cohomological correspondence (6.25) and of new polynomial (more precisely, of its correcting term).

The geometric description of blowup of a component of CR_C^0 is obtained above. Let

$$\sigma: \widetilde{\mathbb{P}\mathcal{E}_{2K}^*} \rightarrow \mathbb{P}\mathcal{E}_{2K}^* \quad (6.64)$$

be the blowup CR_C^0 in $\mathbb{P}\mathcal{E}_{2K}^*$. Then the blowup CR_C^0 in $GA\Theta$ is the intersection

$$\widetilde{GA\Theta} = \bigcap_{i=1}^{h^0(\mathcal{O}_S(2K))} \widetilde{H_{\text{res}(s_i)}} \subset \widetilde{\mathbb{P}\mathcal{E}_{2K}^*}, \quad (6.65)$$

where $\widetilde{H_{\text{res}(s_i)}}$ is a geometric inverse image of the hypersurface $H_{\text{res}(s_i)}$ w.r.t. σ (6.64).

Denote the exceptional divisor of this blowup by $\widetilde{CR}$. Then the projection π' of (6.57) defines the projection

$$(id \times \pi'): S \times \widetilde{CR} \rightarrow S \times \text{Hilb}^{d-K_S \cdot C + C^2} C. \quad (6.66)$$

Let Z' be the inverse image of the universal subscheme Z (6.18) with respect to this morphism. Then on the divisor $S \times \widetilde{CR}$ of the variety $S \times \widetilde{GA\Theta}$ we have the torsion free sheaf $(id \times (\sigma \cdot \pi'))^*(\mathcal{I}_{Z'} \otimes p_S^* \mathcal{O}_S(C))$ and the epimorphism of lifting of the restriction of the universal extension (6.24) to it:

$$(id \times \sigma)^* \mathbb{E}|_{S \times \widetilde{CR}} \rightarrow (id \times (\sigma \cdot \pi'))^*(\mathcal{I}_{Z'} \otimes p_S^* \mathcal{O}_S(C)) \rightarrow 0 \quad (6.67)$$

(see the diagram (6.49)).

Remark. We use here the fact that $c_1(R^0 \pi' \mathbb{E}(C - K_S)|_{S \times CR}) = 0$ but it isn't important for the final computations.

This epimorphism provides the exact sequence:

$$0 \rightarrow \mathbb{E}' \rightarrow (id \times (\sigma \cdot \pi'))^* \mathbb{E} \rightarrow (id \times (\sigma \cdot \pi'))^*(\mathcal{I}_{Z'} \otimes p_S^* \mathcal{O}_S(C)) \rightarrow 0 \quad (6.68)$$

and its kernel $\mathbb{E}'$ is the “elementary transformation” of $\mathbb{E}$ along CR_C^0 . In this new family of torsion free sheaves the vertical extensions in (6.49) were changed by the extensions of the type:

$$0 \rightarrow J_\eta(C) \rightarrow F' \rightarrow \mathcal{O}_S(K_S - C) \rightarrow 0$$

which are $(\frac{1}{2}K_{\min}^2 - K_{\min} \cdot C) \cdot z$ -stable (see Definition 6.1).

The new family $\mathbb{E}'$ (6.68) provides the new cohomological correspondence

$$\mu_{\mathbb{E}'}(\sigma) = \tilde{\sigma} + \frac{1}{2}(K_S \cdot \sigma) \cdot H - \frac{1}{2}((K_S + 2C) \cdot \sigma) \cdot \widetilde{CR}. \quad (6.69)$$

Remark. The exact sequence (6.68) provides immediately

$$c_1(\mathbb{E}') = K_S + H - \widetilde{CR}, \quad (6.69')$$

$$c_1(\mathbb{E}') = Z_d + K_S \cdot H - (K_S + C) \cdot \widetilde{CR},$$

that is, the cohomological correspondence (6.69) doesn't depend on Z' , on the twisting by $-c_1(R^0\pi'\mathbb{E}(C - K_S)|_{S \times CR})$ and so on. This means that the spin polynomials are coarse invariants.

The cohomological correspondence (6.69) gives a new polynomial

$$\gamma P_1(2, K_S, d)(\sigma) \quad (6.70)$$

$$= (\pi^*(\tilde{\sigma}) + \frac{1}{2}(\sigma \cdot K_S) \cdot H - \frac{1}{2}((K + 2C) \cdot \sigma) \cdot \widetilde{CR})^{\text{v.dim}} \cdot (H - \widetilde{CR})^{h^0(\mathcal{O}_S(2K_S))}.$$

Remark. The index 1 of $\gamma P_1(2, K_S, d)$ means that we consider as the next polynomial $\delta = \delta_0 + (\frac{1}{2}K_{\min}^2 - K_{\min} \cdot C) \cdot z$ (see the sequence of polynomials (6.10)).

Moreover the binomial decomposition provides

$$\gamma P_1(2, K_S, d)(\sigma) = \gamma P_0(2, K_S, d)(\sigma), \quad (6.71)$$

$$\begin{aligned} \widetilde{CR} \left[\sum_{i=0}^{p_2(S)} \sum_{j=1}^{\text{v.d.}} \binom{p_2(S)}{i} \binom{\text{v.dim}}{j} ((K_S + 2C) \cdot \sigma)^j \cdot \widetilde{CR}^{i+j-1} \cdot H^{p_2(S)} \times \right. \\ \left. \times (\pi^*(\tilde{\sigma}) + \frac{1}{2}(\sigma \cdot K_S) \cdot H)^{\text{v.dim}} \right], \end{aligned}$$

where $p_2(S) = h^0(\mathcal{O}_S(2K_S))$ and v.dim is the virtual dimension of $\Theta_{d-c_2(S)}$.

Now we can compute the correcting term of polynomial as the intersection of divisors on $\widetilde{CR}$ and using the projection formula, as the intersection of the Segre classes and the Schwarzenberger divisors (6.63') on $\text{Hilb } C$. From this we have

$$\gamma P_1(2, K_S, d) - \gamma P_0(2, K_S, d) \in \mathbb{C}[K_S, C] \quad (6.72)$$

and

$$\gamma P_1(2, K_S, d) \in \mathbb{C}[q_S, K_S, C]. \quad (6.72')$$

Remark. To compute the coefficients of the correcting term (6.72) we assume that C is a smooth curve. Then $\text{Hilb}^d C = C_d$ is the symmetric power of C , that is, the set of effective divisors of degree d on C . Now if $\Delta \subset C \times C_d$ is the universal divisor and p_C and p_H are the projections of the direct product then for any $\xi \in \text{Pic } C$ we have the standard vector bundle $\mathcal{E}_D = R^0 p_H p_C^* \mathcal{O}_C(\xi)$ (see p.339 of [ACGH], and compare (6.19)). It is easy to see that the restriction of the standard bundle (6.19) has the form

$$\mathcal{E}_D|_{\text{Hilb } C} = \mathcal{E}_D \cdot C. \quad (6.73)$$

But the total Chern class of the standard vector bundles on C_d was computed in Lemma 2.5 of Ch.8 of [ACGH]. This provides the coefficients of (6.72).

So the first step of the Geometric Approximation Procedure is over. Next one is to consider the sublocus CR_C^1 where CR_C^n contains all crosses of the type

$$\begin{array}{ccccc}
 & & 0 & & \\
 & & \uparrow & & \\
 & & J_\eta(C) & & \\
 & \nearrow \varphi' & \uparrow & & \\
 0 \rightarrow \mathcal{O}_S & \rightarrow & F & \rightarrow & J_\xi(K_S) \rightarrow 0 \\
 & & \uparrow & \nearrow \varphi & \\
 & & J_{\eta_0}(K_S - C) & & \\
 & & \uparrow & & \\
 & & 0 & &
 \end{array} \quad (6.74)$$

where $\deg \eta_0 = n$. In this case CR_C^n is a small modification (see [T-T]) of the product

$$CR_C^n = \text{Hilb}^n S \times \text{Hilb}^{d-n} C \times_{\text{Pic } C} \text{Hilb}^{d-K_S \cdot C + C^2 - n} C \quad (6.75)$$

and in this case the correcting term is from $\mathbb{Q}[q_S, K_S, C]$.

If all intermediate moduli spaces have the right dimension then continue this procedure step by step, ordering the system of curves by the value $\frac{1}{2}K_{min}^2 - C \cdot K_{min}$ as the first coefficient of δ and n as the second coefficient we get the shape of the spin polynomials:

Shape Theorem 6.2. *For a surface of general type if $k \gg 0$*

$$s\gamma_k \in \mathbb{C}[q_S, K_S, C_1, \dots, C_N], \quad (6.76)$$

where $\{C_i\}$ is the set of d -stable curves (3.14).

From this, by the standard arguments of Corollary 1 of Lemma 3.1, we get the main result

Theorem 6.3. *For a surface of general type S with $p_g > 0$ the proper sublattice (3.17)*

$$kv_2(S) \subset H^2(S, \mathbb{Z}) \quad (6.77)$$

is diffeomorphism invariant, and hence the representation $\text{Mod } S$ is reducible.

To finish the proof we have to add the following remarks:

Remark 1. If the curve C isn't rigid, that is, the complete linear system $|C|$ has positive dimension, then we have to use a slight modification of our constructions. Namely, instead of $\text{Hilb } C$, we consider

$$\text{Hilb } |C| = \bigcup_{C' \in |C|} \text{Hilb } C', \quad (7.78)$$

which is the union of all clusters lying on curves of the linear system $|C|$. Then, as a cohomology class,

$$[\text{Hilb } |C|] = c_{\text{rank } \mathcal{E}_C - h^0(\mathcal{O}_S(C)) - 1}(\mathcal{E}_C) \quad (7.79)$$

and we have the same shape of our polynomials as in (6.72).

Remark 2. For every curve C in the set (3.14) with $C^2 \geq 0$ we have to consider the set $GA\Theta_C$ of all extensions of the type

$$0 \rightarrow \mathcal{O}_S(C) \rightarrow F \rightarrow J_\xi(K_S - C) \rightarrow 0. \quad (7.80)$$

It is easy to see that if k is big enough then

$$\dim GA\Theta_C - \text{v. dim } \Theta = C^2, \quad (6.81)$$

and after all elementary transformations we have to compute the fundamental class of the right moduli space. The recipe of this computation was described in [T1], Section 7 or [P-T], Chapter III. Now $GA\Theta$ is a complete intersection of Grothendieck divisors in $\mathbb{P}^s E_{2K_S-2C}^*$ over $\text{Hilb}^{d-K_S \cdot C+C^2} S$. It is easy to see from the Hodge Index Theorem that $K_S \cdot C - C^2 > 0$. Simple computations show that the fundamental class of the right moduli space is the intersection of C^2 Grothendieck divisors in $GA\Theta_C$ (see Proposition 5 from [T5]). Thus the shape of the spin polynomials for these components is the same as (6.76).

Exercises to Lecture 6.

1. Prove the equality (6.56).

Hint: use the full display of the diagram (6.49).

2. Prove the statements of Proposition 6.2.

Hint: use the cohomology long exact sequence of

$$0 \rightarrow J_\xi(D) \rightarrow \mathcal{O}_S(D) \rightarrow \mathcal{O}_\xi(D) \rightarrow 0$$

3. Using formula (6.43), compute the generating function of the first polynomial when $k \rightarrow \infty$.

4. Compute the difference (6.72) when

a) C is (-1) -curve;

b) C is (-2) -curve.

Hint: use the formula (6.71).

5. Prove the formula (6.81).

Hint: use Exercise 1.

EPILOG

After these lectures were prepared, E. Witten proposed a new approach to realising the Great Program described in Lecture 4. Namely, the following facts have been proved: for non rational surfaces, the class $K_{\min}$ up to sign is diffeomorphism invariant, and the collection of (-1) -curves is diffeomorphism invariant, Conjecture (4.45) is confirmed, the statement of Problem 4.1 is confirmed and so on, but this is another story ...

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ALGEBRA SECTION, STEKLOV MATH INSTITUTE, UL. VAVILOVA 42, MOSCOW 117966 GSP–1, RUSSIA

E-mail address: Tyurin@top.inian.su or Tyurin@CFGauss.Uni-Math.Gwdg.De or Tyurin@Maths.Warwick.Ac.UK

C.I.M.E. Session on "Transcendental Methods in Algebraic Geometry"

List of Participants

F. BATTAGLIA, Dip.to di Matematica Appl., Via S. Marta 3, 50139 Firenze, Italy
I. BAUER, Dip.to di Matematica, Via Buonarroti 2, 56127 Pisa, Italy
R. BRUSSE, Math. Inst., Univ. Bielefeld, Postfach 100131, 35501 Bielefeld, Germany
F. CAPOCASA, Dip.to di Matematica, Via D'Azeglio 85, 46100 Parma, Italy
P. DE BARTOLOMEIS, Dip.to di Matematica Appl., Via S. Marta 3, 50139 Firenze, Italy
S. DEMICHELIS, Via M. Bosso 17, 15030 Villanova (AL), Italy
J. EL GOUL, Residence Ouest, Bat. A, CH 224, 38406 Sain-Martin-d'Hères, France
M. FRANCIOSI, Scuola Normale Superiore, Piazza dei Cavalieri 7, 56126 Pisa, Italy
P. FREDIANI, Dip.to di Matematica, Via Buonarroti 2, 56127 Pisa, Italy
O. GARCIA-PRADA, Univ. de Paris-Sud, Mathématique, 91405 Orsay, France
B. GRAMLICH, Math. Inst., Univ. Bayreuth, D-95440 Bayreuth, Germany
J.-M. HWANG, Dept. of Math. Univ. of Notre Dame, Notre Dame, IN 46556, USA
A. LANTERI, Dip.to di Matematica, Univ., Via C. Saldini 50, 20133 Milano, Italy
M. MANETTI, Scuola Normale Superiore, Piazza dei Cavalieri 7, 56126 Pisa, Italy
S. MANFREDINI, Via Pascoli 12, 56017 S. Giuliano T., Pisa, Italy
M. MELLA, Dip.to di Matematica, Univ. di Trento, 38050 Povo, Italy
H. MIRANDA, Dip.to di Matematica, Univ. di Tor Vergata, Viale della Ricerca Scientifica, 00133 Roma, Italy
P. A. OLIVERIO, Dip.to di Matematica, Univ. della Calabria, 87036 Arcavacata di Rende, Italy
R. PAOLETTI, Via F. Filzi 3, 50019 Sesto Fiorentino, Firenze, Italy
G. PARESCHI, Dip.to di Matematica, Via Machiavelli, 44100 Ferrara, Italy
M. PONTECORVO, SISSA, Via Beirut 2/4, 34014 Trieste, Italy
E. RUBEL, Via IV Novembre 5, 57025 Piombino, Livorno, Italy
A. SCHAFER, Friedrich-Ebert-Str. 33 1/2, 95448 Bayreuth, Germany
M. SCHNEIDER, Dept. of Mathematics, Univ. of Bayreuth, 95448 Bayreuth, Germany
V. SHEVCHISHIN, Ruhr-Univ. Bochum, Fak. f. Math., 44780 Bochum, Germany
M. TEICHER, Math. Dept., Bar-Ilan Univ., Ramat-Gan, Israel
S. TRAPANI, Dip.to di Matematica, Univ. di Tor Vergata, Viale della Ricerca Scientifica, 00133 Roma, Italy
C. TURRINI, Dip.to di Matematica, Univ., Via C. Saldini 50, 20133 Milano, Italy
G. WEILL, Dept. of Math., Polytechnic Univ., 333 Jay Str., Brooklyn, NY 11201, USA
F. ZUCCONI, Via Leonardo da Vinci 151, 55049 Viareggio, Lucca, Italy

FONDAZIONE C.I.M.E.
CENTRO INTERNAZIONALE MATEMATICO ESTIVO
INTERNATIONAL MATHEMATICAL SUMMER CENTER

“Viscosity Solutions and Applications”

is the subject of the Second 1995 C.I.M.E. Session.

The Session, sponsored by the Consiglio Nazionale delle Ricerche (C.N.R.), the Ministero dell'Università e della Ricerca Scientifica e Tecnologica (M.U.R.S.T.) and the Azienda di Promozione Turistica Montecatini Terme/Val di Nievole, will take place, under the scientific direction of Professor ITALO CAPUZZO DOLCETTA (Università di Roma, La Sapienza) and Professor PIERRE LOUIS LIONS (Université Paris-Dauphine) at Montecatini Terme (Pistoia), **from 12 to 20 June, 1995.**

C o u r s e s

- a) **Deterministic optimal control and differential games.** (6 lectures in English)
Prof. M. BARDI (Università di Padova)

The dynamic programming method and Hamilton-Jacobi-Bellman equations
Necessary and sufficient conditions of optimality, synthesis of multivalued optimal feedbacks
Discontinuous viscosity solutions
Convergence of numerical schemes for Hamilton-Jacobi equations, synthesis of approximate optimal feedbacks
Two-players zero-sum differential games, Isaac's equations

References

- L.C. Evans and P.E. Souganidis, Differential games and representation formulas for solutions of Hamilton-Jacobi equations, *Indiana Univ. Math. J.*, 33 (1984).
- G. Barles and B. Perthame, Discontinuous solutions of deterministic optimal stopping time problems, *RAIRO Model. Math. Anal. Num.*, 21 (1987).
- M.G. Crandall, L.C. Evans and P.L. Lions, Some properties of viscosity solutions of Hamilton-Jacobi equations, *Trans. Amer. Math. Soc.*, 282 (1984).
- M. Bardi, M. Falcone and P. Soravia, Fully discrete schemes for the value function of pursuit-evasion games, in *Advances in Dynamic Games and Applications*, T. Basar and A. Haurie eds., Birkhauser (1994).
- M. Bardi and I. Capuzzo Dolcetta, *Optimal Control and Viscosity Solutions of Hamilton-Jacobi-Bellman Equations*, Birkhauser, Boston, to appear.

- b) **General theory of viscosity solutions.** (6 lectures in English)
Prof. M. G. CRANDALL (University of California, Santa Barbara)

Scalar fully nonlinear pde's of first and second order: examples, nonexistence of classical solutions, nonuniqueness of strong solutions
The notion of viscosity solutions, uniqueness in the first order case, examples of existence and uniqueness theorems for viscosity solutions
Uniqueness for second order equations
Closure under limit operations: Perron's method and existence
Generalized boundary conditions
Extensions to infinite dimensions

References

- M.G. Crandall, H. Ishii and P.L. Lions, User's guide to viscosity solutions of second order partial differential equations, *Bull. Amer. Math. Soc.*, 27 (1992).
- A. Pazy, *Semigroups of linear operators and applications to partial differential equations*, Springer-Verlag, New York (1983).
- L.C. Evans, R.F. Gariepy, *Measure theory and fine properties of functions*, Studies in Advanced Math., CRC Press, Ann Arbor (1992).

- c) **Fully nonlinear equations and motion by mean curvature.** (6 lectures in English)
Prof. L. C. EVANS (University of California, Berkeley)

Fully nonlinear elliptic PDE's
Regularity for convex nonlinearities
Viscosity solutions for general nonlinearities
Motion by mean curvature I: introduction
Motion by mean curvature II: level set method
Motion by mean curvature III: properties of the generalized flow

References

- D. Gilbarg and N. Trudinger. Elliptic Partial Differential Equations of the Second Order (chapter 17). Springer (1983).
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- L. C. Evans and J. Spruck. Motion of level sets by mean curvature. *J. Diff. Geom.* 33 (1991)
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- M.G. Crandall, H. Ishii and P.L. Lions. User's guide to viscosity solutions to 2nd order partial differential equations. *Bull. Amer. Math. Soc.*, 27 (1992).

- d) **Optimal control and mathematical finance.** (6 lectures in English)
Prof. M. H. SONER (Carnegie Mellon University)

Optimal control of Markov processes
Dynamic programming equations for controlled Markov diffusion processes
Viscosity solutions of Hamilton-Jacobi-Bellman equations
Investment-consumption and option pricing problems
Scheduling problems in manufacturing

References

- M.G. Crandall, H. Ishii and P.L. Lions. User's guide to viscosity solutions to 2nd order partial differential equations. *Bull. Amer. Math. Soc.*, 27 (1992).
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- e) **Asymptotic problems in front propagation.** (6 lectures in English)
Prof. P. E. SOUGANIDIS (University of Wisconsin)

General discussion and examples from phase transitions
Phase-field theory and asymptotics of reaction-diffusion equations
Macroscopic limits of stochastic Ising models with long range interactions
A general theory about moving fronts and asymptotic limits
Large scale front dynamics for turbulent reactions-diffusion equations

References

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- M.A. Katsoulakis and P.E. Souganidis. Interacting particle systems and generalized evolution of fronts. *Arch. Rat. Mech. Anal.*, in press.
- M.A. Katsoulakis and P.E. Souganidis. Generalized motion by mean curvature as a macroscopic limit for stochastic Ising models with long range interactions and Glauber dynamics. *Comm. Math. Physics*, to appear.
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**FONDAZIONE C.I.M.E.
CENTRO INTERNAZIONALE MATEMATICO ESTIVO
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“Vector Bundles on curves. New Directions”

is the subject of the Third 1995 C.I.M.E. Session.

The Session, sponsored by the Consiglio Nazionale delle Ricerche (C.N.R.), and the Ministero dell'Università e della Ricerca Scientifica e Tecnologica (M.U.R.S.T.) will take place, under the scientific direction of Professor M. S. NARASIMHAN at Grand Hotel San Michele, Cetraro (Cosenza), from 19 to 27 June, 1995.

C o u r s e s

- a) **Kac-Moody Groups, Their Flag Varieties and Moduli Spaces of G-bundles** (8 lectures in English)
Prof. Shrawan KUMAR (University of North-Carolina)

Outline

The basic theory of Kac-Moody groups and their flag varieties will be developed, including their constructions, elementary geometric properties, central extension, homogeneous line bundles, Picard group of the flag varieties and the analog of Borel-Weil-Bott theorem for Kac-Moody groups.

Further, the affine Kac-Moody flag variety will be realized as a parameter space for algebraic G-bundles on a smooth complex projective curve C of any given genus (where G is finite dimensional complex simple simply connected algebraic group). It will be shown that this family plays an important role in connecting the moduli space M of semistable G-bundles on C and the affine flag variety. This connection will be exploited to calculate the Picard group of the moduli space M .

- b) **Drinfeld Stukas**, (8 lectures in English)
Prof. G. LAUMON (Université de Paris-Sud, Orsay)

Outline

In the seventies, Drinfeld has introduced a whole family of new mathematical objects: modules, elliptic sheaves and stukas over a function field F of characteristic p .

These objects have some rank $r \in \mathbb{Z}_{>0}$. In the well-known analogy between function fields and number fields, rank 2 elliptic modules correspond to elliptic curves. The rank r elliptic sheaves and the rank r elliptic modules are equivalent objects. The rank r elliptic sheaves are rank r stukas of a special kind.

Using the étale cohomology of the modular variety of rank 2 stukas, Drinfeld has been able to prove the global Langlands correspondence for the group GL_2 over F in full generality. He also has suggested that the stukas of arbitrary rank r are precisely the objects which are needed to prove the global Langlands correspondence for the group GL_r over F in full generality.

More recently, Stuhler has introduced a D -version of those objects, where D is a global order of a central division algebra D over F . Rapoport, Stuhler and I have used the rank 1 D -elliptic sheaves to prove the local Langlands correspondence in positive characteristic and Lafforgue has studied the D -stukas of arbitrary rank and their modular varieties.

The purpose of these lectures is to explain what the D -stukas are and what they are good for and to give a survey of the recent progresses made on this subject.

Organization of the lectures.

- * **Lecture 1:** Definition of the D -stukas, level structures.
- * **Lecture 2:** Modular stacks, partial Frobenius morphisms, Hecke operators.
- * **Lecture 3:** Adelic description of the D -stukas with “finite” pole and zero.

- * **Lecture 4:** The rank one case, projectivity of the modular varieties and application to Langlands correspondence for D^* .
- * **Lecture 5:** Reducible stukas and horocycles.
- * **Lecture 6:** Harder-Narasimhan filtration of a D -stuka, truncations of the modular stacks.

The main references for these lectures are:

- V.G. Drinfeld, Varieties of modules of F -sheaves, *Funct. Anal. and its Appl.* 21 (1987), 107-122.
- V.G. Drinfeld, Proof of Petersson's conjecture for $GL(2)$ over a global field of characteristic p , *Funct. Anal. and its Appl.* 22 (1988), 28-43.
- L. Lafforgue, D -stukas de Drinfeld, Thèse, Université Paris-Sud, 1994.

- c) **Drinfeld modules and elliptic sheaves.** (8 lectures in English)
Prof. U. STUHLER (Universität Göttingen)

Outline

The above mentioned modules were introduced around 1973 by V. Drinfeld in his fundamental paper [2]. They are an analogue of the classical concept of an elliptic curve in the situation of a function field over a finite field of constants and bear a strong resemblance with this theory. They are an indispensable tool (together with their generalisations, the stukas) to study the Langlands correspondence but have also an interesting life in their own right. Besides this analogy with elliptic curves they have a description in terms of locally free sheaves and Frobenius operations which brings them into contact with the theory of vector bundles over curves. Surprising enough additionally there are close relations with the theory of the Korteweg-de Vries and similar differential as well as difference equations. The purpose of these lectures is to introduce the participants to this beautiful theory with its many facets. Additionally I hope to speak about some variants of this concept, the so called D -elliptic sheaves where D is a sheaf of division algebras over the curve involved.

These are useful in terms of the Langlands program for function fields but otherwise the theory is not as complete so that there are interesting open questions.

Other topics will probably concern questions of uniformisation of the moduli problems (at infinity as well as other primes), compactification questions and results concerning the cohomology and G. Anderson's t -motives.

References for these lectures are:

- P. Deligne, P. Husemoller: Survey of Drinfeld modules. In: Ribet, K. A. (ed.) *Current trends in arithmetical algebraic geometry*, Contemp. Math. vol. 67, pp. 25-91, Providence, R.I., Ann. Math. 1987.
- V.G. Drinfeld, Elliptic modules, *Math. USSR, Sb.* 23, 561-592 (1974).
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- V.G. Drinfeld, Commutative subrings of certain noncommutation rings, *Funct. Anal. Appl.* 11, 9-12, 1977.
- G. Laumon, *Forthcoming book on Drinfeld modules and the cohomology of their moduli spaces*, Cambridge University Press.
- G. Laumon, M. Rapoport, U. Stuhler: D -elliptic sheaves and the Langlands correspondence, *Invent. math.* 113, 217-330 (1993).

Organisation of the lectures:

- * **Lecture 1,2:** Definition of Drinfeld modules, algebraic and analytic theory, level structures, rings of endomorphisms, uniformisation.
- * **Lecture 3:** Description of Drinfeld modules in terms of locally free sheaves, t -motives, the noncommutative projective line, Analogies for rings of differential operators.
- * **Lecture 4:** Coverings of the upper half plane and their cohomology. Rigid analytic geometry.
- * **Lectures 5,6:** D -elliptic sheaves, moduli problems, good and bad fibres, applications to the Langlands program.

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